
MARROW ED8

Ent

Comprehensive Question Bank

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Embryology of Ear and Malformations

Question 1:

Which embryological structure gives rise to the pinna?

- a) 1st branchial arch
- b) 2nd branchial arch
- c) 1st and 2nd branchial arch
- d) 2nd and 3rd branchial arch

Question 2:

A patient is incidentally noted to have the following finding. It is also present on the other ear. Which of the following is the cause for such abnormality?



- a) Persistent opening of first branchial cleft
- b) Improper fusion of auricular tubercle
- c) Infection of haematoma of pinna
- d) Third pharyngeal pouch anomaly

Question 3:

When are reconstructive surgeries for congenital malformations of pinna usually performed?

- a) Soon after birth
- b) Within 1 year of age
- c) 5 or 6 years of age
- d) 3 or 4 years of age

Question 4:

Invagination of which of the following structures will form the external auditory meatus?

- a) First pharyngeal arch
- b) First pharyngeal pouch
- c) Second branchial cleft
- d) Second branchial groove

Question 5:

Examination of a child with otorrhea revealed two discharging sinuses. One was present on the floor of the left external auditory canal and the other on the left neck just below the angle of the mandible. Which of the following structures is abnormal here?

- a) First branchial cleft
- b) First branchial pouch
- c) Second branchial cleft
- d) Second branchial pouch

Question 6:

Which of the following structure develops from all three germ layers?

- a) Tympanic membrane
- b) Pinna
- c) Ear ossicles
- d) Otic capsule

Question 7:

Which of the structures given below forms the eustachian tube?

- a) 1st pharyngeal pouch
- b) 1st and 2nd pharyngeal cleft
- c) 2nd pharyngeal pouch
- d) 2nd and 3rd pharyngeal cleft

Question 8:

Which of the following pharyngeal arches gives rise to malleus and incus?

- a) First arch
- b) Second arch
- c) Third arch
- d) Fourth arch

Question 9:

Which of the following arches gives rise to the stapes?

- a) First arch
- b) Second arch
- c) Third arch
- d) Fourth arch

Question 10:

Which of the following attains adult size before birth?

- a) Ear ossicles
- b) Mastoid
- c) Orbit
- d) Cornea

Question 11:

Which of the following does not reach adult size at birth?

- a) Ear ossicles
- b) Mastoid
- c) Tympanic membrane
- d) Tympanic cavity

Question 12:

An ENT surgeon has difficulty locating the mastoid antrum due to the presence of Korner's septum. Which of the following sutures is persistent in this case?

- a) Frontozygomatic suture
- b) Petrosquamous suture
- c) Temporosquamous suture
- d) Petromastoid suture

Question 13:

From which of the following structure does the saccule develop?

- a) Sacculus anterior
- b) Sacculus posterior
- c) Pars superior
- d) Pars inferior

Question 14:

From how many ossification centre(s) does the inner ear develop?

- a) 1
- b) 2
- c) 7
- d) 14

Question 15:

When does the fetus start hearing?

- a) 10 weeks
- b) 20 weeks
- c) 30 weeks
- d) 32 weeks

Answer Key

Question No.	Correct Option
1	c
2	b
3	c
4	a
5	a
6	a
7	a
8	a
9	b
10	a
11	b
12	b
13	d
14	d
15	b

Detailed Explanations

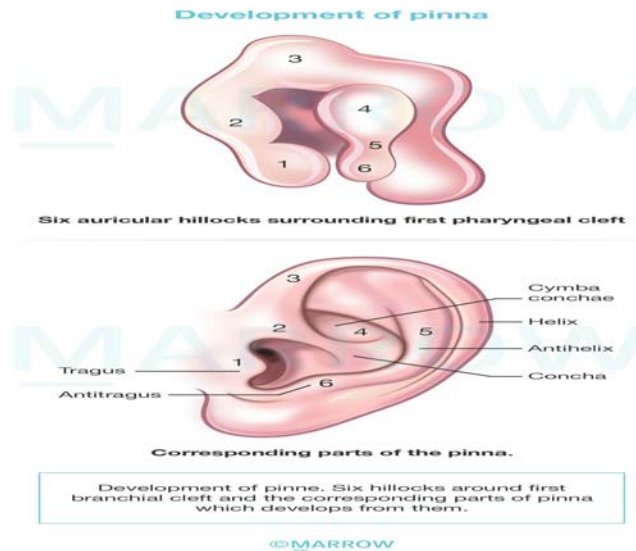
Solution to Question 1:

The 1st and 2nd branchial arches give rise to the pinna (auricle) through six mesenchymal proliferation. These proliferations are called auricular hillocks and they fuse to form the auricle.

The tissue from the second arch forms nearly all of the pinna.

The tissue from the first arch forms the tragus and external auditory meatus.

Image showing the development of pinna:



Solution to Question 2:

The given image showing the presence of sinus seen between the tragus and crus of the helix is suggestive of the preauricular sinus. It is due to improper fusion of the auricular tubercle formed from the first and second arches.

The preauricular sinus is apparent at birth and is usually bilateral. There may be occasional sebaceous discharge from the punctum. It does not regress spontaneously.

Solution to Question 3:

Reconstructive surgeries for congenital malformations of the pinna (auricle) are usually performed around 5 or 6 years of age. Autologous cartilages from ribs are harvested for these purposes and the ideal timing is between 6-10 years.

At around 3 years of age, 85% of the auricular growth is complete and the cartilaginous growth is almost complete by 5 years. Surgery performed on the auricle after 5 years doesn't hinder further growth.

Solution to Question 4:

Invagination of the first pharyngeal arch gives rise to the external auditory meatus.

Previously, it was believed that the external auditory meatus develops from the first pharyngeal cleft. But recent studies suggest that the first pharyngeal arch gives rise to the external auditory meatus.

Solution to Question 5:

Given clinical scenario is suggestive of a collaural fistula, and the first branchial cleft is abnormal here.

The fistula has two openings:

- One situated in the neck just below and behind the angle of mandible and
- The other one in the external canal or the middle ear.

The track of the fistula traverses through the parotid in close relation to the facial nerve.

Solution to Question 6:

Tympanic membrane develops from all three germinal layers:

- Outer epithelial layer is formed by the ectoderm
- Inner mucosal layer by the endoderm
- Middle fibrous layer by the mesoderm

Solution to Question 7:

The eustachian tube develops from the first pharyngeal pouch.

Initially, the first pharyngeal pouch forms the tubotympanic recess. The distal part of this recess widens to form the middle ear cavity. The proximal part forms the eustachian tube.

Solution to Question 8:

The first pharyngeal arch gives rise to malleus and incus.

Stapes superstructure is formed from the second arch whereas the stapes footplate develops from the otic capsule.

Solution to Question 9:

The stapes develop from the second arch, except for its footplate and annular ligament, which develop from the otic capsule.

Solution to Question 10:

Ear ossicles attain adult size before birth.

The development of the middle ear:

- The ossicles are full-sized cartilage models by 15 weeks of gestation.
- Endochondral ossification is complete by 25 weeks.
- The middle ear is adult-sized at birth.

Solution to Question 11:

The mastoid does not reach adult size at birth. It develops fully by 2 years of age.

In infants and children up to 2 years of age, the mastoid process is not developed completely. So, the facial nerve lies just beneath the skin near its exit. Hence, during surgical procedures, the incision is given slanting posteriorly, avoiding the lower part of the mastoid. This spares the facial nerve.

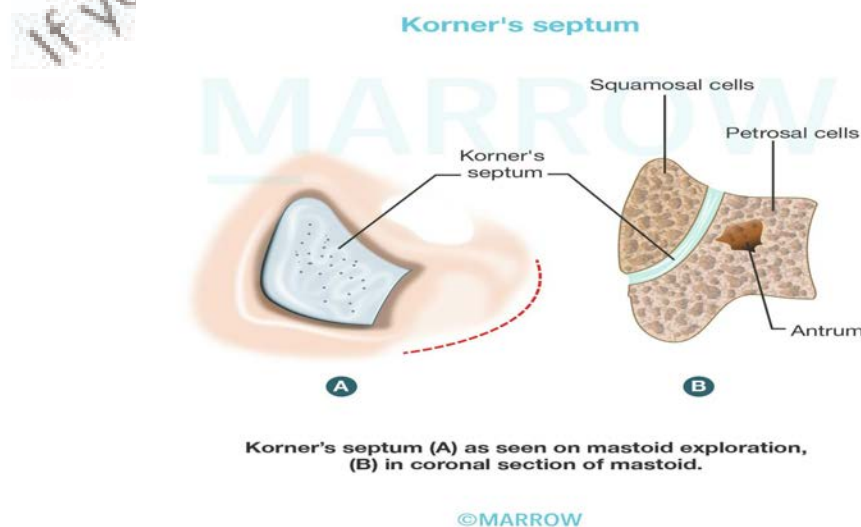
Solution to Question 12:

The persistence of the petrosquamous suture is called Korner's septum and it is responsible for the difficulty in locating the mastoid antrum and the deeper cells.

Korner's septum separates superficial squamosal cells from the deep petrosal cells.

Incomplete removal of disease at mastoidectomy can occur if Korner's septum is not identified. Mastoid antrum cannot be reached unless the Korner's septum has been removed.

Below is an image showing Korner's septum -



Solution to Question 13:

The saccule develops from pars inferior.

The development of the inner ear begins in the 3rd week of fetal life and is completed by the 16th week. The surface ectoderm on each side of the rhombencephalon thickens to form otic/auditory placodes, which then invaginate to form the otic vesicle (otocysts).

The cells of the otic vesicle differentiate to form ganglion cells, which later develop into the vestibulocochlear ganglia. Later each otic vesicle divides into a dorsal component called pars superior and a ventral component called pars inferior.

Pars superior gives rise to the utricle, semicircular canals, and endolymphatic ducts. Pars inferior gives rise to saccule and cochlear duct. The saccule is connected to the cochlea through a narrow duct called ductus reuniens.

Solution to Question 14:

The inner ear develops from 14 ossification centres.

Solution to Question 15:

The fetus starts hearing in the womb of the mother by 20 weeks of gestation due to sufficient development of the cochlea.

Anatomy of External Ear

Question 1:

What is the pinna made up of?

- a) Tympanic part of mastoid
- b) Auricular cartilage
- c) Skin & fat alone
- d) Fat alone.

Question 2:

A 25-year-old woman undergoes a rhinoplasty procedure to correct a deformity of her nose. Auricular cartilage is harvested from her right pinna. Which of the following sites cannot be used to harvest the cartilage?

- a) Cymba concha
- b) Incisura terminalis
- c) Antitragus
- d) Scaphoid fossa

Question 3:

Which of the following statements is true regarding the skin over the pinna?

- a) Firmly adherent on medial side
- b) Loosely adherent on medial side
- c) Firmly adherent on both sides
- d) Loosely adherent on both sides

Question 4:

What is the length of external auditory canal?

- a) 12mm
- b) 24mm

- c) 36mm
- d) 48mm

Question 5:

You are an intern asked to do an otoscopic examination of a young adult. In which direction will you pull the pinna to visualize the tympanic membrane?

- a) Upwards, backwards, and medially
- b) Upwards, backwards, and laterally
- c) Downwards, backwards, and laterally
- d) Downwards, backwards, and medially

Question 6:

Which of the following statements is true about the external auditory canal?

- a) Outer 1/3rd is cartilaginous and inner 2/3rd is bony
- b) Outer 2/3rd is cartilaginous and inner 1/3rd is bony
- c) Outer 1/3rd is bony and inner 2/3rd is cartilaginous
- d) Outer 2/3rd is bony and inner 1/3rd is cartilaginous

Question 7:

A patient with otitis externa develops parotitis. You suspect that the infection has spread through the fissures of Santorini. Which of the following best describes this fissure?

- a) Deficiency in cartilaginous canal
- b) Connection between middle ear and pharynx
- c) Connection between Inner ear and middle ear
- d) Deficiency in bony canal

Question 8:

Which of the following statements is true regarding the external auditory canal?

- a) Outer 2/3rd is cartilaginous and inner 1/3rd is bony part

- b) Furuncles are seen in the cartilaginous part
- c) Fissures of Santorini are present in bony part and permits parotid and superficial mastoid infections to spread to canal
- d) Isthmus is present at junction of cartilaginous and fibrous part

Question 9:

What are ceruminous glands?

- a) Modified Eccrine glands
- b) Modified apocrine glands
- c) Modified Holocrine glands
- d) Modified endocrine glands

Question 10:

Which of the following statements is true regarding isthmus?

- a) It is just lateral to the petrotympanic suture.
- b) It is at the bony cartilaginous junction of EAC.
- c) It is the narrowest part of ear canal.
- d) All of the above

Question 11:

You observe a normal-looking tympanic membrane on otoscopy. What is its colour?

- a) Yellow
- b) Pearly grey
- c) Red
- d) Pink

Question 12:

Which of the following statements is false regarding the tympanic membrane?

- a) It has 3 layers

- b) It is derived from all 3 germ layers
- c) It separates external from middle ear
- d) Central part is more mobile

Question 13:

Which of the following is false about the tympanic membrane?

- a) 0.2 mm thick
- b) 90 mm² in area
- c) Angle of 55 degree with horizontal
- d) Inner surface is convex

Question 14:

What is the other name for Shrapnell's membrane?

- a) Pars tensa
- b) Reissner's membrane
- c) Pars flaccida
- d) Secondary tympanic membrane

Question 15:

During an otoscopic examination of a patient, where will the cone of light be seen on the tympanic membrane?

- a) Antero-superior
- b) Postero-superior
- c) Antero-inferior
- d) Postero-inferior

Question 16:

Which of the following nerves supply a major part of pinna?

- a) Auriculotemporal nerve

- b) Greater auricular nerve
- c) Lesser occipital nerve
- d) Arnold's nerve

Question 17:

A patient develops tingling and numbness along the upper part of the medial surface of the pinna. Which nerve is most likely to be affected?

- a) Great auricular nerve
- b) Lesser occipital nerve
- c) Greater occipital nerve
- d) Auriculotemporal nerve

Question 18:

A patient coughs when removing excess cerumen from the ear. This is due to the stimulation of which of the following nerves?

- a) Facial nerve
- b) Vagus nerve
- c) Trigeminal nerve
- d) Greater auricular nerve

Question 19:

What is Arnold's nerve?

- a) Auricular branch of vagus
- b) Sensory division of facial nerve
- c) Tympanic branch of glossopharyngeal nerve
- d) Greater auricular nerve

Question 20:

Which of the following cranial nerves do not supply the tympanic membrane?

- a) CN V
- b) CN X
- c) CN XI
- d) CN IX

Question 21:

A 25-year-old patient presents with bleeding from his ear following a road traffic accident. Which of the following could be its reason?

- a) Cribriform plate fracture
- b) Temporal bone fracture
- c) Zygomatic fracture
- d) Le Fort's type I fracture

Question 22:

Cough on scratching the external acoustic canal is due to?

- a) Auriculotemporal nerve
- b) Auricular branch of vagus
- c) Great auricular nerve
- d) Facial nerve

Answer Key

Question No.	Correct Option
1	b
2	b
3	b
4	b
5	b
6	a
7	a
8	b

9	b
10	c
11	b
12	d
13	a
14	c
15	c
16	b
17	b
18	b
19	a
20	c
21	b
22	b

Detailed Explanations

Solution to Question 1:

The pinna and the external auditory canal are made up of yellow elastic cartilage. It is also called the auricular cartilage.

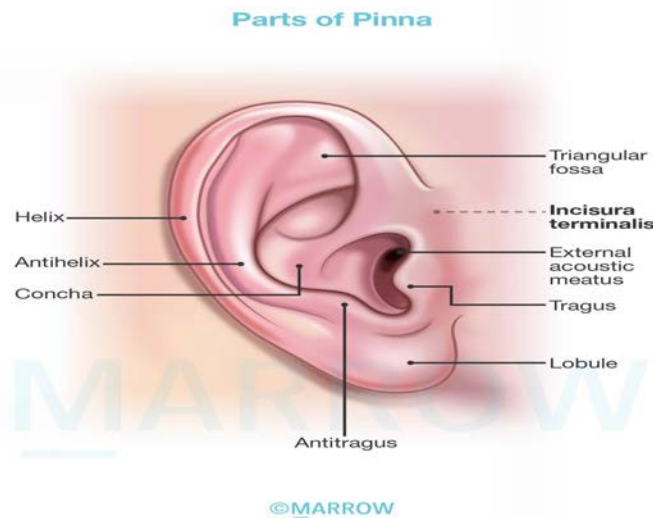
Solution to Question 2:

Incisura terminalis cannot be used to harvest cartilage as it is devoid of cartilage.

It is the part of the pinna between the tragus and the crus of helix. Being devoid of cartilage, the incision for endaural approach for ear surgery can be made here.

The lobule is also devoid of cartilage. Hence, it is the safest site for piercing.

The parts of the pinna are:



Solution to Question 3:

The skin over the pinna is loosely adherent on the medial or cranial side. Hence, it is pinchable.

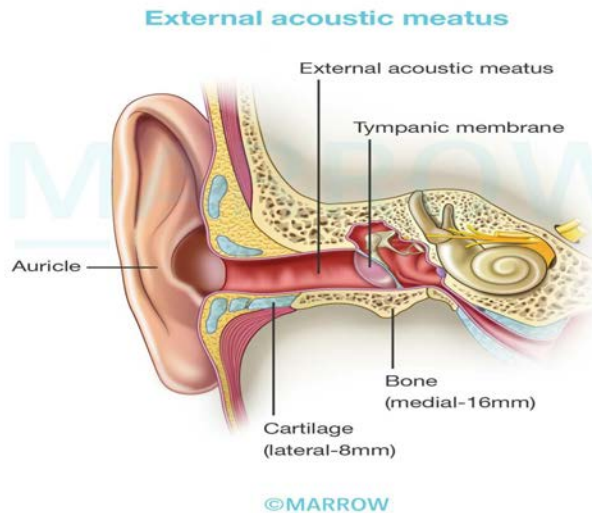
The skin is firmly adherent to the perichondrium on the lateral side and is non-pinchable.

Solution to Question 4:

The length of the external auditory canal is about 24 mm.

It extends from the bottom of the concha to the tympanic membrane. It has two parts:

- Outer cartilaginous part - 8mm
- Inner bony part - 16mm



Solution to Question 5:

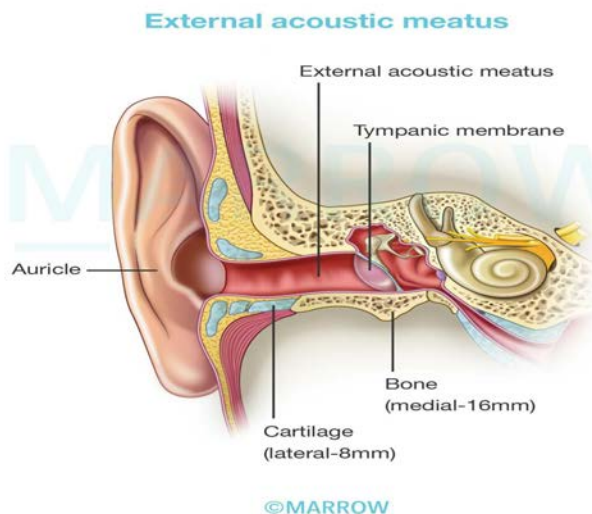
The pinna has to be pulled upwards, backwards and laterally to visualise the tympanic membrane. This is done to bring the bony and cartilaginous parts of the external auditory canal in alignment.

The inner bony part is directed downwards, forwards, and medially.

The outer cartilaginous part is directed upwards, backwards, and medially.

Solution to Question 6:

The outer 1/3rd is cartilaginous (8mm) and the inner 2/3rd is bony (16mm) in the external auditory canal.



Solution to Question 7:

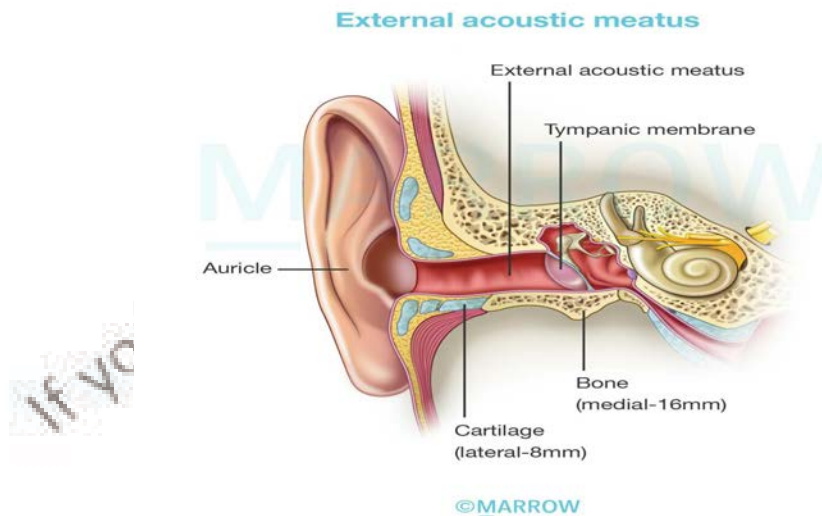
Fissures of Santorini are the two deficiencies in the cartilaginous part of the external auditory canal. They provide a pathway for the spread of parotid or superficial mastoid infections to the canal or vice versa.

The bony canal also has a deficiency called the foramen of Huschke which acts as a pathway for the spread of infections and tumours to the deep lobe of the parotid.

Solution to Question 8:

Furuncles (staphylococcal infection of the hair follicles) are seen only in the outer one-third of the canal since hair is confined to the outer canal.

Skin is also very tightly adherent to the cartilage and hence furuncles are extremely painful.



Solution to Question 9:

Ceruminous glands are modified apocrine sweat glands. They open into hair follicles & participate in wax production.

Apocrine glands release their products by decapitation. It is a process by which membrane-bound cytoplasm from the apical surface of the cells buds off into the lumen of the duct and is secreted.

Some apocrine glands have specific names. Those on the eyelids are referred to as Moll's glands, and those on the external auditory meatus are termed ceruminous glands.

Solution to Question 10:

The isthmus is the narrowest part of the external auditory canal (EAC).

EAC measures about 24 mm long, cartilaginous part forms the outer one-third (8 mm) and bony part forms two-third (16 mm) of the canal. Isthmus is situated 6 mm lateral to the tympanic membrane in the bony part of the EAC. Foreign bodies that go beyond the isthmus are difficult to remove and are called impacted.

Solution to Question 11:

The normal tympanic membrane is pearly grey in color. It is thin and semitranslucent.

**Solution to Question 12:**

Peripheral parts are more mobile than the central part where the handle of the malleus is attached

The tympanic membrane develops from all three germinal layers. It has 3 layers –

- Outer epithelial layer from ectoderm
- Middle fibrous layer from mesoderm
- Inner mucosal layer endoderm.

Solution to Question 13:

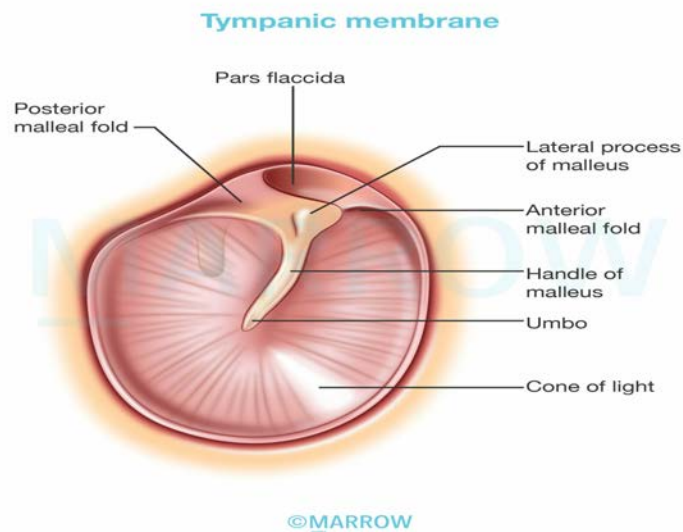
The tympanic membrane has a thickness of 0.1mm.

The dimensions of the tympanic membrane are

- Length: 9-10mm
- Width: 8-9mm
- Area: 90 mm²
- Subtends an angle of 55° with horizontal

The medial (inner) surface is convex and the lateral surface is concave. The most depressed part of its concavity is called the umbo, to which the tip of the malleus is attached.

Image: Right tympanic membrane



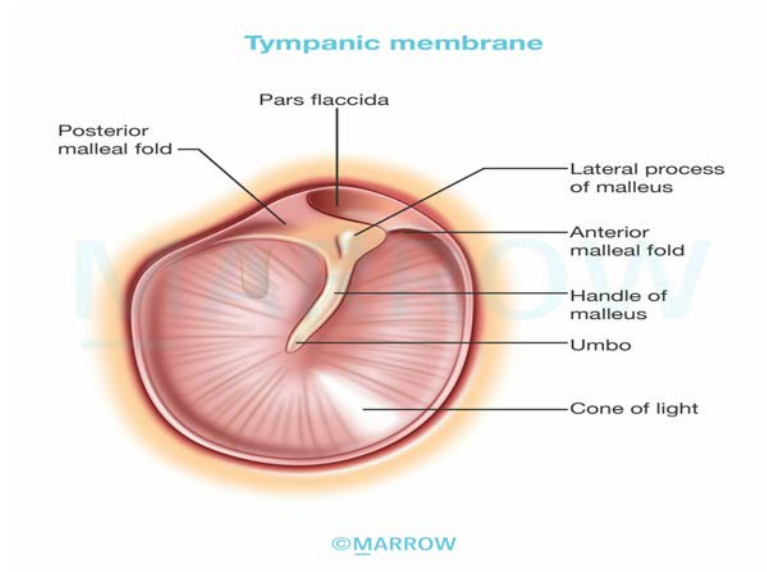
Solution to Question 14:

Pars flaccida is also known as Shrapnell's membrane.

Pars flaccida is situated above the lateral process of malleus between the notch of Rivinus and the anterior and posterior malleal folds.

Solution to Question 15:

A bright cone of light can be seen radiating from the tip of malleus (umbo) to the periphery in the anteroinferior quadrant of the tympanic membrane.

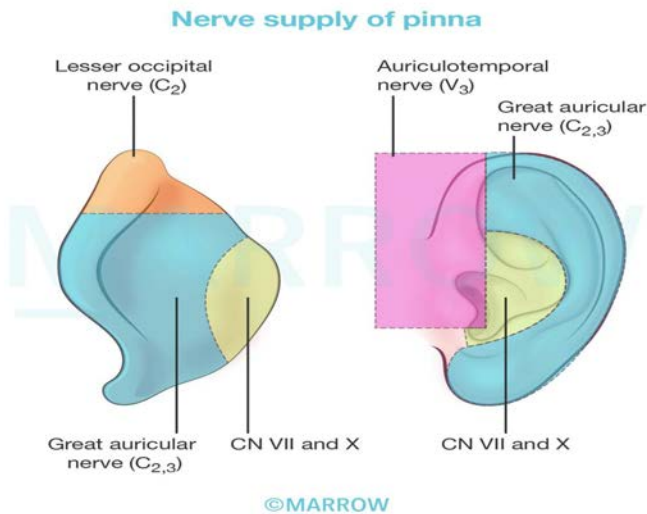


Solution to Question 16:

Greater auricular nerve (C2,C3) supplies most of the medial surface and posterior portion of lateral surface of the external ear.

Nerve	Derivation	Region supplied
Greater/Great auricular nerve	Cervical plexus C2,C3	Medial surface and posterior part of the lateral surface
Lesser occipital nerve	Cervical plexus C2	Superior portion of medial surface

Nerve	Derivation	Region supplied
Auricular branch	Vagus (X)	Concha and antihelixSome supply medial surface (eminetia concha)
Auriculotemporal nerve	Mandibular (V3)	Tragus, crus of helix and adjacent helix
Facial (VII)		Small region in the root of concha and retroauricular groove

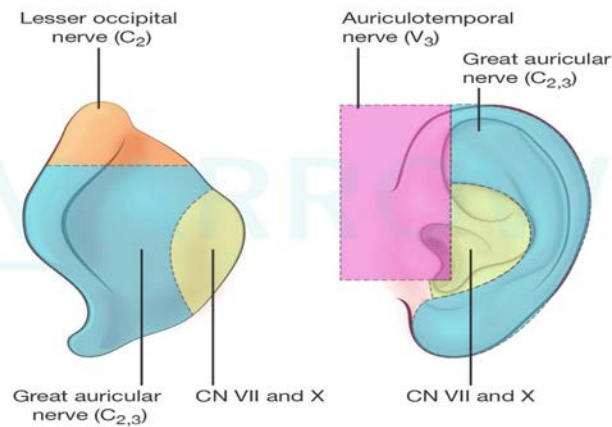


Solution to Question 17:

The upper part of the medial surface of the pinna is supplied by the lesser occipital nerve.

Nerve	Derivation	Region supplied
Greater/Great auricular nerve	Cervical plexus C2,C3	Medial surface and posterior part of the lateral surface
Lesser occipital nerve	Cervical plexus C2	Superior portion of medial surface
Auricular branch	Vagus (X)	Concha and antihelixSome supply medial surface (eminetia concha)
Auriculotemporal nerve	Mandibular (V3)	Tragus, crus of helix and adjacent helix
Facial (VII)		Small region in the root of concha and retroauricular groove

Nerve supply of pinna



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Solution to Question 18:

Reflex cough is due to stimulation of the auricular branch of vagus. It can occur when cleaning or during an examination of the ear.

Nerve	Derivation	Region supplied
Greater/Great auricular nerve	Cervical plexus C2,C3	Medial surface and posterior part of the lateral surface
Lesser occipital nerve	Cervical plexus C2	Superior portion of medial surface
Auricular branch	Vagus (X)	Concha and antihelixSome supply medial surface (eminetia concha)
Auriculotemporal nerve	Mandibular (V3)	Tragus, crus of helix and adjacent helix
Facial (VII)		Small region in the root of concha and retroauricular groove

Solution to Question 19:

Auricular branch of vagus (CN X) is also called Arnold's nerve or Alderman's nerve.

Tympanic branch of CN IX is also called Jacobson's nerve.

Sensory root of the facial nerve is also called the nerve of Wrisberg and carries secretomotor fibres to the lacrimal gland and salivary glands, and brings fibres of taste and general sensation.

Solution to Question 20:

The spinal accessory nerve (CN XI) does not supply the tympanic membrane. It is supplied by:

- Lateral surface -
- Anterior half by auriculotemporal nerve (CN V₃)
- Posterior half by auricular branch of vagus nerve (CN X).
- Medial surface -
- Tympanic branch of the glossopharyngeal nerve (Jacobson's nerve) CN IX.

Solution to Question 21:

A temporal bone fracture can cause bleeding from the ear as it is related to the roof of the middle ear and the external auditory canal.

The relations of the external auditory canal are:

- Superior - Middle cranial fossa
- Inferior - Parotid gland
- Anterior - Temporomandibular joint
- Posterior - Mastoid air cells and facial nerve.

Solution to Question 22:

Cough on scratching the external acoustic canal is due to the auricular branch of the vagus nerve (Arnold's nerve).

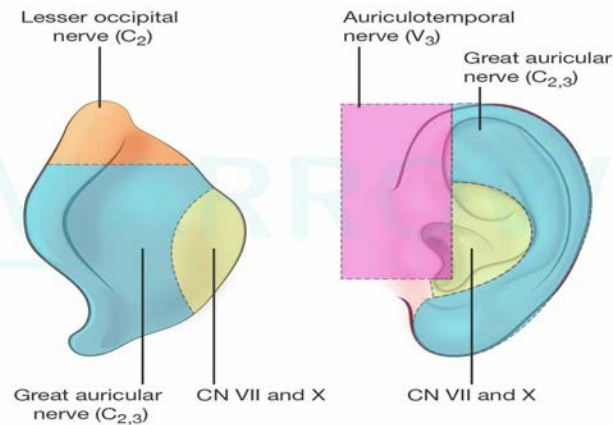
Arnold's nerve supplies the concha and corresponding eminence on the medial surface. It also supplies the posterior wall and floor of the external auditory canal. Scratching or mechanical stimulation of the ear leads to activation of Arnold's nerve, which evokes reflex cough (ear-cough reflex/otorespiratory reflex).

The image below shows the nerve supply of the pinna:

Causes of referred otalgia	Nerve responsible	Site
Temporomandibular joint and dental conditions, parotid infections/tumor, and anterior 2/3rd of tongue	Auriculotemporal nerve	Tragal area, crus of the helix, and anterior wall and roof of the external auditory canal.

Causes of referred otalgia	Nerve responsible	Site
Acute tonsillitis, peritonsillar abscess, carcinoma base of the tongue or tonsils (oropharynx)	The glossopharyngeal nerve (Jacobson's)	Medial surface of the tympanic membrane, and middle ear
Carcinoma of the larynx (cough), hypopharynx, epiglottis, vallecula, esophagus, and thyroid	Auricular branch of the vagus nerve (Arnold's/Alderman's nerve)	Concha, posterior wall, and floor of the external auditory canal
Cervical spondylitis	Great auricular nerve (C _{2,3})	Lateral and medial surfaces of the pinna
Acoustic neuroma	Facial nerve (nervus intermedius/nerve of Wrisberg)	Concha, retro auricular groove, and posterosuperior part of the external auditory meatus. Numbness in the posterosuperior meatal wall known as Hitzelberger's sign, is seen in acoustic neuroma.

Nerve supply of pinna



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Anatomy of Middle Ear

Question 1:

Otoendoscopy of a patient revealed cholesteatoma limited to the protympanum. Which of the following sites is being referred to here?

- a) The developmental predecessor of the middle ear
- b) The anterior part of middle ear
- c) The part of the middle ear foremost in conduction of sound
- d) The part of middle ear around the eustachian tube

Question 2:

Which is the narrowest part of the middle ear?

- a) Epitympanum
- b) Mesotympanum
- c) Hypotympanum
- d) Endotympanum

Question 3:

What is the distance between promontory and tympanic membrane?

- a) 6 mm
- b) 4 mm
- c) 2 mm
- d) 1 mm

Question 4:

In a patient with necrotizing otitis media, CT scans reveal erosions of the structures present in the epitympanum. Which of the following is least likely to be involved?

- a) Head of malleus
- b) Body of incus

- c) Incudomalleolar joint
- d) Footplate of stapes

Question 5:

While performing tympanoplasty, the ENT surgeon notices that the floor of the middle ear is deficient. Which of the following structures is susceptible to injury?

- a) Internal carotid artery
- b) Internal jugular vein
- c) Parotid gland
- d) Mastoid

Question 6:

Which part of middle ear is formed by tegmen tympani?

- a) Floor of middle ear
- b) Roof of middle ear
- c) Lateral wall of middle ear
- d) Medial wall of middle ear

Question 7:

CT scans of a patient with nasopharyngeal carcinoma reveal that the tumor has undergone a local spread, destroying the anterior wall of the middle ear. Which of the following structures is least likely to be involved?

- a) Eustachian tube
- b) Canal for tensor tympani
- c) Internal carotid artery
- d) Prominence of the facial canal

Question 8:

Which of the following structures is not a part of the lateral wall of the middle ear?

- a) Pars tensa

- b) Pars flaccida
- c) Scutum
- d) Subiculum

Question 9:

Which of the following structures is not related to the pyramid?

- a) Facial nerve
- b) Sinus tympani
- c) Stapedius muscle
- d) Round window

Question 10:

You are approaching the middle ear via the mastoid through the facial recess. Regarding anatomical relations of facial recess, which of the following is true?

- a) Bound medially by horizontal part of facial nerve
- b) Bound laterally by chorda tympani
- c) Bound superiorly by chordal ridge
- d) Bound inferiorly by fossa incudis

Question 11:

A child with cholesteatoma undergoes surgical removal via an intact canal wall procedure. Through which of the following structures is the middle ear approached?

- a) Supra pyramidal recess
- b) Infra pyramidal recess
- c) Promontory
- d) Oval window

Question 12:

Which of the following structures forms the promontory?

- a) Basal turn of cochlea
- b) Semicircular canal
- c) Handle of malleus
- d) Short process of incus

Question 13:

Which is the smallest bone in the body?

- a) Malleus
- b) Incus
- c) Scaphoid
- d) Stirrup

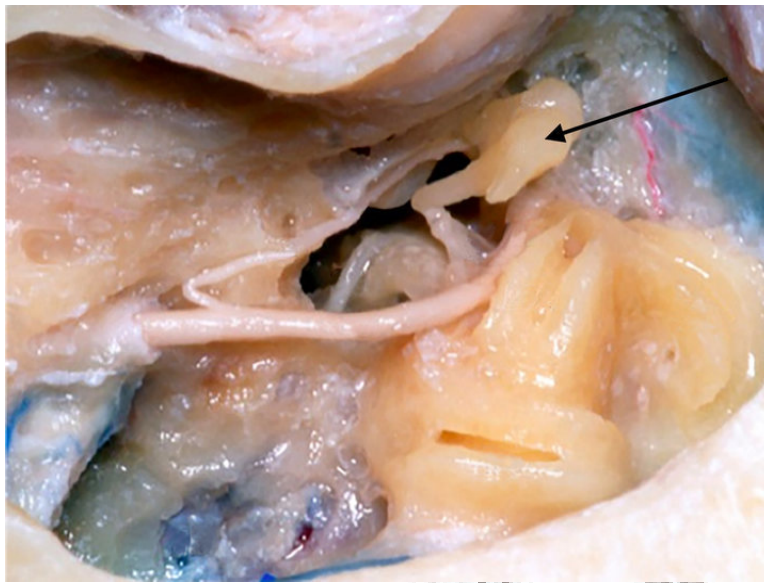
Question 14:

What is the type of joint between the middle ear ossicles?

- a) Primary cartilaginous joint
- b) Secondary cartilaginous joint
- c) Synovial joint
- d) Fibrous joint

Question 15:

What is the structure marked in the given image?



- a) Malleus
- b) Incus
- c) Posterior semicircular Canal
- d) Lateral semicircular Canal

Question 16:

Which of the following is attached to fossa incudis?

- a) Incus head
- b) Incus short process
- c) Incus long process
- d) Lenticular process

Question 17:

Which among the following structures covers the oval window?

- a) Footplate of stapes
- b) Secondary tympanic membrane
- c) Tympanic membrane
- d) Processus cochleariformis

Question 18:

Which of the following structures vibrates when the stapes footplate is moved?

- a) Scala vestibuli
- b) Scala media
- c) Scala tympani
- d) Pyramid

Question 19:

Which of the following covers the round window?

- a) Footplate of stapes
- b) Secondary tympanic membrane
- c) Tympanic membrane
- d) Reissner's membrane

Question 20:

Which of the following is related to processus cochleariformis?

- a) Tendon of tensor tympani
- b) Tendon of stapedius
- c) Handle of malleus
- d) Incus

Question 21:

Impedance audiometry performed in a patient showed absent stapedial reflex. Which of the following nerves is most likely to be injured?

- a) Facial nerve
- b) Anterior division of mandibular nerve
- c) Posterior division of mandibular nerve
- d) Maxillary nerve

Question 22:

Which of the following nerves supply the tensor tympani?

- a) Facial nerve
- b) Branch from the medial pterygoid nerve
- c) Posterior division of mandibular nerve
- d) Maxillary nerve

Question 23:

Which of the following cranial nerves provide sensory supply to middle ear cavity?

- a) Facial
- b) Vagus
- c) Glossopharyngeal
- d) Trigeminal

Question 24:

Which of the following nerves does not contribute to the tympanic plexus?

- a) Tympanic branch of Glossopharyngeal nerve
- b) Superior Caroticotympanic nerve
- c) Inferior Caroticotympanic nerve
- d) Tympanic branch of Facial nerve

Question 25:

Which of the following is true regarding the facial nerve course in the middle ear?

- a) Horizontal in the medial wall, vertical in the posterior wall
- b) Vertical in medial wall, horizontal in posterior wall
- c) Horizontal in the medial wall, vertical in the anterior wall
- d) Vertical in medial wall, horizontal in anterior wall

Question 26:

Which of the following is true regarding the course of chorda tympani in the middle ear?

- a) Travels anterior to posterior
- b) Enters middle ear lateral to tympanic membrane
- c) Runs in between the malleus and incus
- d) Enters middle ear via the canal of Huguier

Question 27:

Which of the following is false about the lesser petrosal nerve?

- a) It is formed over the promontory of the middle ear
- b) It is a branch of the tympanic plexus
- c) It supplies parotid glands.
- d) Its fibers synapse in the pterygopalatine ganglion

Answer Key

Question No.	Correct Option
1	d
2	b
3	c
4	d
5	b
6	b
7	d
8	d
9	d
10	b
11	a
12	a
13	d
14	c
15	b
16	b

17	a
18	a
19	b
20	a
21	a
22	b
23	c
24	d
25	a
26	c
27	d

Detailed Explanations

Solution to Question 1:

The part of the middle ear around the eustachian tube is referred to as the protympanum.

The middle ear is divided into 3 main parts:

- Epitympanum/attic- above the level of pars tensa, medial to pars flaccida, and scutum.
- Mesotympanum- opposite the pars tensa
- Hypotympanum- below the level of TM

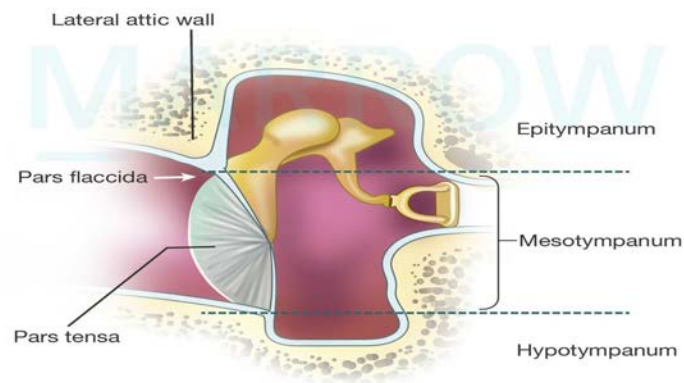
Solution to Question 2:

Mesotympanum is the narrowest part of the middle ear.

Transverse diameters of middle ear at various levels are:

- Epitympanum- 6 mm
- Mesotympanum- 2 mm
- Hypotympanum- 4 mm.

Divisions of the middle ear



Solution to Question 3:

The distance between the promontory and tympanic membrane, i.e., the transverse diameter at the level of mesotympanum, is 2 mm.

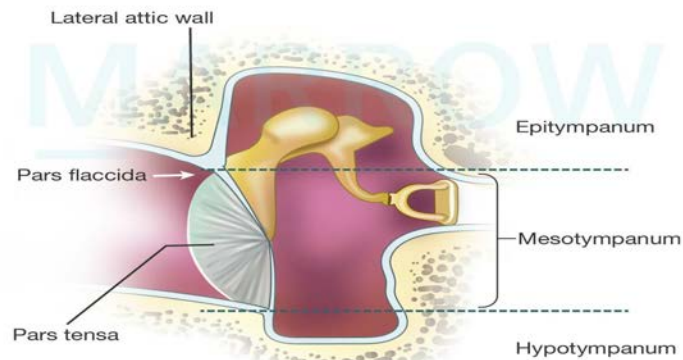
Solution to Question 4:

Footplate of stapes is least likely to be involved in the given scenario because it is not a part of epitympanum.

Epitympanum contains:

- Head of the malleus
- Incudomalleolar joint
- Short process and body of the incus

Divisions of the middle ear



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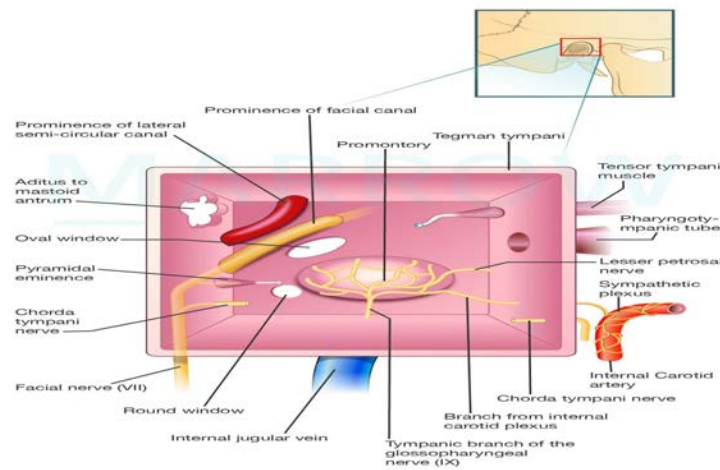
Solution to Question 5:

The internal jugular vein is susceptible to injury in this given clinical scenario.

The floor of the middle ear is related to the internal jugular vein. Occasionally, the floor is deficient and the jugular bulb is then covered only by fibrous tissue and a mucous membrane making it susceptible to injury during middle ear surgery.

The middle ear can be likened to a room with a roof, floor, and 4 walls (anterior, posterior, medial & lateral), as shown in the image:

Boundaries of the middle ear



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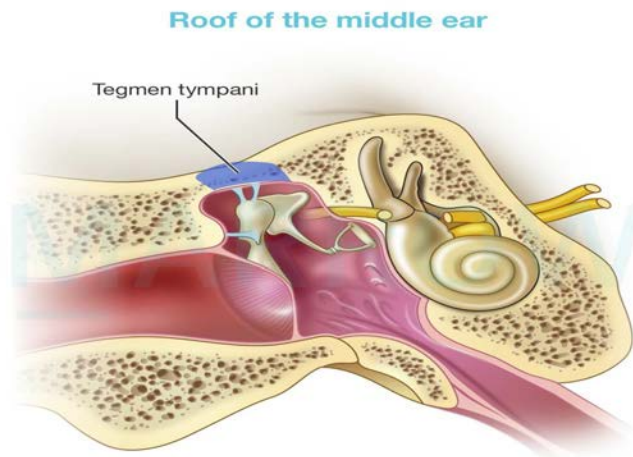
Solution to Question 6:

The tegmen tympani forms the roof of the middle ear.

It separates the tympanic cavity from the middle cranial fossa.

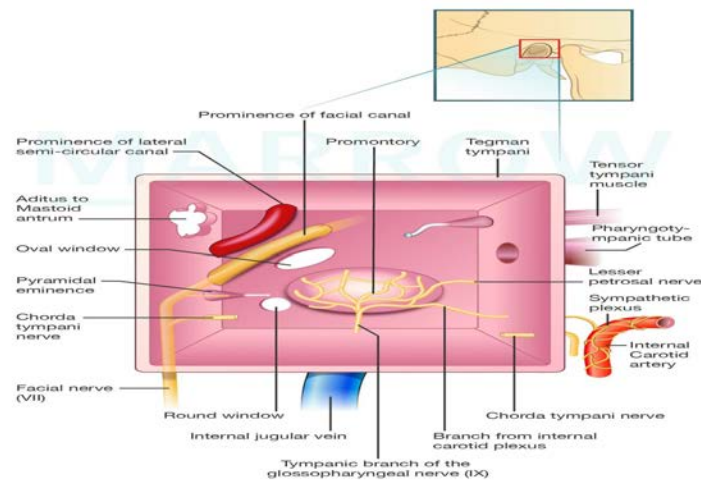
It also extends posteriorly to form the roof of the aditus and antrum as tegmen mastoidea.

The following images shows the tegmen tympani and relations of middle ear:



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Boundaries of the middle ear



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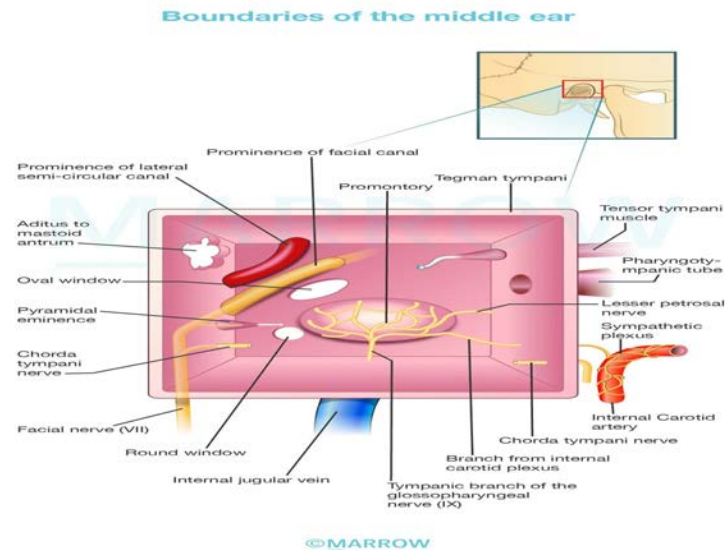
Solution to Question 7:

The prominence of the facial canal is least likely to be injured in this given clinical scenario because it is a part of the medial wall of the middle ear and not the anterior wall.

The anterior wall of the middle ear is related to the following structures:

- Eustachian tube

- The canal for tensor tympani
- Internal carotid artery



Solution to Question 8:

The subiculum is not a part of the lateral wall of the middle ear. It is present in the medial wall, below the sinus tympani.

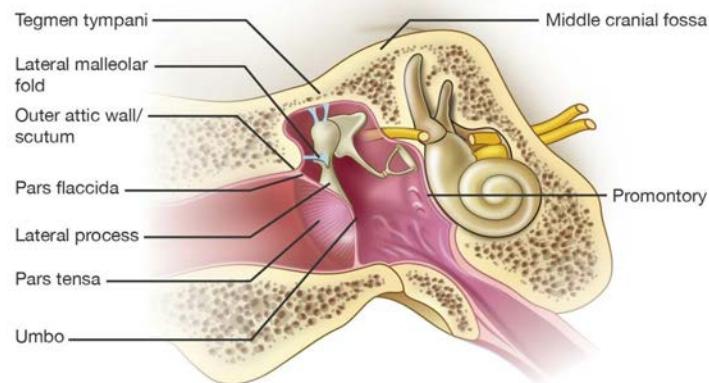
The lateral wall of the middle ear separates it from the external ear. It contains:

- Pars tensa lying opposite to the mesotympanum
- Pars flaccida and scutum lying opposite to the epitympanum.

Scutum or lateral attic wall is a wedge-shaped bone that separates the epitympanum from the external auditory canal. It can be eroded in attic cholesteatomas.

Image showing the boundaries of middle ear:

Boundaries of the middle ear

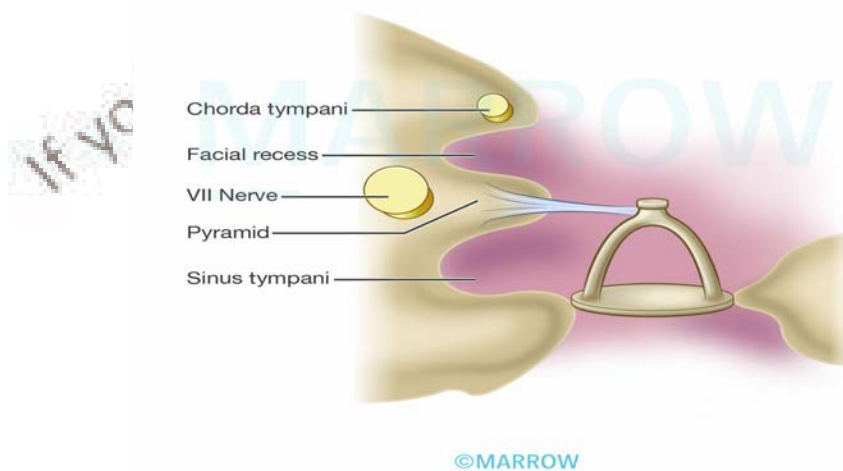


Solution to Question 9:

The round window is not related to the pyramid because it is a medial wall structure whereas the pyramid is a projection present in the posterior wall.

Image showing pyramidal eminence:

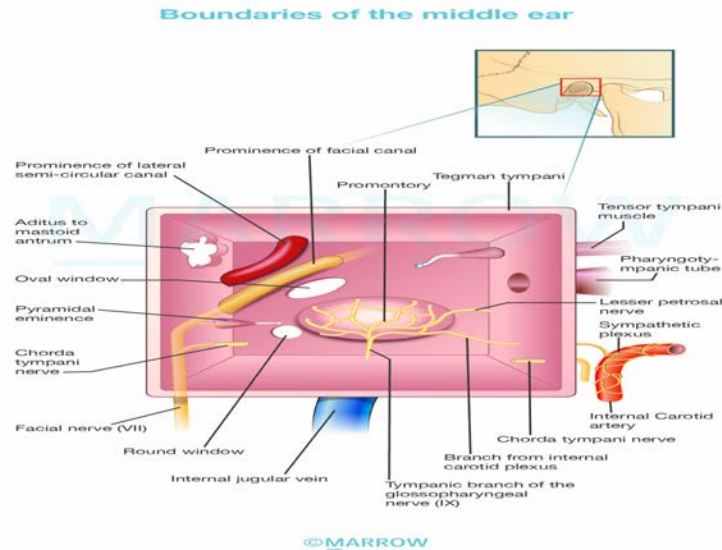
The facial recess and sinus tympani



The pyramid has the following features:

- It contains stapedius muscle and the tendon leaves the pyramid to attach the neck of the stapes
- It divides the posterior space into a medial sinus tympani and a lateral facial recess.
- It serves as the landmark for the second genu of the facial nerve and houses the vertical segment of the facial nerve & the nerve to stapedius.

- The aditus (opening to the mastoid) lies just above it.

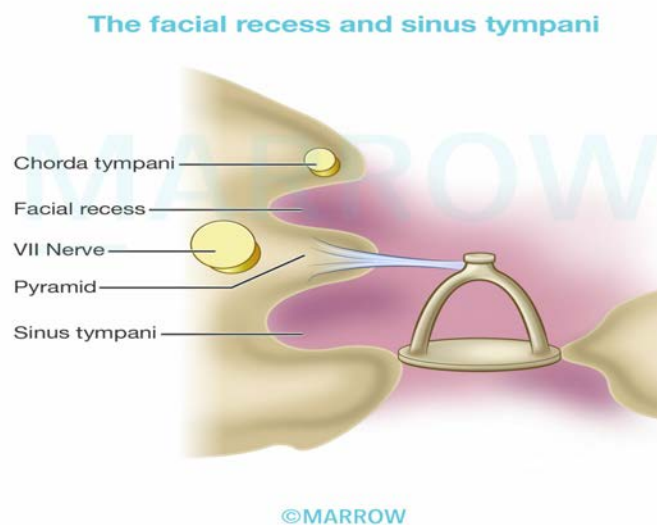


Solution to Question 10:

Facial recess or the posterior sinus is a depression in the posterior wall lateral to the pyramid. It is bounded medially by the vertical part of the facial nerve, laterally by the chorda tympani and above, by the fossa incudis.

Facial recess is important surgically, as direct access can be made through this into the middle ear without disturbing the posterior canal wall. This is called intact canal wall technique.

Image showing the facial recess:

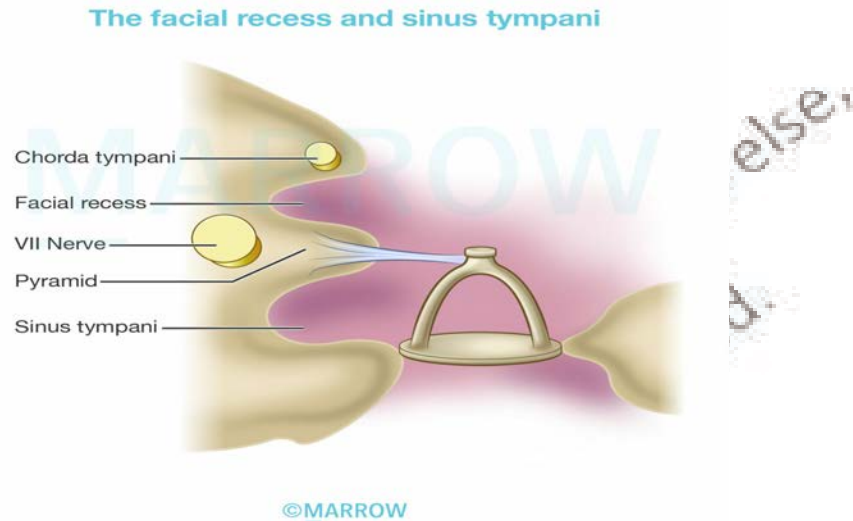


Solution to Question 11:

In intact canal wall surgeries, the middle ear is approached through the suprapyramidal recess.

Surgically, facial recess/posterior sinus/supra pyramidal recess is important, as direct access can be made through this into the middle ear without disturbing the posterior canal wall (intact canal wall technique).

Image showing the facial recess:



Solution to Question 12:

The basal turn of the cochlea forms the promontory.

The promontory is a bulge in the middle ear's medial wall, which is formed by the labyrinth.

It gives 2 extensions:

- Subiculum below to the round window.
- Ponticulus to the posterior wall.

Solution to Question 13:

The smallest bone in the body is the stapes. It is also called a stirrup as it looks like the stirrup/footrest for jockeys while riding a horse.

The malleus is called the hammer.

The incus is called the anvil (large metal striking surface for shaping metals).

Solution to Question 14:

The type of joint between the middle ear ossicles is the synovial joint.

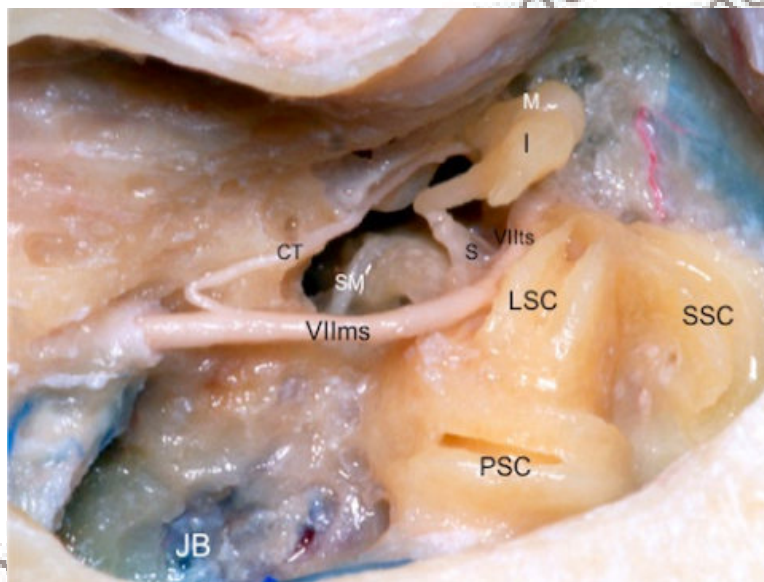
The malleo-incudal joint is a saddle type.

The incudostapedial joint is a ball and socket type.

Solution to Question 15:

The marked structure is incus.

This is an image taken during mastoidectomy. The incus articulates with the malleus and has a short process and a long process. The long process articulates with the head of the stapes.



CT: chorda tympani; I: incus; JB: jugular bulb; LSC: lateral semicircular canal; M: malleus; PSC: posterior semicircular canal; S: stapes; SM: stapedius muscle; SSC: superior semicircular canal; VIIms: mastoid segment of VII cranial nerve; VIIts: tympanic segment of VII cranial nerve.

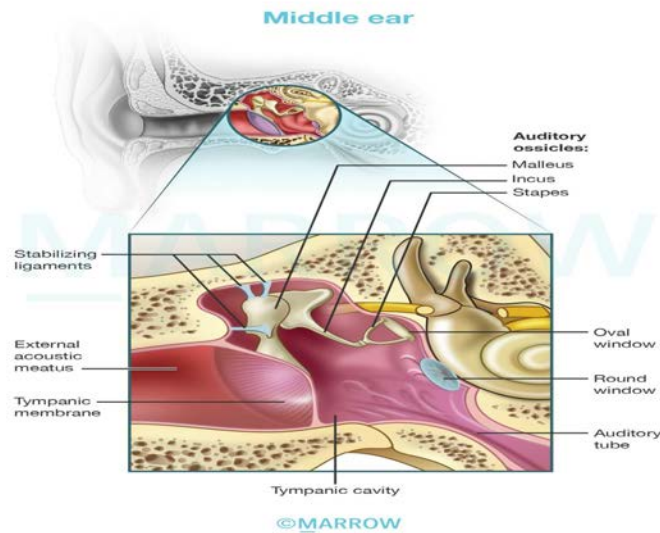
Solution to Question 16:

Fossa incudis gives attachment to the short process of incus.

Fossa incudis is a depression on the posterior wall of the middle ear. It lies lateral to aditus ad antrum.

Solution to Question 17:

The oval window is covered by the footplate of stapes.

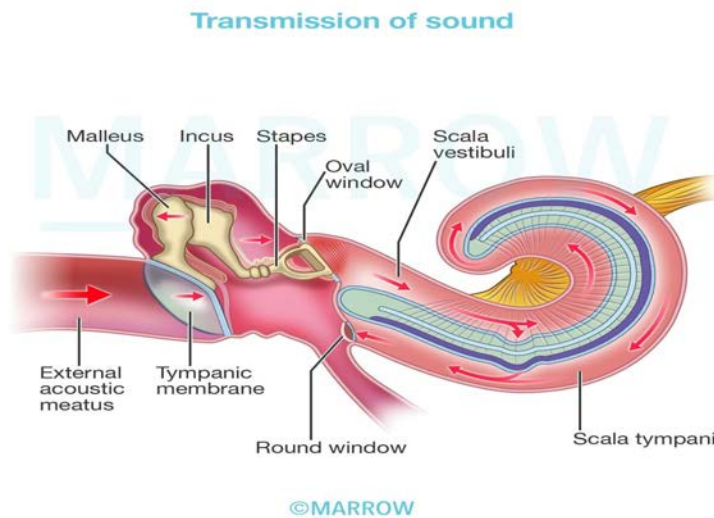


Solution to Question 18:

The movement of the stapes footplate causes vibration in the scala vestibuli. This is because the scala vestibuli, which opens at the oval window, is closed by the footplate of stapes.

The scala tympani opens at the round window.

Image depicting the transmission of sound:



Solution to Question 19:

The round window or the fenestra cochlea is covered by the secondary tympanic membrane.

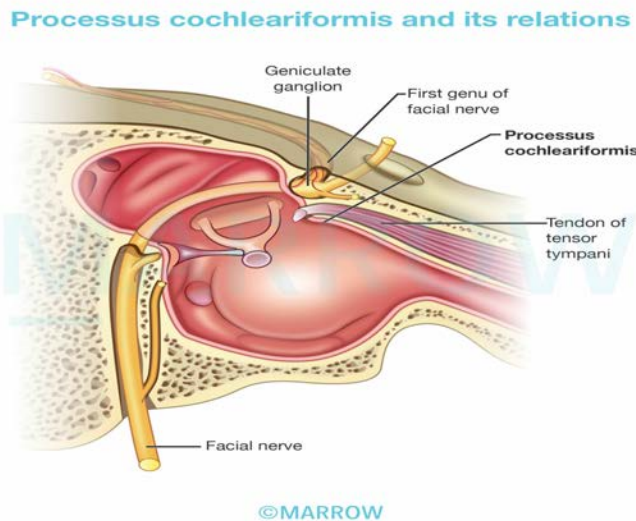
Solution to Question 20:

Processus cochleariformis is related to the tendon of tensor tympani

Just anterior to the oval window, the medial wall presents a hook-like projection called the processus cochleariformis.

The tendon of tensor tympani takes a turn here to get the attachment to the neck of the malleus.

It is also the landmark for the 1st genu of the facial nerve and the geniculate ganglion.



Solution to Question 21:

The absence of stapedial reflex is suggestive of facial nerve injury among the given options. The facial nerve forms the efferent arm of the stapedial or acoustic reflex.

Anatomy of stapedius muscle:

Stapedius is a second arch muscle and it arises from the pyramid and attaches to the neck of the stapes.

Its action is to lateralize the stapes & dampen loud sounds.

It is supplied by a branch of the facial nerve.

Solution to Question 22:

Tensor tympani is supplied by a branch from the medial pterygoid nerve which is a direct branch of the mandibular division of the trigeminal nerve (and not the posterior division of the mandibular nerve).

Anatomical features of tensor tympani muscle:

- The tensor tympani develops from the first arch.
- Arises from the semicanal for tensor tympani in the anterior wall of the middle ear.
- Hooks around the processus cochleariformis in the medial wall
- Attaches the neck of the malleus in the lateral wall.
- Action is to medialize the malleus & conduct sound.

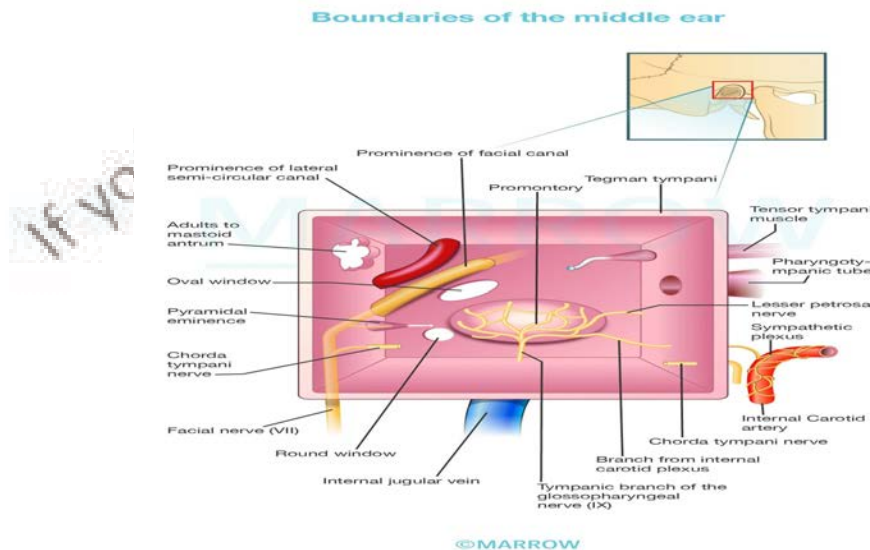
The posterior division of the mandibular nerve is mainly sensory and divides into three branches:

- The auriculotemporal nerve.
- The lingual nerve.
- The inferior alveolar (dental) nerve.

The motor component of the posterior division of the mandibular nerve supplies the mylohyoid and anterior belly of the digastric.

Solution to Question 23:

The sensory supply to the middle ear cavity is provided by the tympanic branch of the glossopharyngeal nerve. It forms the tympanic plexus with the caroticotympanic nerves from the internal carotid plexus of the sympathetic fibers.



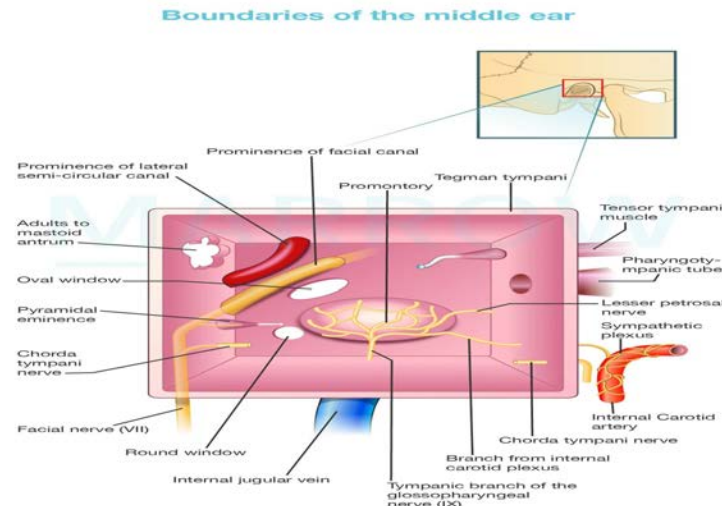
Solution to Question 24:

The tympanic branch of the facial nerve does not contribute to the tympanic plexus.

The tympanic plexus is formed by 3 nerves :

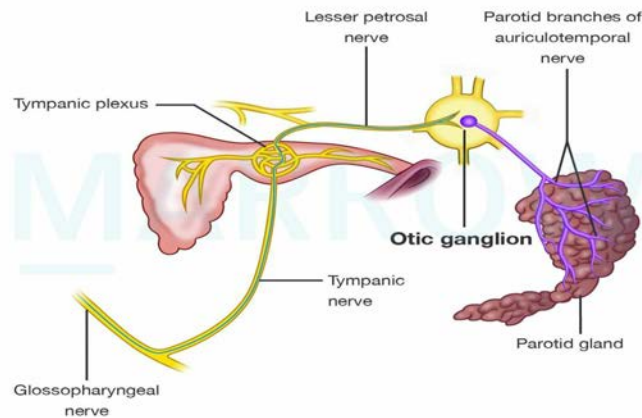
- The glossopharyngeal nerve forms the tympanic branch
- The sympathetic fibers from internal carotid plexus forms the superior and inferior caroticotympanic nerve.

The tympanic plexus innervates the tympanic membrane's medial surface, the tympanic cavity, mastoid air cells, and the bony eustachian tube. It also carries secretory fibres for the parotid gland, and in Frey's syndrome, the tympanic branch of the glossopharyngeal nerve can be sectioned.



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Otic ganglion



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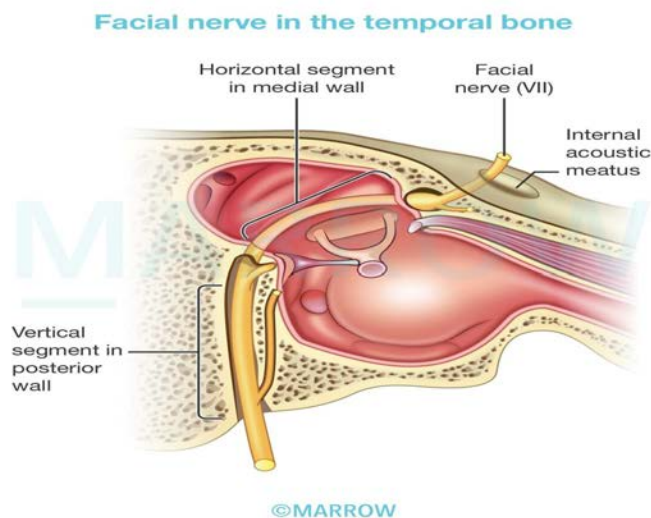
Solution to Question 25:

The facial nerve runs horizontally in the medial wall and vertically in the posterior wall of the middle ear.

The medial wall is related to the 3rd part, which runs from 1st genu (processus cochleariformis) to 2nd genu (pyramid). It is called the tympanic or horizontal part.

The posterior wall is related to the 4th part, which runs from the genu (pyramid) to the stylomastoid foramen where it exits the temporal bone. It is called the mastoid or the vertical part.

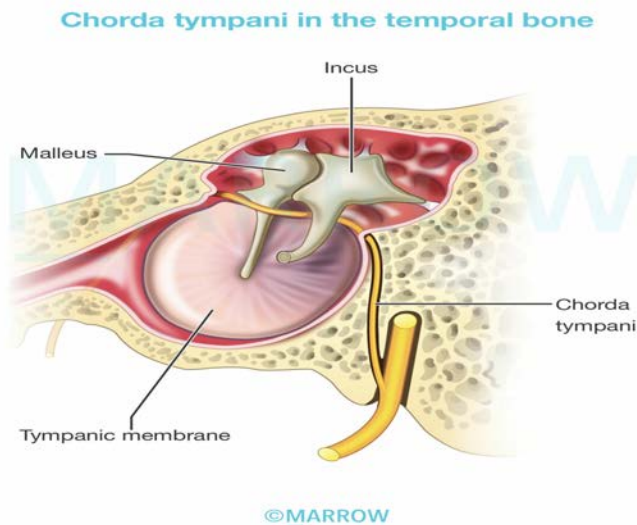
Image showing course of facial nerve in ear:



Solution to Question 26:

In the middle ear, the chorda tympani runs in between the malleus and incus.

Image showing the course of chorda tympani:

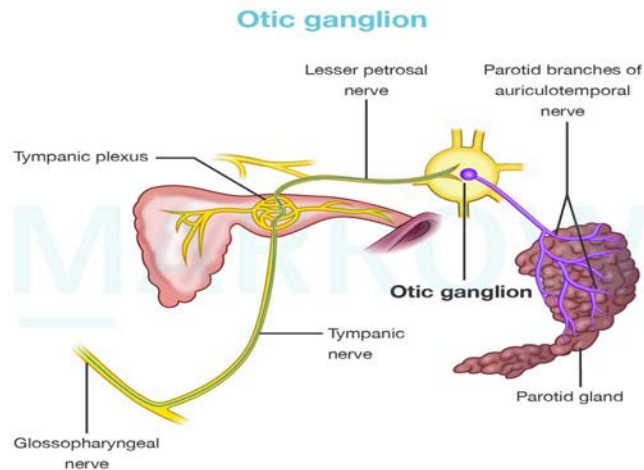


Solution to Question 27:

The fibers of the lesser petrosal nerve synapse in the otic ganglion and not pterygopalatine ganglion.

The lesser petrosal nerve is the parasympathetic or secretomotor nerve for the parotid glands. It arises over the promontory of the middle ear from the tympanic plexus (glossopharyngeal nerve).

Image showing lesser petrosal nerve synapsing with otic ganglion:



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Anatomy of Mastoid

Question 1:

Where does one find flat non-ciliated epithelium in the middle ear cleft?

- a) Eustachian tube
- b) External auditory meatus
- c) Mesotympanum
- d) Mastoid air cells

Question 2:

Which is the largest mastoid air cell?

- a) Antrum
- b) Aditus
- c) Attic
- d) Atrium

Question 3:

A patient with acute coalescent mastoiditis undergoes cortical mastoidectomy. Which of the following is used as a surgical landmark to locate the mastoid antrum?

- a) Trautmann's triangle
- b) Macewen's triangle
- c) Donaldson's line
- d) Citelli's angle

Question 4:

Which of the following structures does not form the boundary of the suprameatal triangle?

- a) Temporal line
- b) Posterosuperior segment of bony external auditory canal

- c) Promontory
- d) Tangent drawn to external auditory canal

Question 5:

Which of the following is not a pneumatic bone?

- a) Maxillary
- b) Ethmoid
- c) Mandible
- d) Mastoid

Question 6:

You are evaluating a patient with past history of chronic suppurative otitis media. The CT is suggestive of sclerosis and loss of aeration of mastoid air cells. What is this type of mastoid pneumatization called?

- a) Acellular mastoid
- b) Pneumatized mastoid
- c) Diploetic mastoid
- d) Conchal mastoid

Question 7:

Which of the following is not a part of the temporal bone?

- a) Mastoid
- b) Squamous
- c) Parietal
- d) Tympanic

Question 8:

While performing cortical mastoidectomy, the head and neck surgeon drills the bone up to the Citelli's angle. This angle is seen between which of the following structures?

- a) Tympanic membrane and the horizontal

- b) 3rd and 4th part of facial nerve
- c) Styloid and the vertical
- d) Sigmoid sinus and tegmen

Question 9:

Endolymphatic sac surgery is performed on a patient who has Ménière disease with intractable vertigo. Which of the following is the surgical landmark to identify the sac?

- a) Trautmann's triangle
- b) MacEwen's triangle
- c) Donaldson's line
- d) Citelli's angle

Question 10:

A head and neck surgeon enters the cranial cavity via the Trautmann's triangle. Which fossa can be approached through this triangle?

- a) Anterior cranial fossa
- b) Middle cranial fossa
- c) Posterior cranial fossa
- d) Hypophyseal fossa

Answer Key

Question No.	Correct Option
1	d
2	a
3	b
4	c
5	c
6	a
7	c
8	d

9	c
10	c

Detailed Explanations

Solution to Question 1:

The flat non-ciliated epithelium is found in the mastoid air cells and the epitympanum.

The mucosal lining of the middle ear is continuous with that of the nasopharynx but shows variations:

- Bony eustachian tube- ciliated columnar
- Cartilaginous eustachian tube- ciliated pseudostratified columnar
- Protympanum- ciliated columnar
- Most of the middle ear space- cuboidal
- Epitympanum & mastoid air cells- flat non-ciliated.

Solution to Question 2:

The mastoid antrum is the largest mastoid air cell.

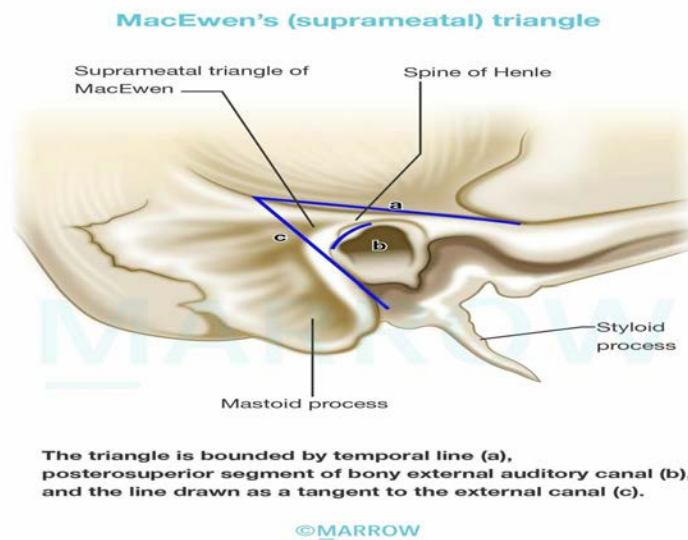
It is the most constant mastoid air cell because it is present even in a sclerotic mastoid.

The attic or the epitympanum is the part of the middle ear adjacent to its roof.

Aditus ad antrum is the pathway or the connection between mastoid antrum and attic.

Solution to Question 3:

Macewen's (suprameatal) triangle is used as a surgical landmark to locate the mastoid antrum. In an adult, the antrum lies 12–15 mm from the surface. This triangle is also the site of tenderness in acute mastoiditis.



Solution to Question 4:

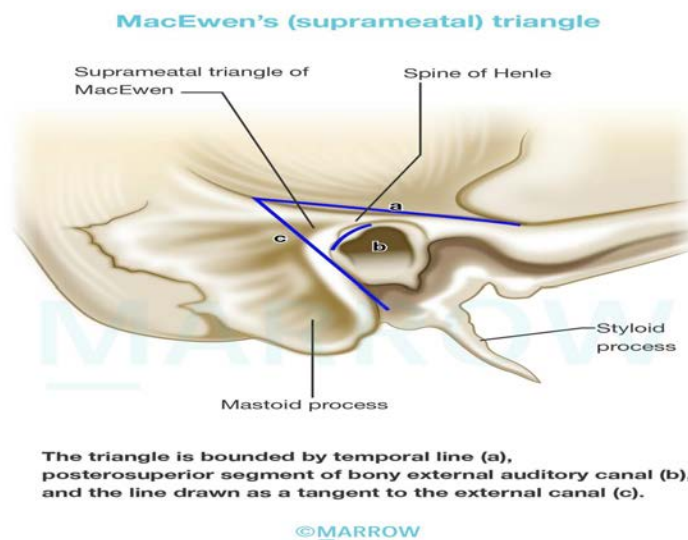
Promontory does not form the boundary of the suprameatal triangle.

Boundaries of MacEwen's (suprameatal) triangle:

- Superior- temporal line
- Anterior- posterosuperior segment of the bony external auditory canal
- Posterior- the line drawn as a tangent to the external canal.

It is an important landmark to locate the mastoid antrum in mastoid surgery.

Image showing the suprameatal triangle:



Solution to Question 5:

The mandible is not a pneumatic bone.

Some bones of the skull, notably the mastoid process of the temporal bone and the paranasal sinuses of the frontal, maxilla, sphenoid and ethmoid bones, many of the internal cavities are filled with air, i.e. they are variably pneumatized.

The maxillae are the largest of the pneumatized bones of the midface. They contain the maxillary air sinuses and bear the upper teeth.

The mastoid consists of a bone cortex with a 'honeycomb' of air cells underneath.

Solution to Question 6:

The type of mastoid pneumatization in the given clinical scenario is acellular or sclerotic mastoid.

Here, sclerosis and loss of aeration of mastoid air cells are seen. Sclerotic mastoids contain only mastoid antrum which is small with anteposed sigmoid sinus. It can occur due to eustachian or middle ear disease in childhood preventing mastoid development.

Other types of mastoids pneumatization:

- Well-pneumatized or cellular- contains multiple air spaces
- Diploetic- contains air cells and marrow spaces.

Solution to Question 7:

The parietal is not a part of the temporal bone.

The temporal bone consists of the following parts:

Squamous part- anterosuperiorly

It articulates with the parietal bone. The persistence of petrosquamous suture is called Korner's septum.

Mastoid part- posteroinferiorly.

Its outer surface forms a conical projection called the mastoid process. Sternocleidomastoid, splenius capitis and longissimus capitis attach here.

Petrous part- medially

- The arcuate eminence is related to the superior semicircular canal below
- The posterior surface contains the internal auditory meatus (IAM). The vestibular aqueduct is present behind the IAM.
- The inferior surface contains the carotid canal and the jugular fossa (jugular notch).

Tympanic part- centrally

- It contains the external auditory meatus (EAM), and the bony EAM is 16 mm long.

- The suprameatal triangle is present above the EAM

Styloid process- anteroinferior

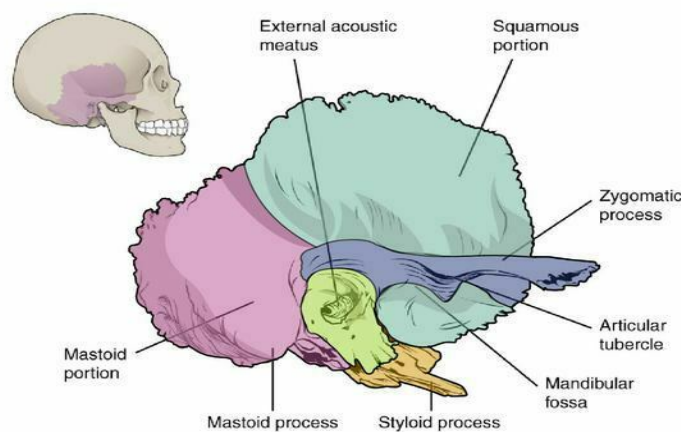
it gives attachment to:

- 3 muscles - stylohyoid, styloglossus, stylopharyngeus
- 2 ligaments - stylohyoid ligament, stylomandibular ligament

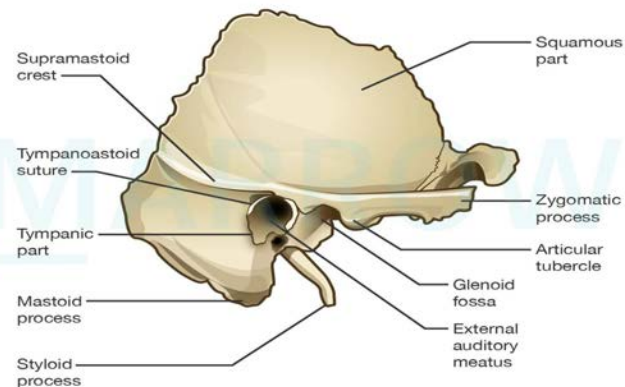
Zygomatic process

The temporal bone gives the zygomatic process which articulates with the temporal process of the zygomatic bone to form the zygomatic arch.

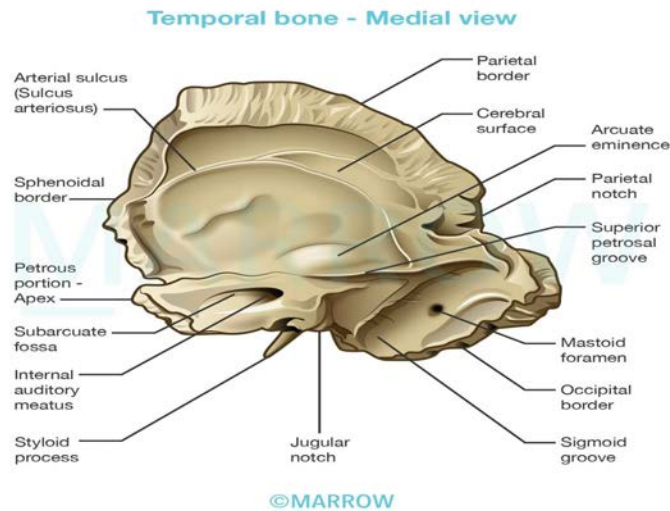
Parts of temporal bone



Temporal bone - Lateral view



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Solution to Question 8:

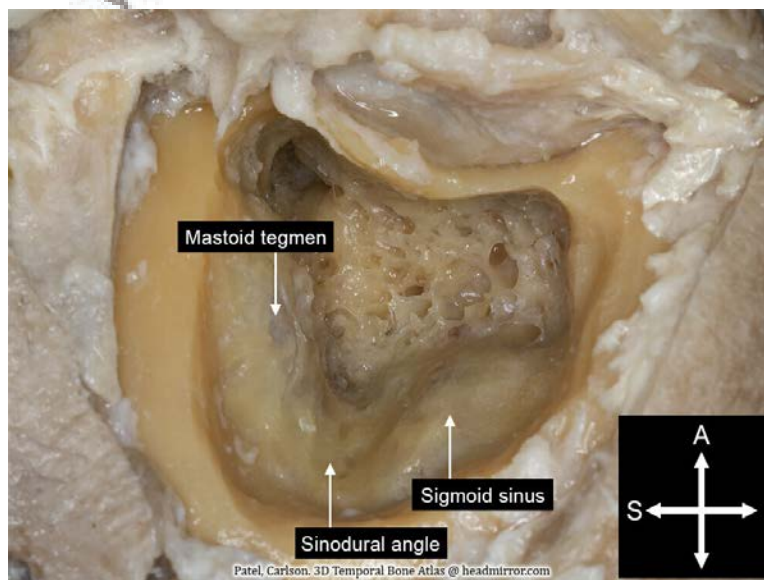
Citelli's angle or sinodural angle is the angle between the sigmoid sinus plate and tegmen or dural plate.

The boundaries of cortical mastoidectomy are:

- Tegmen plate superiorly
- Sigmoid sinus plate posteriorly
- Posterior bony external auditory canal wall anteriorly

This is also called the triangle of attack.

Image showing sinodural angle:

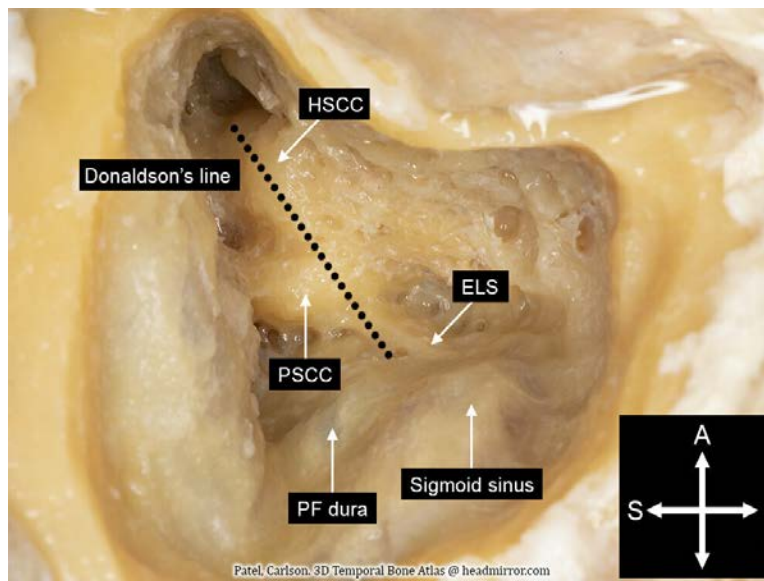


Solution to Question 9:

Donaldson's line is the surgical landmark to locate the endolymphatic sac.

It is an imaginary line drawn from the horizontal semicircular canal, which bisects the posterior semicircular canal. The endolymphatic sac lies just below it.

Image showing Donaldson's line:



ELS: Endolymphatic sac; HSCC: Horizontal semicircular canal, PSCC: Posterior semicircular canal; PF dura: Posterior fossa dura.

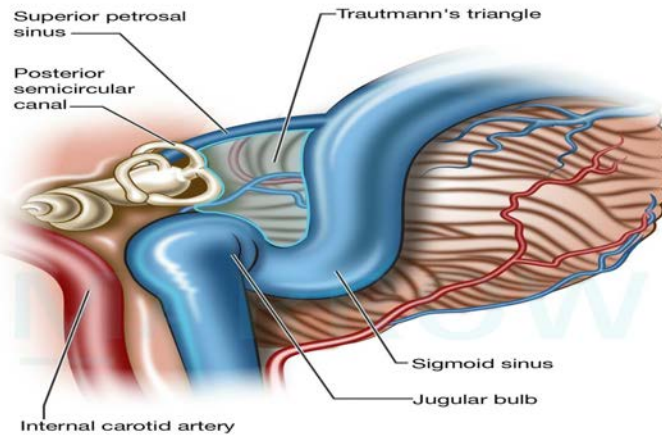
Solution to Question 10:

Trautmann's triangle is a landmark to approach the posterior cranial fossa.

- It is a triangular area on the medial wall of the mastoid antrum.
- It is bounded superiorly by the dura or superior petrosal sinus, posteriorly by the sigmoid sinus, and anteriorly by the bony labyrinth.
- Since here the bone is thin, it can lead to the spreading of infections from the mastoid to the cerebellum.

Image showing the Trautmann's triangle:

Trautmann's triangle



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Anatomy of Inner Ear

Question 1:

You are assisting an ENT surgeon who places a cochlear implant into the scala tympani. Which part of the bone is the implant placed in?

- a) Parietal bone
- b) Occipital bone
- c) Petrous part of temporal bone
- d) Petrous part of sphenoid bone

Question 2:

Where is the electrode placed in cochlear implant procedures?

- a) Round window
- b) Oval window
- c) Scala vestibuli
- d) Scala tympani

Question 3:

You are examining the CT scan images of the temporal bone of a patient. Which of the following structures can not be visualized on the CT scan?

- a) Cochlea
- b) Vestibule
- c) Semicircular canal
- d) Organ of Corti

Question 4:

The bony cochlea is a coiled tube making _____ turns around a bony pyramid called _____

- a) 1.5, modiolus
- b) 2.5, modiolus

- c) 1.5, helicotrema
- d) 2.5, helicotrema

Question 5:

Which two structures does the helicotrema connect?

- a) Scala media and scala vestibuli
- b) Scala media and scala tympani
- c) Scala vestibuli and scala tympani
- d) Scala tympani and subarachnoid space

Question 6:

Which of the following is not a part of the bony cochlea?

- a) Scala tympani
- b) Scala vestibule
- c) Scala media
- d) Perilymph

Question 7:

Which of the following statements is true regarding the vestibule?

- a) It is a part of the membranous labyrinth
- b) Contains spherical recess for utricle
- c) Contains spherical recess for saccule
- d) It is part of the auditory pathway

Question 8:

A patient has undergone a high-resolution CT of the temporal bone which confirmed a diagnosis of semicircular canal dehiscence syndrome. Which of the following will not be affected?

- a) Posterior semicircular canal

- b) Medial semicircular canal
- c) Superior semicircular canal
- d) Lateral semicircular canal

Question 9:

Which of the following structures is the crus commune a part of?

- a) Cochlea
- b) Semicircular canal
- c) Vestibule
- d) Middle ear

Question 10:

A 45-year old construction worker presents to you with hearing loss. His audiogram reveals a diagnosis of noise-induced hearing loss. Which organ is most likely to be affected?

- a) Macula
- b) Cristae
- c) Organ of Corti
- d) Semicircular canals

Question 11:

The organ of Corti is situated over which membrane?

- a) Basilar membrane
- b) Reissner's membrane
- c) Shrapnell's membrane
- d) Stria vascularis

Question 12:

What does the stria vascularis secrete?

- a) Endolymph

- b) Perilymph
- c) Cerebrospinal fluid
- d) Both endolymph and perilymph

Question 13:

Where is the endolymph is seen?

- a) Scala vestibuli
- b) Scala media
- c) Scala tympani
- d) Helicotrema

Question 14:

Which of the following fluids have high potassium and low sodium content?

- a) Endolymph
- b) Perilymph
- c) CSF
- d) ECF

Question 15:

A 20-year-old man diagnosed with meningitis later develops sensorineural hearing loss. Which structure is responsible for the spread of his infection to the cochlea?

- a) Endolymphatic sac
- b) Cochlear aqueduct
- c) Scala media
- d) Vestibular aqueduct

Question 16:

Which of the following statements is true about membranous cochlea?

- a) Higher frequencies heard at the base

- b) Lower frequencies are heard at the base
- c) Higher frequencies are heard at the apex
- d) All frequencies are heard at the apex

Question 17:

Which of the following sentence is false?

- a) Cristae are present in semicircular canal
- b) Maculae are present in utricle and saccule
- c) Organ of corti rests over basilar membrane
- d) Reissner's membrane overlies the organ of Corti

Question 18:

Angular movements are sensed by

- a) Utricle
- b) Saccule
- c) Semicircular canals
- d) Cochlea

Question 19:

The endolymphatic duct is a continuation of which of the following?

- a) Ductus re-unions
- b) Parotid duct
- c) Utriculo-saccular duct
- d) Cochlear aqueduct

Question 20:

Endolymphatic duct connects

- a) Scala vestibuli and scala media
- b) Scala media and subdural space

- c) Scala vestibuli with aqueduct of cochlea
- d) Utriculosaccular duct to endolymphatic sac

Question 21:

Where is the endolymphatic sac is present?

- a) Between two layers of dura
- b) Extradural space in frontal bone
- c) Subdural space in petrous part of temporal bone
- d) Subdural space in frontal bone

Question 22:

Blood supply to the labyrinth is from which of the following arteries?

- a) Anterior inferior cerebellar artery
- b) Posterior inferior cerebellar artery
- c) Middle meningeal artery
- d) Auriculotemporal artery

Answer Key

Question No.	Correct Option
1	c
2	d
3	d
4	b
5	c
6	c
7	c
8	b
9	b
10	c
11	a

12	a
13	b
14	a
15	b
16	a
17	d
18	c
19	c
20	d
21	a
22	a

Detailed Explanations

Solution to Question 1:

The implant is placed in the cochlea which is a part of the inner ear. The inner ear lies in the petrous part of the temporal bone.

The inner ear is well protected between the medial wall of the middle ear and the internal auditory canal.

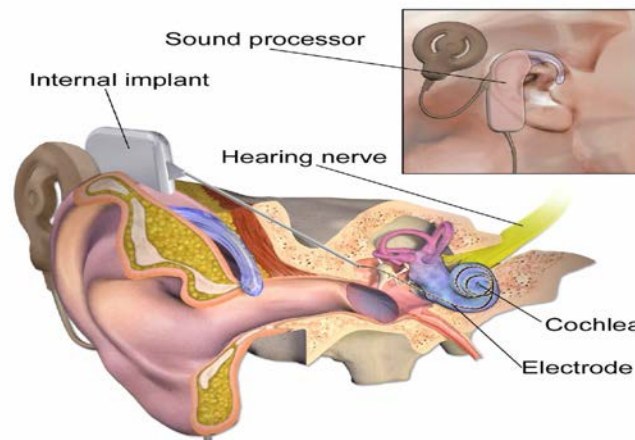
Solution to Question 2:

The electrode of a cochlear implant is placed in the scala tympani. It allows for the electrodes to be in close proximity to the ganglion cells and their dendrites which are located in the modiolus and bony spiral lamina of the cochlea.

A cochlear implant is an electronic device for severe to profound SNHL and who cannot benefit from a hearing aid. It converts sound to an electrical signal (transduction) and delivers this to the auditory nerve. Prerequisites include an intact cochlear nerve and higher auditory pathway.

In the auditory brainstem implant, the site of the implant is the lateral recess of the fourth ventricle. It is used when the cochlear nerve is not intact.

Cochlear implant



Solution to Question 3:

The organ of Corti present in the scala media is a part of the membranous labyrinth. It cannot be visualized on a CT scan.

Bony labyrinth consists of:

- Vestibule
- Cochlea
- Semicircular canals

These structures can be seen on a CT scan of the temporal bone.

Membranous labyrinth consists of:

- Saccule and utricle
- Cochlear duct/ scala media
- Semicircular ducts
- Endolymphatic duct and sac

Solution to Question 4:

The bony cochlea is a coiled tube making 2.5 to 2.75 turns round a central pyramid of bone called modiolus.

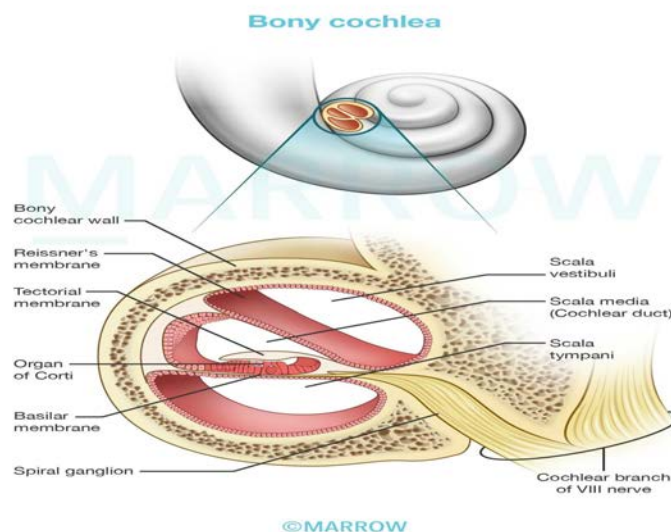
Solution to Question 5:

The scala vestibuli and scala tympani communicate with each other at the apex of the cochlea through an opening called the helicotrema.

Scala tympani and subarachnoid space are connected by the aqueduct of cochlea.

Solution to Question 6:

Scala media is part of the membranous cochlea.



The bony cochlea is a coiled tube with 2.5 to 2.75 turns around a central pyramid of bone called modiolus.

From the modiolus laterally into the tube winding like the thread of a screw is the osseous spiral lamina.

Spiral lamina along with membranes attached to it divide the tube into 3 parts:

- Scala vestibuli above
- Scala media in between
- Scala tympani below.

The scala vestibule & tympani are part of the bony labyrinth and are filled with perilymph. They communicate with each other at the apex called helicotrema.

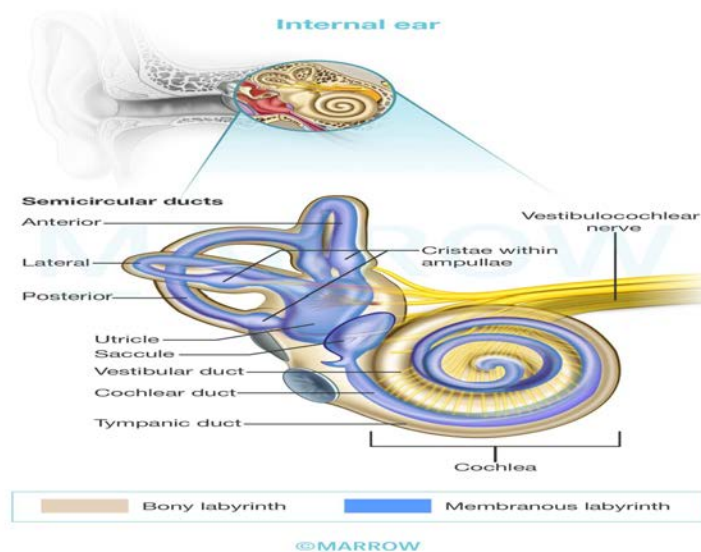
Scala media is part of the membranous cochlea and contains the sensory organ of Corti with hair cells.

Solution to Question 7:

Vestibule contains the spherical recess for the saccule.

It is part of the bony labyrinth and not part of the auditory pathway. The inside of the medial wall presents with recesses & openings:

- A spherical recess, which lodges the saccule
- An elliptical recess, which lodges the utricle
- An opening for aqueduct of vestibule
- An opening for the semicircular canals



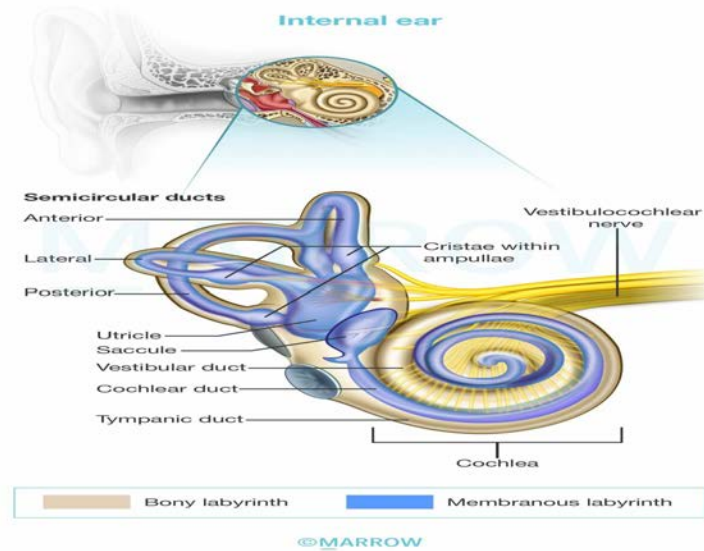
Solution to Question 8:

The medial semicircular canal does not exist.

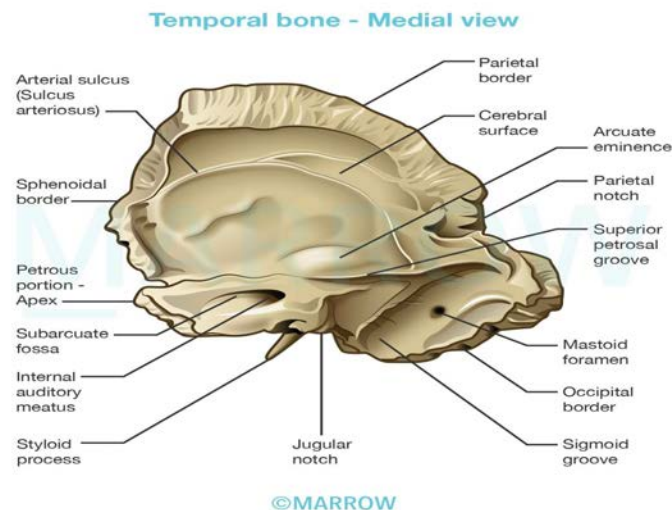
The semicircular canals are part of the bony labyrinth. They house the semicircular ducts which are the end organs for angular momentum.

They are 3 in number:

- Superior also called anterior in the X-axis
- Posterior in Y-axis
- Lateral also called horizontal in Z-axis.



Semicircular canal dehiscence syndrome is an inner ear abnormality where the bony covering of the semicircular canals is very thin or absent. It usually affects the superior semicircular canal but can rarely involve the posterior or lateral semicircular canals. The superior semicircular canal is present inferior to the arcuate eminence of the temporal bone.



Solution to Question 9:

Crus commune is a part of the semicircular canals.

Each semicircular canal has 2 ends, ampullated & non-ampullated which open into the vestibule. The non-ampullated ends of posterior and superior canals unite to form a common channel called crus commune.

Solution to Question 10:

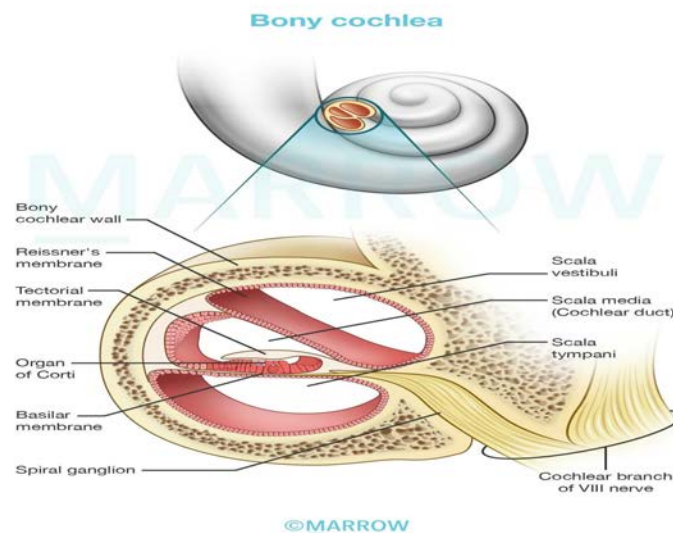
The organ of Corti is the sensory organ of hearing and is situated on the basilar membrane.

Chronic exposure to loud noises may mechanically damage the organ of Corti and result in noise induced hearing loss.

Solution to Question 11:

The organ of Corti is situated over the basilar membrane in scala media.

The membranes of the cochlear duct:



- The basilar membrane separates scala media from tympani and supports the organ of Corti.
- Reissner's membrane separates scala media from scala vestibuli.
- Stria vascularis is the lateral part of scala media which secretes endolymph.

Shrapnell's membrane is another name for pars flaccida of the tympanic membrane.

Solution to Question 12:

The stria vascularis contains vascular epithelium and is concerned with secretion of endolymph.

Solution to Question 13:

Endolymph is present in the scala media.

Solution to Question 14:

Endolymph is akin to intracellular fluid and hence high in potassium.

Perilymph and cerebrospinal fluid (CSF) are both extracellular fluids (ECF) that are high in sodium.

Solution to Question 15:

Infections can spread from the CNS to the inner ear or vice versa through the aqueduct of cochlea. Scala tympani is connected with the subarachnoid space through this structure.

Solution to Question 16:

Higher frequencies of sound are heard at the basal coil while lower frequencies are heard at the apical coil of membranous cochlea.

This is because the length of basilar membrane increases as we proceed from the basal coil to the apical coil.

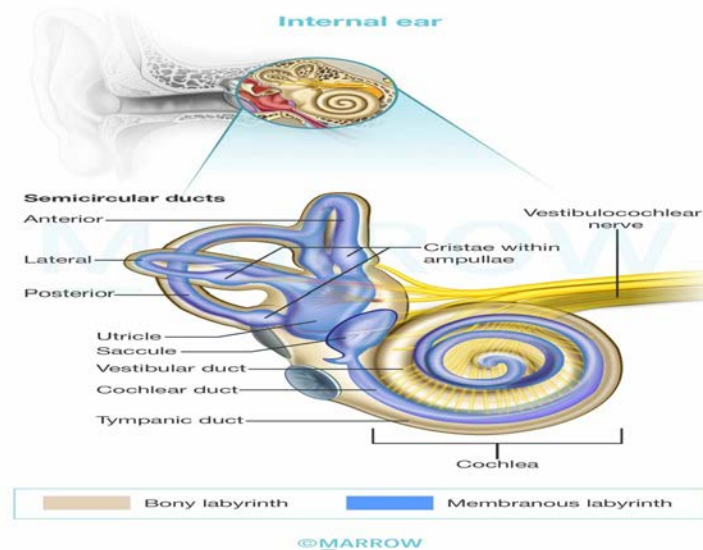
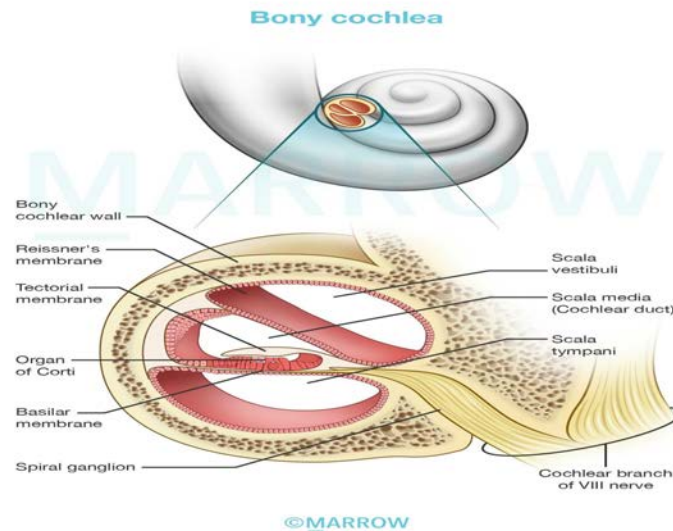
Solution to Question 17:

The tectorial membrane overlies the organ of Corti. Reissner's membrane separates scala vestibuli and scala media.

The basilar membrane supports the organ of Corti.

The sensory epithelium of the utricle and saccule is called macula and is concerned with linear acceleration and deceleration.

The ampullated end of each semicircular canal contains a thickened ridge of neuroepithelium called crista ampullaris.



Solution to Question 18:

Semicircular canals respond to angular acceleration and deceleration. The receptors are the cristae, located in the ampullated ends of the three semicircular ducts.

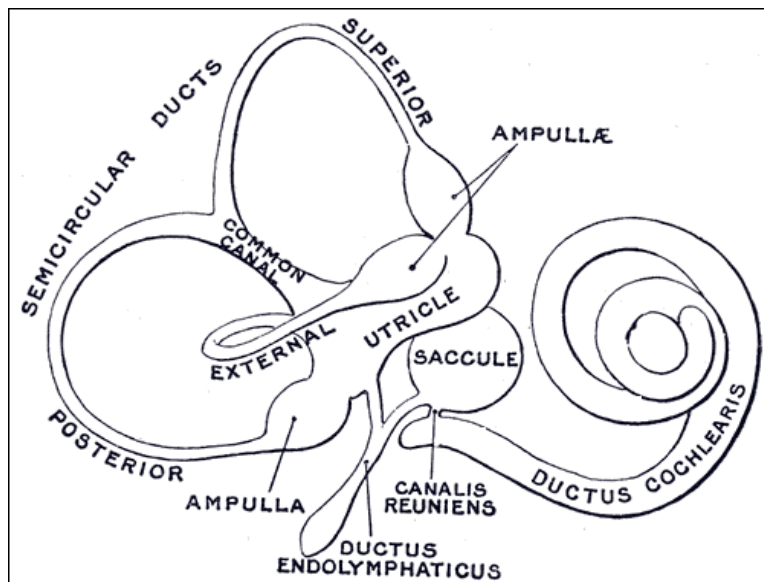
Their utricle and saccule which contain macula, are stimulated by linear acceleration and deceleration or gravitational pull during the head tilts.

Solution to Question 19:

The endolymphatic duct is formed by the union of two ducts, one each from the saccule and the utricle.

Ductus reunions connect the cochlear duct to the saccule.

The aqueduct of cochlea (cochlear aqueduct) connects scala tympani to the subarachnoid space.



Solution to Question 20:

The endolymphatic duct is formed by the union of two ducts, one each from the saccule and the utricle.

It passes through the vestibular aqueduct. Its terminal part is dilated to form an endolymphatic sac, which lies between the two layers of dura on the posterior surface of the petrous bone.

The scala vestibuli and scala tympani are filled with perilymph and communicate with each other at the apex of cochlea through an opening called helicotrema.

Aqueduct of cochlea connects scala tympani with subarachnoid space.

Solution to Question 21:

The terminal part of the endolymphatic duct is dilated to form an endolymphatic sac, which lies between the two layers of dura on the posterior surface of the petrous bone.

Endolymphatic sac is surgically important. It is exposed for drainage or shunt operation in Meniere's disease.

Solution to Question 22:

Blood supply to the inner ear is by a branch of the anterior inferior cerebellar artery.

The inner ear is supplied by labyrinthine artery, a branch of the anterior inferior cerebellar artery, which is a branch of the basilar artery.

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Disorders of External Ear

Question 1:

In a 7-year-old child with hearing loss and intellectual disability, you notice the following external ear anomaly. What could the anomaly in this case be?



- a) Preauricular sinus
- b) Preauricular tag
- c) Ganglion
- d) Neurodermatitis

Question 2:

A patient presents to the OPD with the following finding and history of pus discharge from the same area. What is the likely etiology of this condition?



- a) Instrumental injury
- b) Congenital anomaly
- c) Accessory opening of the mastoid
- d) Idiopathic

Question 3:

An 8-month-old child was brought to your OPD with the following finding. What is your diagnosis?



- a) Wildermuth ear

- b) Mozart ear
- c) Microtia
- d) Anotia

Question 4:

A child with protruding ears is brought to the OPD by his worried parents. They opt for surgical correction of the defect due to cosmetic reasons. After what age would you proceed with the surgery?

- a) 3 months
- b) 6 months
- c) 3 years
- d) 6 years

Question 5:

A 5-year-old girl was brought to you with complaints of serous discharge from a cutaneous opening on the right side of her neck. A fistulogram confirms your diagnosis of collaral fistula. Which of the following is a feature of this condition?

- a) Abnormality of second branchial cleft
- b) Opens in front of the tragus
- c) Acquired secondary to infection
- d) Traverses close to the facial nerve

Question 6:

A professional wrestler presents to your OPD with the deformity shown below. What is the most likely cause of the same?



- a) Auricular hematoma
- b) External ear exostosis
- c) Auricular pseudocyst
- d) Avulsion of pinna

Question 7:

A young man is rushed to the ER following a road traffic accident with severe injuries and a completely avulsed pinna. How would you preserve the avulsed pinna until the patient is stable enough for reconstructive surgery?

- a) By storing it in a refrigerator in isotonic saline
- b) By suturing it to buccal mucosa after careful cleaning
- c) By peeling its skin and inserting cartilage into postauricular area
- d) By implanting it into the abdominal cavity

Question 8:

What is the most likely diagnosis in a patient who presents with a painful ear swelling as shown below?



- a) Cauliflower ear
- b) Perichondritis
- c) Chondrodermatitis chronica helicis
- d) Otitis externa

Question 9:

A middle-aged man presents with a lesion as shown below. He is unable to sleep on the affected side. There is no history of trauma. What is the most probable diagnosis?



- a) Acute suppurative perichondritis of pinna

- b) Furuncle
- c) Chondrodermatitis nodularis chronica helices
- d) Relapsing polychondritis

Question 10:

A patient is planned for excision of the following lesion. Which of the following measures will not prevent its recurrence post-excision?



- a) Preoperative radiotherapy
- b) Postoperative radiotherapy
- c) Neoadjuvant chemotherapy
- d) Intralesional steroid

Question 11:

Which of the following is true about furunculosis of ear?

- a) Patients with repeated attacks should be investigated for diabetes mellitus
- b) Most common cause is streptococcal infection of hair follicle
- c) Can occur anywhere in the external auditory canal
- d) Tragal sign is negative in majority of cases

Question 12:

Which of the following is synonymous with swimmer's ear?

- a) Hematoma of pinna
- b) Exostosis
- c) Diffuse otitis externa
- d) Malignant otitis externa

Question 13:

A 40-year-old male presents with right-sided ear discharge, itching and pain for the past 4 days. He gives a history of insect entry into the external auditory canal which he removed himself. On examination, there is purulent discharge in a narrow canal and the tympanic membrane is normal. Which of the following is the most effective single treatment for this condition?

- a) Medicated wicks
- b) Aural toilet
- c) Antibiotics
- d) Analgesics

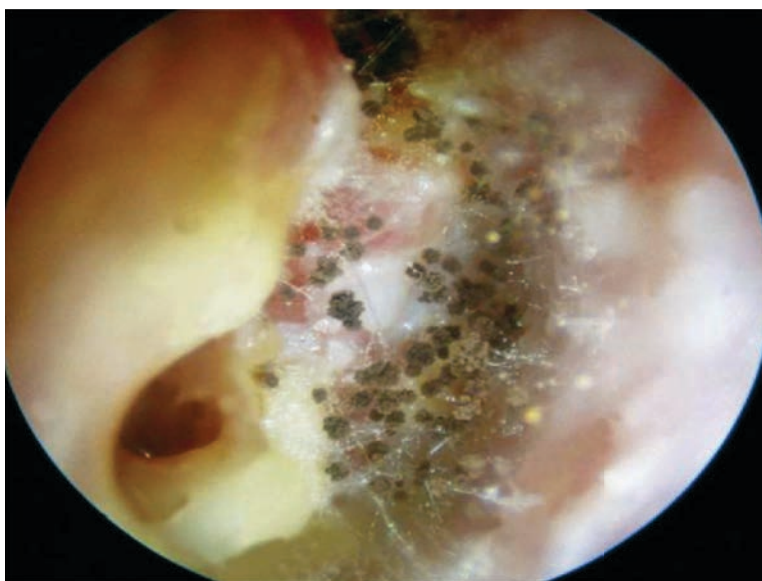
Question 14:

Which of the following is the most common cause of otomycosis?

- a) *Mucor velutinosus*
- b) *Candida albicans*
- c) *Aspergillus fumigatus*
- d) *Aspergillus niger*

Question 15:

A young man presented with irritation and itching in the ear, associated with watery discharge and a musty odor. Otoscopic imaging is shown below. What is the causative organism?



- a) Aspergillus niger
- b) Rhizopus oryzae
- c) Candida albicans
- d) Aspergillus fumigatus

Question 16:

A 30-year-old woman presents to the OPD with right-sided facial weakness. On examination, she has vesicles in the external auditory canal and over the tympanic membrane. What is the diagnosis?

- a) Bell's palsy
- b) Ramsay Hunt syndrome
- c) Möebius syndrome
- d) Melkersson syndrome

Question 17:

A 60-year-old diabetic presents to the ER with fever, excruciating left ear pain, and otorrhoea which is unresponsive to antibiotics. On examination, she has a deviation of the angle of mouth to the right side, and granulation tissue is seen in the left external auditory canal. What is the likely diagnosis?

- a) Otomycosis
- b) Serous otitis media

- c) Malignant otitis externa
- d) Eczematous otitis externa

Question 18:

Which of the following statements is false about malignant otitis externa?

- a) It is due to pseudomonas infection
- b) It mainly affects immunocompromised individuals
- c) Technetium 99 bone scan is used for follow up
- d) Facial nerve is the most common affected nerve

Question 19:

A young man presents to the OPD with complaints of intense irritation in the ear following the use of ear drops. Examination reveals vesicle formation, oozing, and crusting in the canal. Which of the following is responsible?

- a) Clotrimazole
- b) Paradichlorobenzene
- c) Dexamethasone
- d) Neomycin

Question 20:

Which of the following statements is false about wax?

- a) Contains secretions of glands in the external auditory canal
- b) Formed by dried up middle ear discharge
- c) Has antibacterial activity
- d) Is extruded out on its own during jaw movements

Question 21:

Ceruminous glands are modifications of which of the following?

- a) Apocrine glands

- b) Eccrine glands
- c) Holocrine glands
- d) Cytocrine glands

Question 22:

While performing syringing for a patient with impacted wax, along which wall of the meatus would you direct the jet of water?

- a) Anterosuperior wall
- b) Posterosuperior wall
- c) Anteroinferior wall
- d) Posteroinferior wall

Question 23:

After the extraction of the mass in a patient with keratosis obturans, it is sent for histopathological examination. Which of the following is seen here?

- a) Keratin in random pattern
- b) Cholesterol crystals
- c) Desquamated epithelial cells
- d) Squamous epithelial metaplasia

Question 24:

Which of the following is not seen in a patient with keratosis obturans?

- a) Ulceration and granuloma formation
- b) Isolation of Staphylococcus aureus
- c) Association with bronchiectasis
- d) Narrowing of meatus causing facial nerve palsy

Question 25:

What is the surgery done to widen the cartilaginous part of the external auditory canal called?

- a) Meatoplasty
- b) Tympanoplasty
- c) Myringoplasty
- d) Otoplasty

Question 26:

In which of the following is meatoplasty not performed?

- a) Congenital atresia of canal
- b) Chronic stenotic otitis externa
- c) Tortuous canal during tympanoplasty
- d) Canal wall up procedure

Question 27:

While performing otoscopy, which of the following would not indicate a normal tympanic membrane?

- a) Pearly grey in colour
- b) Primarily translucent
- c) Convex to external auditory canal
- d) Mobile with a change in middle ear pressures

Question 28:

Which of the following is false regarding retracted tympanic membrane?

- a) Shiny, pearly white tympanic membrane
- b) Foreshortened handle of malleus
- c) Prominent lateral process of malleus
- d) Sickling of anterior and posterior malleal folds

Question 29:

While performing routine otoscopy, a patient is found to have tympanic membrane retraction with the pars tensa draping over the incudostapedial joint. What is the grade of retraction?

- a)** Grade I Sade
- b)** Grade II Sade
- c)** Grade I Tos
- d)** Grade II Tos

Answer Key

Question No.	Correct Option
1	b
2	b
3	c
4	d
5	d
6	a
7	c
8	b
9	c
10	c
11	a
12	c
13	b
14	d
15	a
16	b
17	c
18	c
19	d
20	b
21	a
22	b
23	c
24	d
25	a
26	d
27	c

28	a
29	b

Detailed Explanations

Solution to Question 1:

The external ear anomaly seen here is pre-auricular tag/appendage.

Pre-auricular tags/appendages are congenital external ear anomalies. They are seen as skin-covered pieces of cartilage anywhere between the tragus and the angle of the mouth.

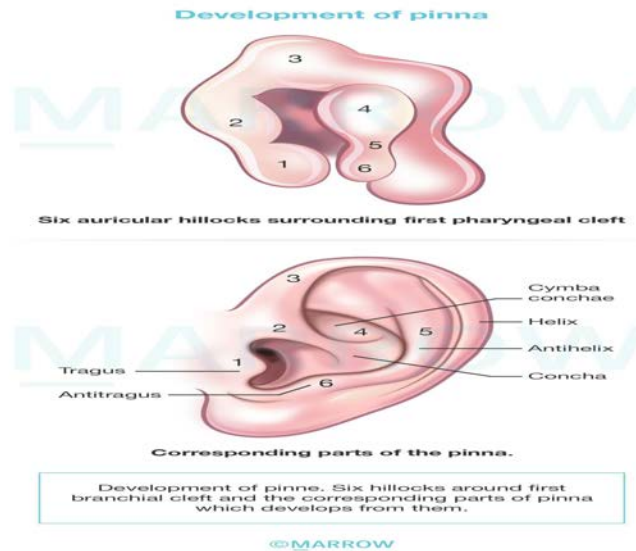
The clinical features of facial asymmetry, intellectual disability, limbal dermoid, and pre-auricular tags and mixed or conductive hearing loss point to a diagnosis of Goldenhar syndrome (oculoauriculovertebral spectrum). Clinical features include:

- Hypoplasia of the malar, maxillary, and mandibular regions
- Macrostomia and microtia
- Preauricular and facial skin tags
- Hemivertebrae (usually cervical)
- Intellectual disability
- Cardiac, renal, and central nervous system (CNS) anomalies
- Ocular features- Limbal dermoids, upper lid notching or coloboma, microphthalmos, disc coloboma

Solution to Question 2:

The given clinical scenario and the image are suggestive of preauricular sinus seen between the tragus and crus of the helix. It is the most common congenital anomaly of the 1st branchial arch, due to the incomplete fusion of the hillocks of His that develop to form the pinna.

It can get repeatedly infected and form an abscess, which would then require excision.



Solution to Question 3:

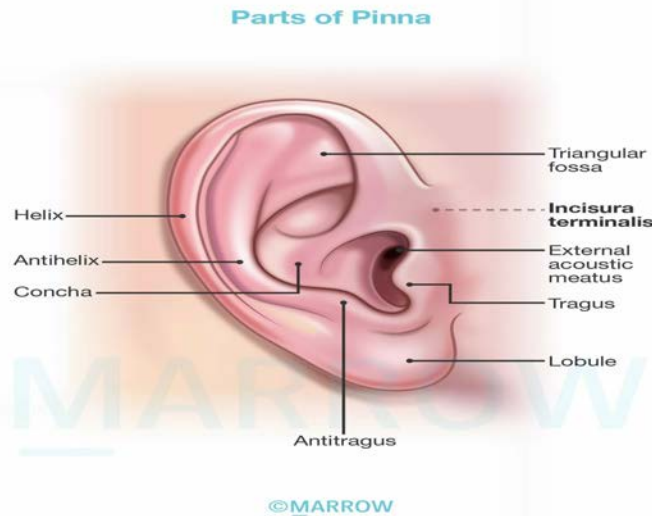
The given image showing malformed small pinna appearing like tubercles are suggestive of microtia. It is frequently accompanied by hearing loss.

Solution to Question 4:

Protruding ears or bat ears can be corrected anytime after 6 years of age if cosmetic appearance demands. This is due to the fact that the pinna is completely developed and achieves adult configuration by the age of 6 years.

Protruding ears are caused by a large concha with poorly developed antihelix and scapha.





Solution to Question 5:

Collaural fistula traverses close to the facial nerve within the substance of the parotid. It is a congenital anomaly of 1st branchial cleft.

The fistula is between the external auditory canal and neck, where the upper opening is on the floor of the external auditory canal and the lower opening is on the neck between mandible and sternocleidomastoid.

Solution to Question 6:

The image shows a cauliflower ear deformity, which is likely due to auricular hematoma.

Auricular hematoma is a collection of blood between the auricular cartilage and its perichondrium. If left untreated, the extravasated blood may clot and then organize, resulting in a typical deformity called cauliflower ear. It is also called pugilistic or boxer's ear.

It is often seen in boxers, wrestlers, and rugby players due to blunt trauma.

Treatment comprises of aspiration of the hematoma under strict aseptic precautions and a pressure dressing, to prevent reaccumulation. Aspiration may need to be repeated. If aspiration fails, incision and drainage should be done. All cases should receive prophylactic antibiotics.

Solution to Question 7:

The avulsed pinna is preserved by peeling its skin and inserting the cartilage into the postauricular area which can be preserved for delayed reconstruction.

However, immediate reimplantation using microvascular techniques would be ideal.

If the pinna is still attached to the head by a small pedicle of skin, primary reattachment should be considered.

Solution to Question 8:

The given clinical scenario of painful, erythematous ear swelling is suggestive of perichondritis.

The most common cause is trauma either iatrogenic (surgical incision) or non-iatrogenic (laceration, hematoma). It can be due to the extension of infection from diffuse otitis externa or furuncle of the meatus. The most common organism responsible is *Pseudomonas aeruginosa*.

The image given below shows a patient with a cauliflower ear deformity. It is also known as perichondrial hematoma or wrestler's ear. It is usually caused by repeated episodes of trauma.



Chondrodermatitis chronica helices is a small, nodular, tender, chronic inflammatory lesion occurring on the helix of the ear. It is shown in the image below.



Otitis externa, also called swimmer's ear, is the inflammation of the ear canal. It has the following appearance:



Solution to Question 9:

A small tender nodule in the helix in a middle-aged patient is suggestive of chondrodermatitis nodularis chronica helicis. The patient will have difficulty sleeping on the affected side.

Treatment includes excision of the lesion.

Solution to Question 10:

The given image shows a keloid of the pinna and neoadjuvant chemotherapy does not reduce its recurrence post excision.

Keloid commonly occurs following trauma. The common sites are lobule and helix.

There is a high chance of recurrence after excision, which can be prevented by:

- Intralesional steroids
- Pre and post-operative radiotherapy

Solution to Question 11:

Patients with repeated attacks of furunculosis should be investigated for diabetes mellitus.

A furuncle is a staphylococcal infection of the hair follicle. As the hairs are confined only to the cartilaginous part of the meatus, furuncle is seen only outer one-third of EAC.

Symptoms are severe compared to signs as the skin over the EAC is tightly adherent. The tragal sign is positive and there is an obliteration of the retro auricular groove.

Treated by incision and drainage or 10% ichthammol glycerine wicks and antibiotics.

Solution to Question 12:

Swimmer's ear is synonymous with diffuse otitis externa.

The normal pH of the external auditory canal is acidic. Any condition that makes the pH alkaline predisposes to otitis externa. This includes situations such as:

- Hot & humid climate (excessive sweating) and
- Prolonged swimming

Solution to Question 13:

The clinical scenario is suggestive of diffuse otitis externa following a trauma. Aural toilet remains the most effective single treatment for this condition.

All exudate and debris should be carefully removed, with special emphasis on the anteroinferior meatal recess, which forms a blind pocket and acts as a reservoir of infective secretion. The aural toilet can be done by dry mopping, suction clearance, or irrigating the canal with warm, sterile normal saline.

Further treatment modalities:

- After the aural toilet is done, wicks medicated with antibiotic-steroid preparation can be used. These wicks have to be changed daily for 2 to 3 days, which thereafter can be substituted by ear drops.

- Broad-spectrum systemic antibiotics are used when there is cellulitis and acute tender lymphadenitis.
- Analgesics are usually a part of management, as otitis externa is associated with severe pain.
- Gauze wick soaked in 10% ichthammol glycerine is used to reduce swelling.
- Astringents such as aluminum acetate (8%) or silver nitrate (3%) are used to dry up an oozing meatus.

Solution to Question 14:

Otomycosis is a fungal infection of the ear canal that often occurs due to *Aspergillus niger* (most common), *Aspergillus fumigatus*, or *Candida albicans*.

Solution to Question 15:

The given clinical scenario along with the otoscopic image showing black-headed filamentous growth is suggestive of otomycosis by *Aspergillus niger*.

Irritation and itching are the most predominant symptoms of *Aspergillus niger* infection. Pain is predominant in *Aspergillus flavus* infection.

The fungal mass in otomycosis is likened to a wet blotting paper/wet newspaper appearance.

The appearances of otomycosis on otoscopy and the causative organisms are:

- Black-headed filamentous growth - *Aspergillus niger* infection
- Green or brown - *Aspergillus fumigatus* infection
- White and creamy deposit - *Candida* infection

Rhizopus oryzae is one of the causative agents of mucormycosis. The common clinical form of this opportunistic infection is rhino-orbital-cerebral mucormycosis (ROCM). It is rarely implicated in otomycosis.

Solution to Question 16:

The given clinical scenario of facial paralysis along with vesicular eruptions in the external auditory canal and tympanic membrane is suggestive of Ramsay Hunt syndrome (herpes zoster oticus).

Ramsay Hunt Syndrome is caused by the reactivation of Herpes zoster in the geniculate ganglion. In most patients, there is an involvement of the trunk of the facial nerve, which results in severe facial palsy.

Often, the 8th cranial nerve is also involved, which causes hearing loss and vertigo. There may also be anesthesia of the face due to the involvement of the 5th cranial nerve.



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Bell's palsy is an idiopathic, peripheral facial paralysis or paresis of acute onset due to a lower motor neuron lesion of the facial nerve.



Moebius syndrome is a congenital unilateral/bilateral lower motor neuron facial nerve paralysis with abducens palsy causing limited ocular abduction.

Melkersson syndrome is an idiopathic disorder consisting of a triad of:

- Facial paralysis
- Swelling of lips
- Fissured tongue.



Solution to Question 17:

The clinical scenario of an elderly diabetic with severe otalgia, otorrhoea, granulation tissue in the external auditory canal, and lower motor neuron facial nerve palsy indicates malignant otitis externa.

It is termed malignant, even though it is an infection, due to the high mortality rate and poor response to treatment.

Solution to Question 18:

Technetium 99 bone scan remains positive for a year or so and cannot be used to follow up response to treatment. It is particularly useful when the clinical suspicion for malignant otitis externa is high but the CT scan is negative.

Features of malignant otitis externa:

- Pseudomonas infection in the elderly diabetics and those on immunosuppressive drugs.
- Excruciating pain, profuse discharge & granulations on the floor of the external auditory canal.
- Necrosis of bone aids in the spread of infection.
- Infection may spread to the skull base and jugular foramen causing multiple cranial nerve palsies.
- Facial paralysis is the most common nerve palsy seen.
- Mitotic figures will be normal since it is an infective and not a malignant condition.
- Gallium-67 is more useful in the follow-up of the patient.

Solution to Question 19:

The given clinical scenario is suggestive of eczematous otitis externa and the drug implicated is neomycin.

Eczematous otitis externa is due to hypersensitivity to infective organisms or topical ear drops such as chloromycetin or neomycin, etc. The presence of intense irritation, vesicle formation, oozing, and crusting in the canal is observed.

Management involves stopping the implicated ear drops, changing to a different combination and application of steroid cream.

Note - Otitis externa hemorrhagica is a condition characterized by the formation of haemorrhagic bullae on the tympanic membrane and deep meatus. This causes severe pain in the ear and blood-stained discharge when the bullae rupture. It is probably viral in origin.

Solution to Question 20:

Wax or cerumen is not made up of middle ear discharge.

Composition of wax:

- Secretions of ceruminous glands (modified apocrine sweat glands)
- Secretions of the sebaceous gland (fatty secretion)
- Desquamated epithelial cells
- Hair

Wax is normally expelled by the movement of the jaw during eating, talking.

Functions include antibacterial action and trapping of foreign bodies and dirt. Impacted wax causes a sensation of blockage of the ear, reduced hearing, vertigo, tinnitus.

Treatment:

- Softening with ceruminolytics like 2% paradichlorobenzene, 5% sodium bicarbonate in equal parts of glycerine and water.
- Syringing or instrumentation or suction clearance.

Solution to Question 21:

Ceruminous glands are modified apocrine sweat glands.

Ceruminous and sebaceous glands open into the space of the hair follicle in the external auditory canal.

Solution to Question 22:

During syringing, the pinna is pulled upwards and backward and the water jet is directed along the posterosuperior wall of the meatus. This is preferred because of two anatomical reasons:

- On maneuvering the pinna to bring the EAC into a straight configuration, the posterior wall is straighter and easily accessible than the anterior wall.
- The posteroinferior wall of EAC is supplied by the vagus. Direct stimulation of this can lead to sudden cardiac arrest.

Complications of syringing:

- Vertigo– caloric stimulation of lateral semicircular canal if water is too hot or too cold
- Rupture of the tympanic membrane
- Reactivation of quiescent otitis media, if there is pre-existing perforation

Contraindications for syringing:

- Presence of ear infection, tympanic membrane perforation
- Presence of grommet or history of ear surgery

Solution to Question 23:

Keratitis obturans is the collection of a pearly white mass of desquamated epithelial cells in the deep meatus.

The keratin plug is arranged in a lamellar fashion, not at a random pattern.

Solution to Question 24:

Keratitis obturans, by pressure effect, causes absorption of bone which causes widening (not narrowing) of the meatus leading to facial nerve palsy.

Normally, epithelium from the surface of the tympanic membrane migrates onto the posterior meatal wall. Failure of this migration, or obstruction to migration caused by wax, may lead to keratitis obturans which is the accumulation of a large plug of desquamated keratin in the external auditory meatus.

Features of keratitis obturans:

- Severe otalgia
- Conductive hearing loss
- Thickened tympanic membrane
- Keratin plug/pearly white mass occluding the canal
- Widened ear canal- may lead to facial nerve paralysis
- Hyperemia of canal skin, sometimes with ulcerations and granulomas

Staphylococcus aureus is the most frequent isolate.

Keratitis obliterans may be associated with systemic conditions like bronchiectasis and chronic sinusitis.

Solution to Question 25:

The surgery done to widen the cartilaginous part of the external auditory canal is meatoplasty.

Meatoplasty is a surgical procedure that involves the removal of a crescent-shaped portion of conchal cartilage to widen the external auditory meatus. It is performed in combination with all canal wall-down procedures like modified radical and radical mastoidectomies, for easy access to the mastoid cavity for periodic inspection and cleaning.

It is also done as an isolated procedure in elderly people with sagging auricles. A sagging auricle causes obstruction to the meatus leading to the retention of wax and difficulty in hearing.

Other options:

Option B: Tympanoplasty is the surgical reconstruction to correct damage in the middle ear and restore the integrity of the tympanic membrane for improved hearing. It is ossicular reconstruction only or ossicular reconstruction with myringoplasty.

Option C: Myringoplasty refers to the reconstruction of the tympanic membrane only.

Option D: Otoplasty is the correction of external ear/deformities of the pinna.

Solution to Question 26:

Meatoplasty is not a part of canal wall-up (CWU) procedures as the meatus appears normal.

However, meatoplasty is performed in combination with all canal wall down procedures like modified radical and radical mastoidectomies, for the treatment of middle ear disease such as cholesteatoma. The widened meatus helps to easily visualize and access the mastoid cavity for periodic inspection and cleaning.

It is also done in atresia/stenosis to recanalize the ear and in tortuous canals post-tympanoplasty to improve hearing.

Solution to Question 27:

The normal tympanic membrane is concave to the EAC with the maximum concavity at the umbo.

Features of normal TM:

- Pearly grey in color
- Shiny

- Degree of transparency varies
- Mobile with change in middle ear pressures

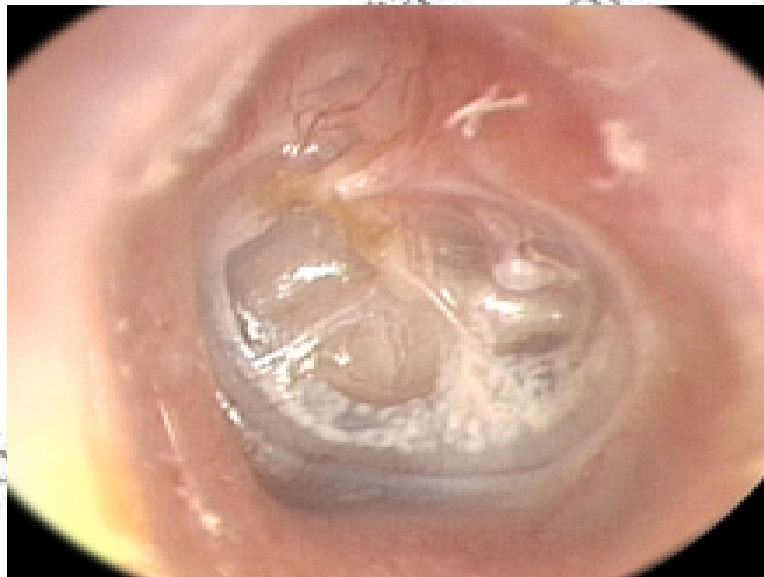
Solution to Question 28:

A retracted TM is dull and lustreless, not shiny and pearly white.

Features of retracted TM:

- The cone of light is absent or interrupted.
- The handle of the malleus appears foreshortened.
- The lateral process of the malleus becomes more prominent.
- Anterior and posterior malleal folds become sickle-shaped.

The image below shows a retracted TM with the above mentioned findings:



Solution to Question 29:

Draping of pars tensa over the incudostapedial joint is classified as Sade's grade II retraction.

Sade classification of retraction of Pars tensa:

Tos classification of retraction of Pars flaccida:

Grade	Description
Grade 1	Mild retraction not touching the long process of the incus

Grade	Description
Grade 2	Retracted drum touching the long process of the incus
Grade 3	Touching promontory but mobile on sealization - atelectasis of the middle ear
Grade 4	Plastered to promontory and immobile - adhesive middle ear

Grade	Description
Grade 1	Mild attic retraction not touching the neck of the malleus.
Grade 2	Touching the neck of the malleus
Grade 3	Limited outer attic wall erosion
Grade 4	Severe outer attic wall erosion

Disorders of Middle Ear

Question 1:

What is the most common route of infection in acute suppurative otitis media?

- a) Hematogenous
- b) Eustachian tube
- c) External ear
- d) Cutaneous lymphatics

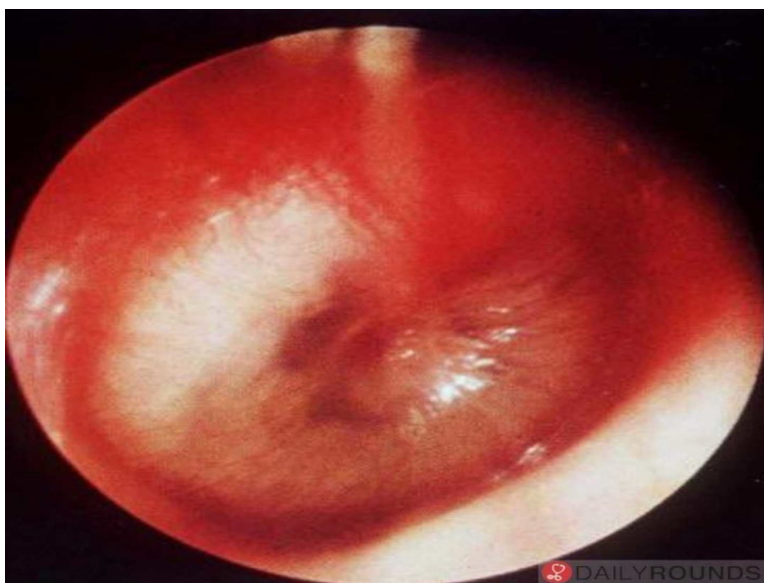
Question 2:

Which of the following conditions does not predispose to eustachian tube obstruction?

- a) Choanal atresia
- b) Adenoid hypertrophy
- c) Allergic rhinitis
- d) Cleft palate

Question 3:

A worried mother brings her febrile infant to the ER. He is crying incessantly and clutching his ear. He had the following finding on otoscopy. What is the most likely organism implicated in this condition?



- a) Beta haemolytic streptococcus
- b) Streptococcus pneumoniae
- c) Haemophilus influenza
- d) Staphylococcus

Question 4:

Otologic examination of a febrile child with otalgia is shown below. What is the stage of acute suppurative otitis media seen here?



- a) Pre-suppurative

- b) Complication
- c) Resolution
- d) Suppuration

Question 5:

Pain is most severe in which stage of acute suppurative otitis media?

- a) Stage of tubal occlusion
- b) Stage of suppuration
- c) Stage of pre-suppuration
- d) Stage of resolution

Question 6:

What is the most common site of perforation of the tympanic membrane in acute suppurative otitis media?

- a) Superior half of pars tensa
- b) Anterior half of pars flaccida
- c) Posterior half of pars tensa
- d) Superior half of pars flaccida

Question 7:

How will you manage a patient who is diagnosed to have acute suppurative otitis media in the pre-suppuration stage?

- a) Antibiotics and decongestants
- b) Myringotomy with antibiotics
- c) Myringotomy with grommet insertion
- d) Antihistaminics and decongestants

Question 8:

A 23-year-old woman presented to the OPD with complaints of excruciating pain in the right ear, deafness and decreased sleep. On examination, she had right-sided LMN facial nerve

paralysis and bulging tympanic membrane on otoscopy. Which of the following best describes the procedure to be done?

- a) Surgical opening in tympanic membrane
- b) Removal of disease and reconstruction of hearing
- c) Surgical opening in eustachian tube
- d) Facial nerve decompression

Question 9:

A 4-year-old boy is brought to the OPD with fever and earache. On examination, the tympanic membrane appears congested and bulging with a loss of landmarks. What is the next line of management?

- a) Myringotomy and grommet insertion
- b) Antibiotics and myringotomy
- c) Antibiotics and decongestants
- d) Tympanoplasty

Question 10:

You are an ENT resident assisting on a myringotomy in a patient with serous otitis media. In which quadrant would you make the incision?

- a) Posteroinferior
- b) Anterosuperior
- c) Anteroinferior
- d) Posterosuperior

Question 11:

In which of the following conditions would you expect to observe the light-house effect?

- a) Otosclerotic lesion
- b) Acute mastoiditis
- c) Intra-tympanic glomus tumor
- d) Chronic suppurative otitis media

Question 12:

A 6-year-old child with measles is brought to the OPD with complaints of profuse otorrhea. Otoscopy is suggestive of acute necrotizing otitis media. What is the most likely causative organism?

- a) Staphylococcus aureus
- b) β - hemolytic streptococcus
- c) Streptococcus pneumoniae
- d) Haemophilus influenzae

Question 13:

What is glue ear also called as?

- a) Acute otitis media
- b) Secretory otitis media
- c) Chronic suppurative otitis media
- d) Recurrent otitis media

Question 14:

What is the most common cause of hearing loss in children?

- a) Acute Otitis Media
- b) Otitis media with effusion
- c) Chronic Suppurative Otitis Media
- d) Prematurity

Question 15:

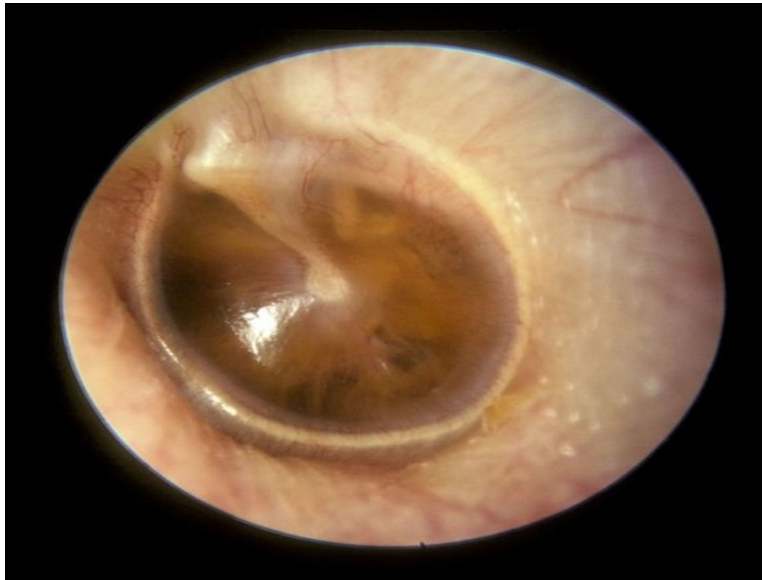
Which of the following statements is false regarding serous otitis media?

- a) Function of the eustachian tube is abnormal in most cases
- b) Adenoid hypertrophy is the most common cause in children causing bilateral serous otitis media
- c) Tumors of nasopharynx is the most common cause in adults causing unilateral serous otitis media

- d) The effusion is thick, viscid and unsterile in most cases

Question 16:

You see a child with delayed development of speech as part of a health visit in a local school. The otoscopic appearance of the ear is shown below. Hearing loss of 20 dB is detected in audiometric tests. What is your diagnosis?



- a) Acute suppurative otitis media
- b) Serous otitis media
- c) Chronic suppurative otitis media
- d) Glomus tumor

Question 17:

What type of tympanogram would you expect to see in a patient with serous otitis media?

- a) A type
- b) Ad type
- c) As type
- d) B type

Question 18:

Which of the following procedures is not used to treat a patient with glue ear?

- a) Myringotomy and aspiration
- b) Grommet insertion
- c) Tympanotomy
- d) Radical mastoidectomy

Question 19:

A child with hearing difficulty is diagnosed to have glue ear. Which of the following is not a possible sequela of this condition?

- a) Middle ear atelectasis
- b) Otosclerosis
- c) Cholesteatoma
- d) Tympanosclerosis

Question 20:

Which of the following is not a feature of tubercular otitis media?

- a) Ear ache
- b) Multiple perforations
- c) Pale granulation
- d) Foul smelling ear discharge

Question 21:

A 42-year-old farmer presented to your clinic after developing painless otorrhoea and hearing loss in his right ear. Examination revealed multiple perforations in the pars tensa in addition to a purulent, foul-smelling discharge with right facial nerve palsy. Audiometry indicated a 35 dB conductive hearing loss. Which of the following does not constitute an essential part of the treatment?

- a) Corticosteroids
- b) Antitubercular therapy
- c) Aural toilet
- d) Mastoid surgery

Answer Key

Question No.	Correct Option
1	b
2	a
3	b
4	a
5	b
6	c
7	a
8	a
9	b
10	c
11	b
12	b
13	b
14	b
15	d
16	b
17	d
18	d
19	b
20	a
21	a

Detailed Explanations

Solution to Question 1:

The eustachian tube is the most common route of infection in acute suppurative otitis media (ASOM).

The infection travels via the lumen of the tube or along subepithelial peritubal lymphatics. It can also spread from the external ear through a traumatic perforation or via the blood but it is rare.

Solution to Question 2:

Choanal atresia does not predispose to eustachian tube obstruction.

Eustachian tube obstruction is caused by:

- Extrinsic causes:
- Adenoid hypertrophy
- Nasopharyngeal tumors
- Nasal pack
- Intrinsic causes:
- Allergic rhinitis
- Rhinosinusitis
- Upper respiratory tract infection
- Defective functioning of the tube:
- Cleft palate
- Down syndrome
- Submucous cleft palate

Eustachian tube obstruction can lead to complications such as:

- Acute suppurative otitis media
- Otitis media with effusion
- Chronic suppurative otitis media
- Atelectasis and ossicular necrosis

Solution to Question 3:

The clinical scenario along with the image showing congested tympanic membrane with purulent effusion and absent cone of light is suggestive of acute suppurative otitis media. The most likely organism implicated is *Streptococcus pneumoniae*.

Organisms causing ASOM in infants and young children:

- *Streptococcus pneumoniae* (most common)
- *Haemophilus influenzae*
- *Moraxella catarrhalis*
- *Streptococcus pyogenes*
- *Staphylococcus aureus*
- *Pseudomonas aeruginosa*

Solution to Question 4:

The clinical picture and image show the stage of pre-suppurative of acute suppurative otitis media (ASOM).

During this stage, due to prolonged tubal occlusion, pyogenic organisms invade the tympanic cavity. An inflammatory exudate appears in the middle ear and there is congestion of pars tensa.

A leash of blood vessels appears along with the handle of malleus and at the periphery of the tympanic membrane imparting it a cart-wheel appearance. Later, the whole of the tympanic membrane including pars flaccida becomes uniformly red.

Symptoms of this stage include marked earache, a high degree of fever, hearing loss, and tinnitus (seen only in adults). Tuning fork tests show the conductive type of hearing loss.

Solution to Question 5:

Pain is most severe in stage of suppuration in acute suppurative otitis media.

Solution to Question 6:

The most common site of perforation of the tympanic membrane in ASOM is the posterior half of pars tensa where usually, a small perforation is seen.

Note: As per Dhingra Diseases of Ear Nose and Throat 7th edition, the perforation most commonly occurs in the anteroinferior quadrant. However, as Scott-Brown's Otorhinolaryngology: Head and Neck Surgery, 7th Edition mentions the most common site as the posterior half of pars tensa, we have picked it as the best option in this question.

Solution to Question 7:

Management of ASOM in the pre-suppurative stage involves the usage of antibiotics and decongestants.

Myringotomy is indicated only when there is an impending rupture of tympanic membrane.

Solution to Question 8:

The above clinical scenario points to a diagnosis of acute suppurative otitis media (ASOM) complicated by right-sided LMN facial nerve palsy. Otoscopic finding of bulging tympanic membrane is an indication to perform myringotomy i.e., surgical opening in the tympanic membrane.

Treatment of facial nerve paralysis secondary to otitis media

- Conservative management is preferred, using antibiotics and corticosteroids.
- Myringotomy and a ventilation tube should be added when spontaneous perforation of the tympanic membrane has not occurred.
- Facial nerve decompression is not usually necessary.

Tympanoplasty is the removal of disease from the middle ear and reconstruction of hearing with or without myringoplasty.

Solution to Question 9:

The scenario points towards a diagnosis of acute suppurative otitis media (ASOM). Antibiotics form the mainstay of treatment in acute suppurative otitis media, and if the tympanic membrane is bulging, myringotomy should be done.

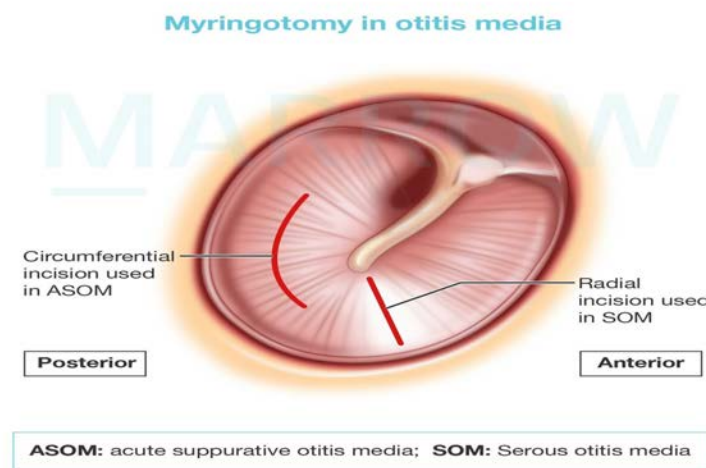
Myringotomy is the procedure involving incising the tympanic membrane to evacuate pus. It is indicated when:

- The tympanic membrane is bulging and there is acute pain.
- There is an incomplete resolution despite antibiotics and the drum remains full with persistent conductive deafness.
- There is persistent effusion beyond 12 weeks.

Solution to Question 10:

In serous otitis media, a radial incision is made in the anteroinferior quadrant of the tympanic membrane.

A curvilinear incision is used in acute suppurative otitis media in the posteroinferior quadrant (smiling incision).



Note: Although an anteroinferior incision is the first step, we resort to an anterosuperior incision in the case of thick secretions that are difficult to aspirate. The 'beer-can' principle supports the said mechanism.

Solution to Question 11:

The light-house effect is seen in acute mastoiditis, which is a complication of otitis media. This effect is described as a pulsatile mucopurulent or purulent discharge, coming through a central pin-hole perforation of pars tensa. This suggests a reservoir of infection in the mastoid and middle ear.

Note: Otoscopic signs in intratympanic glomus tumors include:

- Red reflex through an intact tympanic membrane
- Pulsation sign or Brown sign - when ear canal pressure is raised with Siegel's speculum, the tumor pulsates vigorously and then blanches

Solution to Question 12:

β -hemolytic streptococcus is the most common organism causing acute necrotizing otitis media.

Acute necrotizing otitis media is a variety of acute suppurative otitis media. This is often seen in children suffering from measles, scarlet fever, or influenza.

It leads to the destruction of the entire tympanic membrane, middle ear mucosa, and air cells. There is profuse otorrhoea. In such cases, healing is followed by fibrosis or ingrowth of squamous epithelium from the meatus leading to a secondary acquired cholesteatoma.

Treatment is early antibiotic therapy for at least 7- 10 days. Cortical mastoidectomy may be indicated if medical treatment fails to control or the condition gets complicated by acute mastoiditis.

Solution to Question 13:

Glue ear is also called secretory otitis media. It refers to otitis media with effusion.

Solution to Question 14:

Otitis media with effusion, or glue ear, is the most common cause of hearing loss in children.

Clinical features of glue ear include:

- Bilateral, painless, and fluctuating hearing loss. Conductive hearing loss not more than 40 dB is observed.
- Defective and delayed development of speech.
- Occasional mild earaches are present.

Solution to Question 15:

The effusion is thick, viscid and sterile in most cases of serous otitis media.

Etiology of serous otitis media:

- Malfunctioning of eustachian tube:
- Adenoid hyperplasia in children - causes bilateral serous otitis media
- Benign and malignant tumors of the nasopharynx - should always be excluded in unilateral serous otitis media in an adult
- Cleft lip, cleft palate and chronic tonsillitis - interference with ET opening
- Otitic barotrauma
- Allergy - increased middle ear secretions
- Unresolved ASOM - due to inadequate treatment dosage or duration
- Viral infections - increased secretory activity due to stimulation of middle ear mucosa

Solution to Question 16:

The given clinical scenario and image suggest a diagnosis of serous otitis media.

The golden yellow discoloration of the tympanic membrane is caused by the clear watery straw-colored effusion filling the middle ear. Since the fluid is clear, one can see the structures of the middle ear quite clearly on otoscopy.

Patients having serous otitis media will have hearing loss that rarely exceeds 40 dB. This deafness can pass unnoticed, and maybe accidentally discovered during audiometric screening tests. Because of hearing loss, the development of speech is delayed or defective.

Otoscopic findings in serous otitis media:

- Fluid level and air bubbles are seen when fluid is thin and the tympanic membrane transparent.
- In chronic cases, the fluid becomes very sticky. Hence, the name glue ear.
- The tympanic membrane is often dull and opaque with a loss of light reflex.
- The mobility of the tympanic membrane is restricted.

Solution to Question 17:

B-type tympanogram is expected in a case of serous otitis media.

Here, compliance decreases i.e., mobility of tympanic membrane is restricted due to the effusion, usually secondary to negative middle ear pressure (eustachian tube obstruction). This results in a B-type tympanogram.

Hearing tests suggestive of conductive hearing loss as seen in serous otitis media:

- Rinne negative (BC > AC).
- Weber - lateralized towards the worse ear
- Audiogram - AC abnormal, BC normal, air-bone gap present
- Impedance audiometry - flat curve (reduced compliance, shift to the negative side)

Note: Tympanograph is Flat when there is Fluid in the middle ear.

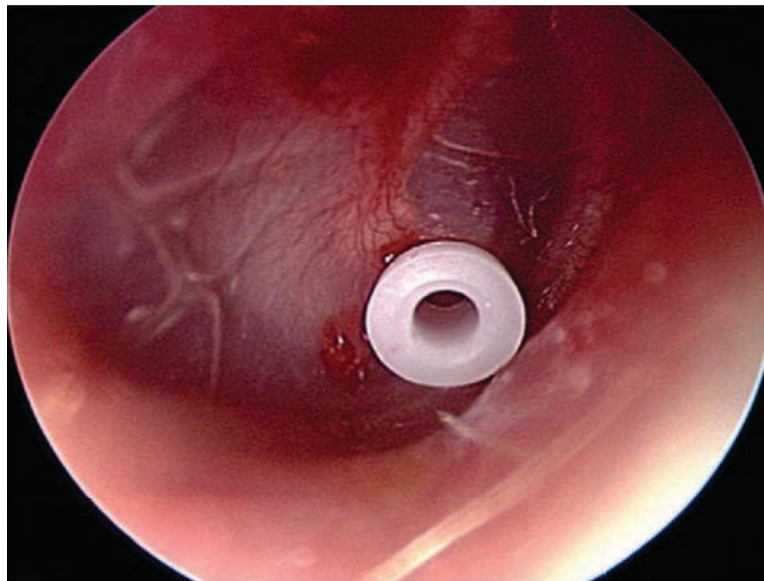
Solution to Question 18:

Serous otitis media can be managed by cortical mastoidectomy. It is sometimes required for removing loculated thick fluid or other associated pathologies such as cholesterol granuloma.

Management of glue ear:

- Medical:
 - Antihistaminics
 - Decongestants
 - Antibiotics
 - Valsalva maneuver.
- Surgical:
 - Myringotomy and aspiration
 - Grommet insertion
 - Tympanotomy
 - Cortical mastoidectomy

A grommet is placed in the middle ear to aerate and drain the middle ear. It acts as a substitute for the blocked eustachian tube. It is placed through radial myringotomy in the anteroinferior quadrant. The device is shown in the image below:



Solution to Question 19:

Otosclerosis is not a possible sequela of glue ear. Otosclerosis is a hereditary disease of the bone of the inner ear, which mostly affects the stapes footplate area. It is not related to other ear pathologies.

The sequelae of long-standing glue ear include:

- Atrophy of tympanic membrane (TM) and myringosclerosis (fibrosis of TM)
- Atelectasis of middle ear - when the tympanic membrane gets retracted over the promontory
- Ossicular necrosis
- Tympanosclerosis - collagen deposits within the middle ear
- Retraction pockets in TM and cholesteatoma
- Cholesterol granulomas - due to stasis of secretions in the mastoid and middle ear

Solution to Question 20:

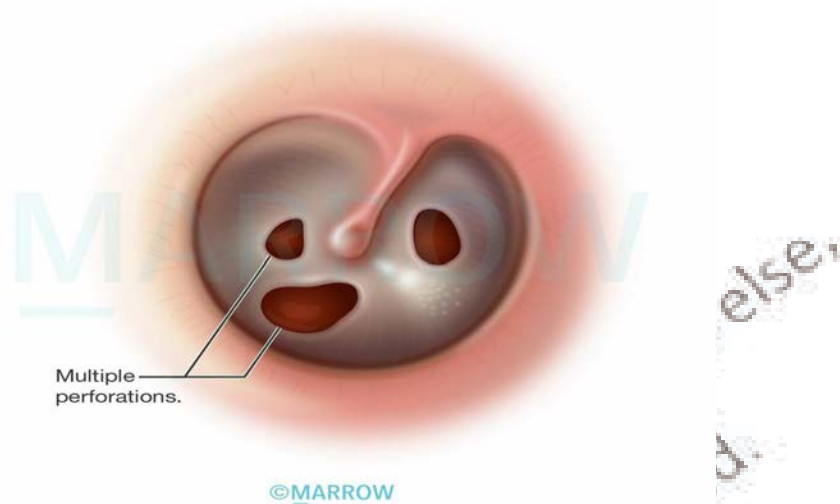
Earache is not a feature of tubercular otitis media. It is a painless condition.

Tuberculosis of the ear is usually secondary to pulmonary tuberculosis. It occasionally spreads hematogenously, from a tubercular focus in the lungs, tonsils, and cervical or mesenteric lymph nodes

It presents as a painless condition with foul-smelling discharge from the ear, that does not respond to standard antibiotic treatment. Patients have severe hearing loss that is out of proportion to symptoms. Although predominantly conductive, sensorineural hearing loss may also occur if the labyrinth is involved.

A thickened tympanic membrane with multiple perforations in pars tensa of the tympanic membrane and pale granulations in the middle ear cleft is seen.

The image below shows the tympanic membrane in tuberculous otitis media:



Histopathology shows submucosal tubercles with Langhans giant cells and caseous necrosis. Smears and culture of ear discharge may be positive for acid-fast bacilli (AFB). Evidence of primary pulmonary tuberculosis may also be seen on chest radiography.

Facial nerve palsy is a common complication. Mastoiditis, postauricular fistula, and osteomyelitis are also seen.

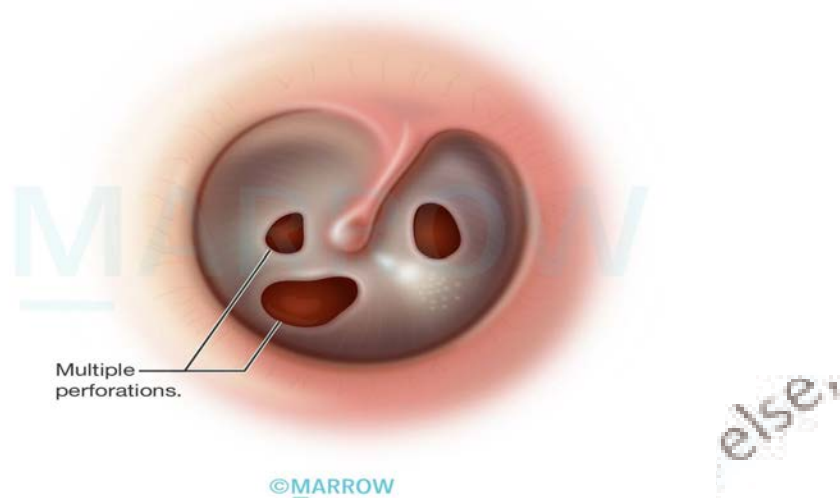
Treatment includes aural toilet, treatment of secondary bacterial infections, and a full course of antitubercular therapy followed by reconstructive surgery.

Solution to Question 21:

The clinical stem consisting of painless otorrhoea, conductive hearing loss with multiple perforations in the pars tensa, and right facial nerve palsy point to a diagnosis of tuberculous otitis media, for which corticosteroids do not form an essential part of the treatment.

Tuberculosis of the ear usually occurs secondary to pulmonary tuberculosis. It is occasionally blood-borne, spreading from a tubercular focus in the lungs, tonsils, cervical or mesenteric lymph nodes. Tuberculous otitis media is mostly seen in children and young adults.

It presents as a painless condition with foul-smelling discharge from the ear, that does not respond to standard antibiotic treatment. A thickened tympanic membrane with multiple perforations in pars tensa of the tympanic membrane and pale granulations in the middle ear cleft is seen.



Facial nerve palsy is the most common complication. Mastoiditis, postauricular fistula, osteomyelitis are also seen.

Histopathology shows submucosal tubercles with Langhans giant cells and caseation necrosis. Smears and culture of ear discharge may be positive for AFB. Chest X-ray features of tuberculosis along with other evidence of tuberculosis in the body help in confirmation of the diagnosis. DNA probe and PCR from the ear discharge can provide an early diagnosis.

Treatment includes:

- Antitubercular therapy
- Aural toilet
- Treatment of secondary infection
- Mastoid surgery for complications.

Reconstructive middle ear surgery is delayed till antitubercular therapy has been completed.

Cholesteatoma and Types of CSOM

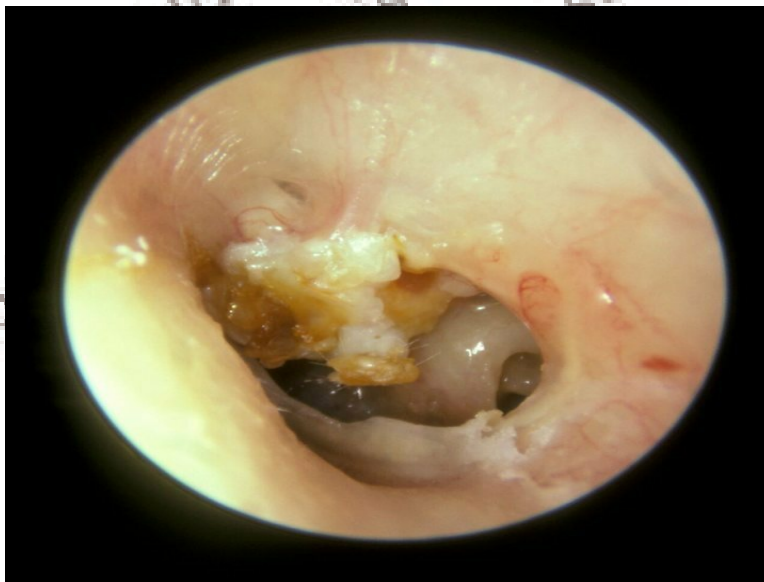
Question 1:

Which of the following epithelia does not line the middle ear?

- a) Ciliated columnar
- b) Keratinizing squamous
- c) Cuboidal
- d) Pavement-like

Question 2:

Otoscopic image of a patient with a history of repeated middle ear infections is shown below. Which one of the following statements is true regarding this condition?



- a) It is a benign tumor
- b) It contains cholesterol crystals
- c) It is also known as epidermosis
- d) It has lymphatic permeation

Question 3:

You see a patient with a history of painless foul-smelling ear discharge and loss of hearing for the past year. He had undergone tympanoplasty on the same ear 3 years back. What is the most likely diagnosis?

- a) Primary acquired cholesteatoma
- b) Tertiary acquired cholesteatoma
- c) Secondary acquired cholesteatoma
- d) Congenital cholesteatoma

Question 4:

Which of the following theories does not explain the origin of cholesteatoma?

- a) Wittmaack's theory of retraction pocket
- b) Ruedi's theory of basal cell hyperplasia
- c) Habermann's theory of epithelial invasion
- d) von Bekesy theory of mucosal metaplasia

Question 5:

Which of the following is the most accepted theory for secondary cholesteatoma?

- a) Presence of congenital cell rests
- b) Basal cell hyperplasia
- c) Squamous metaplasia
- d) Migration of squamous epithelium

Question 6:

What is the most common site for primary cholesteatoma?

- a) Mastoid antrum
- b) Sinus tympani
- c) Facial Recess
- d) Prussack's space

Question 7:

An 8-year-old boy was referred to an otolaryngologist for subjective left-sided hearing loss. A hearing test revealed mild conductive hearing loss at low and middle frequencies. Otoendoscopy showed a mass with the appearance as shown below. Which of the following criteria is used to diagnose this condition?



- a) AAO-HNS criteria
- b) Levenson criteria
- c) Bent and Kuhn criteria
- d) Taverner criteria

Question 8:

Which of the following statements is false regarding the expansion of cholesteatoma and destruction of bone?

- a) Release of enzymes like collagenase and acid phosphatase at the site of destruction
- b) Destruction of bone mediated by pressure necrosis
- c) Osteoclasts and mononuclear infiltrates play a role in bone destruction
- d) Erosion of ossicles is more common in children compared to adults

Question 9:

A patient presented to your OPD for a second look surgery following a canal wall-up procedure for cholesteatoma removal. Where is the residual disease most likely to be seen?

- a) Facial recess
- b) Hypotympanum
- c) Sinus tympani
- d) Prussack's space

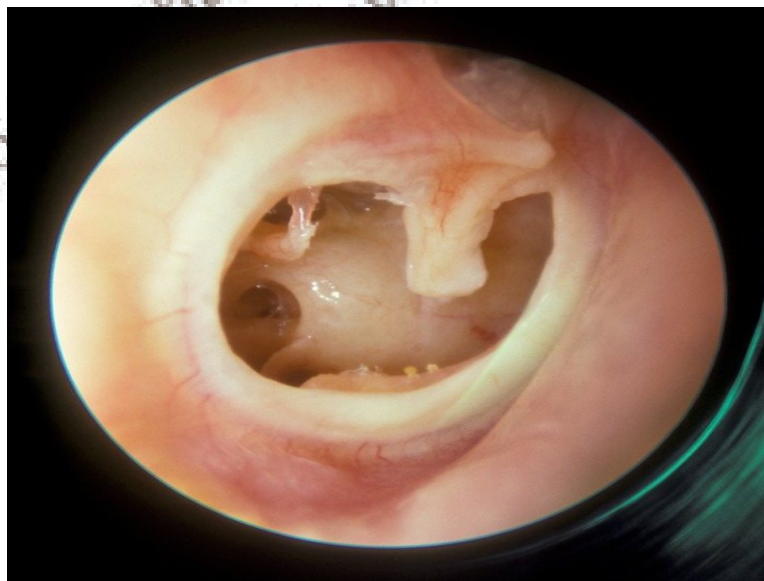
Question 10:

Patients with tubotympanic type of chronic suppurative otitis media are considered safe compared to atticoantral disease. What is the reason?

- a) Does not progress into malignancy usually
- b) Medical treatment can be tried
- c) Does not involve the mastoid air cells
- d) Rarely lead to complications

Question 11:

Which of the following best describes the given otoscopy finding in a child with poor scholastic performance?



- a) Large central perforation
- b) Subtotal perforation
- c) Total perforation
- d) Attic perforation

Question 12:

A young adult presents with intermittent non-foul smelling mucoid discharge from the left ear for the past few years. Otologic examination revealed a central perforation of the tympanic membrane. Which of the following statements is false regarding his condition?

- a) No evidence of bone destruction in CT scans
- b) Mixed aerobic and anaerobic infections are seen
- c) Aural polyp if present, should never be avulsed
- d) Often associated with cholesteatoma

Question 13:

A known case of tubotympanic disease was taken up for surgery and on examining the middle ear, few chalky white deposits were found between the ossicles. What is the diagnosis?

- a) Otosclerosis
- b) Tympanosclerosis
- c) Myringosclerosis
- d) Candidal infection of the middle ear

Question 14:

A first-year ENT postgraduate sees a child with permanent perforation of the tympanic membrane. Which of the following is a feature of such perforation?

- a) It is seen in acute suppurative otitis media
- b) It is caused by necrotizing otitis externa
- c) It has squamous epithelium at its edges
- d) Medical management can be tried initially

Question 15:

A young adult presents with a 3-year history of intermittent ear discharge. He complains of poor hearing, which improves during discharge periods, but is worse when the ear is dry. Which of the following explains this phenomenon?

- a) Bone conduction
- b) Cholesteatoma hearer
- c) Round window shielding effect
- d) Patient is malingering

Question 16:

A patient with chronic suppurative otitis media is posted for surgery. Preoperatively, the surgeon wants to assess the status of the ossicles so that he can be prepared for ossicular construction if required. Which of the following tests is useful here?

- a) Nine-step test
- b) Side-step test
- c) Patch test
- d) Unterberger test

Question 17:

Which of the following is true regarding the tubotympanic type of chronic suppurative otitis media?

- a) Central perforation in antero-inferior quadrant is commonly seen
- b) Blood stained ear discharge is seen
- c) The discharge is scanty, purulent and foul smelling
- d) Tinnitus is a common symptom

Question 18:

Which of the following is true about the atticotympanic type of chronic suppurative otitis media?

- a) Most commonly involves the protympanum
- b) Polyps are rarely present
- c) The discharge is usually profuse
- d) Involvement of Shrapnell's membrane is seen

Answer Key

Question No.	Correct Option
1	b
2	c
3	c
4	d
5	d
6	d
7	b
8	b
9	c
10	d
11	b
12	d
13	b
14	c
15	c
16	c
17	a
18	d

Detailed Explanations

Solution to Question 1:

The middle ear is not lined by keratinizing squamous epithelium, and hence its presence anywhere within the middle ear cleft is considered cholesteatoma.

Various kinds of epithelia lining the middle ear are:

- Ciliated columnar- around the eustachian tube opening (Protympanum)
- Pavement like- attic/epitympanum
- Cuboidal- rest

Solution to Question 2:

The image shows the presence of a large mass of white keratin debris in the left upper quadrant of the left tympanic membrane. This along with clinical scenario is suggestive of cholesteatoma which is also known as epidermosis or keratoma

Cholesteatoma is the presence of stratified squamous epithelium in middle ear where it is not usually present, i.e., skin in wrong place.

A central white mass, consisting of keratin debris is produced by the matrix. For this reason, it has also been named epidermosis or keratoma.

The term cholesteatoma is a misnomer because it neither contains cholesterol crystals nor it is a tumor to merit the suffix "oma."

Solution to Question 3:

The given clinical scenario suggests a diagnosis of secondary acquired cholesteatoma. The history of middle ear infection and hearing loss after surgery of the tympanic membrane makes the diagnosis more likely.

Classification of cholesteatoma:

1. Congenital cholesteatoma is that which has its nidus of trapped squamous epithelium present at birth.
2. Primary acquired cholesteatoma presents with no history of previous otitis media or a pre-existing perforation. It is associated with a defect in pars flaccida.
3. Secondary acquired cholesteatoma presents with a history of previous otitis media or a pre-existing perforation. It is associated with a defect in pars tensa.

Solution to Question 4:

von Békésy proposed the traveling wave theory, whereas Sade's metaplasia theory explains the origin of cholesteatoma.

Cholesteatoma can be of 2 main types:

- Congenital cholesteatoma - explained by the presence of congenital cell rests within the middle ear and other parts of the temporal bone.
- Acquired cholesteatoma - explained by a pathology either within the mastoid air cell system and eustachian tube drainage, or the tympanic membrane epithelium.

Solution to Question 5:

The most accepted theory for secondary cholesteatoma is Habermann's theory. It proposes that the skin of the external auditory meatus grows or migrates into the middle ear through the posterosuperior marginal perforation or attic perforation forming cholesteatoma.

Solution to Question 6:

Prussak's space is the most common site of primary cholesteatoma. It is the space in the epitympanum lying in between pars flaccida and the neck of the malleus.

Solution to Question 7:

The given clinical scenario with the otoendoscopy showing a white pearly mass behind the intact eardrum in the anterosuperior quadrant is suggestive of congenital cholesteatoma. Levenson criteria should be satisfied in order to diagnose this condition.

Levenson criteria:

- A white mass behind the normal tympanic membrane
- Normal pars tensa and pars flaccida
- No prior history of otorrhoea
- No prior history of otological procedure
- Episodes of otitis media with effusion should not be considered as exclusion criteria.

Another popular guideline used is Derlaci and Clemis which includes all the above except for the 4th criteria.

Solution to Question 8:

Cholesteatoma does not cause bone destruction by pressure necrosis.

Bone destruction by cholesteatoma has been attributed to enzymes liberated by osteoclasts and mononuclear inflammatory cells such as:

- Collagenase
- Acid phosphatase and
- Proteolytic enzymes.

Pediatric cholesteatoma (PC) has more inflammation and greater chances of ossicular erosion compared to adults. However, PC has lesser chances of petrous labyrinth invasion and lateral semicircular canal fistula formation.

Solution to Question 9:

Residual cholesteatoma is most likely to be seen in the sinus tympani.

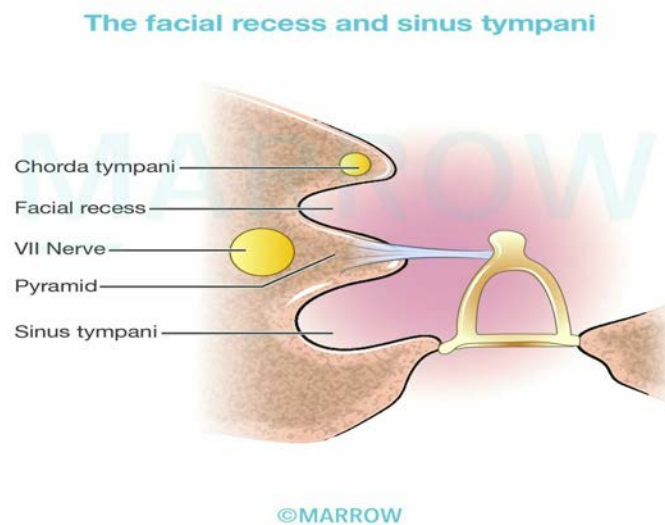
The involvement of sinus tympani, along with posterior mesotympanum, is a recognized risk factor for residual cholesteatoma. This is because it is particularly difficult to remove the disease from these areas. Canal wall-up procedures have a higher risk of leaving residual disease behind than the canal wall-down procedures.

Sinus tympani or infra pyramidal recess is the medial of the two posterior spaces in the middle ear.

It is bounded by:

- Pyramid and vertical segment of facial nerve laterally
- The stapes & oval window medially
- Ponticulus superiorly
- Subiculum inferiorly

Image showing sinus tympani:



Solution to Question 10:

Tubotympanic type (TTD) of CSOM is considered the safe type in relative to atticoantral disease (AAD) as TTD has less chance of complications than AAD.

Note: TTD can also lead to complications but the incidence is much less than that in AAD.

Solution to Question 11:

The given otoscopic image shows a subtotal perforation. It involves all four quadrants of the tympanic membrane but is covered by remnant pars tensa/annulus all around.

Chronic suppurative otitis media (CSOM) is the single most important cause of hearing impairment in rural populations. Children with CSOM have poor academic performance.

Solution to Question 12:

The clinical scenario of intermittent mucoid discharge and central perforation of the tympanic membrane points to a diagnosis of tubotympanic type of CSOM, which is not associated with cholesteatoma. Atticoantral type is associated with cholesteatoma and it leads to dangerous complications due to its bone-eroding properties.

Etiology of tubotympanic (benign or safe) type of CSOM:

- As a sequela of acute otitis media
- Ascending infection via the eustachian tube
- Allergic conditions

Bone destruction is not present in this type.

Culture reveals mixed aerobic and anaerobic infection.

Aural polyp if present should be removed and not avulsed before starting local treatment. This is due to the fact that the polyp may be arising from the stapes, facial nerve, or horizontal canal and its avulsion can lead to facial paralysis or labyrinthitis.

Solution to Question 13:

The given clinical scenario is suggestive of tympanosclerosis. It is a sequela of long-standing middle ear disease where there are hyaline deposits within the middle ear, which calcify leading to impaired mobility of ossicles.

The common sites are the tympanic membrane, promontory, ossicles, joints, tendons, and oval and round windows.

When hyalinization and calcification involve only the tympanic membrane, it is known as myringosclerosis.

Solution to Question 14:

A permanent perforation is lined by squamous epithelium at its edges, hindering spontaneous healing.

Features of a permanent perforation:

- Seen in chronic suppurative otitis media (CSOM)
- Is similar to an epithelial lined fistulous tract
- Cannot be medically managed, thus mandating surgery

Solution to Question 15:

The given scenario describes the round window-shielding effect that is seen in tubotympanic type of CSOM.

The round window-shielding effect is where the ear discharge shields the round window from sound waves. During the inactive phase (no discharge), the sound strikes both the oval window and round window together and the noise cancels itself out. So, the hearing is worse when the ear is dry.

Cholesteatoma hearer is a patient with a discontinuous ossicular chain but the sound is conducted through the cholesteatoma to the footplate. On removing the cholesteatoma by surgery, the patient has worse hearing.

Solution to Question 16:

The patch test is useful in assessing the status of the ossicles preoperatively so that the surgeon can be prepared for ossicular reconstruction if required. It is done in tubotympanic type of CSOM.

A thin paper-like cigarette foil is soaked in vaseline and cut into the size of perforation and is used to close the perforation. Pure tone audiometry is done before and after the closure of perforation. Results are as follows:

- Improved hearing – intact ossicular chain
- Decreased hearing – ossicular fixity or discontinuity
- No improvement – technical fault or improper hearing.

Another test done in the tubotympanic type of CSOM is Bonsin's test. When the eardrum is painted with 1 in 20% solution of silver nitrate, the perforation stands out as a black spot against a white background.

Option A: Nine-step test is used to predict the ears at risk of middle ear barotrauma.

Options B and D: The side-step test and Unterberger test are vestibular tests and are positive in perilymphatic fistula.

Solution to Question 17:

Central perforation in the antero-inferior quadrant of tympanic membrane is commonly seen in tubotympanic type, while marginal perforation in the postero-superior quadrant is commonly seen in atticoantral type of CSOM.

Bloody, scant, foul-smelling ear discharge, ear pain, hearing loss, and tinnitus are often seen in the atticoantral type of CSOM.

Solution to Question 18:

Atticoantral type also called unsafe or dangerous type involves perforation of Shrapnell's membrane or pars flaccida (attic perforation).

The disease is often associated with a bone-eroding process such as cholesteatoma, granulations, or osteitis. The risk of complications is high in this variety.

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CSOM - Treatment and Complications

Question 1:

Which of the following is the most common surgery done for tubotympanic type of chronic suppurative otitis media?

- a) Simple mastoidectomy
- b) Radical mastoidectomy
- c) Modified radical mastoidectomy
- d) Tympanoplasty

Question 2:

As a first-year ENT resident, you are preparing a patient for tympanoplasty. Which of the following would not be in your pre-operative checklist?

- a) Pure tone audiometry
- b) Hemoglobin estimation
- c) CT scan of temporal bone
- d) Informed consent

Question 3:

Wullstein's classification is used for which of the following procedures?

- a) Tympanoplasty
- b) Ossiculoplasty
- c) Myringoplasty
- d) Meatoplasty

Question 4:

A 30-year-old man presents with complaints of painless, odourless and profuse discharge from the right ear. An otoscopic image of the affected ear is given below. Which of the following would not be a part of the management?



- a) Aural toilet
- b) Topical and systemic antibiotics
- c) Tympanoplasty
- d) Canal wall down mastoidectomy

Question 5:

Which of the following is the most common graft material used for tympanoplasty?

- a) Temporalis fascia
- b) Dura mater
- c) Periosteum
- d) Vein

Question 6:

A patient having chronic otitis media with perforation of pars tensa is scheduled for myringoplasty to restore hearing. Which of the following is true regarding the techniques used here?

- a) In underlay technique, graft is placed medial to annulus
- b) In underlay technique, graft is placed lateral to malleus
- c) In overlay technique, graft is placed medial to annulus
- d) In overlay technique, graft is placed lateral to malleus

Question 7:

A patient with mucosal disease of the middle ear is set to undergo myringoplasty. Which of the following incisions is used here?

- a) Killian's incision
- b) Freer's incision
- c) Wilde's incision
- d) Gluck–Sorenson's incision

Question 8:

Austin's classification for anatomical defects in the ossicular chain is based on which of the following structures?

- a) Malleus head and stapes suprastructure
- b) Malleus head and stapes footplate
- c) Malleus handle and stapes footplate
- d) Malleus handle and stapes suprastructure

Question 9:

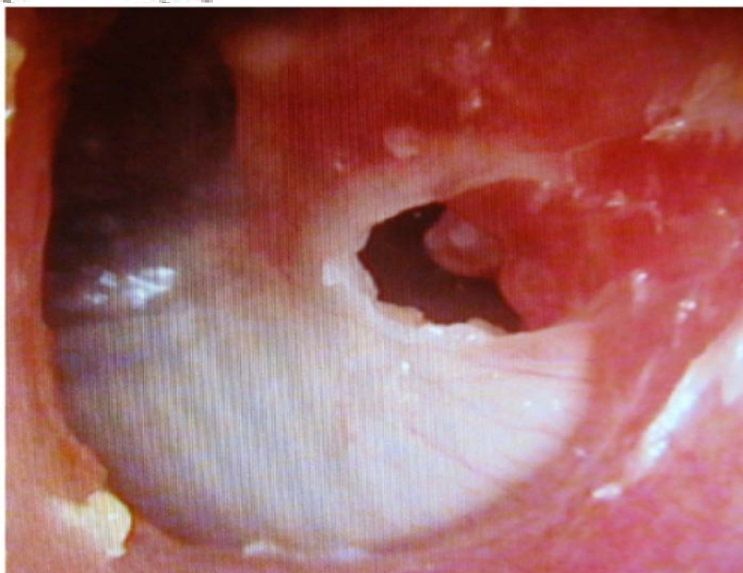
A patient presents to your OPD with hearing difficulty and scanty, foul-smelling discharge from the left ear. Otologic examination reveals the following finding. What is the treatment of choice in this condition?



- a) Myringotomy
- b) Mastoidectomy
- c) Tympanoplasty
- d) Myringoplasty

Question 10:

A 7-year-old child presents with scanty, foul-smelling ear discharge. Otologic examination shows the following finding. Which of the following is an appropriate management plan?



- a) Topical antibiotics with decongestants

- b) IV antibiotics and follow-up
- c) Tympanoplasty
- d) Tympanomastoid exploration

Question 11:

What would be the primary aim of surgery in a patient presenting with the following otoscopic finding?



- a) To restore hearing
- b) To clear the disease
- c) To reconstruct middle ear
- d) To close the perforation

Question 12:

A 12-year-old child with chronic suppurative otitis media and cholesteatoma is planned for a mastoidectomy. He is part of the school swimming club and wants to pursue the sport. Which of the following would you mention while counselling the parents regarding canal wall up (CWU) and canal wall down (CWD) mastoidectomies?

- a) Incidence of residual disease is very low not requiring second look surgery in CWU procedure
- b) Easy to wear a hearing aid if need in CWD procedure
- c) Post surgery healing is rapid and periodic cleaning is unnecessary in CWU procedure
- d) The patient is allowed to swim after CWD procedure

Question 13:

Which of the following is not seen in a case of otitis media?

- a) Bezold abscess
- b) Griesinger's sign
- c) Battle sign
- d) Delta sign

Question 14:

A patient complains of severe pain in his right ear associated with high-grade fever and profuse ear discharge. Clinical examination reveals mastoid tenderness and perforation of the tympanic membrane. If this infection were to burst out through the medial aspect of the tip of the mastoid, which of the following conditions would ensue?

- a) Zygomatic abscess
- b) Bezold abscess
- c) Luc abscess
- d) Citelli's abscess

Question 15:

A 30-year-old woman presented with a history of ear discharge in the left ear for the past 4 weeks, deep-seated ear pain for the past 1 week, and diplopia for the past 2 days. What is the most probable diagnosis in this scenario?

- a) Burst Luc's abscess
- b) Nasopharyngeal carcinoma
- c) Schwannoma of petrous apex
- d) Petrous apicitis

Question 16:

In which of the following conditions would you expect to see Griesinger's sign?

- a) Petrositis

- b) Meningitis
- c) Sigmoid sinus thrombosis
- d) Cerebellar abscess

Question 17:

A patient with a history of foul-smelling ear discharge, for which he did not complete treatment now presents with complaints of fever with chills and swelling over the mastoid. On examination, there is tenderness along the jugular vein. What is the diagnosis?

- a) Petrous apicitis
- b) Labyrinthitis
- c) Lateral sinus thrombophlebitis
- d) Acute mastoiditis

Question 18:

A patient with hearing loss has the following otoscopic finding. What is the treatment modality of choice?



- a) Medical management
- b) Surgery
- c) Radiotherapy
- d) Chemotherapy

Answer Key

Question No.	Correct Option
1	d
2	c
3	a
4	d
5	a
6	a
7	c
8	d
9	b
10	d
11	b
12	c
13	c
14	b
15	d
16	c
17	c
18	b

Detailed Explanations

Solution to Question 1:

Tympanoplasty is the most common surgery done in the tubotympanic type of CSOM.

It involves ossiculoplasty (the eradication of the disease from the middle ear and repair of ossicles) with or without myringoplasty (repair of perforation of the tympanic membrane, which is done to check repeated infections from external ear).

Solution to Question 2:

CT scan of the temporal bone is not required in tympanoplasty procedures. It is required only in cases of mastoidectomy.

For tympanoplasty, routine blood investigations, pure tone audiometry, and an X-ray of the mastoid are sufficient.

Solution to Question 3:

Zollner and Wullstein's classification is used to classify tympanoplasty.

Solution to Question 4:

The given clinical scenario and image showing a medium central perforation of the pars tensa is suggestive of tubo-tympanic type (TTD) of chronic suppurative otitis media (CSOM). Canal wall down mastoidectomy is not a part of the management plan.

This is only performed in attico-antral type and in cases of tubo-tympanic type where a reservoir of infection in the middle ear is suspected.

Management of tubo-tympanic type of CSOM:

- Medical management:
 - Aural toilet
 - Topical antibiotics with steroid drops
 - Systemic antibiotics
 - Treatment of causative factors.
- Surgical management:
 - It is done once the ear is dry for 6 weeks without antibiotics.
 - Tympanoplasty (ossiculoplasty with or without myringoplasty).

Solution to Question 5:

Temporalis fascia is the most common graft material used for tympanoplasty

Other graft materials used are:

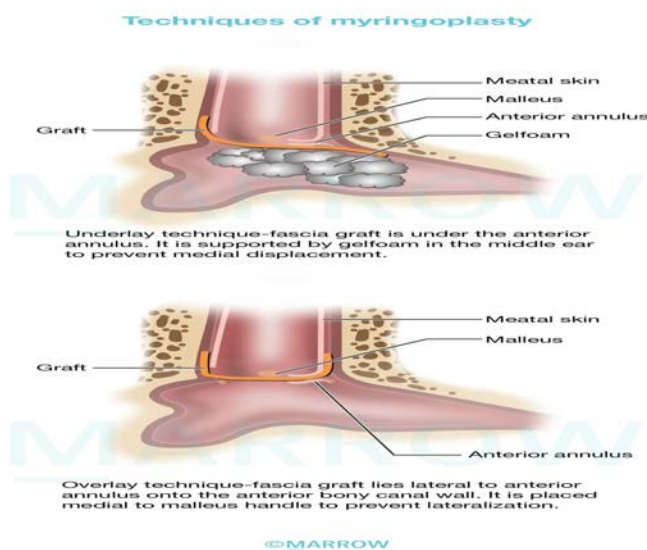
- The areolar fascia overlying the temporal fascia
- Perichondrium from the tragus
- Conchal cartilage
- Vein
- Periosteum

Dura mater was used earlier but then discontinued due to the risk of prion diseases like Cruetzfeldt Jacob disease.

Solution to Question 6:

In the underlay technique, the graft is placed medial to the annulus. Whereas, in the overlay technique, the graft is placed lateral to the annulus. The graft is medial to malleus in both techniques.

Myringoplasty techniques are based on the relation of the graft with lamina propria of tympanic membrane remnants and fibrous annulus.



Solution to Question 7:

Wilde's (or post-aural) incision is used in myringoplasty.

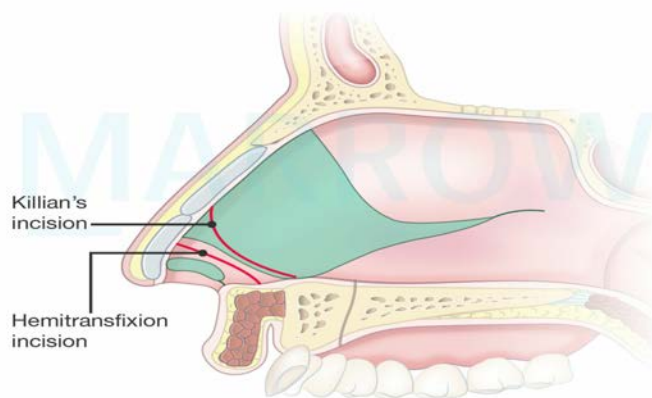
Wilde's incision is used in the following:

- Cortical mastoidectomy
- Modified radical and radical mastoidectomy
- Tympanoplasty - when perforation extends anterior to handle of malleus
- Exposure of CN VII in the vertical segment
- Surgery of endolymphatic sac.

Option A: Killian's incision is used in submucous resection surgery for deviated nasal septum. In cases of a deviated septum, a slightly curvilinear incision is made, 2–3 mm behind the caudal end of septal cartilage at the mucocutaneous junction, always on the left side.

Option B: Freer's hemitransfixion incision is used in septoplasty done for deviated nasal septum. In case of caudal dislocation, a transfixion or hemitransfixion incision is made. It is a septocolumellar incision between the caudal end of septal cartilage and columella on the concave side.

Killian's and Hemitransfixion incision



Option D: Gluck–Sorenson's incision used for laryngectomy with radical neck dissection.

Solution to Question 8:

Austin's classification of anatomical defects in the ossicular chain is based on the presence or absence of handle of malleus and stapes suprastructure.

Incus is the most common bone to be eroded and assuming it to be absent in most cases, the remaining 2 ossicles, i.e., malleus and stapes are considered for the classification.

Solution to Question 9:

The given clinical scenario describes a case of atticoantral type of CSOM and modified radical mastoidectomy is the treatment of choice here.

Option A: Myringotomy is done in cases of acute suppurative otitis media (ASOM) and serous otitis media.

Options C and D: Medical management followed by myringoplasty or tympanoplasty once the ear is dry for 6 weeks without antibiotics is the treatment of choice for tubotympanic type of CSOM.

Solution to Question 10:

The given clinical scenario along with the image showing marginal perforation is suggestive of atticotympanic type of chronic suppurative otitis media (CSOM), and the appropriate management plan is surgical intervention with tympanomastoid exploration.

Solution to Question 11:

The given image shows a white mass located in the anterosuperior quadrant behind the intact eardrum suggestive of cholesteatoma. The primary aim is to clear the disease and render the ear safe. The second priority is to preserve or reconstruct the hearing, but never at the cost of the primary aim.

Solution to Question 12:

Post-surgery healing is rapid and periodic cleaning is unnecessary in case of canal wall up procedure (CWU). This procedure has a low predisposition to infection but carries the risk of leaving behind residual disease requiring the need for second-look surgery.

Canal wall up procedure (CWU)	Canal wall down procedure (CWD)
Meatus appears normal	Meatus opening is wide communication with mastoid
Does not require routine cleaning	Cleaning once or twice a year required.
High rate of recurrent or residual cholesteatoma	Low rate of recurrence
Requires second-look surgery after 6 months	Not required
The patient is allowed to swim. No limitation.	Swimming can lead to infection of the mastoid cavity
Easy to wear a hearing aid if needed	Problems in fitting a hearing aid due to large meatus and mastoid cavity

Solution to Question 13:

Battle sign refers to retro auricular or mastoid ecchymosis that typically results from a fracture of the middle cranial fossa following head trauma. It is prominent when there is a fracture of the temporal bone. Battle sign is not seen in the case of otitis media.

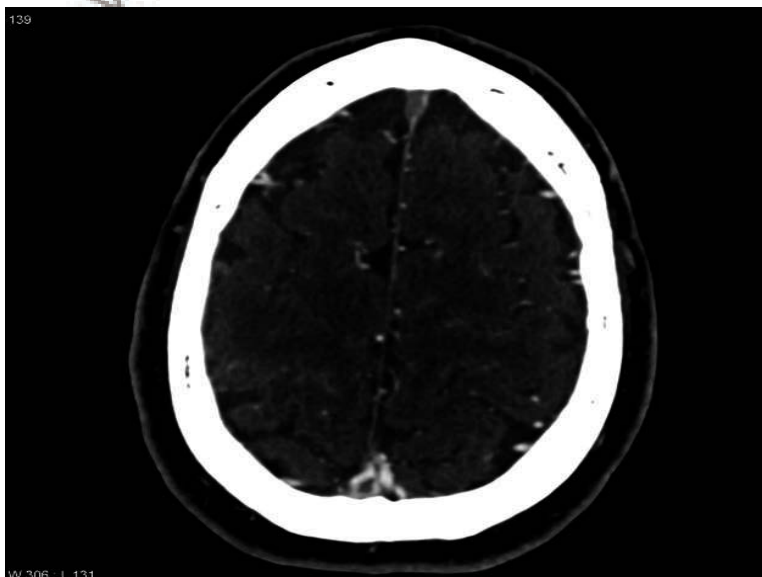


Option A: Bezold abscess is a complication of acute otomastoiditis where the infection/pus breaks through the thin medial side of the tip of the mastoid and presents as a swelling in the upper part of the neck.

Option B: Griesinger sign refers to edema of the postauricular soft tissues overlying the mastoid process as a result of thrombosis of the mastoid emissary vein seen in sigmoid sinus thrombosis. It is a complication of acute otomastoiditis.

Option D: Delta sign is a contrast-enhanced CT finding seen in sagittal sinus thrombosis, which can occur as a complication of acute mastoiditis or chronic suppurative otitis media. It is a triangular area with rim enhancement and the central low-density area is seen in the posterior cranial fossa on axial cuts. MR venography is useful to assess the progression or resolution of a thrombus.

The image below depicts a filling defect (thrombus) outlined by contrast medium in the sagittal sinus.



Solution to Question 14:

The clinical scenario of severe pain in his right ear associated with high-grade fever, profuse ear discharge, mastoid tenderness, and perforation of the tympanic membrane is suggestive of acute mastoiditis, which has most likely occurred as a complication of acute infection of the middle ear. If this infection is to burst out through the medial side of the tip of the mastoid, it results in the formation of Bezold abscess.

Acute mastoiditis is used to refer to an acute infection that spreads from the mucosa, lining the mastoid air cells, to involve the bony walls of the mastoid air cell system.

The image below shows a picture of acute mastoiditis:



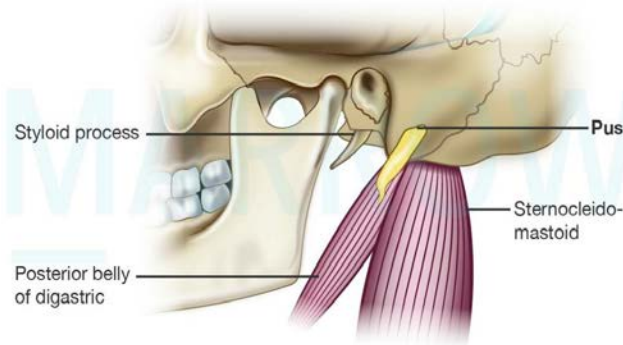
Blood leukocytes and ESR is raised. CT scan of the temporal bone shows clouding of air cells due to the collection of exudate in them. In later stages, a cavity may be seen in the mastoid. Treatment is done by administration of intravenous antibiotics and if indicated, surgical procedures such as myringotomy and cortical mastoidectomy.

Abscesses in Relation to mastoid infection are:

- Bezold abscess lies deep to the sternocleidomastoid and presents as a swelling in the tip of the mastoid or the angle of the jaw and is seen in the upper part of the posterior triangle or the parapharyngeal space

The images below show a picture of Bezold abscess:

Bezold abscess

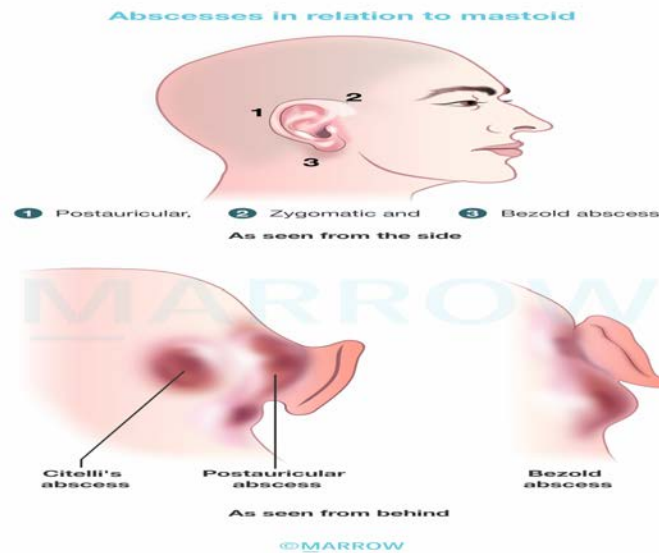


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Other abscesses in relation to mastoid infection:

- Post auricular abscess - it is the most common abscess that forms over the mastoid. In infants and children, this presents over the MacEwen's triangle.
- Zygomatic abscess - infection of the zygomatic air cells. It is situated at the posterior root of the zygoma
- Luc abscess (meatal abscess) - the pus breaks through the bony wall between the mastoid antrum and external auditory meatus
- Citelli's abscess - the abscess is formed posterior to the mastoid, occupying a position more towards the occipital bone



Solution to Question 15:

The given clinical scenario is suggestive of Gradenigo's syndrome, which occurs secondary to petrous apicitis. This is a complication of otitis media.

The triad of Gradenigo's syndrome consists of:

- Ipsilateral abducens nerve palsy (6th nerve palsy) - external rectus palsy
- Deep ear/ facial pain (5th nerve involvement)
- Persistent ear discharge (otitis media)

The incidence of this syndrome has reduced with the advent of antibiotics.

Option A- Following acute coalescent mastoiditis the pus breaks through the bony wall between the mastoid antrum and the external meatus and forms Luc's abscess. The swelling can burst into the meatus causing otorrhea.

Options B and C- Nasopharyngeal carcinoma and trigeminal schwannoma can present with cranial nerve palsy, but they do not follow the above clinical course.

Solution to Question 16:

Griesinger's sign is seen in sigmoid sinus thrombosis. It is described as mastoid tenderness and edema, which is secondary to thrombophlebitis of the mastoid emissary vein.

However, Gradenigo's syndrome/triad is a complication of otitis media and is seen when the infection extends to the petrous apex (petrositis). The triad consists of:

- Ipsilateral abducens nerve palsy (6th nerve palsy) - external rectus palsy
- Deep ear/ facial pain (5th nerve involvement)

- Persistent ear discharge (otitis media)

Solution to Question 17:

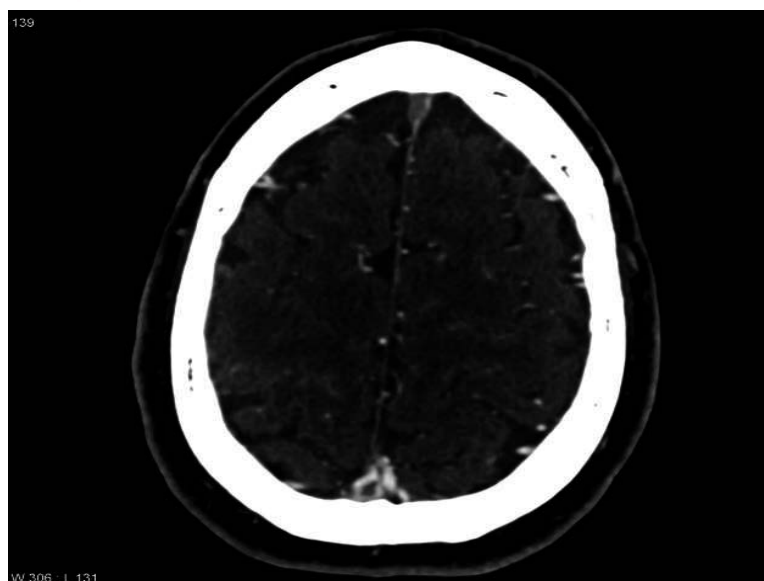
The given clinical scenario is suggestive of lateral sinus thrombophlebitis. It is an inflammation of the inner wall of the lateral venous sinus with the formation of an intrasinus thrombus.

Clinical features of lateral sinus thrombophlebitis:

- Hectic picket fence fever with chills and rigors. This is due to septicemia. The clinical picture resembles malaria but lacks regularity.
- Headache - In the early stage, it may be due to peri-sinus abscess and is mild. Later, it may be severe when intracranial pressure rises due to venous obstruction.
- Griesinger's sign - this is due to thrombosis of the mastoid emissary vein. Edema appears over the posterior part of the mastoid.
- Papilloedema - its presence depends on obstruction to venous return. It is often seen with the right sinus (which is larger than the left).
- Tenderness along the jugular vein - This is seen when thrombophlebitis extends along the jugular vein. There may be associated enlargement and inflammation of the jugular chain of lymph nodes and torticollis.

Investigations:

- Blood culture
- CSF analysis - normal except for increased CSF pressure that helps in ruling out meningitis.
- X-ray mastoid shows:
 - Clouding of air cells seen in acute mastoiditis.
 - Destruction of bone seen in cholesteatoma.
- CECT shows a typical delta sign.



Treatment:

- Intravenous antibacterial therapy
- Mastoidectomy and exposure of sinus
- Ligation of internal jugular vein
- Anticoagulant therapy

Solution to Question 18:

The image shows a cherry-red small mass of granulation tissue arising from the pars flaccida of the left tympanic membrane strongly suggesting that there is an underlying cholesteatoma in the attic area. Surgery is the treatment of choice in case of cholesteatoma.

The continuing discharge from the granulation tissue has caused the surface epithelia of the tympanic membrane to become edematous and thickened. This gives the surface of the tympanic membrane a thickened whitish appearance.

The primary aim in surgical treatment is to remove the disease and render the ear safe, and second in priority is to preserve or reconstruct hearing but never at the cost of the primary aim. Mastoidectomy is done.

Otosclerosis

Question 1:

Which of the following structures does otosclerosis affect?

- a) Otic labyrinth
- b) Periotic labyrinth
- c) Otic capsule
- d) Periotic capsule

Question 2:

Which layer of the otic capsule is affected in otosclerosis?

- a) Endosteal
- b) Enchondral
- c) Periosteal
- d) Perichondral

Question 3:

What is the most common site of origin of otosclerosis?

- a) Fissula ante fenestram
- b) Fissula post fenestram
- c) Footplate of stapes
- d) Round window

Question 4:

What is the type of otosclerotic focus when the disease process involves the entire footplate except for the annular ligament?

- a) Circumferential
- b) Obliterative

- c) Biscuit
- d) Footplate

Question 5:

You are an OBGYN resident and your patient is one of your staff nurses. She is a primigravida who has just been diagnosed with otosclerosis and wants to know its inheritance pattern. Which of the following is the inheritance pattern for otosclerosis?

- a) Autosomal dominant
- b) Autosomal recessive
- c) X linked dominant
- d) X linked recessive

Question 6:

For research on pharmacological treatment in otosclerosis, a researcher wanted to set the age limit of the study to the most common age group when the hearing loss usually manifests. Which of the following will be a suitable set?

- a) 5-10 years
- b) 10-20 years
- c) 20-30 years
- d) 30-40 years

Question 7:

Which of the following is a false statement regarding otosclerosis?

- a) Females are affected twice as often as males.
- b) Stapedotomy should be avoided in children.
- c) HRCT shows double ring sign in stapedial otosclerosis.
- d) It may be associated with measles.

Question 8:

Which of the following hearing defects is not seen in otosclerosis?

- a) Paracusis willisii
- b) Presbycusis
- c) Progressive conductive hearing loss
- d) Sensorineural hearing loss

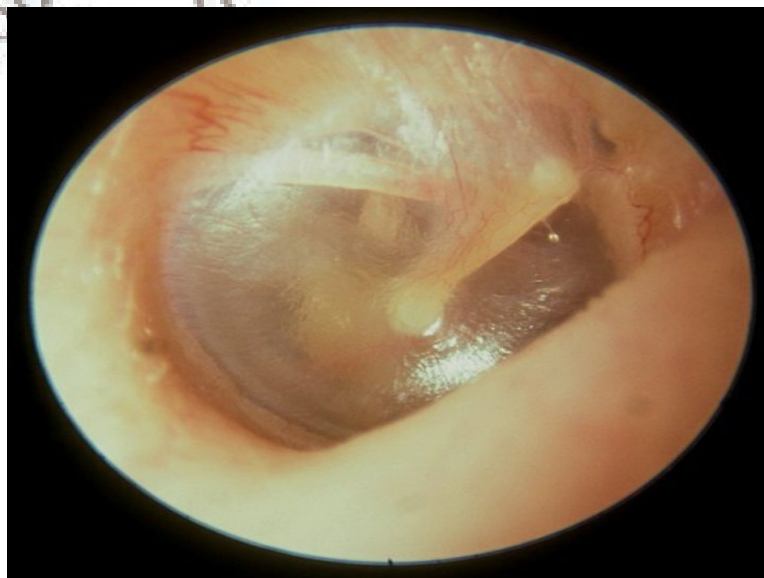
Question 9:

In a stapedectomy specimen of a patient with otosclerosis, a pathologist noted that a few areas of the otic capsule appeared more basophilic than normal. What is this appearance called?

- a) Blue mantle sign
- b) Blue dot sign
- c) Blue cement sign
- d) Islets in ocean sign

Question 10:

A 30-year-old Caucasian woman came with a history of progressive hearing impairment which improves in a noisy environment. Tuning fork tests revealed a conductive hearing loss in both ears. The otoscopic finding is given in the image below. What is the most probable diagnosis among the given options?

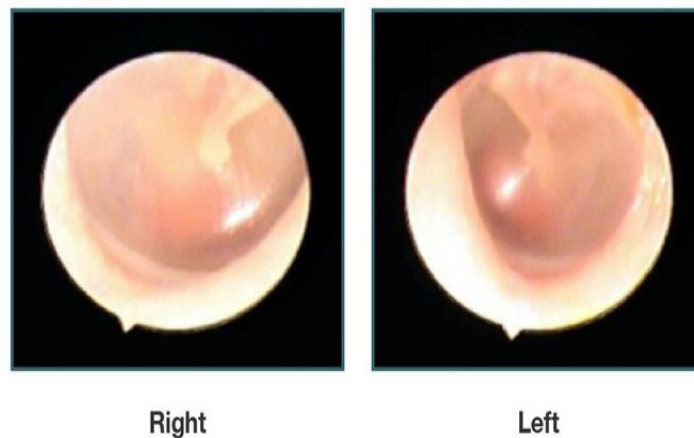


- a) Otosclerosis with matured focus
- b) Chronic suppurative otitis media

- c) Otosclerosis with active focus.
- d) Otitis media with effusion

Question 11:

A 20-year-old female patient presents with features of bilateral conductive hearing loss, not associated with tinnitus. The following image shows the otoscopic findings of her tympanic membranes. What is this sign called?



- a) Rising sun sign
- b) Schwartz sign
- c) Setting sun sign
- d) Cart wheel appearance

Question 12:

An ENT senior resident posted in a remote village comes across a patient in whom he suspects otosclerosis. Which of the following tests can be used if he wants to clinically confirm the diagnosis?

- a) Gelle's Test
- b) Stenger's test
- c) Schwabach Test
- d) Rinne's Test

Question 13:

Which of the following symptoms is not included in the triad of Van der Hoeve syndrome?

- a) Osteogenesis imperfecta
- b) Otosclerosis
- c) Blue sclera
- d) Icterus

Question 14:

A 25-year-old woman with a family history of hearing loss in her mother developed a hearing problem during pregnancy. Hearing loss is bilateral and progressive. Which of the following is the single best investigation to establish a diagnosis?

- a) Impedance audiometry
- b) PTA showing the Carhart's notch
- c) MRI
- d) Schwabach's test

Question 15:

Carhart's notch typically dips at which of the following frequencies?

- a) 4000 Hz in both BC and AC
- b) 2000Hz in both BC and AC
- c) 2000 Hz in AC
- d) 2000 Hz in BC

Question 16:

An elderly woman with a history of otosclerosis has now come with worsening deafness. She attained menopause 1 year ago. What would be the most likely report if impedance audiometry was done for her?

- a) As graph alone
- b) Ad graph with increased stapedial reflex
- c) As graph with loss of stapedial reflex

- d) As graph with increased stapedial reflex

Question 17:

A patient was diagnosed with otosclerosis and Schwartz's sign was positive on otoscopy. How will you manage this patient?

- a) Steroids
- b) Sodium fluoride
- c) Stapedotomy
- d) Stapedectomy

Question 18:

A 25-year-old woman presents with bilateral progressive conductive hearing loss. On examination, the tympanic membrane is normal and the audiogram shows bilateral conductive deafness. Impedance audiometry shows As type of curve and acoustic reflexes are absent. Which of the following treatment options is not used in patients with the above diagnosis?

- a) Stapedectomy
- b) Stapedotomy
- c) Hearing aids
- d) Gentamicin

Question 19:

A primipara is being evaluated for hearing impairment which had worsened during her pregnancy. Rinne's test is negative in both the ears and Weber's lateralized to the right side. On otoscopy, tympanic membranes appear normal. Audiometry shows a dip in BC at 2000 Hz. What is the treatment of choice for the above condition?

- a) Sodium fluoride
- b) Tympanoplasty
- c) Stapedectomy
- d) Stapedotomy

Question 20:

You are assisting in a stapedotomy procedure for a patient with otosclerosis. After appropriate dissection, where will you place the piston?

- a) Between incus and stapes head
- b) Between incus and oval window
- c) Between malleus and incus
- d) Between malleus and oval window

Question 21:

You are an ENT surgeon and the following are few patients with otosclerosis. For which of the following patients will you advise stapedotomy?

- a) 1,3,4 only
- b) 1 and 4 only
- c) 2 & 5 only
- d) 4 & 5 only

Question 22:

An ENT surgeon has referred a patient with otosclerosis posted for stapedotomy to your anesthesia OPD. She has no known co-morbidities. Which of the following techniques will you recommend for the above patient?

- a) General anesthesia
- b) Local anesthesia
- c) Spinal anesthesia
- d) Nerve block

Question 23:

In a patient with otosclerosis who underwent stapedotomy, which of the following complications is least likely?

- a) Dead ear
- b) Sensorineural hearing loss

- c) Cholesteatoma
- d) Facial palsy

Question 24:

In Lempert's fenestration, an alternative window is created in which of the following structures?

- a) Oval window
- b) Lateral semicircular canal
- c) Round window
- d) Tympanic membrane

Answer Key

Question No.	Correct Option
1	c
2	b
3	a
4	c
5	a
6	c
7	c
8	b
9	a
10	a
11	b
12	a
13	d
14	a
15	d
16	c
17	b
18	d
19	d

20	b
21	d
22	b
23	c
24	b

Detailed Explanations

Solution to Question 1:

Otosclerosis affects the otic capsule.

The otic capsule is the dense bony labyrinth of the inner ear that surrounds the cochlea, the vestibule, and the semicircular canals. In otosclerosis, one or more foci of irregularly laid spongy bone replace part of the normally dense bone.

The inner ear is divided into:

- The membranous labyrinth is also called the otic labyrinth. It is filled with endolymph.
- The bony labyrinth, which is also called the otic capsule.
- Periotic labyrinth, which is the space between the bony and membranous labyrinth and is filled with perilymph.

Solution to Question 2:

Otosclerosis affects the enchondral layer of the otic capsule.

Within the enchondral layer, there are sometimes islands of unossified cartilage rests that later give rise to otosclerosis.

The otic capsule is made of 3 layers:

- Endosteal: Innermost
- Enchondral: Middle layer derived from cartilage that ossifies
- Periosteal: Outermost layer

Solution to Question 3:

The most common site of origin of otosclerosis is the fissula ante fenestram lying in front of the oval window. It is seen in the stapedial type of otosclerosis.

Types of otosclerosis :

- Stapedial

- Cochlear
- Histological.

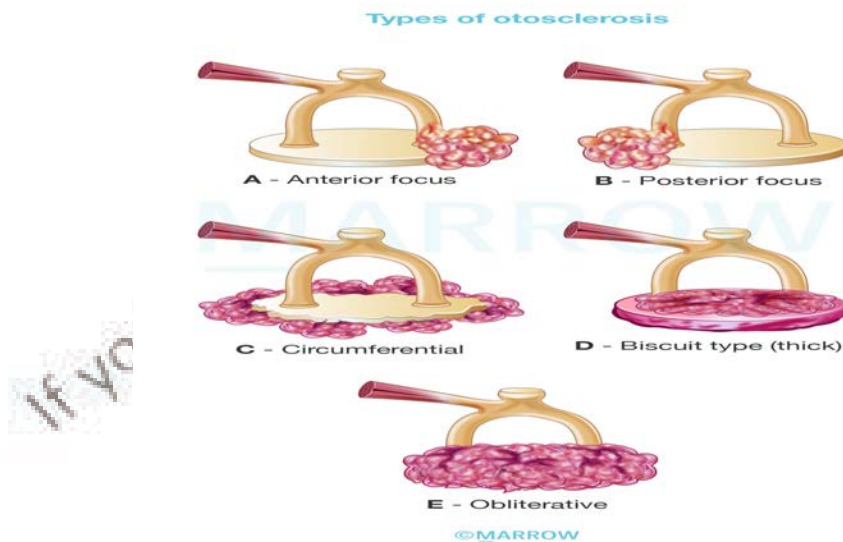
Stapedial otosclerosis is the most common type of otosclerosis.

Solution to Question 4:

The otosclerotic focus is biscuit type when the disease process involves the entire footplate except for the annular ligament.

The otosclerotic foci in the stapedial type of otosclerosis can be of various types:

- Anterior focus - involving only the fissula ante fenestrum
- Posterior focus- involving only the fissula post fenestrum
- Circumferential - affects the margin of the footplate only (annular ligament alone)
- Biscuit type - affects the entire footplate except for the annular ligament
- Obliterative- affects the entire footplate & its annular ligament.



Solution to Question 5:

Otosclerosis has an autosomal dominant type of inheritance.

About 50% of the cases of otosclerosis have a positive family history while the rest are sporadic. Genetic studies reveal that it is an autosomal dominant trait with incomplete penetrance and variable expressivity.

Solution to Question 6:

The most common age group when the hearing loss in otosclerosis usually manifests is 20-30 years of age. It is rare before 10 and after 40 years.

Solution to Question 7:

HRCT shows the double-ring sign in cochlear otosclerosis, not in the stapedial type.

Otosclerosis affects females twice as often as males. But in India, otosclerosis seems to predominate in males.

Stapedotomy is contraindicated in children because recurrent eustachian tube dysfunction is common in them. This can displace the prosthesis or cause acute otitis media.

Measles RNA has been found in otosclerotic foci implicating its possible role in pathogenesis.

Solution to Question 8:

Presbycusis is not seen in otosclerosis.

The following hearing defects are seen in otosclerosis:

- Painless progressive bilateral conductive hearing loss most commonly.
- Sensorineural hearing loss may be seen in cochlear type of otosclerosis.
- Paracusis willisii: Patient hears better in noisy surroundings as other people raise their voice to be heard.

Solution to Question 9:

The blue mantle sign of Manasseh is a histopathological finding seen in otosclerosis.

On histopathological examination of an otosclerotic foci, the osteoblasts and osteoclasts lie in an abundance of cement substance, which stains blue with hematoxylin and eosin stain.

Option B - Blue dot sign is seen in the torsion of the testicular appendage.

Blue cement sign and islets in ocean sign do not exist.

Solution to Question 10:

The given image is of a normal tympanic membrane. There is bilateral conductive deafness, paracusis willisi and, a normal tympanic membrane which suggest a diagnosis of otosclerosis with matured focus.

Paracusis willisi - Patient hears better in noisy surroundings as other people raise their voice to be heard.

The tympanic membrane has a normal appearance in most cases. This is indicative of a matured or inactive focus of otosclerotic bone.

Option B - A case of CSOM will have a perforated tympanic membrane.

Option C - Otosclerosis with an active focus will show the Schwartz sign.

Option D - In the case of otitis media with effusion the tympanic membrane is often dull and opaque with loss of light reflex.

Solution to Question 11:

The otoscopic picture of both the tympanic membranes shows the Schwartz sign. This finding along with the history of bilateral conductive deafness is seen in active otosclerosis.

A reddish hue may be seen on the promontory through the tympanic membrane (Schwartz sign). This is indicative of an active focus with increased vascularity.

The tympanic membrane is quite normal and mobile.

Option A: Rising sun sign is seen in glomus tumor.

Option C: Setting sun sign is an eye sign seen in hydrocephalus.

Option D: Cartwheel appearance is seen in acute suppurative otitis media.

Solution to Question 12:

Gelle's is a tuning fork test done to clinically confirm otosclerosis in the olden days.

It is based on the fact that in a normal ear if pressure is increased in the external ear the stapes footplate will get medialized and will stop conducting sound. But, in an ear with otosclerosis of the footplate, there will be no movements and the patient will continue to hear the sound. Thus, Gelle's test is considered positive in normal persons and in those with sensorineural hearing loss. It is negative when the ossicular chain is fixed or disconnected.

Stenger's test is done to rule out malingering.

Schwabach is a tuning fork test that compares the bone conduction of the patient with that of the examiner.

Rinne's is a tuning fork test that compares bone conduction with air conduction.

Solution to Question 13:

Icterus is not included in the triad of Van der Hoeve syndrome

The triad of symptoms is called Van der Hoeve syndrome:

- Osteogenesis imperfecta
- Otosclerosis
- Blue sclera

Solution to Question 14:

The above clinical scenario is suggestive of otosclerosis and the single best investigation to diagnose otosclerosis is impedance audiometry.

Option B- In pure tone audiometry (PTA), Carhart's notch with an air-bone gap indicates otosclerosis.

Option C- MRI is usually not done. HRCT is the preferred imaging modality for otosclerosis.

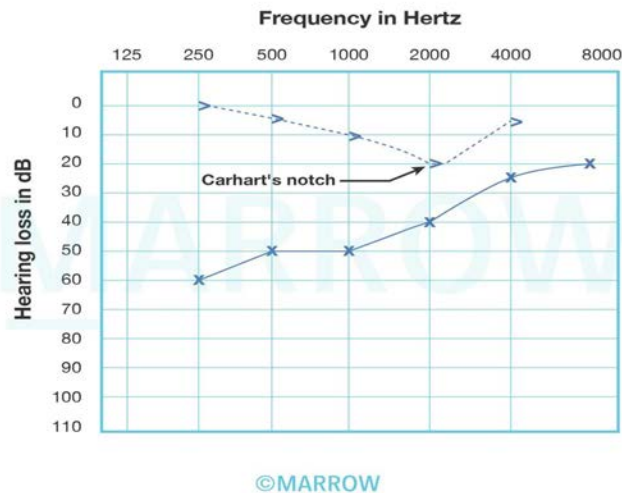
Option D- Schwabach's test is a tuning fork test that shows conductive hearing loss.

Solution to Question 15:

Carhart's notch typically dips at 2000 Hz in bone conduction.

Pure tone audiometry in otosclerosis:

- Air conduction is poor & worse for lower frequencies.
- Bone conduction is normal.
- There is a large air-bone gap
- In some cases, there is a dip in the bone conduction curve. It is different at different frequencies but the maximum dip is at 2000 Hz and is called Carhart's notch.



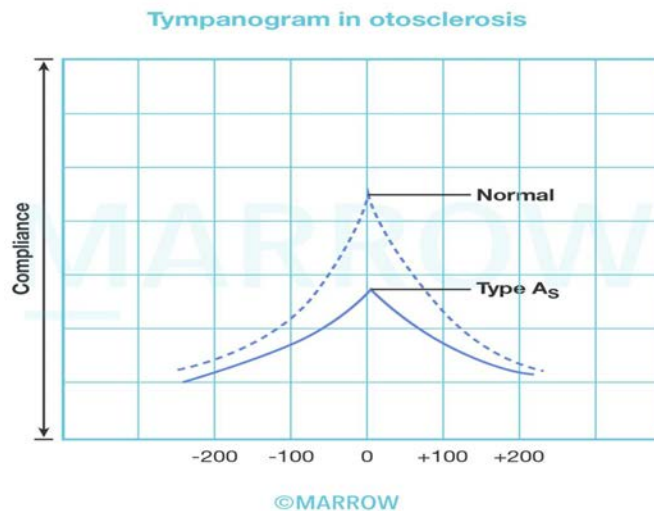
Note: Boiler's notch is a similar dip in BC and AC curve but at 4000 Hz seen in noise-induced hearing loss.

Solution to Question 16:

The above case is suggestive of a long-standing case of otosclerosis and impedance audiometry will show As graph and loss of stapedial reflex as a result of fixation of ossicles.

Impedance audiometer measures 2 parameters in the middle ear:

- Tympanogram - Type As is seen in otosclerosis due to decreased compliance at or near ambient air pressure
- Stapedial reflex - Stapedial reflex is lost due to immobile ossicles.



Solution to Question 17:

Sodium fluoride is used in the medical management of otosclerosis. This is an active case of otosclerosis evidenced by Schwartz's sign and in an active case of otosclerosis medical management is preferred.

Surgery is contraindicated if Schwartz's sign is positive.

Schwartz sign - A reddish hue is seen on the promontory through the tympanic membrane. This is indicative of an active focus with increased vascularity.

In an active case of otosclerosis, medical management with Sodium fluoride (NaF) is preferred because NaF hastens the maturity of active focus in otosclerosis. It acts by inhibiting the osteoclasts and decreasing the proteolytic enzymes.

Solution to Question 18:

The above clinical scenario of bilateral progressive conductive deafness with As type of tympanogram and absent acoustic reflexes is suggestive of otosclerosis. Gentamicin is not used in the management of otosclerosis.

Medical management includes the use of sodium fluoride and bisphosphonates in active cases.

Surgical management includes stapedectomy and stapedotomy, though the latter is preferred due to decreased risk of fistula formation.

Hearing aids may be used as an alternative to surgery.

Solution to Question 19:

The above clinical scenario is suggestive of otosclerosis and the treatment of choice for otosclerosis is stapedotomy.

Stapedotomy is the excision of the superstructure of stapes and drilling a hole in the footplate & placing a piston in the hole. This is the treatment of choice.

Stapedectomy is the excision of the entire stapes (superstructure + footplate) and placing a piston in its place. It is outdated.

NaF is used in an active case of otosclerosis to hasten maturation.

Note:

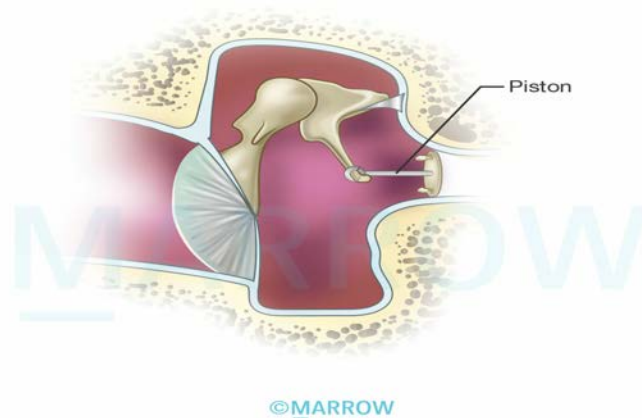
- Both stapedectomy and stapedotomy have been described for the treatment of otosclerosis. Even in the stapedectomy procedure, stapedotomy (drilling) has been practiced. Hence stapedotomy is preferred over stapedectomy and is considered the treatment of choice.
- In the above case since Rinne's test is negative on both ears, it indicates bilateral conductive deafness. Weber's test will lateralize to the ear with greater conductive loss.

Solution to Question 20:

The piston is placed between the incus and oval window.

Stapedotomy is done to bypass the immobile stapes using a piston which will take over the function of conducting sound. For this reason, the piston replaces the removed stapes superstructure.

Stapedotomy piston



Solution to Question 21:

A stapedotomy will be indicated for patients 4 and 5 as bronchial asthma and diabetes without schwartz sign are not contraindications for the procedure.

Contraindications for Stapedotomy:

- Only hearing ear
- Positive Schwartze sign
- Otitis externa
- Meniere's disease: can damage the inner ear while drilling a hole in the footplate
- Young children
- Professional athletes, pilots, construction workers, and divers: Sudden changes in pressure can displace the piston and cause vertigo.
- Those who work in noisy surroundings: Surgery makes them more vulnerable to noise-induced hearing loss.

Solution to Question 22:

Stapedotomy is preferred to be done under local anesthesia because -

- Immediate on-table improvement of hearing can be assessed
- Damage to the inner ear can be assessed by the patient's symptoms of vertigo/ nystagmus
- Damage to the facial nerve can be checked.

Solution to Question 23:

Cholesteatoma is not a complication of stapedotomy.

Complications of stapedotomy:

- Sensorineural hearing loss
- Injury to the facial nerve
- Floating/ submerged footplate
- Endolymph gush due to damage to the inner ear
- Dislodgement of piston
- Post-op vertigo
- Dead ear.

Solution to Question 24:

Lempert's fenestration, an alternative window is created in the lateral semicircular canal to function for the obliterated oval window.

Lempert's fenestration operation for otosclerosis is outdated now. It has the disadvantage of a postoperative mastoid cavity and an inherent hearing loss of 25 dB which cannot be corrected.

Meniere's Disease

Question 1:

Which of the following is not Ménière's disease?

- a) Vestibular hydrops
- b) Cochlear hydrops
- c) Perilymphatic hydrops
- d) Endolymphatic hydrops

Question 2:

Which of the following statements are true about Ménière's disease?

- a) 1,2, and 4
- b) 3,4, and 5
- c) 1,3, and 5
- d) 2,3, and 4

Question 3:

Which of these symptoms is least likely to be present in a patient diagnosed with Ménière's disease?

- a) Episodic vertigo
- b) Otagia
- c) Fluctuating hearing loss
- d) Tinnitus

Question 4:

A 35-year-old patient presented with complaints of intolerance to loud sounds and mild giddiness associated with ringing in the ear. What is the probable cause of intolerance to loud sounds in this patient?

- a) Recruitment phenomenon

- b) Rollover phenomenon
- c) Diplacusis
- d) Tullio phenomenon

Question 5:

A middle-aged man presented with a history of vertigo, tinnitus, and intolerance to loud sounds. He constantly gets vertiginous episodes when exposed to loud sounds or noise. This phenomenon is not seen in which of the following conditions?

- a) Meniere's disease
- b) Superior semicircular canal dehiscence
- c) Perilymph fistula
- d) ASOM with a perforated tympanic membrane

Question 6:

Which of the following would be the most likely audiometric finding in a patient who is diagnosed with early Ménière's disease?

- a) Higher frequencies affected
- b) Lower frequencies affected
- c) Middle frequencies affected
- d) All frequencies affected

Question 7:

What is the summing potential: action potential ratio seen in Ménière's disease on electrocochleography?

- a) ≈ 0.3
- b) ≈ 0.1
- c) ≈ 0.4
- d) ≈ 0.2

Question 8:

A middle-aged patient presented with a history of recurrent episodes of whirling vertigo, which usually are preceded by a variable sense of ear pressure and fullness, decreased hearing, and low-tone roaring tinnitus. Which of the following is a useful test to diagnose this patient's condition?

- a) Otoacoustic emission
- b) Glycerol test
- c) Caloric test
- d) Dix-Hallpike test

Question 9:

A 37-year-old man presented with a history of progressive hearing loss over a year, followed by an episode of sudden onset vertigo. He noticed an improvement in hearing after the vertiginous episode. What is the most likely diagnosis?

- a) Lermoyez syndrome
- b) Lemiere's syndrome
- c) Leriche syndrome
- d) Leigh syndrome

Question 10:

A patient diagnosed with Meniere's disease presented with a history of sudden fall without loss of consciousness and no associated neurological or cardiovascular symptoms. On examination, there were no neurological deficits. What is the most likely diagnosis?

- a) Tullio phenomenon
- b) Tumarkin's otolithic crisis
- c) Lermoyez syndrome
- d) Vestibular hydrops

Question 11:

Which of the following is not a differential diagnosis for a patient with suspected Ménière's disease?

- a) Vestibular neuronitis
- b) Vertebrobasilar insufficiency

- c) CSOM
- d) Perilymph fistula

Question 12:

A 27-year-old woman diagnosed with Ménière's disease presented with complaints of sudden onset dizziness and sensation of surroundings swirling. Which of the following drugs cannot be used as a vestibular sedative to treat this condition?

- a) Dimenhydrinate
- b) Prochlorperazine
- c) Promethazine
- d) Loratadine

Question 13:

Which of the following drugs is a vasodilator of the inner ear?

- a) Nicotine
- b) Betahistine
- c) Antihistaminics
- d) Serotonin

Question 14:

An ENT surgeon is going to perform a chemical labyrinthectomy on a patient with Meniere's disease, that is refractory to treatment. Which of the following drugs will he be using for this procedure?

- a) Cinnarizine
- b) Gentamicin
- c) Glucocorticoids
- d) Amoxicillin

Question 15:

Microwick and microcatheter devices are used in which of the following diseases?

- a) 2 only
- b) 1, 3, and 5
- c) 2 and 4
- d) 2, 4, and 5

Question 16:

A patient with Meniere's disease has debilitating vertigo episodes which are causing him multiple daily falls. On examination, mild hearing loss is present. Which of the following is not a treatment option for this patient?

- a) Labyrinthectomy
- b) Decompression of endolymphatic sac
- c) Sacculotomy
- d) Ultrasonic destruction of vestibular labyrinth

Question 17:

A 61-year-old woman is on treatment for Ménière's disease for two years. Unfortunately, there is no improvement in her symptoms despite taking medications and intratympanic gentamicin injections. Which of the following treatment modalities would you not recommend for her?

- a) Fick's operation
- b) Labyrinthectomy
- c) Meniett's device
- d) Vestibulectomy

Question 18:

What is the definitive treatment of intractable vertigo due to Meniere's disease?

- a) Acetazolamide
- b) Labyrinthectomy
- c) Vestibular neurectomy
- d) Intratympanic gentamicin

Question 19:

Which of the following features is not associated with superior semicircular canal dehiscence syndrome?

- a) Gaze dependent nystagmus
- b) Positive Fistula test
- c) Sensorineural hearing loss
- d) Positive Tulio phenomenon

Question 20:

A patient complains that he feels like the room is spinning when he gets up from lying down or turning his head. He has no history of loss of consciousness. Which of the following could be the probable diagnosis?

- a) Benign paroxysmal positional vertigo
- b) Meniere's disease
- c) Labyrinthitis
- d) Syncope

Question 21:

A 52-year-old woman presents to the ENT surgeon with a history of vertigo whenever she moves sideways on the bed while lying supine. She feels nauseous and as if the room is spinning. The symptoms subside after a few minutes but return when she moves her head. Which of the following tests is most appropriate to diagnose this condition?

- a) Dix-Hallpike's manoeuvre
- b) Glycerol test
- c) Caloric test
- d) Epley's manoeuvre

Answer Key

Question No.	Correct Option
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1	c
2	d
3	b
4	a
5	d
6	b
7	c
8	b
9	a
10	b
11	c
12	d
13	b
14	b
15	d
16	a
17	d
18	b
19	c
20	a
21	a

Detailed Explanations

Solution to Question 1:

Ménière's disease is also called endolymphatic hydrops.

It is a disorder of the inner ear where the endolymphatic system is distended with endolymph. This occurs due either due to excessive production or faulty absorption of endolymph.

It has 2 variants:

- Cochlear hydrops - the main feature is hearing loss. Vertigo is absent.
- Vestibular hydrops - repeated episodes of vertigo without loss of hearing.

Solution to Question 2:

Ménière's disease is commonly seen in the age group of 35-60 years and incidence peaks in the fifth decade. It is a disorder of the inner ear causing a sensorineural type of hearing loss. Usually, it is unilateral but the other ear may be affected after a few years.

Symptoms of the disease occur in attacks (episodes) and there is not a constant presence of the symptoms.

Solution to Question 3:

Otalgia is least likely to be present in a patient diagnosed with Ménière's disease.

The characteristic triad of Ménière's disease includes:

- Episodic vertigo- attacks come in clusters, with periods of spontaneous remission lasting for weeks, months, or years
- Fluctuating hearing loss- hearing is affected during episodes of vertigo, improves after the attack, and maybe normal during remissions
- Tinnitus- low-pitched whistling or roaring type

Warning symptoms preceding an attack of vertigo in Ménière's:

- Sense of fullness in the ear
- Change in pitch or increasing intensity of tinnitus
- Decrease in hearing

Solution to Question 4:

Intolerance to loud sounds in this patient is due to the recruitment phenomenon (sound is amplified to be heard better). It is a feature of all cochlear pathologies.

The characteristic presentation of a middle-aged man with vertigo, tinnitus, and intolerance to loud sounds points to a diagnosis of Ménière's disease. It is a cochlear pathology, hence shows the recruitment phenomenon.

The rollover phenomenon is a feature of retro-cochlear pathologies like an acoustic neuroma.

Diplacusis is the perception of a single auditory stimulus as sounds of a different pitch in two ears.

Tullio phenomenon is a condition where loud sounds precipitate a vertiginous attack.

Solution to Question 5:

The phenomenon described above is the Tullio phenomenon and is not seen in ASOM with a perforated tympanic membrane.

The Tullio phenomenon refers to sound-induced vertigo. People with this condition experience unsteadiness, vertigo, nausea and rapid involuntary eye movements (nystagmus) when exposed to

loud sounds or noise. It is due to the distended saccule lying against the stapes footplate.

Some cases of Ménière's disease show the Tullio phenomenon.

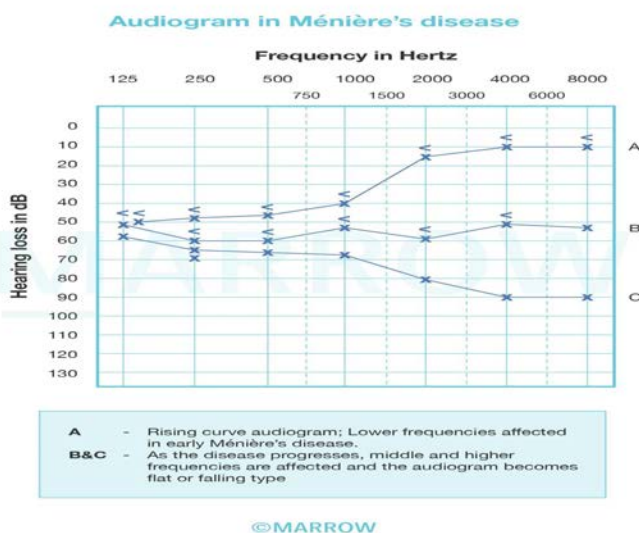
It is also seen whenever there is a third mobile window lesion such as otic syphilis, perilymph fistula and dehiscence of semicircular canals.

Solution to Question 6:

Early Ménière's disease affects lower frequencies. As the disease progresses, all the frequencies become affected.

There is no classical audiogram pattern that can be used to identify Ménière's disease. However, there are certain features that have been commonly observed. These include:

- A varying degree of sensorineural hearing loss.
- This hearing loss is usually low frequency.



Solution to Question 7:

Summating potential: action potential ratio (SP: AP) > 0.4 or 0.5 are consistent with a diagnosis of Ménière's disease.

Normally, the ratio of SP to AP is 0.3 (suggestive of normal cochlear function).

Solution to Question 8:

The given clinical history points to a diagnosis of Ménière's disease. The glycerol test is useful for the diagnosis of Ménière's disease.

Glycerol is an osmotic diuretic which when given orally reduces hydrops and improves symptoms. Electrocochleography done along with a glycerol test. An improvement after administration of glycerol is diagnostic of Ménière's disease.

The caloric test used in Ménière's disease is not diagnostic as it is abnormal in many other conditions like ototoxicity, vestibular schwannoma, vestibular neuritis, etc.

Dix-Hallpike test is performed to diagnose BPPV. A positive test would result in a vertiginous episode and nystagmus observed by the clinician.

Solution to Question 9:

The most likely diagnosis would be Lermoyez syndrome.

It is a variant of Ménière's wherein the symptoms appear in reverse order. The patient presents with a progressive hearing loss followed by an attack of vertigo which causes improvement in hearing.

Lemierre's syndrome is a condition characterized by infectious thrombophlebitis of the internal jugular vein and bacteremia caused by primarily anaerobic organisms, following a recent oropharyngeal infection.

Leriche syndrome, commonly referred to as an aortoiliac occlusive disease, is due to thrombotic occlusion of the abdominal aorta just above the site of its bifurcation.

Leigh syndrome (subacute necrotizing encephalomyelopathy) is a rare genetic neurometabolic disorder characterized by degeneration of the CNS.

Solution to Question 10:

The given clinical presentation is suggestive of Tumarkin's otolithic crisis.

Tumarkin's otolithic crisis (drop attacks) is when a patient with Meniere's develops sudden drop attacks without loss of consciousness. It is an uncommon manifestation. Its pathogenesis is not well understood.

Tullio phenomenon: It is a condition seen with some cases of Meniere's disease, where loud sounds or noise produce vertigo. This phenomenon is due to the distended saccule lying against the stapes footplate.

Lermoyez syndrome: Here symptoms of Meniere's disease are seen in reverse order. Initially, there is a progressive deterioration of hearing, followed by an attack of vertigo.

Solution to Question 11:

CSOM is a middle ear disease with symptoms of ear discharge and is not a differential for Ménière's.

Ménière's is a cochlear pathology with symptoms of vertigo, hearing loss & tinnitus.

Differentials of Ménière's disease:

- Vestibular neuronitis
- BPPV
- Cervical spondylosis
- Acoustic tumours
- Vertebrobasilar insufficiency
- Labyrinthitis
- Perilymph fistula
- Migraine induced vertigo

Solution to Question 12:

Among the given drugs, Loratadine is not a vestibular sedative.

Loratadine is a second generation antihistamine which lacks CNS depressant effects. It is used for allergic rhinitis, chronic idiopathic urticaria, and exercise-induced bronchospasm prophylaxis.

The first line of treatment of an acute episode of Ménière's is symptomatic and hence vestibular sedatives are the mainstay in the treatment. These are

- Dimenhydrinate
- Prochlorperazine
- Promethazine
- Rarely- diazepam and atropine

Solution to Question 13:

Betahistine is a vasodilator of the inner ear. It is used in the treatment of Meniere's disease.

It is reported to act by:

- Improving microvascular circulation in the stria vascularis of the cochlea
- Inhibiting vestibular nuclei activity

Solution to Question 14:

Gentamicin is the drug used in chemical labyrinthectomy.

Gentamicin is mainly vestibulotoxic. It has been used in daily or biweekly injections into the middle ear (Intratympanic gentamicin therapy). The drug is absorbed through the round window

and destroys the vestibular labyrinth.

Solution to Question 15:

Microwick and microcatheter devices are used to treat Meniere's disease, sudden sensorineural hearing loss (sudden SNHL), and autoimmune inner ear disease.

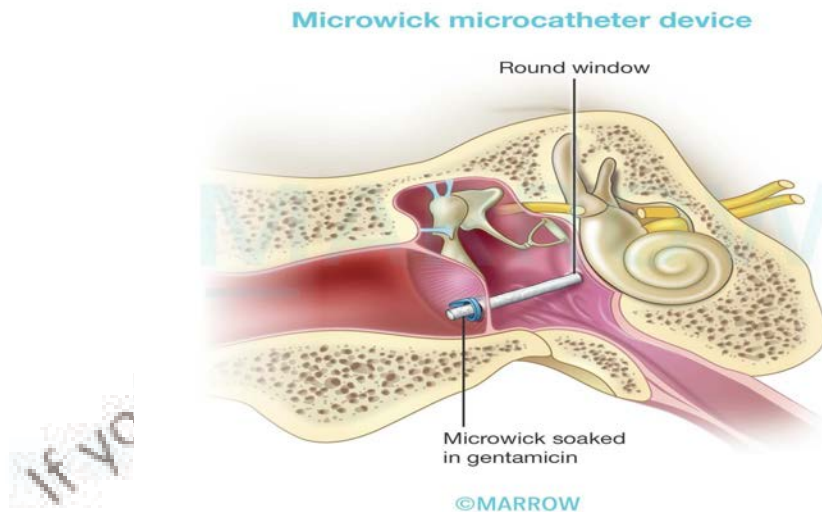
These devices are used to deliver drugs from the external canal to the inner ear thereby avoiding repeated intratympanic injections. When soaked with the drug, wick delivers it to the round window to be absorbed into the inner ear. It has been used to deliver :

-

Steroids in sudden sensorineural deafness and autoimmune inner ear disease.

-

Gentamicin to destroy the vestibular labyrinth in Ménière's disease.



Solution to Question 16:

Labyrinthectomy is a destructive procedure that totally destroys cochlear and vestibular function. Thus, It is used only when the cochlear function is not serviceable. This patient has mild hearing loss with disabling vertigo, so conservative procedures are to be offered. In these procedures, hearing is preserved.

Conservative procedures done for Ménière's include:

- Decompression of endolymphatic sac
- Endolymphatic shunt operation
- Sacculotomy (Fick's operation)

- Section of the vestibular nerve
- Ultrasonic destruction of vestibular labyrinth.

Solution to Question 17:

A vestibulectomy is not done in Ménière's disease. It is a surgical procedure that can be used to treat vulvar pain, specifically in cases of provoked vestibulodynia.

Ménière's disease, unresponsive or refractory to medical management (including intratympanic gentamicin) is treated with surgery. They can either be:

- Conservative (hearing is preserved):
- Decompression of endolymphatic sac
- Fick's Sacculotomy
- Endolymphatic shunt
- Vestibular neurectomy
- Ultrasonic destruction of the vestibule
- Destructive: Labyrinthectomy
- Meniett's device- provides intermittent positive pressure to the inner ear which improves hearing and reduces vertigo in these patients

Solution to Question 18:

The definitive treatment of intractable vertigo due to Ménière's disease is labyrinthectomy.

Labyrinthectomy involves uniform destruction of hearing and vestibular function. Ideal candidates have no functional hearing and have failed more conservative treatments, such as gentamicin injection. Despite this morbidity, labyrinthectomy has the highest rate of vertigo control (>95%), more than vestibular neurectomy (78-90%), and is considered the single definitive treatment.

The step-by-step management approach for Ménière's disease consists of dietary modifications and vestibular suppressants as the initial measures, followed by intratympanic gentamicin ablation, and finally, surgical intervention.

Primarily, the management of Menier's disease involves a sodium restricted diet and diuretics like acetazolamide (Option A). They are more useful for short-term control.

Gentamicin (Option D) is a selective vestibulotoxic aminoglycoside antibiotic that is preferentially taken up by type I hair cells. It is opted for only if dietary modifications and vestibular suppressants have failed to control the symptoms. If used judiciously, hearing is preserved.

The options for surgical management include - Endolymphatic sac surgery, vestibular neurectomy, and labyrinthectomy. Endolymphatic sac surgery is currently declining in popularity due to lack of evidence of vertigo control.

Vestibular neurectomy (Option C) refers to selective ablation of the vestibular nerve. Even though it is considered a hearing sparing surgery and carries comparable rates of vertigo control as labyrinthectomy, it is an intradural procedure and carries a greater risk of complications. Hearing loss will still progress as it would have in the natural course of illness.

Solution to Question 19:

Sensorineural hearing loss is not seen in superior semicircular canal dehiscence syndrome (SSCDS). They may experience mild to moderate hearing conductive loss.

Conductive hearing loss is due to the abnormal transmission and dissipation of acoustic energy through the defect (third window), leading to reduced conduction to the cochlea for lower frequencies. This can manifest as a low-frequency air-bone gap at multiple frequencies. There is no defect in the cochlea or the auditory pathway.

Superior semicircular canal dehiscence (SSCD) is a condition in which bone between the apex of the superior semicircular canal and the middle cranial fossa is absent. This causes endolymph in the labyrinthine system to move with sound or pressure, causing an activation of the vestibular system.

Patients may have vestibular symptoms like vertigo triggered by loud sounds (Tullio phenomenon) or pressure such as coughing and can also experience hyperacusis for bone-conducted sounds. They may perceive the sounds of their own eye movements or footsteps in the affected ear.

The patients may exhibit nystagmus when exposed to loud sounds or during the Valsalva maneuver or pressure in the ear canal (positive fistula test). The direction of the nystagmus aligns with the affected superior semicircular canal, and some patients may tilt their heads in the plane of the affected superior semicircular canal in response to loud sounds.

Noncontrast HRCT of the temporal bone can demonstrate the defect. The click-evoked VEMP (vestibular evoked myogenic potential) response and vestibuloocular reflex are pronounced in SSCD patients, in which the evoked eye movements align with the affected superior canal.

For mild symptoms, avoidance of triggers is recommended. In case of severe disabling symptoms, surgical resurfacing of the superior semicircular canal or plugging of the superior semicircular canal may be attempted.

Solution to Question 20:

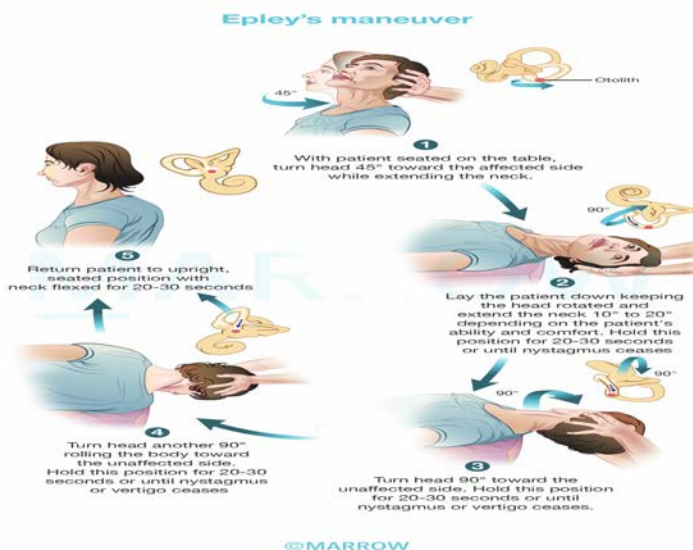
The given clinical scenario, with complaints of dizziness associated with postural change and no history of loss of consciousness is suggestive of benign paroxysmal positional vertigo (BPPV).

BPPV is one of the common causes of recurrent vertigo, and the episodes are brief (<1min) and are always provoked by changes in head position. The attacks are caused by free-floating calcium carbonate crystals usually dislodged from the posterior canal.

A positive Dix-Hallpike maneuver confirms the diagnosis of BPPV. In this maneuver, the patient begins sitting up with their head turned 45 degrees toward the ear to be tested. Then the patient is

made to lie down quickly with their head past the end of the bed. Then, the patient's neck is extended 20 degrees below the horizontal. The clinician then watches the patient's eyes for torsional and up-beating nystagmus, which should persist for no more than one minute. This would indicate a positive test.

Epley's maneuver is performed to reposition the crystals into the macula from the semicircular canals. This is only a temporary cure and there is a high rate of recurrence.



Solution to Question 21:

The given clinical scenario is suggestive of benign paroxysmal positional vertigo (BPPV). The diagnostic maneuver used in BPPV is the Dix-Hallpike maneuver, the gold standard for diagnosing BPPV. The therapeutic maneuver is Epley's maneuver.

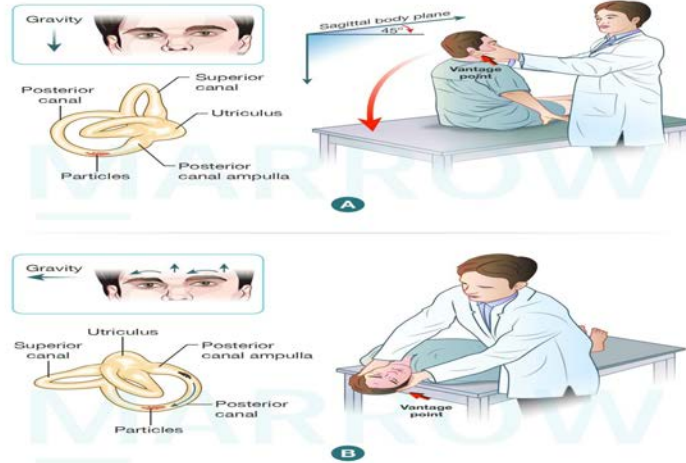
BPPV is one of the common causes of recurrent vertigo, and the episodes are brief (<1min) and are always provoked by changes in head position. The attacks are caused by free-floating calcium carbonate crystals usually dislodged from the posterior canal.

A patient presenting with vertigo should be evaluated for ruling out the central lesions (like a tumor in the fourth ventricle, multiple sclerosis, etc) before confirming the diagnosis of BPPV. Dix-Hallpike maneuver helps rule out central lesions and also diagnose BPPV.

Dix-Hallpike maneuver:

- The maneuver starts with the patient in a sitting position.
- Their head is turned 45 degrees towards the ear to be tested.
- The examiner quickly brings the patient to a supine position with their head 30 degrees below the horizontal.
- The 45-degree head rotation is maintained during the procedure.
- Now the examiner observes the patient for nystagmus.

Dix-Hallpike manoeuvre



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Tumors of Ear

Question 1:

Which of the following is the most common site for ear carcinoma?

- a) Cartilaginous external auditory meatus
- b) Isthmus of canal
- c) Pinna
- d) Mastoid

Question 2:

Where are sebaceous cysts most commonly located in the ear?

- a) Helix
- b) Cymba concha
- c) Pre auricular area
- d) Post auricular area

Question 3:

What is shown in the given image?



- a) Dermoid cyst
- b) Sebaceous cyst
- c) Meningioma
- d) Post auricular fistula

Question 4:

A surfer presented with hearing loss and a feeling of fullness in his ear. On otoscopy, multiple bony swellings were present in the external auditory canal. Which of the following features is diagnostic of exostosis?

- a) Unilateral
- b) Irregular
- c) Sessile
- d) Seen in cartilaginous canal

Question 5:

A middle-aged woman presents with foul-smelling ear discharge and hearing loss for 1 year. On examination, a hemorrhagic polypoidal mass was seen in the middle ear. Biopsy revealed squamous cell carcinoma of the middle ear. Where is it least likely to spread to?

- a) Dura
- b) Facial canal
- c) Eustachian tube
- d) Petrous apex

Question 6:

Which of the following is the most common benign tumor of the middle ear?

- a) Glomus tumour
- b) Acoustic neuroma
- c) Exostosis
- d) Osteoma

Question 7:

A post-menopausal woman with progressive hearing loss and tinnitus is diagnosed to have a glomus tumor. Which cells do her tumor originate from?

- a) Perivascular cells
- b) Peritubal cells
- c) Paraganglionic cells
- d) Paraneoplastic cells

Question 8:

Which of the following statements is false about paragangliomas of the head and neck?

- a) Slow growing tumor
- b) More common in men
- c) Usually seen in middle age
- d) Highly vascular tumour

Question 9:

What is the earliest symptom of glomus tumors?

- a) Sensory-neural deafness
- b) Headache
- c) Tinnitus
- d) Vertigo

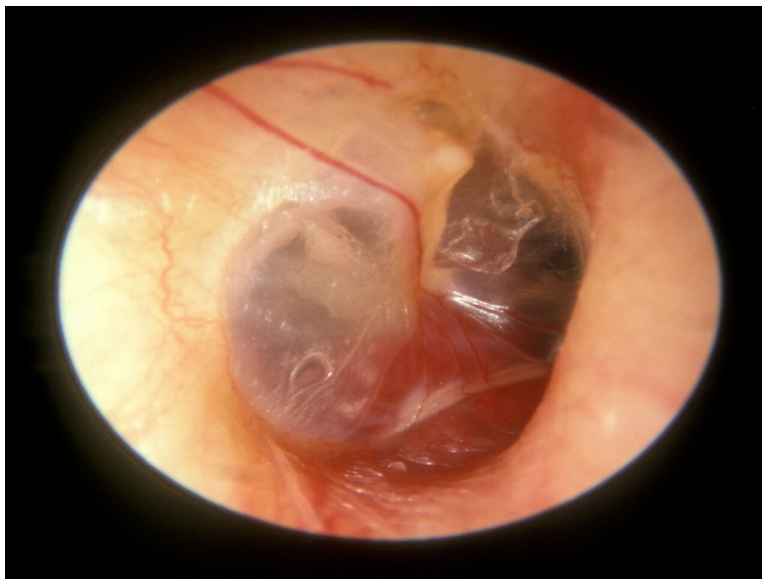
Question 10:

You are a junior resident in the ENT department. A patient presents to you with pulsatile tinnitus. Which of the following will be your first differential diagnosis?

- a) CSOM
- b) ASOM
- c) Mastoiditis
- d) Glomus tumour

Question 11:

You are evaluating a patient with a glomus tumor and on otoscopy, see the following image. Which of the following signs is seen here?



- a) Cartwheel appearance
- b) Lighthouse sign
- c) Rising sun sign
- d) Schwartze sign

Question 12:

While evaluating a patient with a suspected aural polyp, you apply Siegel's speculum and observe that the mass pulsates vigorously and then blanches. What is this sign called?

- a) Rising sun sign
- b) Schwarze's sign
- c) Lighthouse sign
- d) Brown's sign

Question 13:

A patient presents with a history of hearing loss and tinnitus. On examination, a polypoidal mass is seen in the middle ear, which bleeds heavily on touch. What is the diagnosis?

- a) Angiofibroma
- b) Exostosis

- c) Glomus tympanicum
- d) Acoustic neuroma

Question 14:

A 60-year-old woman presented with hearing loss, pulsatile ringing in the right ear, and deviation of the tongue to the right side. MRI revealed a mass in the right jugular fossa. On examination, a pulsatile reddish mass was seen behind the intact tympanic membrane. Which of the following structures is invaded by the mass?

- a) Epitympanum
- b) Mesotympanum
- c) Hypotympanum
- d) Mastoid

Question 15:

A 45-year-old is referred to your ENT practice with pulsatile tinnitus, and Brown's sign is positive. The patient gives you her CT films. Which of the following CT signs will help you differentiate between glomus jugulare and glomus tympanicum?

- a) Battle sign
- b) Lyre sign
- c) Phelps sign
- d) Griesinger sign

Question 16:

A 68-year old woman presents with complaints of right-sided facial weakness for 15 days and progressive hearing loss for 2 years. On examination, there is tachycardia and carotid bruit on the right side. Otoscopy shows a red mass behind the tympanic membrane and 24-hour urine collection shows the presence of metanephrines. Which of the following classification systems is used for this condition?

- a) Lederman's classification
- b) Fisch classification
- c) Austin's classification
- d) Sade's classification

Question 17:

A middle-aged woman presents with deafness, tinnitus, and bleeding from the ear. On examination, a reddish mass is seen behind the tympanic membrane which blanches on applying pressure with the pneumatic speculum. Which of the following is not a part of its management?

- a) Preoperative biopsy
- b) Radiotherapy
- c) Preoperative embolisation
- d) Surgery

Question 18:

What is the most common tumor at the cerebellopontine angle?

- a) Acoustic neuroma
- b) Cholesteatoma
- c) Glomus tumor
- d) Meningioma

Question 19:

Where does acoustic neuroma most commonly arise from?

- a) Facial nerve
- b) Vestibular division of 8th cranial nerve
- c) Cochlear division of 8th cranial nerve
- d) Auricular nerve

Question 20:

Which of the following features is common to vestibular schwannomas and glomus tumors?

- a) Encapsulated
- b) Common in middle aged females
- c) Vascular

- d) Slow growing

Question 21:

Which of the following is the most common symptoms that acoustic neuroma presents with?

- a) Conductive hearing loss
- b) Sensorineural hearing loss
- c) Vertigo
- d) Headache

Question 22:

Which of the following is the earliest cranial nerve to be involved in acoustic neuroma?

- a) 5th
- b) 6th
- c) 7th
- d) 8th

Question 23:

A 50-year-old patient presents with progressive unilateral hearing loss, decreased lacrimation, and loss of taste. On examining the external auditory meatus, there is decreased sensation on the posterior side. You suspect acoustic neuroma in this case. What sign was elicited here?

- a) Brown's sign
- b) Pulsation sign
- c) Hitzelberger's sign
- d) Rising sun sign

Question 24:

What is the gold standard investigation to confirm acoustic neuroma?

- a) CECT
- b) Gadolinium enhanced MRI

- c) X-Ray
- d) Biopsy

Question 25:

A middle-aged-man presents with progressive sensorineural hearing loss, imbalance, and tinnitus. Which of the following features rules out Meniere's disease and is diagnostic of acoustic neuroma?

- a) Stapedial reflex is present
- b) Recruitment phenomenon is present
- c) High SISI score
- d) Rollover phenomenon is seen

Question 26:

A 50-year-old man presents with difficulty hearing. Pure tone audiometry shows rollover phenomenon. An MRI with Gadolinium contrast shows a mass over the cerebellopontine angle. What is the treatment of choice?

- a) Surgery
- b) Radiotherapy
- c) Chemotherapy
- d) Conservative

Question 27:

Which of the following is not used in the treatment of vestibular schwannomas?

- a) X-Knife
- b) Gamma knife
- c) Cold knife
- d) CyberKnife

Answer Key

Question No.	Correct Option
1	c
2	d
3	b
4	c
5	a
6	a
7	c
8	b
9	c
10	d
11	c
12	d
13	c
14	c
15	c
16	b
17	a
18	a
19	b
20	d
21	b
22	d
23	c
24	b
25	d
26	a
27	c

Detailed Explanations

Solution to Question 1:

The most common site for ear carcinoma is the pinna, followed by the external auditory meatus, the middle ear, and mastoid.

Solution to Question 2:

The most common site of a sebaceous cyst is postauricular sulcus or below and behind the ear lobule. Treatment is total surgical excision.

Solution to Question 3:

The image shows a dark-colored summit over a swelling, usually seen in a sebaceous cyst.

Sebaceous cysts are the retention cysts of the sebaceous gland that are usually present on the face, scalp, or scrotum. They contain yellowish-white cheesy material with fat and epithelium.

Clinical features:

- The swelling is painless, soft, and freely mobile.
- Though non-transilluminating, they are Paget's test positive (fluctuant).
- Punctum is seen in 70% of the patients with sebaceous cysts. They are dark in color due to the accumulation of denuded epithelium.
- They also have an unpleasant sebum odor.
- The swelling can ulcerate and become painful (Cock's peculiar tumor).
- Some cysts over the scalp may also harden over time (Sebaceous horn).
- Many may get infected and lead to abscess formation.

Treatment:

- Excision including skin adjacent to punctum using elliptical incision-dissection method.
- Incision and avulsion of cyst wall.
- In the presence of an abscess, an initial treatment with an incision later followed by excision of the cyst

Other options:

Option A: Dermoid cysts present as firm painless subcutaneous nodules and are of 4 types:

- Sequestration dermoid - sequestration of epithelium during embryological development
- Tubulodermoid - accumulation of secretions from ectodermal tube/duct
- Implantation dermoid - implantation of the epidermis due to trauma
- Teratomatous dermoid - having all 3 germ layers (ecto, meso, and endoderm)

Option C: Meningiomas are benign tumors arising from the meningotheial cells of the dura. They present with headaches, seizures, or focal neurological deficits.

Option D: Postauricular cutaneous mastoid fistula is a rare complication of chronic otitis media that heals very slowly and needs surgical closure.

Solution to Question 4:

Exostoses are bilateral, multiple, smooth, sessile bony swellings in the deeper canal seen in divers/swimmers and surfers. It is also called as surfer's ear.

Osteoma	Exostosis/Surfer's Ear
Single	Multiple
Unilateral	Bilateral
Pedunculated, hard	Sessile, smooth
Outer canal	Inner, deeper canal

Solution to Question 5:

Squamous cell carcinoma is least likely to spread to the dura, as it is resistant to invasion. It commonly affects middle-aged women. Other clinical features include-

- Chronic foul-smelling discharge that is usually blood-stained.
- Friable, hemorrhagic granulations
- Polyp
- Hearing loss

It usually spreads to

- Ossicles
- Facial canal
- Internal ear
- Jugular bulb & carotid artery
- Petrous pyramid
- Parotid & temporomandibular joint
- Eustachian tube, Nasopharynx & Infratemporal fossa.

Lymph node involvement is a late finding.

Solution to Question 6:

Glomus tumor is the most common benign neoplasm of the middle ear and is so-named because of its origin from the glomus bodies.

Solution to Question 7:

Glomus tumors originate from paraganglionic cells that are derived from neural crest cells. They are found in glomus bodies present in the dome of the jugular bulb or promontory.

These cells are found in 2 sites giving rise to 2 similar yet distinct clinical tumors:

- Glomus jugulare: in the jugular bulb
- Glomus tympanicum: over the promontory along Jacobson's nerve (IX nerve).

Solution to Question 8:

Paragangliomas of the head and neck, also known as glomus tumors, affect women more than men.

Glomus tumors are benign, non-encapsulated but vascular. They are extremely slow-growing, locally invasive, and often seen in middle age (40–50 years).

Solution to Question 9:

Earliest symptoms of glomus tumor are tinnitus and hearing loss.

Hearing loss is conductive and slowly progressive.

Solution to Question 10:

Glomus tumour will be your first differential diagnosis. Tinnitus in glomus tumors is:

- Pulsatile
- Of swishing character
- Synchronous with pulse and
- Can be temporarily stopped by carotid pressure.

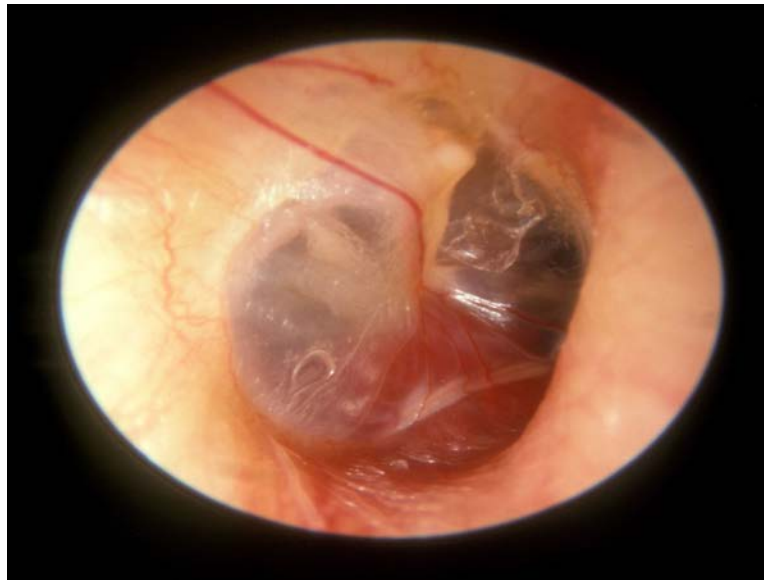
Solution to Question 11:

Rising sun sign is seen here, as it is a case of glomus tumor. The rising sun appearance is seen when a tumor arises from the floor of the middle ear. Otoscopy shows a red reflex through an intact tympanic membrane.

Option A - Cartwheel appearance is the leash of blood vessels appear along the handle of malleus and at the periphery of tympanic membrane.

Option B - Lighthouse sign is a pulsatile discharge from a pin-hole perforation of pars tensa seen in mastoiditis.

Option D - Schwartze sign is the reddish hue seen on the promontory through the tympanic membrane in otosclerosis.



Solution to Question 12:

The sign is called Brown's sign or pulsation sign which is suggestive of glomus tumor.

When ear canal pressure is raised with Siegel's speculum, the tumor pulsates vigorously and then blanches. The reverse happens with the release of pressure.

Option A - Rising Sun sign is seen when glomus tumor arises from the floor of the middle ear.

Option B - Schwarze sign is the reddish hue seen on the promontory through the tympanic membrane in otosclerosis.

Option C - Lighthouse sign is a pulsatile discharge from a pin-hole perforation of pars tensa seen in mastoiditis.

Solution to Question 13:

The given clinical scenario is suggestive of glomus tympanicum, a subtype of glomus tumors.

Symptoms of glomus tumors:

- Hearing loss
- Tinnitus
- Bleeding from the ear
- Vertigo and facial palsy

- Dysphagia, hoarseness of voice (cranial nerve palsies).

Examination findings:

- Intratympanic tumor: rising sun appearance with a positive Brown sign
- Red, vascular polyp that bleeds readily on manipulation or biopsy
- Once the tympanic membrane is breached: bleeding polyp in the external ear
- Audible bruit over the mastoid.

Solution to Question 14:

The given case scenario of hearing loss, pulsatile tinnitus, 12th cranial nerve involvement, and red mass on otoscopy is suggestive of glomus jugulare. Hypotympanum is invaded by the tumor.

It arises in the adventitia of the dome of the jugular bulb and invades the hypotympanum, jugular foramen, and jugular vein.

In addition to ear signs and symptoms, cranial nerve involvement is commonly seen of the IXth to XIIth cranial nerves in glomus jugulare.

Solution to Question 15:

Phelp sign helps differentiate between glomus jugulare and glomus tympanicum on CT.

It is the erosion of the spine of the carotico-jugular crest, seen in glomus jugulare tumors. It is intact in glomus tympanicum.

Option A - Battle's sign is ecchymosis over the mastoid seen in fractures of the temporal bone.

Option B - Lyre sign- splaying apart of internal and external carotid arteries on angiogram in cases of carotid body tumor of the neck.

Option D - Griesinger's sign is edema over the mastoid and is seen in lateral sinus thrombosis.

The image given shows Battle's sign.



Solution to Question 16:

The given case scenario with right-sided facial palsy, hearing loss, red mass behind the tympanic membrane, tachycardia, carotid bruit and positive urinary metanephrines is suggestive of a functional glomus tumor.

10% of all glomus tumors are functional and secrete catecholamines. This is diagnosed by testing the patient's 24-hour urine collection for urinary VMA (vanillylmandelic acid) or metanephrines.

Glomus tumors are classified based on Fisch classification

Option A: Lederman's classification is used in maxillary carcinoma.

Option C: Austin's classification is used to classify anatomical defects in the ossicular chain.

Option D: Sade's classification is used to classify atelectasis or retraction of the pars tensa.

Class	Extent
Class A	Tumor limited to the middle ear (mesotympanum)
Class B	Tumor in the middle ear (meso- and hypotympanum) and the mastoid
Class C	Tumour originates in the dome of the jugular bulb, destroying the overlying bone and has extensive local spread.
Class D	Intracranial extension. It can be intradural or extradural

Solution to Question 17:

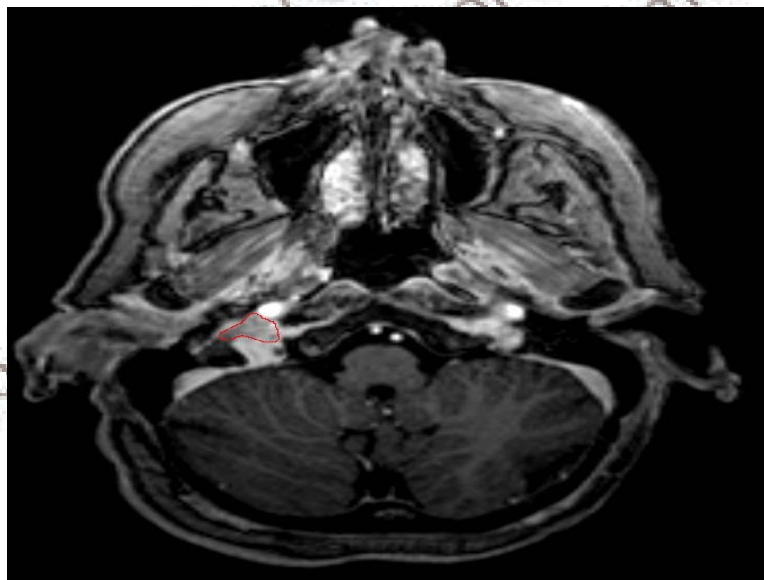
A pre-operative biopsy is not done in the management of glomus tumor due to the following reasons.

- Tumour is highly vascular and since the vessels lack the contractile musculature it bleeds profusely.
- Risk of injury to
- High jugular bulb
- Aberrant internal carotid artery

Clinical and radiological features are enough to arrive at a diagnosis. Treatment consists of:

- Surgical removal.
- Radiation.
- Embolization.
- Combination of the above

Glomus tumor: Circled in red:



Note: Although CT is the test of choice for the evaluation of bone involvement, gadolinium-enhanced MRI provides a superior definition of the relationship of the tumor to adjacent vascular and skull base structures. Hence Both MRI and CT are typically needed for a suspected jugulotympanic paraganglioma given that the information derived from each study is complementary.

Solution to Question 18:

Acoustic neuroma is the most common tumor at the cerebellopontine angle. It arises from the vestibular division of the VIIIth nerve within the internal auditory canal, widens and erodes the

canal and appears at the cerebellopontine angle.

Other tumors arising at the cerebellopontine angle are:

- Meningioma
- Cholesteatoma
- Arachnoid cysts
- Facial nerve schwannomas
- Glomus tumor.

Solution to Question 19:

Acoustic neuroma or vestibular schwannoma almost always arises from the Schwann cells of the vestibular division of 8th nerve.

It arises from the superior and the inferior division of the vestibular nerve in the medial internal auditory canal at the site of transition of central to peripheral myelin i.e., Obersteiner-Redlich zone.

Rarely, it can arise from the cochlear division of the 8th nerve within the internal auditory canal.

Solution to Question 20:

Glomus tumors and vestibular schwannomas are both slow-growing, benign tumors of the temporal bone with locally invasive properties.

Vestibular schwannomas are encapsulated tumors. They are seen in middle aged individuals and have equal incidence in both sexes.

Glomus tumors are vascular paragangliomas and are non-encapsulated tumors. They are seen more commonly in middle-aged women.

Solution to Question 21:

Progressive unilateral sensorineural hearing loss (SNHL), often accompanied by tinnitus, is the presenting symptom in the majority of cases.

Early symptoms of acoustic neuroma:

- Progressive unilateral SNHL
- Gross inability to understand speech out of proportion to the PTA average
- Tinnitus
- Unsteadiness and imbalance

The involvement of other cranial nerves leads to other deficits.

Solution to Question 22:

The 8th cranial nerve is the earliest to be involved in acoustic neuroma.

Cochleovestibular symptoms are the earliest symptoms when the tumor is still intracanalicular and are caused by pressure on cochlear or vestibular nerve fibers or on the internal auditory artery.

Once the tumor erodes the canal, it presents in the cerebellopontine angle and can expand to involve other nearby cranial nerves such as the 5th and 7th cranial nerves.

Note: Acoustic neuroma arises from the 8th cranial nerve. Hence, it is the first nerve involved. However, if the 8th cranial nerve is not given as an option, then the answer would be the 5th cranial nerve.

Solution to Question 23:

Hitzelberger's sign is the sign elicited in the given clinical scenario of acoustic neuroma.

Hitzelberger's sign is the hypoaesthesia of the posterior meatal wall because the tumor has affected the sensory fibers of the facial nerve. In this case, sensory fibers of the facial nerve are affected, as the patient has decreased lacrimation and loss of taste. Motor fibers are more resistant and are affected late. A delayed blink reflex may be an early manifestation.

Solution to Question 24:

MRI with gadolinium contrast is the gold standard for the diagnosis of acoustic neuroma.

It is superior to a CT scan. Intracanalicular tumors measuring even a few millimeters can be easily diagnosed by this method.

Solution to Question 25:

Rollover phenomenon is diagnostic of acoustic neuroma. It is a feature of retro-cochlear/neural lesions wherein the speech discrimination steadily increases with intensity but suddenly drops significantly beyond a particular intensity of sound.

Meniere's disease is an important differential for acoustic neuroma as the patients suffer from similar symptoms. Meniere's is a cochlear pathology and acoustic neuroma is retrocochlear.

Option A - Stapedial reflex is present in cochlear lesions (Meniere's) and absent in retrocochlear lesion (acoustic neuroma)

Option B - Recruitment phenomenon is the abnormal growth in loudness in the affected ear, seen in cochlear pathologies.

Option C - SISI (short increment sensitivity score) is a test of intensity discrimination which is very high in cochlear pathologies but low in retro-cochlear.

Solution to Question 26:

The given clinical scenario is suggestive of acoustic neuroma. Treatment of choice in this case is surgery.

The approach depends upon site and size of tumor:

- Middle cranial fossa approach
- Translabyrinthine approach
- Retrosigmoid/Suboccipital approach
- Combined.

Solution to Question 27:

Cold knife is not used in the treatment of vestibular schwannomas. It is used for the conization of the cervix in cervical carcinoma.

Stereotactic radiosurgery is a reasonable alternative for people who refuse or cannot tolerate surgery. Here, beams are modified and focused only to concentrate on the tumor and have little effect on surrounding tissue. It can be done by using

- X-knife using a linear accelerator (LINAC)
- Gamma-knife using Cobalt 60
- CyberKnife - image-guided linear accelerator using real-time image guidance to deliver radiation from multiple angles.

External beam radiation is not recommended as it causes significant damage to the surrounding brain and neural structures.

Anatomy of Facial Nerve

Question 1:

What type of nerve is the facial nerve?

- a) Mixed peripheral nerve
- b) Sensory cranial nerve
- c) Motor peripheral nerve
- d) Mixed cranial nerve

Question 2:

Which of the following is not a component of the facial nerve?

- a) Special visceral efferent
- b) Special visceral afferent
- c) General visceral efferent
- d) General somatic efferent

Question 3:

In a patient with an acoustic neuroma involving the sensory division of the facial nerve, the Hitzelberger sign was positive. What is the other name for this nerve?

- a) Jacobson's nerve
- b) Arnold's nerve
- c) Nerve of Wrisberg
- d) Herring's nerve

Question 4:

You are performing the cranial nerve examination in a patient with a stroke. You noticed a loss of taste sensation in the anterior 2/3rd of the tongue. What is this sensation carried by?

- a) Special visceral efferent
- b) General visceral efferent

- c) Special visceral afferent
- d) General somatic afferent

Question 5:

A patient with the following findings is presented to you. Where is the motor nucleus of the involved cranial nerve located?



- a) Medulla
- b) Pons
- c) Midbrain
- d) Cerebellum

Question 6:

Which of the following best describes the course of the facial nerve in the pons?

- a) It hooks around the 8th nerve nucleus
- b) It hooks around the 6th nerve nucleus
- c) It exits the pons dorsally
- d) It exits the pons at ponto-pyramidal junction

Question 7:

Which of the following is false about the first genu of the facial nerve?

- a) Contains a ganglion which supplies the lacrimal gland
- b) Gives the first branch of the facial nerve
- c) Lies on the posterior wall of the middle ear
- d) Between the fundus of the internal acoustic meatus and the geniculate ganglion.

Question 8:

Which of the following is true for the 2nd genu of the facial nerve?

- a) It contains gasserian ganglion
- b) It gives the second branch of the facial nerve
- c) It lies on the inferior wall of the middle ear
- d) It is at the junction of 3rd and 4th intratemporal segments

Question 9:

Which of the following is the shortest and narrowest segment of the facial nerve?

- a) Meatal segment
- b) Labyrinthine segment
- c) Tympanic segment
- d) Mastoid segment

Question 10:

In the examination of a patient with facial nerve palsy, the Schirmer test was positive. Involvement of which of the following branches of the facial nerve is responsible for this?

- a) Greater superficial petrosal nerve
- b) Nerve to stapedius
- c) Chorda tympani
- d) Lesser petrosal nerve

Question 11:

A young man who underwent surgery for chronic suppurative otitis media presented with a loss of taste sensation. Which segment of the facial nerve gives rise to the involved branch?

- a) Labyrinthine
- b) Tympanic
- c) Mastoid
- d) Extratemporal

Question 12:

Which of the following is the most common site for congenital dehiscence of the fallopian canal?

- a) Meatal
- b) Labyrinthine
- c) Tympanic
- d) Mastoid

Question 13:

Which of the following is not a terminal branch of the facial nerve?

- a) Temporal
- b) Frontal
- c) Buccal
- d) Cervical

Question 14:

A patient with an acoustic neuroma at the cerebellopontine angle presents with a facial nerve palsy. Which of the following muscles will not be paralyzed in this case?

- a) Stylohyoid
- b) Anterior belly of digastric
- c) Orbicularis oculi
- d) Occipital belly of occipitofrontalis

Question 15:

The facial nerve is not supplied by which of the following vessels?

- a) Anterior inferior cerebellar artery
- b) Labyrinthine artery
- c) Superficial petrosal artery
- d) Facial artery

Question 16:

You are an ENT resident assisting on a modified radical mastoidectomy in a patient with cholesteatoma. Which of the following landmarks will not help you to identify the facial nerve?

- a) Processus cochleariformis
- b) Pyramid
- c) Short process of incus
- d) Posterior semicircular canal

Question 17:

While performing a parotidectomy, which of the following landmarks would you use to identify the facial nerve?

- a) Tip of mastoid
- b) Temporomandibular joint
- c) Internal jugular vein
- d) Tragus

Question 18:

While performing mastoid exploration, which of the following is not useful in preventing the injury to the facial nerve?

- a) Constant saline irrigation
- b) Using a smaller burr
- c) Drilling parallel to course of nerve

d) Good magnification

Answer Key

Question No.	Correct Option
1	d
2	d
3	c
4	c
5	b
6	b
7	c
8	d
9	b
10	a
11	c
12	c
13	b
14	b
15	d
16	d
17	d
18	b

Detailed Explanations

Solution to Question 1:

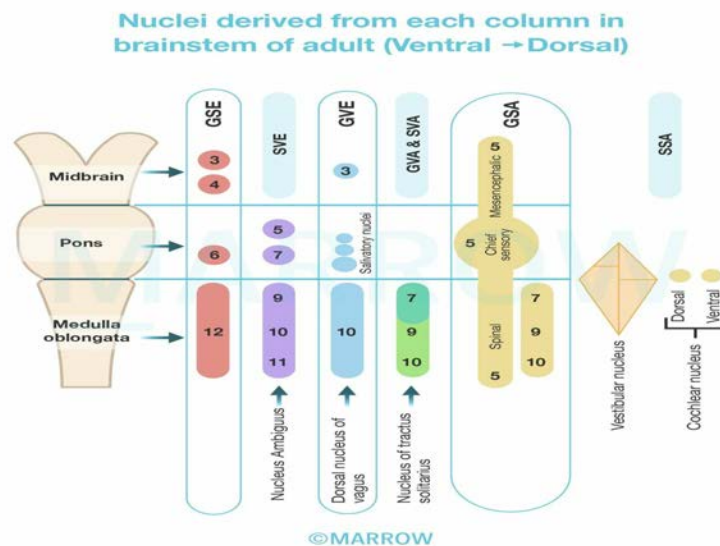
The facial nerve is a mixed cranial nerve which contains both sensory and motor part. It is the 7th cranial nerve.

Solution to Question 2:

The general somatic efferent is not a component of the facial nerve.

Components of the facial nerve:

- Special visceral efferent - motor to muscles of facial expression (derived from branchial arches, hence special visceral and not general somatic like the other muscles)
- General visceral efferent - secretomotor to lacrimal and submandibular glands
- Special visceral afferent - taste from anterior 2/3rd tongue
- General somatic afferent - general sensations from pinna, external auditory canal, and tympanic membrane.



Solution to Question 3:

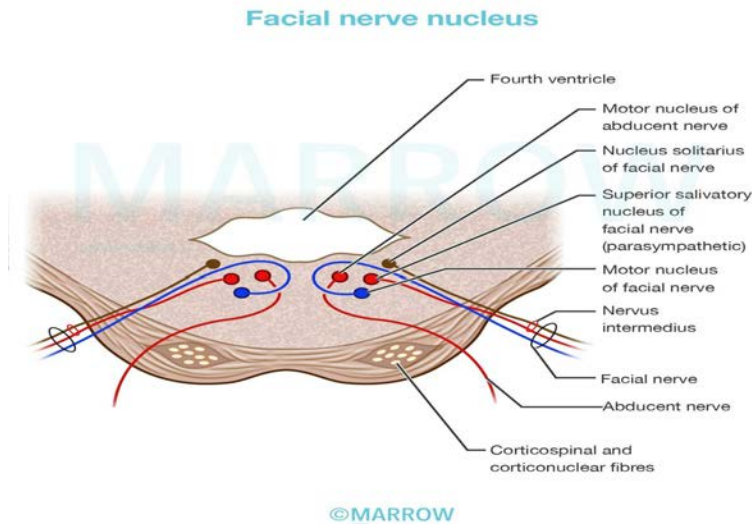
The sensory division of the facial nerve is also called the nerve of Wrisberg. The facial nerve is a mixed nerve having the motor and a sensory root.

Hitzelberger sign refers to hypoesthesia of the posterior meatal wall in acoustic neuroma due to the compression of the facial nerve.

Option A: Jacobson's nerve - the tympanic branch of the glossopharyngeal nerve (9th cranial nerve). It is associated with glomus tympanicum.

Option B: Arnold's nerve - the auricular branch of the vagus nerve (10th cranial nerve).

Option D: Herring's nerve - the carotid sinus branch of the glossopharyngeal nerve (9th cranial nerve).



Solution to Question 4:

The taste sensation from the anterior 2/3rd of the tongue is carried by the special visceral afferent of the facial nerve via chorda tympani nerve.

The taste sensation from the hard and soft palate is also carried by the facial nerve via the greater superficial petrosal nerve to the nucleus tractus solitarius.

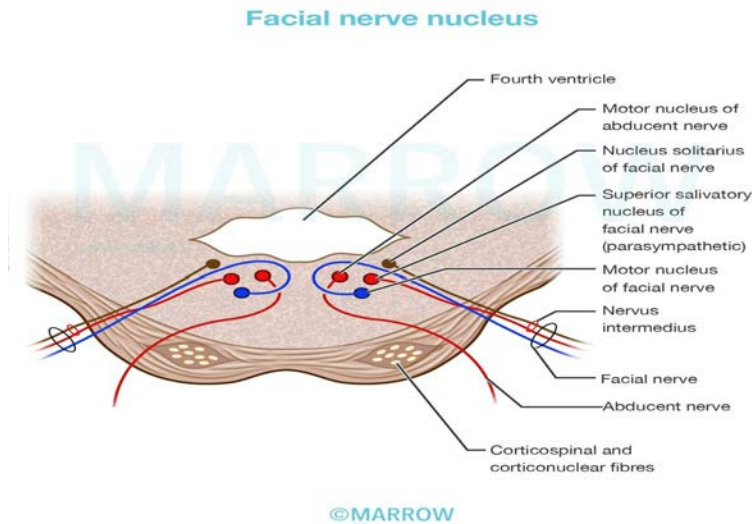
Solution to Question 5:

The given image shows left-sided lower motor neuron facial nerve palsy and the motor nucleus is located in the pons. It is located along with abducens and vestibulocochlear nerve nuclei.

Solution to Question 6:

The facial nerve in the pons hooks around the 6th (abducens) nerve nucleus.

It exits the pons ventrally at the pontomedullary (ponto-olivary) junction. It then runs in the cerebellopontine angle to reach the internal auditory meatus.

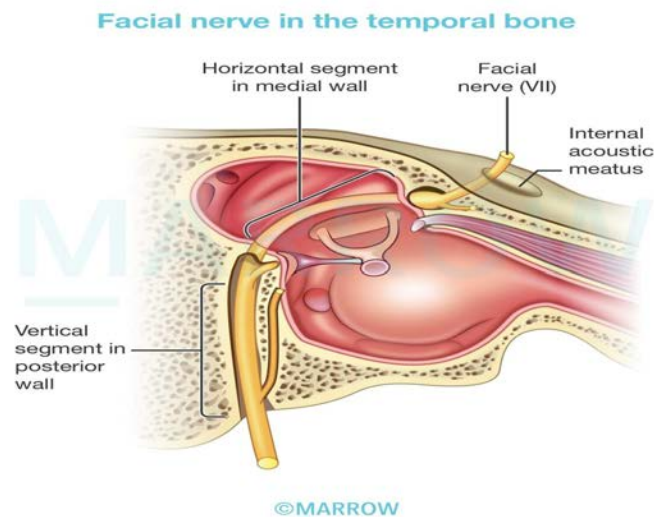


Solution to Question 7:

The 1st genu of the facial nerve lies on the medial wall of the middle ear and not posterior wall.

It is formed at the geniculate ganglion when the facial nerve from the fundus of the internal acoustic meatus takes a turn posteriorly. Then the facial nerve continues as a tympanic or horizontal segment.

It homes the geniculate ganglion which supplies the lacrimal gland (via greater superficial petrosal nerve). It gives off the first branch of the facial nerve.



Solution to Question 8:

The 2nd genu of the facial nerve lies at the junction of the 3rd and the 4th intratemporal segments, between the tympanic and the mastoid segments.

The 2nd genu of the facial nerve is formed where the horizontal facial nerve takes a turn inferiorly at the junction between the medial and posterior walls of the middle ear.

It continues as the vertical segment of the facial nerve behind the posterior wall of the middle ear. It does not contain a ganglion and does not give any branches.

The course of the facial nerve has 3 parts-

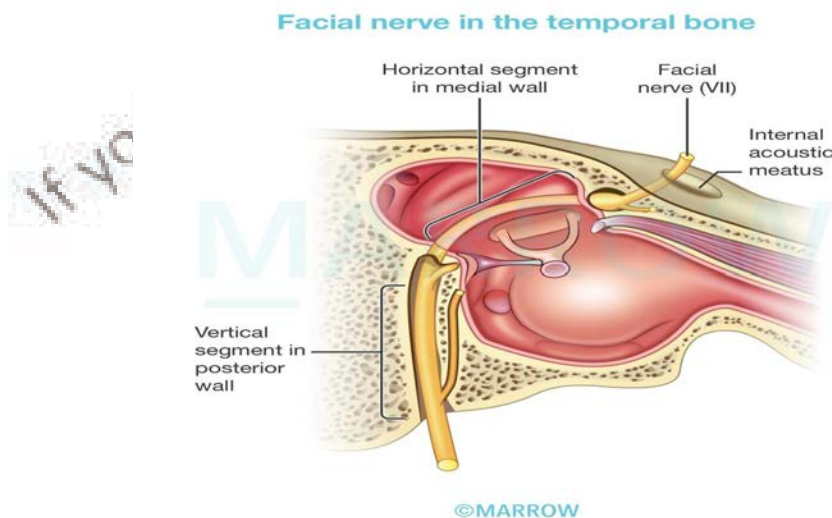
- Intracranial
- Intratemporal
- Extracranial.

Intratemporal part of the facial nerve has 4 segments-

- Meatal
- Labyrinthine
- Tympanic/horizontal
- Mastoid/vertical segments.

Option A: Gasserian ganglion is present in the trigeminal nerve.

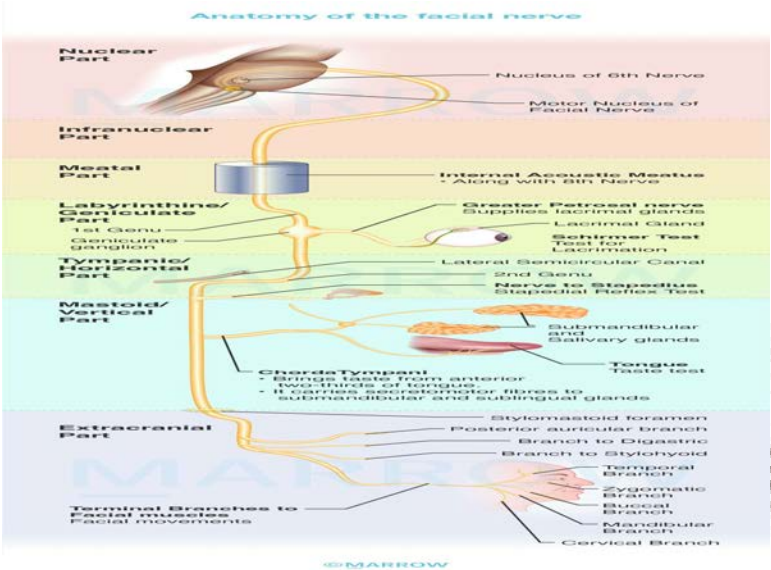
Option B: The temporal segment of the facial nerve ends with the 2nd genu. The nerve then passes vertically downward to form the mastoid segment. The 2nd branch of facial nerve (nerve to stapedius) arises from this segment close to the level of 2nd genu and not from the 2nd genu.



Solution to Question 9:

The shortest and narrowest segment of the facial nerve is the labyrinthine segment (length 4.0 mm, diameter 0.61–0.68 mm).

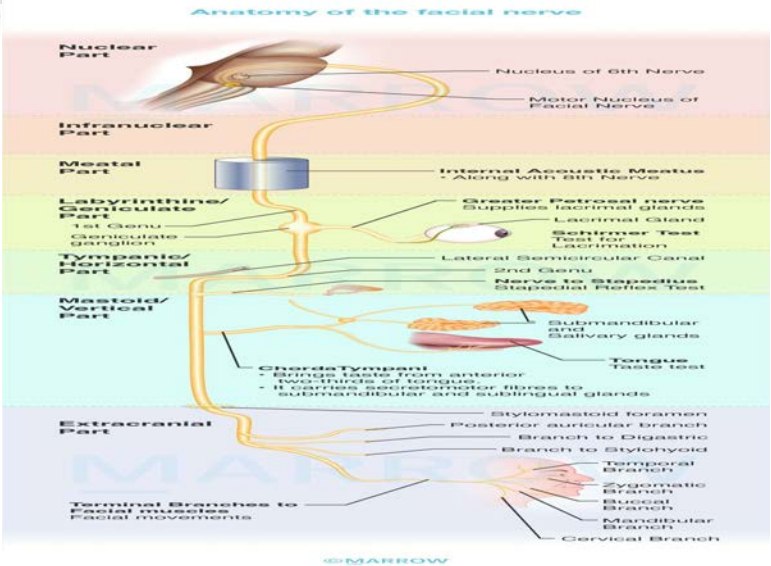
The labyrinthine segment is from the fundus of meatus to the geniculate ganglion where the nerve takes a turn posteriorly forming a genu. The nerve in the labyrinthine segment has the narrowest diameter and the bony canal in this segment is also the narrowest. Thus edema or inflammation can easily compress the nerve and cause paralysis.



Solution to Question 10:

Involvement of the greater superficial petrosal nerve causes a decrease in lacrimation, detected by the Schirmer test.

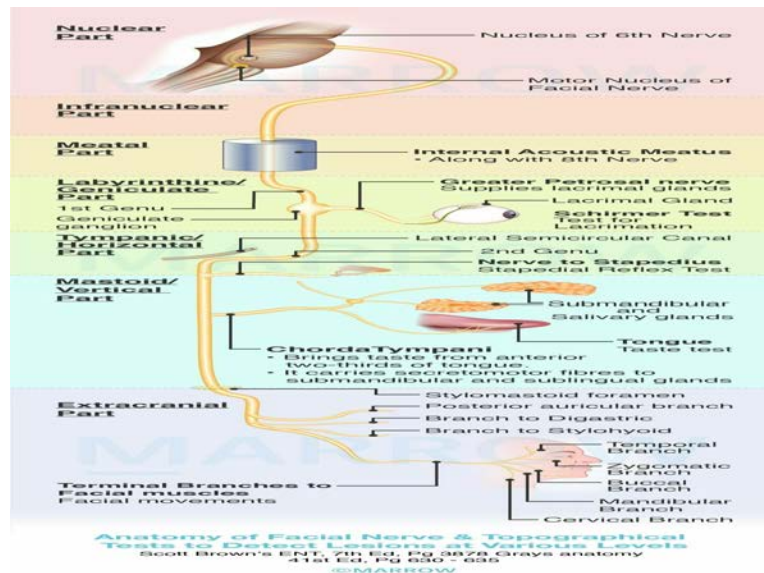
It is the first branch of the facial nerve and arises from the geniculate ganglion. It carries secretomotor fibres to the lacrimal gland and the glands of nasal mucosa and palate.



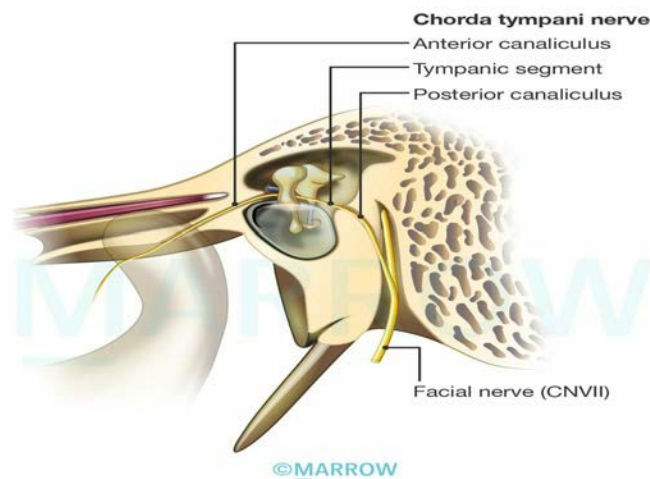
Solution to Question 11:

The injury to chorda tympani nerve produces loss of taste sensation in the anterior 2/3rd of the tongue and it arises from the mastoid or vertical segment of the facial nerve.

Chorda tympani nerve carries the taste sensation from anterior 2/3rd of the tongue. It is a branch of the facial nerve arising from the mastoid/vertical segment after the nerve to stapedius before exiting the stylomastoid foramen. It relays in the nucleus of tractus solitarius.



Course of chorda tympani nerve



Solution to Question 12:

The congenital dehiscence (absence of bony cover) of the fallopian canal of the facial nerve occurs most commonly in the tympanic or horizontal segment over the oval window.

It is the most common anomaly of the facial canal. A dehiscent nerve is prone to injury at the time of surgery. It gets easily compressed or inflamed in mastoid and middle ear infections.

Solution to Question 13:

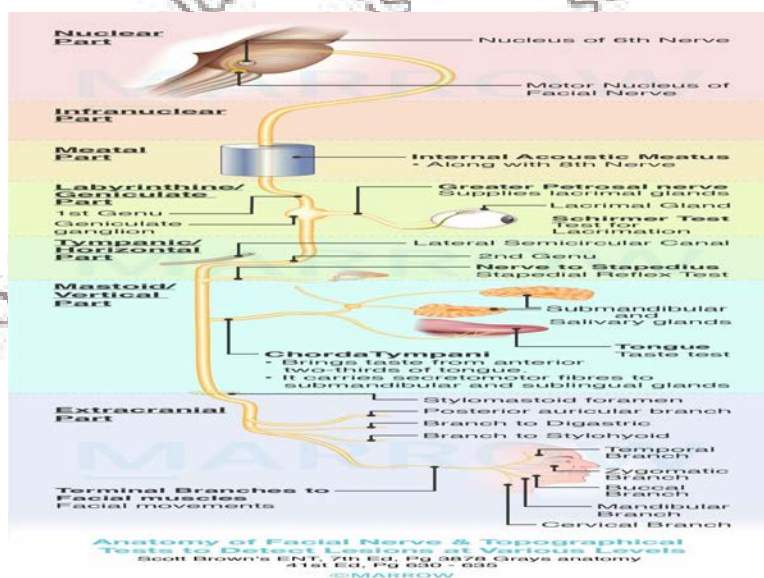
The frontal branch is not a terminal branch of the facial nerve.

The five terminal branches of the facial nerve are:

- Temporal
- Zygomatic
- Buccal
- Mandibular
- Cervical

The extratemporal course of the facial nerve starts at the stylomastoid foramen where it exits the temporal bone and runs anteriorly to enter the substance of the parotid gland.

Here it divides into 2 main terminal trunks, the temporofacial and the cervicofacial. These further divide and re-anastomose to give the five terminal branches.



Solution to Question 14:

Anterior belly of digastric is not paralyzed in facial nerve palsy.

Muscles supplied by facial nerve:

- Nerve to stapedius - stapedius muscle.
- Posterior auricular nerve - muscles of the pinna and the occipital belly of occipitofrontalis.

- Muscular branches - stylohyoid and posterior belly of digastric.
- Terminal branches - muscles of facial expression.

Solution to Question 15:

Facial artery does not supply the facial nerve

Vascular Supply of facial nerve:

- Anterior-inferior cerebellar artery supplies the nerve in the cerebellopontine angle
- Labyrinthine artery, a branch of the anterior-inferior cerebellar artery supplies the nerve in the internal auditory canal
- Superficial petrosal artery, a branch of the middle meningeal artery supplies geniculate ganglion and the adjacent region
- Stylomastoid artery, a branch of the posterior auricular artery supplies the mastoid segment.

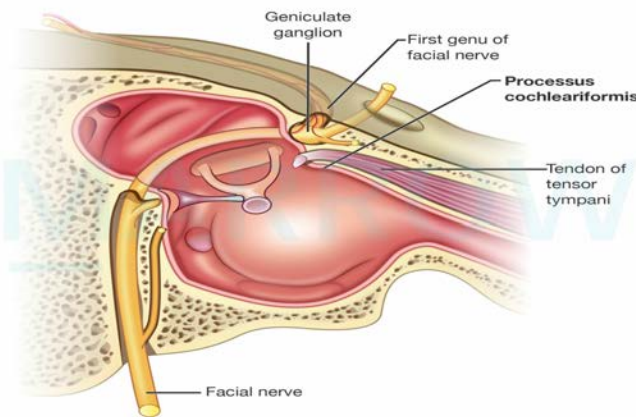
Solution to Question 16:

Posterior semicircular canal is not used as a surgical landmark to identify the facial nerve.

Surgical Landmarks of the facial nerve for middle ear and mastoid surgery:

- Processus cochleariformis - It demarcates the geniculate ganglion which lies just anterior to it. The tympanic segment of the nerve starts at this level.
- Oval window and horizontal semicircular canal - The horizontal segment of the facial nerve runs above the oval window (stapes) and below the horizontal/lateral semicircular canal.
- Short process of incus - The vertical segment of the facial nerve lies medial to the short process of incus at the level of aditus.
- Pyramid - The vertical segment of the facial nerve runs behind the pyramid and the posterior tympanic sulcus.

Processus cochleariformis and its relations



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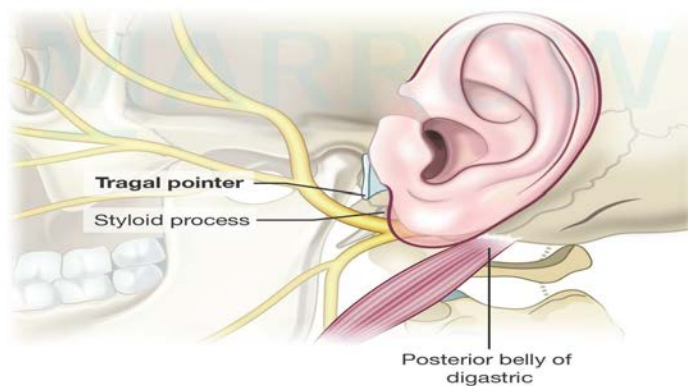
Solution to Question 17:

Tragus is used as a surgical landmark to identify facial nerve in parotid surgery.

Surgical landmarks of the facial nerve for parotid surgery:

- Tragal pointer - triangular inferior portion of the tragus. The main trunk lies 1 cm deep to it.
- Tympanomastoid suture - the main trunk lies 6-8 mm deep to it.
- Styloid process - the nerve lies lateral to it
- Posterior belly of digastric attachment at the digastric groove - the nerve can be found emerging from the stylomastoid foramen under the muscle but lateral to the styloid process.

Surgical landmarks of facial nerve in parotid surgery



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Solution to Question 18:

A larger burr is used to prevent facial nerve injury in the mastoid exploration and not a smaller burr.

Measures to reduce facial nerve injury during mastoid exploration:

- Good anatomical knowledge
- Working parallel to the nerve, never at an angle.
- Use the biggest burr available. Smaller burrs are more harmful.
- Constant irrigation prevents thermal injury
- Granulations present over the facial nerve are best left alone.
- Good magnification.

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Facial Nerve Disorders

Question 1:

Which of the following is the most common type of facial nerve injury?

- a) Neuropraxia
- b) Axonotemesis
- c) Neurotemesis
- d) Transection

Question 2:

You are a final-year student examining a patient with a right supranuclear facial palsy. Which of the following findings you will see?

- a) Loss of taste from anterior 2/3rd of tongue
- b) Paralysis of lower facial muscle on left side
- c) Paralysis of lower facial nerve on right side
- d) Loss of corneal reflex on left side

Question 3:

While examining a patient with upper motor neuron facial nerve palsy, you notice emotional expressions such as smiling and crying are unaffected. How will you explain this?

- a) Incomplete paralysis of facial muscles
- b) Cross regeneration of fibres
- c) Innervation by fibres from thalamus
- d) Innervation by fibres from prefrontal cortex

Question 4:

Which of the following is the most common cause of LMN facial nerve palsy?

- a) Bell's palsy
- b) Ramsay hunt syndrome

- c) Cerebellopontine angle tumors
- d) Postoperative

Question 5:

A 35-year-old housewife presents with facial weakness on the right side, difficulty in closing the right eye, and loss of taste sensation. A diagnosis of Bell's palsy is made. Which of the following is false regarding this condition?

- a) Herpes virus is a cause
- b) It is an idiopathic condition
- c) Recurrence is uncommon
- d) Males are affected more commonly than females

Question 6:

While examining a patient with LMN facial palsy, you observe Bell's phenomenon. In which direction would you expect the eyeball to move?

- a) Upward and inward
- b) Upward and outward
- c) Downward and inward
- d) Downward and outward

Question 7:

You are an ENT resident examining a patient with Bell's palsy. Which of the following findings is unlikely to be present in this patient?

- a) Ipsilateral facial muscle weakness
- b) Ipsilateral ptosis
- c) Ipsilateral ageusia
- d) Hyperacusis

Question 8:

A patient with facial nerve palsy is intolerant to loud sounds. Involvement of which muscle will cause this?

- a) Tensor tympani
- b) Stapedius
- c) Tensor veli palatini
- d) Levator veli palatini

Question 9:

In the evaluation of a patient with facial nerve palsy, what is the ideal time to do electroneuronography?

- a) Immediately following injury
- b) Within 2 days of injury
- c) 4-21 days following injury
- d) 2 months following injury

Question 10:

A 2nd year ENT resident is evaluating a patient with Bell's palsy. Which of the following tests he will not do to identify the site of the lesion?

- a) Stapedial reflex
- b) Schirmer test
- c) Electrogustometry
- d) Electromyography

Question 11:

A patient with facial nerve palsy presented with dry eye. Injury to which part of the facial nerve will produce this symptom?

- a) Geniculate ganglion
- b) Tympanic segment
- c) Mastoid segment
- d) Chorda tympani

Question 12:

A middle-aged man presents with facial asymmetry. On examination, there is loss of taste from anterior 2/3rd of the tongue, absent acoustic reflex, and normal Schirmer test. What is the site of the lesion?

- a) Proximal to geniculate ganglion
- b) Mastoid segment
- c) Tympanic segment
- d) Chorda tympani

Question 13:

A 30-year-old man presented with the following findings which he noticed when he woke up in the morning. There was no history of associated fever or rash. A panel of investigations was done and were found to be normal. Which of the following is false regarding the treatment of this condition?



- a) 2,4
- b) 2,3,4
- c) 2,4,5
- d) 1,2,4

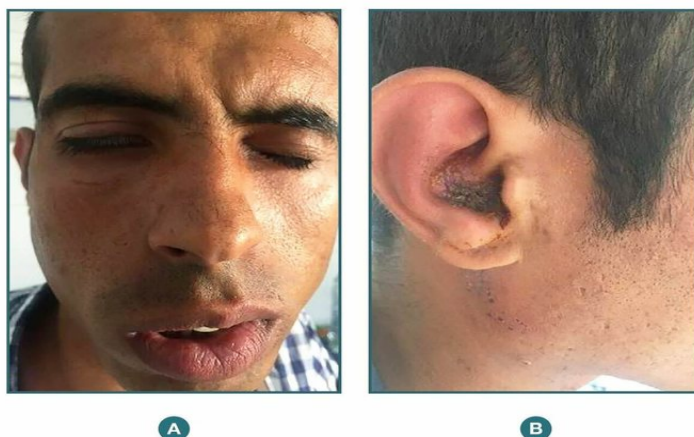
Question 14:

Which of the following would you not expect to see in a patient with Melkersson-Rosenthal syndrome?

- a) Facial nerve palsy
- b) Swelling of lips
- c) Cleft palate
- d) Fissured tongue

Question 15:

A patient with the following findings in the image comes to the OPD. What is the most likely causative agent for this condition?



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- a) Herpes simplex virus
- b) Herpes zoster virus
- c) Epstein barr virus
- d) Cytomegalovirus

Question 16:

A 30-year-old woman presented with left-sided facial weakness. On examination, she has vesicles in the external auditory canal and over the tympanic membrane. Which of the following is false about this condition?

- a) Herpes zoster is the cause

- b) Involves 7th cranial nerve
- c) 8th cranial nerve may also be involved
- d) Surgery is the mainstay of treatment

Question 17:

An anxious woman presents to you with complaints of watering of the right eye while eating. She gives a history of transient facial weakness a few months back. Which of the following is the most likely cause for this condition?

- a) Faulty regeneration of facial nerve fibres
- b) Faulty regeneration of trigeminal nerve fibres
- c) Faulty regeneration of vagal nerve fibres
- d) Faulty regeneration of auriculotemporal nerve fibres

Question 18:

A 50-year-old woman presented with sweating over the right cheek while eating. She underwent parotid surgery 6 months back. The finding of the starch iodine test is shown in the image. Which of the following best explains the pathophysiology of this condition?



- a) Aberrant reinnervation of parasympathetic fibres of auriculotemporal nerve with sympathetic sweat glands
- b) Aberrant cross innervation of the facial nerve with greater auricular nerve

- c) Aberrant reinnervation of sympathetic fibres of auriculotemporal nerve with parasympathetic sweat glands
- d) Aberrant cross innervation of the facial nerve with auriculotemporal nerve

Question 19:

Which of the following is the most common cause of hemifacial spasm?

- a) Acoustic neuroma
- b) Glomus tumor
- c) Vascular loop
- d) Congenital cholesteatoma

Question 20:

Which temporal bone fracture is most commonly associated with facial nerve palsy?

- a) Longitudinal fracture
- b) Transverse fracture
- c) Mastoid fracture
- d) Styloid fracture

Question 21:

Following a blow to the head by a metal rod during a gang fight, a patient presented with clear watery discharge from the ear. The discharge is positive for beta-2 transferrin. Which of the following is the most likely cause for this condition?

- a) Fracture of cribriform plate
- b) Fracture of petrous temporal bone
- c) Rupture of tympanic membrane
- d) Fracture of parietal bone

Question 22:

You suspect a temporal bone fracture in a patient who presented with right side facial weakness, decreased hearing, vertigo, and bleeding from the ear. What is the investigation of

choice to confirm the diagnosis?

- a) NCCT
- b) CECT
- c) HRCT
- d) MRI

Question 23:

An intern is assisting the surgeon in total parotidectomy. During the procedure, he accidentally injured the facial nerve. Which of the following cannot be used as a graft for repair in this case?

- a) Sural nerve
- b) Mandibular nerve
- c) Greater auricular nerve
- d) Lateral cutaneous nerve of thigh

Answer Key

Question No.	Correct Option
1	a
2	b
3	c
4	a
5	d
6	b
7	b
8	b
9	c
10	d
11	a
12	c
13	c
14	c

15	b
16	d
17	a
18	a
19	c
20	b
21	b
22	c
23	b

Detailed Explanations

Solution to Question 1:

The most common type of nerve injury, in general, is neuropraxia. It is also true for facial nerve injury.

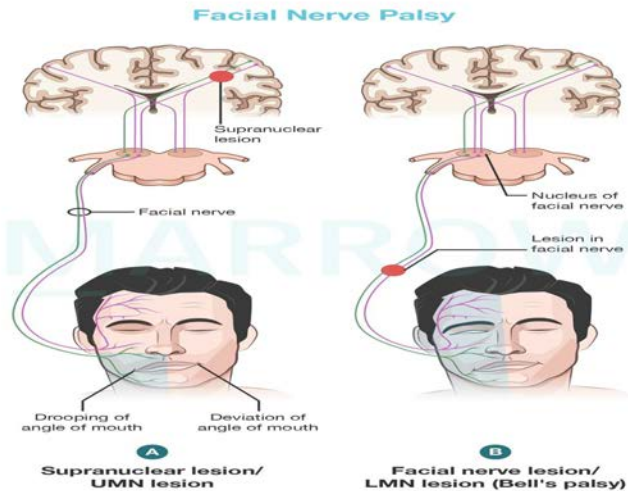
Solution to Question 2:

Right upper motor neuron (UMN) facial nerve palsy causes paralysis of lower facial muscles on the left side.

The upper part of the facial nerve's motor nucleus, which innervates forehead muscles, receives fibers from both the cerebral hemispheres. The lower part of the nucleus, which supplies the lower face, only receives crossed fibers from the contralateral hemisphere. Therefore, the upper part of the face gets bilateral innervation and the lower part of the face gets crossed innervation from the opposite side.

Hence, in supranuclear lesion (UMN palsy), forehead is spared and muscles of the lower half of the opposite side of the face are affected.

In LMN palsy, all muscles of the same side are affected.

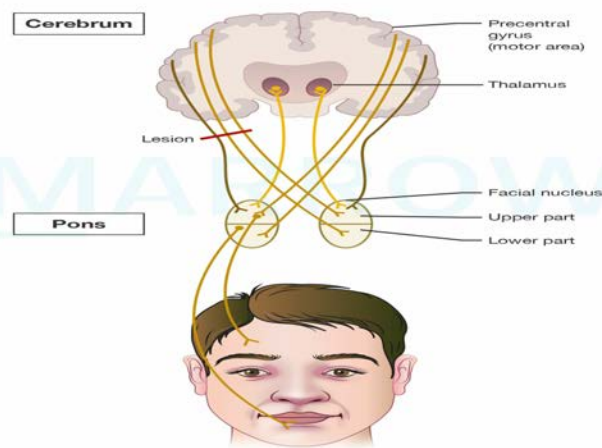


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Solution to Question 3:

The fibres from the thalamus are responsible for the involuntary movements of facial muscles in emotional states. These fibres reach the facial nerve nucleus via alternate pathways and are unaffected by UMN causes.

Supranuclear innervation of facial nerve



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Solution to Question 4:

Bell's palsy is the most common cause of LMN facial nerve palsy. It is defined as idiopathic, ipsilateral lower motor neuron (LMN) facial paralysis, or paresis of acute onset.

Causes of facial nerve palsy:

	UMN	LMN
Unilateral	Vascular (infarct) Cerebral tumor Multiple sclerosis	Bell's palsy (most common overall) Parotid tumor Head injuries Skull base tumors Diabetes Hypertension
Bilateral	Vascular (infarct) Motor neuron disease	Guillain-Barre syndrome Sarcoidosis Leukemia Lymphoma Leptospirosis

Solution to Question 5:

Bell's palsy affects both males and females equally. It is an acute onset idiopathic ipsilateral LMN facial nerve palsy.

Most of the evidence supports the viral etiology due to herpes simplex, herpes zoster, or the Epstein-Barr virus.

Positive family history is present in 6–8% of patients. Most of the patients recover fully and the remaining require surgery. Recurrent facial palsy is seen in 3–10% of cases.

Solution to Question 6:

In Bell's phenomenon, the eyeball turns upwards and outwards on the affected side while attempting to close the eyes.

This phenomenon is seen in LMN facial palsy.

The following image shows Bell's phenomenon.



Solution to Question 7:

Ptosis is not seen in patients with Bell's palsy.

Clinical presentation of Bell's palsy:

- Sudden onset
- Bell's phenomenon
- Asymmetrical face
- Some may complain of noise intolerance (stapedial paralysis) or loss of taste (involvement of chorda tympani).

The following image shows a patient with Bell's palsy.



Solution to Question 8:

Noise intolerance in a patient with facial nerve palsy is due to stapedius muscle paralysis.

The stapedius muscle is supplied by the facial nerve via nerve to stapedius. It acts by dampening the loud sounds and the loss of this function results in hyperacusis.

Tensor veli palatini helps in opening the eustachian tube, hence also called dilator tubae muscle.

Tensor tympani, supplied by the mandibular branch of the trigeminal nerve, also helps in dampening the loud sounds. It is not involved in facial nerve palsy.

Solution to Question 9:

Electroneuronography is most useful when done between 4 and 21 days of the onset of complete paralysis.

Degeneration of the nerve fibers cannot be detected earlier than 48–72 h of its commencement. Hence electrodiagnostic tests are done after 72 hours following injury.

Solution to Question 10:

Electromyography is not a topodiagnostic test used in facial palsy. It is used to monitor nerve degeneration in facial palsy.

Topodiagnostic tests are done to identify the site of lesion in the intratemporal part of the facial nerve. They are:

- Schirmer test - quantification of lacrimation
- Stapedial reflex
- Taste test by electrogustometry
- Submandibular salivary flow test.

Solution to Question 11:

Injury at the level or proximal to the geniculate ganglion will produce dry eye due to decreased lacrimation.

Geniculate ganglion gives the secretomotor fibers to the lacrimal gland via the greater superficial petrosal nerve.

Solution to Question 12:

The site of lesion in this patient with loss of taste from anterior 2/3rd of tongue and absent acoustic reflex is at the tympanic segment.

Injury to the greater superficial petrosal nerve leads to loss of lacrimation which is detected by the Schirmer test. It is the first branch given at the level of the geniculate ganglion.

Injury to the nerve to stapedius, the first motor branch of the facial nerve, leads to loss of stapedial reflex and hyperacusis, which can be detected by tympanometry.

Injury to chorda tympani nerve leads to loss of taste from anterior 2/3rd of the tongue which is detected by electrogustometry and reduced submandibular and sublingual salivary flow detected by salivary flow test.

So in this patient, the lesion is proximal to the origin of nerve to stapedius i.e. tympanic segment, and not proximal to geniculate ganglion as the lacrimation is normal.

Solution to Question 13:

The given clinical scenario is suggestive of Bell's palsy and statements 2,4 and 5 are false with respect to the treatment.

Management of Bell's palsy:

Steroids are the drug of choice for Bell's palsy. If the patient reports within 1 week, the adult dose of prednisolone 1 mg/kg/day in 2 divided doses for 5 days should be given.

Steroids can be combined with acyclovir for herpes zoster oticus or if viral etiology is suspected in Bell's palsy.

General measures include reassurance, physiotherapy, or massage of facial muscles, and care of the eye to prevent exposure keratitis. It includes artificial tears drops, eye ointment, and taping of eyes during the night. If the recovery is prolonged, temporary tarsorrhaphy is indicated.

Surgical treatments like nerve decompression are done only when the medical management fails.

Solution to Question 14:

Cleft palate is not seen in a patient with Melkersson-Rosenthal syndrome.

Melkersson-Rosenthal Syndrome is an idiopathic disorder consisting of a triad of:

- Facial paralysis
- Swelling of lips and
- Fissured tongue.

The following image shows a patient with Melkersson-Rosenthal syndrome.



Solution to Question 15:

The given clinical scenario is suggestive of Ramsay-Hunt syndrome and it is caused by the Herpes zoster virus.

The image shows right-sided LMN facial paralysis along with vesicular rash in the external auditory canal and pinna which is a classical presentation of this syndrome.

Solution to Question 16:

The given clinical scenario with facial nerve palsy and vesicles in the external auditory canal is suggestive of Ramsay-Hunt syndrome (Herpes Zoster oticus). Surgery is not the mainstay of treatment. Steroids and antivirals are the mainstays of treatment.

There may also be anesthesia of the face, giddiness, and hearing impairment due to the involvement of the 5th and 8th cranial nerves.

Solution to Question 17:

The given clinical scenario is crocodile tear syndrome or Bogard syndrome (gustatory lacrimation) and is due to faulty regeneration of facial nerve fibres. It presents as unilateral lacrimation during eating and drinking.

It is seen in cases where the facial nerve is damaged proximal to the geniculate ganglion. During regeneration, the fibres supplying the salivary gland aberrantly supply the lacrimal gland.

It can be treated by section of the greater superficial petrosal nerve.

Solution to Question 18:

The given clinical scenario is Frey's syndrome or gustatory sweating. It is due to aberrant reinnervation (or cross innervation) of parasympathetic fibers of the auriculotemporal nerve (which supplies parotid) with sympathetic sweat glands.

The patient usually presents with sweating and flushing over the parotid area during mastication. This usually occurs after parotid surgery.

It is diagnosed by Minor's test or starch iodine test. The affected area is painted with iodine and is allowed to dry. Then dry corn starch is applied. The starch turns blue on exposure to iodine in the presence of sweat.

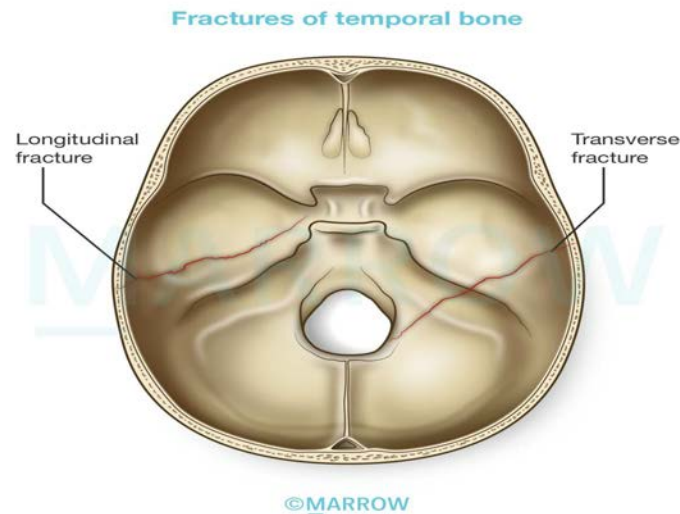
Solution to Question 19:

A vascular loop abutting the facial nerve at the cerebellopontine angle is the most common cause of hemifacial spasm. This usually presents as a unilateral involuntary uncontrollable twitching of the face.

Treatment is microvascular decompression or removal of any other inciting cause like acoustic neuroma, congenital cholesteatoma, glomus tumor. In refractory cases, botulinum toxin and selective nerve sectioning are done.

Solution to Question 20:

The transverse fracture of the petrous temporal bone is most commonly associated with facial nerve palsy.

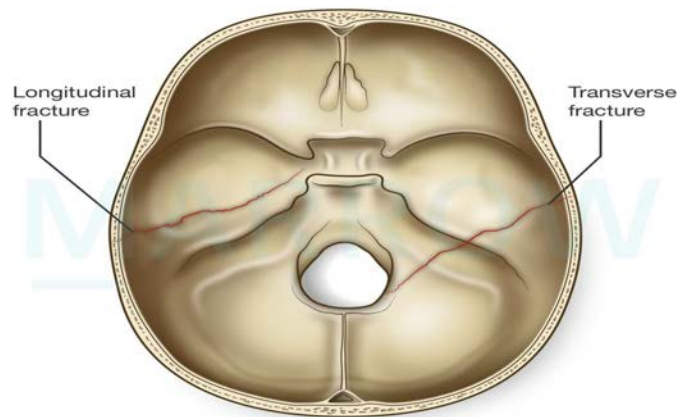


Solution to Question 21:

The given clinical scenario is suggestive of CSF otorrhea and it is seen in longitudinal fracture of the petrous part of the temporal bone.

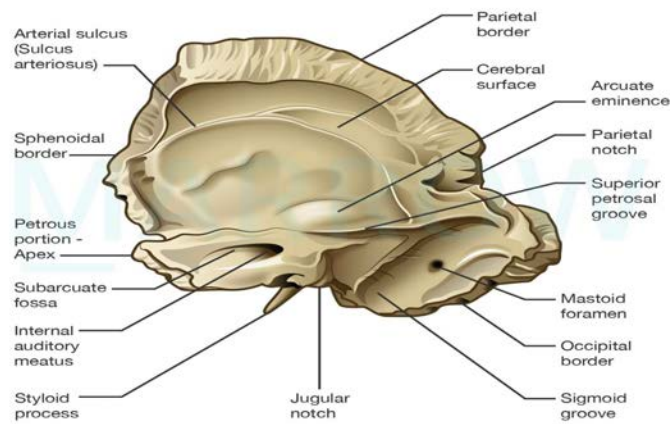
The transverse fracture of petrous temporal bone and fracture of cribriform plate leads to CSF rhinorrhea.

Fractures of temporal bone



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Temporal bone - Medial view

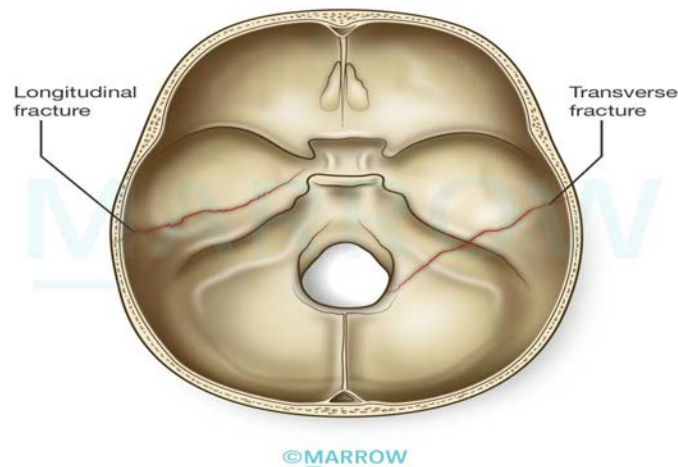


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Solution to Question 22:

HRCT (High-resolution CT) is the investigation of choice for the temporal bone fractures complicated by facial nerve palsy or CSF leak.

Fractures of temporal bone



Solution to Question 23:

The mandibular nerve cannot be used as a graft for facial nerve surgeries.

Only those nerves whose functions can be sacrificed are chosen as grafts and these are usually sensory nerves.

The most commonly used grafts for facial nerve are:

- Greater auricular nerve (can be extracted from the same incision)
- Lateral cutaneous nerve of thigh
- Sural nerve (when a large segment is required).

Eustachian Tube

Question 1:

What is the length of the eustachian tube?

- a) 12 mm
- b) 24 mm
- c) 36 mm
- d) 48 mm

Question 2:

Which of the following is false regarding the eustachian tube?

- a) Also known as pharyngotympanic tube
- b) 36 mm in length
- c) Outer 1/3rd is bony and inner 2/3rd is cartilaginous
- d) Connects oropharynx with middle ear

Question 3:

Which of the following is the narrowest part of the eustachian tube?

- a) Tympanic end
- b) Pharyngeal end
- c) Isthmus
- d) Infundibulum

Question 4:

You are an intern asking a patient to perform a Valsalva manoeuvre for the relief of an ear block. Which of the following structure helps in opening the eustachian tube during this manoeuvre?

- a) Tensor tympani
- b) Tensor veli palatini

- c) Stylopharyngeus
- d) Palatopharyngeus

Question 5:

Which of the following supplies the tensor veli palatini muscle?

- a) Facial nerve
- b) Trigeminal nerve
- c) Pharyngeal plexus
- d) Glossopharyngeal nerve

Question 6:

A child presented with recurrent attacks of acute otitis media. Which of the following properties of the eustachian tube predisposes to this?

- a) More angulated at isthmus
- b) Rigid cartilage
- c) Dense elastin in the cartilage
- d) Wider and shorter tube

Question 7:

Which of the following is a cause for functional eustachian tube obstruction?

- a) Adenoid hypertrophy
- b) Allergy
- c) Cleft palate
- d) Deviated nasal septum

Question 8:

You are evaluating the patency and function of the Eustachian tube in a patient with a right earache and ear block. Which of the following test you will not do?

- a) Valsalva manoeuvre

- b) Stenger test
- c) Tympanometry
- d) Toynbee's manoeuvre

Question 9:

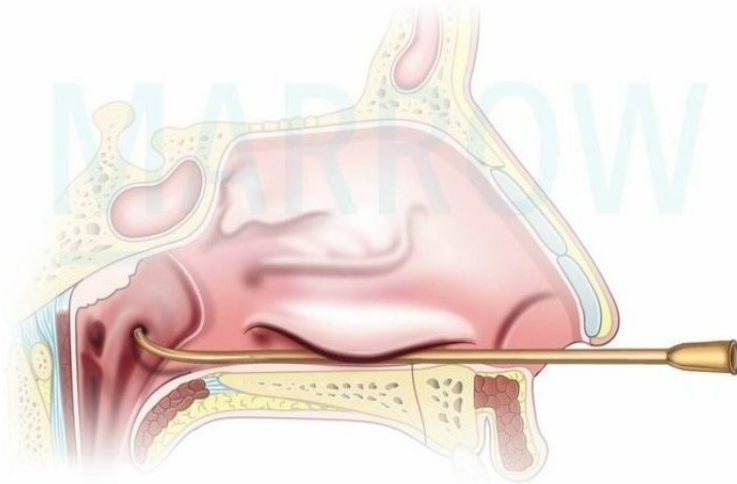
An ENT resident is performing the following maneuver in a patient as shown in the image. Which of the following is false regarding the same?



- a) It builds negative pressure in the nasopharynx
- b) Tympanic membrane bulges out
- c) Assesses eustachian tube patency
- d) Avoided in times of URTI

Question 10:

An otolaryngologist is performing the following procedure in a patient. Which of the following is true about this procedure?



- a) Uses Politzer's bag
- b) Methylene blue dye is used
- c) Easy and safe procedure
- d) Used to instill medications into middle ear

Question 11:

A 3-year-old child was brought to the OPD with fever, earache, and poor feeding after an episode of URTI. You suspect this as acute suppurative otitis media due to an eustachian tube defect. Which is the most accurate test to detect this?

- a) Valsalva maneuver
- b) Toynbee maneuver
- c) Tympanometry
- d) Politzerisation

Question 12:

A middle-aged retropositive woman presented with complaints of hearing her own voice and breath sounds with ear fullness. She has lost significant weight in the past 6 months. What is the likely diagnosis in this case?

- a) HIV-related psychosis
- b) Superior canal dehiscence syndrome
- c) Patulous eustachian tube

- d) Blocked eustachian tube

Question 13:

For a patient to get barotraumatic otitis media, what is the critical level of the atmospheric pressure required?

- a) 30 mmHg
- b) 60 mmHg
- c) 90 mmHg
- d) 100 mmHg

Question 14:

Which of the following activities is least likely to cause otitic barotrauma?

- a) Pilot performing take-off
- b) Skydiver performing a jump
- c) Deep sea diving
- d) Patient undergoing hyperbaric oxygen therapy

Question 15:

Which of the following is not a feature of otitic barotrauma?

- a) Conductive hearing loss
- b) Hyperacusis
- c) Severe earache
- d) Vertigo and sensorineural hearing loss

Answer Key

Question No.	Correct Option
1	c
2	d

3	c
4	b
5	b
6	d
7	c
8	b
9	a
10	a
11	c
12	c
13	c
14	a
15	b

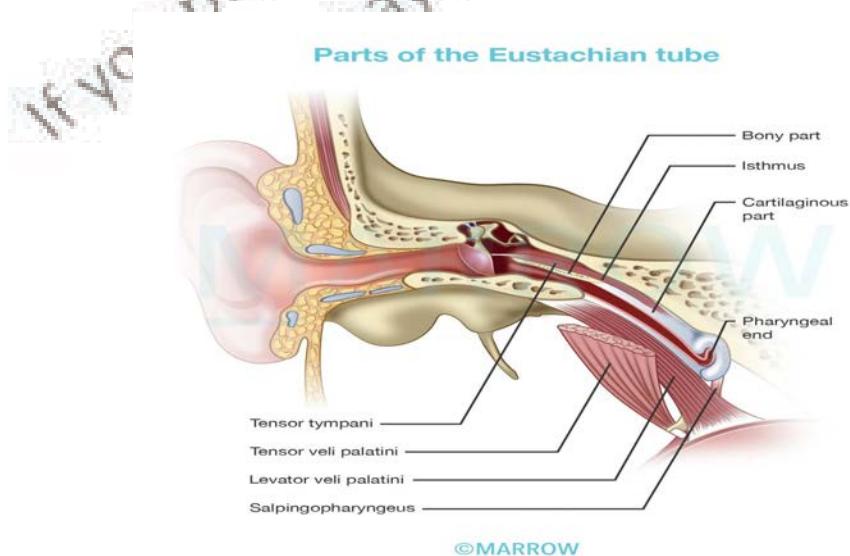
Detailed Explanations

Solution to Question 1:

The length of the eustachian tube is 36 mm (Bony part: 12 mm and Cartilaginous part: 24 mm).

It is directed forward, downward, and medially with an angle of 45° to the horizontal axis.

Note: The eustachian tube in infants is 18 mm, i.e. about half the size of an adult eustachian tube.



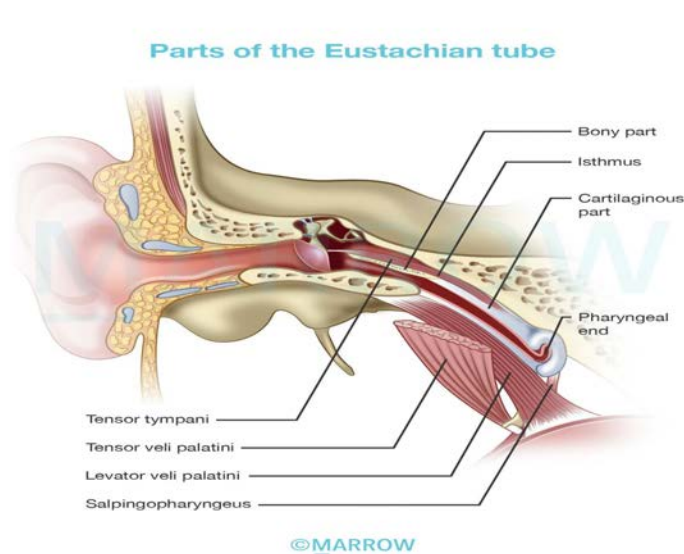
Solution to Question 2:

The eustachian tube also called the auditory or pharyngotympanic tube connects the nasopharynx with the tympanic cavity.

It is 36 mm long in adults and is divided into two parts:

- Bony part - forms the outer one-third of the tube as it is located near the middle ear and is 12 mm in size.
- Fibrocartilaginous part - forms the inner two-third as it is located near the nasopharynx and is 24 mm in size.

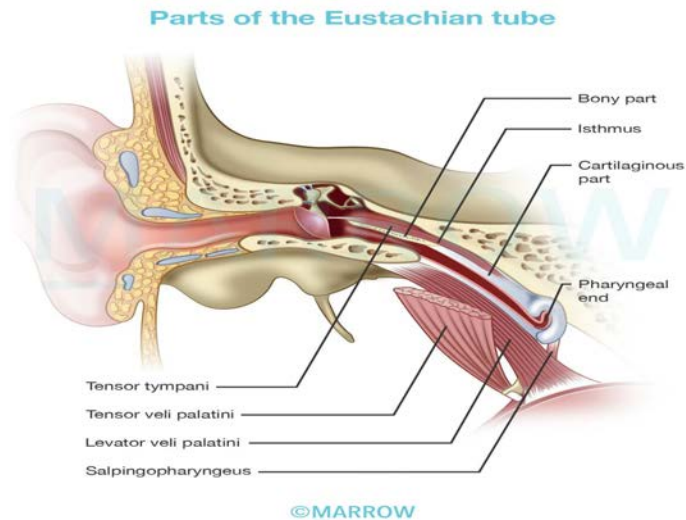
In the middle ear it opens in the anterior wall and in the nasopharynx it opens 1-1.25 cm posterior to the posterior end of the inferior turbinate.



Solution to Question 3:

The narrowest part of the eustachian tube is the isthmus. It is the junction of the bony and cartilaginous parts.

Infundibulum is not a part of eustachian tube.



Solution to Question 4:

Tensor veli palatini helps in opening the Eustachian tube during the Valsalva manoeuvre.

It is closed at rest and opens intermittently during swallowing, yawning and sneezing.

Opening of the tube

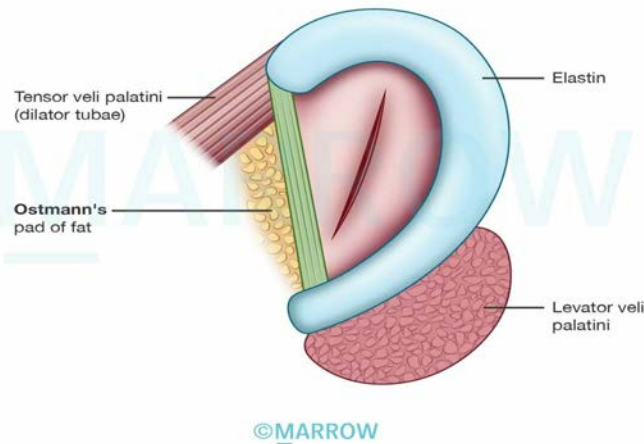
- The medial fibres of the tensor veli palatini are attached to the lateral lamina of the tube and contract to open the tubal lumen. It is also called the dilator tubae muscle.
- Accessory muscles for opening are levator veli palatine & salpingopharyngeus

Closure of the tube

- It is mainly due to the elastic recoil of the cartilage aided by the high degree of elastin on the roof.

Ostmann's pad of fat helps to keep the tube closed.

Eustachian tube (closed)



Solution to Question 5:

The mandibular branch of the trigeminal nerve supplies the tensor veli palatini which opens the Eustachian tube.

Muscles of the Eustachian tube:

Sensory & parasympathetic supply to ET: Glossopharyngeal Nerve

Muscle	Nerve supply	Action
Tensor veli palatine/ Dilator tubae	Mandibular branch of trigeminal nerve	Opens the tube
Levator veli palatine	Pharyngeal plexus	Opens & lifts the tube
Salpingopharyngeus	Pharyngeal plexus	Opens the tube

Solution to Question 6:

The Eustachian tube in children is wider, shorter (13-18mm), and more horizontal. The cartilaginous part is flaccid due to less elastin.

The above differences make children more prone to retrograde reflux of nasopharyngeal secretions and middle ear infections. In infants along with the secretions, the milk also refluxes back while feeding in the supine position.

Solution to Question 7:

Cleft palate is a cause for functional Eustachian tube (ET) obstruction.

Eustachian tube obstruction can be mechanical, functional or both. Mechanical obstruction can result from intrinsic causes such as inflammation or allergy or extrinsic causes such as tumour or adenoids. Functional obstruction is caused by collapse of the tube due to increased cartilage compliance or failure of active tubal opening mechanism due to poor function of tensor veli palatini.

In the cleft palate, the obstruction is due to high elastin content at torus tubaris and the poor function of tensor veli palatini.

Solution to Question 8:

Stenger test is not used for the assessment of patency or function of the eustachian tube. It is used to detect malingering of hearing loss.

Tests of Eustachian tube patency:

- Pneumatic otoscopy
- Nasopharyngoscopy
- Valsalva manoeuvre
- ET catheterization
- Politzer test.

Tests of Eustachian tube function:

- Toynbee's test
- Tympanometry
- Saccharine/dye tests in perforated ears
- Sonotubometry.

Solution to Question 9:

The given image depicts the Valsalva manoeuvre and it builds up positive pressure in the nasopharynx and not negative pressure.

In this test, the patient is asked to close the mouth and nostrils and blow. This produces positive pressure within the nasopharynx which opens the patent eustachian tube and is confirmed by the tympanic membrane bulge via otoscopy.

It is used to assess the Eustachian tube patency and to auto-inflate the middle ear.

It is contraindicated in:

- Atrophic tympanic membrane as it might get ruptured.
- URTI as the infection can spread into the middle ear.

Solution to Question 10:

The procedure depicted in the image is Eustachian tube (ET) catheterization and it uses Politzer's bag and an ET catheter.

The catheter is passed along the nasal floor and manoeuvred into the ET opening and the bag is insufflated. The outward movement of the tympanic membrane is then noted with an otoscope.

It is an unsafe and outdated ET function test. It can lead to complications like TM rupture, scarring of ET, epistaxis, spread of nasal/pharyngeal infection to the middle ear.

In the methylene blue dye test, the dye is instilled into the middle ear and the time taken by it to stain the pharynx is noted.

Politzer test is a non-invasive alternative for this and works on the same principle. It uses a nasal olive instead of a catheter and is useful in children who can't perform the Valsalva or Toynbee test.

Solution to Question 11:

Tympanometry is the most accurate test to assess eustachian tube (ET) dysfunction. It is an objective measure of ET function and does not rely on patient cooperation.

In this test, the movement of TM is visualized to the variation in external canal pressure. The disadvantage is middle ear pathologies can overshadow ET function.

Solution to Question 12:

The given clinical scenario describes autophony which is seen patulous eustachian tube.

Autophony is a symptom where the patient hears his own voice and breath sounds.

Patulous eustachian tube is usually idiopathic where the tube is abnormally patent. But it is also seen in rapid weight loss, pregnancy especially the third trimester and patients with multiple sclerosis.

Treatment includes weight gain, administration of potassium iodide solution. In long-standing cases, specific treatment could include cauterization of tubes or insertion of the grommet.

Option A: In HIV-related psychosis, there will be persecutory delusion, hallucination, and no ear fullness.

Option B: The autophony of superior canal dehiscence syndrome usually involves hearing their own voice but not their breathing when compared with patients who have patulous eustachian tube.

Option D: Patient with blocked eustachian tube does not present with autophony.

Solution to Question 13:

The critical level of the atmospheric pressure required to cause barotraumatic otitis media is 90 mm Hg.

At this point, the atmospheric pressure is higher than the middle ear and it causes the eustachian tube blockage.

Solution to Question 14:

A pilot performing take-off (i.e. rapid ascent in the air) is least likely to cause otitic barotrauma compared to other options.

The usual causes for otitic barotrauma are

- Rapid descent during air travel (skydiver)
- Underwater diving (scuba diving)
- Compression in a pressure chamber (hyperbaric oxygen therapy).

Solution to Question 15:

Hyperacusis is not a feature of otitic barotrauma.

Otitic barotrauma is a non-suppurative condition resulting from the failure of the eustachian tube to maintain middle ear pressure at ambient atmospheric level. It is usually caused by rapid descent during air flight, underwater diving or compression in the pressure chamber.

- Severe earache, deafness, and tinnitus are common complaints.
- The TM appears retracted and congested. It can get ruptured in severe cases.
- The middle ear may show air bubbles or hemorrhagic effusion.
- Hearing loss is usually conductive.
- Rarely, there is a rupture of labyrinthine membranes causing vertigo and sensorineural hearing loss.

Physiology of hearing and Tuning fork tests

Question 1:

Which of the following is not a function of the middle ear?

- a) Impedance matching
- b) Protection of inner ear
- c) Transduction of sound
- d) Amplification of sound intensity

Question 2:

What is the lever ratio of the ear ossicles?

- a) 14:1
- b) 1.3:1
- c) 18:1
- d) 22:1

Question 3:

The effective vibratory area of the tympanic membrane is much larger than the area of stapes footplate. Which of the following is the most accurate value of the average ratio between the two?

- a) 17:1
- b) 1.3:1
- c) 18:1
- d) 22:1

Question 4:

What is the transformer ratio of the middle ear?

- a) 1.3:1
- b) 14:1

- c) 17:1
- d) 22:1

Question 5:

What is the natural resonance of the tympanic membrane?

- a) 3000 Hz
- b) 800-1600 Hz
- c) 1500-2000 Hz
- d) 2000-3000 Hz

Question 6:

Which of the following factors do not contribute to the impedance matching mechanism of the middle ear?

- a) Lever action of ossicles providing mechanical advantage of 1.3
- b) Phase difference between the oval and round windows
- c) Relatively large area of the tympanic membrane compared to stapes footplate
- d) More movements at the centre of the tympanic membrane than at periphery

Question 7:

Who proposed the travelling wave theory of cochlear mechanics?

- a) von Békésy
- b) Young
- c) Helmholtz
- d) Sade

Question 8:

What is the endocochlear potential recorded from the scala media?

- a) +80 mV
- b) -80 mV

- c) +40 mV
- d) -40 mV

Question 9:

You are an intern performing the tuning fork tests on a patient who presented to the ENT OPD with complaints of decreased hearing in the left ear. Which of the following tuning forks is most commonly used for the routine clinical assessment for such patients?

- a) 1024 Hz
- b) 512 Hz
- c) 256 Hz
- d) 126 Hz

Question 10:

A 45-year-old man presents to the ENT OPD with hearing loss in the right ear. A 512 Hz tuning fork is used, which highlights the Rinne test is positive bilaterally. What is the likely inference?

- a) Air conduction is better than bone conduction
- b) Bone conduction is better than air conduction
- c) Air conduction is equal to bone conduction
- d) Bone conduction is worse than that of examiner's

Question 11:

What should be the minimum hearing loss in a patient for the Rinne test to be negative?

- a) 5-10dB
- b) 15-20dB
- c) 25-30dB
- d) 35-40dB

Question 12:

In a patient presenting with a conductive type of hearing loss, a negative Rinne test with a 512 Hz tuning fork indicates a minimum air-bone gap of?

- a) 15 dB
- b) 30 dB
- c) 45 dB
- d) 60 dB

Question 13:

You are a first-year resident performing the tuning fork tests on the following patients to assess their hearing loss. Which one of them will you expect to have a positive Rinne test?

- a) A child with chronic suppurative otitis media
- b) An adolescent man with impacted wax
- c) An elderly patient with presbycusis
- d) A pregnant woman with otosclerosis

Question 14:

A middle-aged man diagnosed with Ménière's disease presented with severe unilateral hearing loss. Which of the following would be the most likely finding on performing the Rinne test on this patient?

- a) Positive
- b) Negative
- c) False negative
- d) False positive

Question 15:

A young lady presented to the ENT OPD with a history of sudden onset unilateral hearing loss, tinnitus, and dizziness following an episode of acute otitis media two weeks back. Rinne test and fistula test were positive. She had refused treatment then and currently came with complaints of worsened hearing loss in the affected ear. On examination, the fistula test was negative. What would be the most likely finding on the Rinne test in this scenario?

- a) True positive
- b) True negative

- c) False positive
- d) False negative

Question 16:

Which of the following is false about the test being performed on a patient by an ENT resident as shown below?



- a) It is lateralized to worse ear in superior semicircular canal dehiscence syndrome
- b) It is heard equally on both sides in normal individual
- c) It is lateralized towards better ear in cochlear otosclerosis
- d) It is lateralized towards worse ear in Ménière's disease

Question 17:

A vibrating tuning fork is placed on the mastoid while the examiner alternately closes and opens the ear canal of the patient by pressing the tragus inwards to examine the effect of occlusion on hearing. What is the test being performed here?

- a) Absolute bone conduction test
- b) Schwabach test
- c) Bing test
- d) Gille's test

Question 18:

A young female diagnosed with otosclerosis presented to the ENT OPD with worsening hearing loss. Which of the following will be the most likely observation on the Bing test?

- a) Hears louder when canal is occluded
- b) Appreciates no change
- c) Hears softer when canal is occluded
- d) The test is said to be positive

Question 19:

In which of the following diseases would you find a negative Gelle's test?

- a) Acoustic neuroma
- b) Presbycusis
- c) Otosclerosis
- d) Meniere's disease

Question 20:

A second-year resident performs the tuning fork tests on a patient with hearing loss. The Rinne test is positive in both ears and Weber is lateralized towards the right. What is the interpretation?

- a) Right-sided conductive deafness
- b) Left-sided conductive deafness
- c) Right-sided sensorineural deafness
- d) Left-sided sensorineural deafness

Question 21:

A metal factory worker suffers from noise-induced hearing loss. Which of the following structures will be affected?

- a) Crista ampullaris
- b) Macula
- c) Cupola

d) Outer hair cells

Answer Key

Question No.	Correct Option
1	c
2	b
3	a
4	d
5	b
6	d
7	a
8	a
9	b
10	a
11	b
12	b
13	c
14	c
15	d
16	d
17	c
18	b
19	c
20	d
21	d

Detailed Explanations

Solution to Question 1:

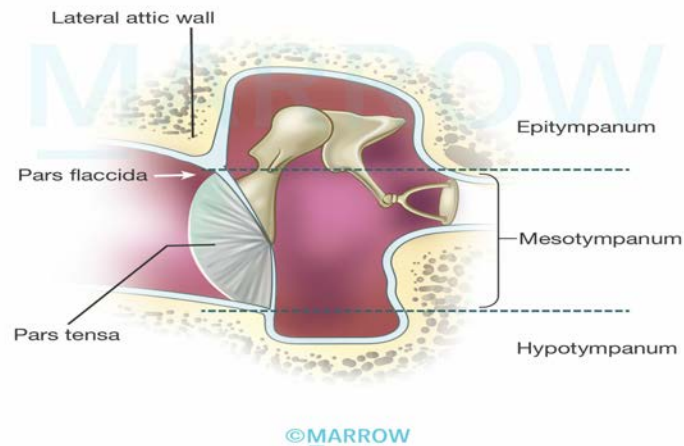
Transduction of sound (mechanical energy to electrical impulses) is done by the sensory system of the cochlea. It is not a function of the middle ear.

The functions of the middle ear are:

- Conduction of sound through impedance matching or transformer action

- Protection of inner ear
- Amplification of sound intensity

Divisions of the middle ear

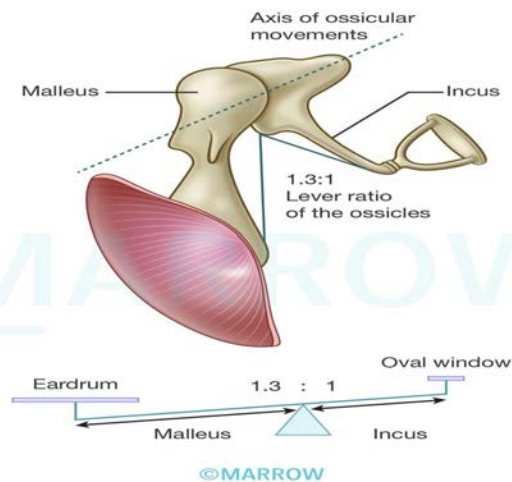


Solution to Question 2:

The lever ratio of the ear ossicles is 1.3:1.

The handle of the malleus is 1.3 times longer than the long process of the incus, providing a mechanical advantage of 1.3.

Transformer ratio



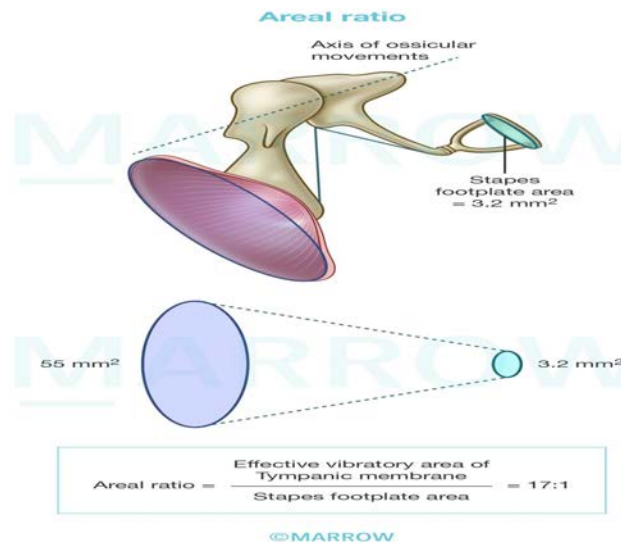
Solution to Question 3:

The ratio of effective vibratory area of the tympanic membrane to the area of footplate of stapes (areal ratio) is 17:1 .

Effective vibratory area of TM = 55 mm²

Footplate area = 3.2 mm²

Areal ratio = Effective vibratory area of tympanic membrane ÷ stapes footplate area

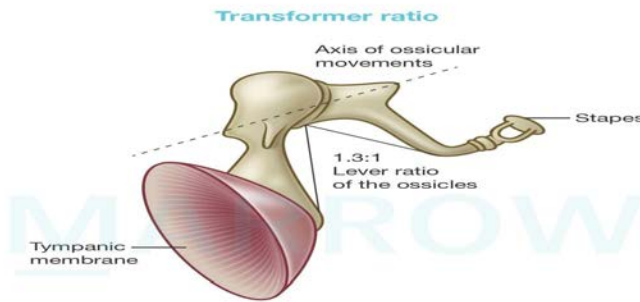


Solution to Question 4:

The transformer ratio of the middle ear is 22:1

The middle ear has to conduct sound waves from a low impedance tympanic membrane to a high impedance oval window. This is done by matching their impedances. It is called transforming action.

Transformer ratio = Areal ratio × Lever ratio



Effective vibratory area of	: 55 mm ²
Tympanic membrane (TM)	
Footplate area	: 3.2 mm ²
Areal ratio = $\frac{\text{Vibratory area of TM}}{\text{Stapes footplate area}} = \frac{55}{3.2} = 17:1$	
Lever ratio = 1.3:1	
Total transformer ratio = Areal ratio × Lever ratio	
= 17 × 1.3	
= 22:1	

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Solution to Question 5:

The natural resonance of the tympanic membrane is 800 – 1600 Hz.

Aging causes a decrease in the external ear resonance amplitude due to the collapse of the external auditory canal and the tympanic membrane becoming senile. This factor has to be considered to appropriately diagnose hearing loss and in the selection of hearing aids.

Natural resonance and efficiency of auditory apparatus:

- External auditory canal- 3000 Hz
- Tympanic membrane- 800–1600 Hz
- Middle ear- 800 Hz
- Ossicular chain- 500–2000 Hz

Solution to Question 6:

Movements of the tympanic membrane are more at the periphery than at the center, where the malleus handle is attached. This is called the curved membrane effect and it contributes to the impedance matching mechanism of the ear.

Other factors responsible for the impedance matching mechanism of the middle ear are:

- The lever action of the ossicles- handle of malleus is 1.3 times longer than the long process of incus, providing a mechanical advantage of 1.3.
- Hydraulic action of the tympanic membrane- relatively large area of the tympanic membrane compared to the stapes footplate.

- The medial side of the tympanic membrane is multiphasic whereas the lateral side is uniphasic. This provides a multiple integrated multilever mechanical advantage.

Solution to Question 7:

The traveling wave theory was proposed by von Békésy.

It states that a sound wave, depending on its frequency, reaches maximum amplitude on a particular place on the basilar membrane and stimulates that segment.

Higher frequencies are represented in the basal turn of the cochlea and the progressively lower ones towards the apex.

Solution to Question 8:

Endocochlear potential is a direct current (DC) potential recorded from scala media. It is +80 mV and is generated from the stria vascularis by Na⁺/K⁺-ATPase pump and provides a source of energy for cochlear transduction. It is present at rest and does not require a sound stimulus.

Solution to Question 9:

A tuning fork of 512 Hz is most commonly used for routine clinical assessment in ENT.

Tuning forks of lower frequencies produce a sense of bone vibration while those of higher frequencies have a shorter decay time and are thus not routinely preferred.

Solution to Question 10:

Rinne test is positive when the air conduction (AC) is longer or louder than bone conduction (BC).

Solution to Question 11:

The minimum hearing loss for the Rinne test to be negative is 15-20dB.

Negative Rinne (BC > AC) is seen in conductive deafness.

Solution to Question 12:

The negative Rinne test for 512 Hz indicates a minimum air-bone gap of 30 dB.

Rinne's test is a tuning fork test performed commonly with tuning forks of frequencies such as 128, 256, 512, and 1024 Hz.

Rinne's test is done by placing a vibrating tuning fork on the patient's mastoid and when he stops hearing, it is brought beside the external auditory meatus (EAC). If he still hears the sound, air conduction (AC) is better than bone conduction (BC) i.e. $AC > BC$. The inferences are as follows:

- Rinne's test is considered positive (normal) when the air conduction $AC > BC$ is seen in normal people and people with sensorineural deafness.
- It is considered negative when the $BC > AC$, and is seen in patients with conductive deafness.

As a general prediction, a negative Rinne's test with any tuning fork frequency suggests a minimum hearing loss of at least 15 - 20 dB. The table below indicates a rough minimum air-bone gap for negative Rinne's test at different frequencies:

A prediction of the air-bone gap can be made based on the combination of tuning fork tests of 256, 512, and 1024 Hz:

- A Rinne test negative for 256 Hz but positive for 512 Hz indicates an air-bone gap of 20-30 dB.
- A Rinne test negative for 256 and 512 Hz but positive for 1024 Hz indicates an air-bone gap of 30-45 dB.
- A Rinne test negative for 256, 512, and 1024 Hz indicates an air-bone gap of 45-60 dB.

Negative Rinne test	Minimum air-bone gap
256 Hz	15 dB
512 Hz	30 dB
1024 Hz	45 dB

Solution to Question 13:

A positive Rinne test is seen in presbycusis, which is a type of sensorineural deafness.

A positive Rinne test indicates that air conduction is longer than bone conduction, as seen in normal ear and sensorineural deafness.

A negative Rinne test indicates that bone conduction is longer than air conduction, as seen in conditions causing conductive deafness such as chronic suppurative otitis media, impacted wax, and otosclerosis.

Solution to Question 14:

Patients diagnosed with Ménière's disease present with a sensorineural type of hearing loss.

A false negative Rinne test is seen in severe unilateral sensorineural hearing loss.

The patient doesn't respond to air conduction but is able to hear when a tuning fork is placed over the mastoid process i.e. bone conduction > air conduction. Hence reported as negative. This is a false negative because this response to bone conduction is from the opposite ear due to transcranial transmission of sound.

Solution to Question 15:

The given clinical scenario indicates severe sensorineural hearing loss as a consequence of untreated suppurative labyrinthitis and Rinne's test is a false negative in this case.

Initially, the patient had a positive fistula test, which implies that the labyrinth is still functioning. Rinne's test is positive which could be due to normal hearing or sensorineural hearing loss.

Currently, the patient has a negative fistula test which implies that the labyrinth is dead. This is an example of severe sensorineural hearing loss and here, Rinne test is a false negative.

If intermittent pressure applied to the external ear produces vertigo and nystagmus, it is a positive fistula test. The pressure changes in the external ear are transmitted to the labyrinth. This is suggestive of abnormal communication between the middle ear and the inner ear.

Solution to Question 16:

The image shows the Weber test being performed. Ménière's disease causes sensorineural hearing loss and hence Weber test is lateralized towards the better ear.

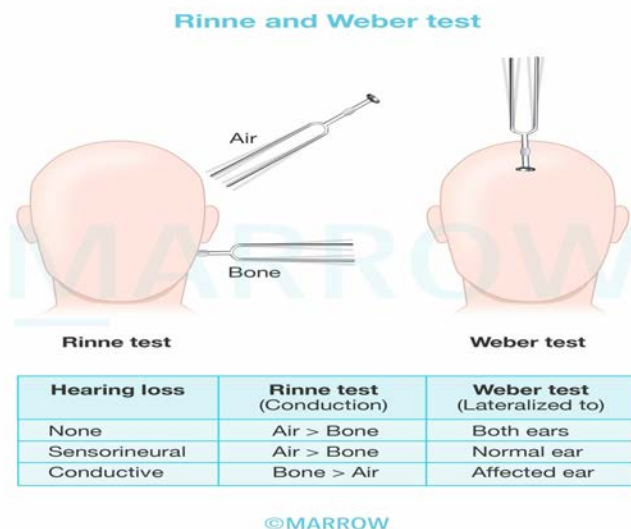
In Weber's test, a vibrating tuning fork is placed in the middle of the forehead or the vertex and the patient is asked in which ear the sound is heard.

It is lateralized to the worse ear in conductive deafness and to the better ear in sensorineural deafness.

Option A: Superior semicircular canal dehiscence syndrome causes conductive hearing loss. Thus, Weber is lateralized to the worse ear.

Option B: In normal individuals, the sound is heard equally in both ears. So Weber is not lateralized.

Option C: Cochlear otosclerosis causes sensorineural deafness, whereas stapedial otosclerosis causes conductive hearing loss. Thus, Weber is lateralized to the better ear in case of cochlear otosclerosis.



Solution to Question 17:

Bing test is being performed in the given clinical scenario.

It is a test of bone conduction and examines the effect of occlusion of the ear canal on hearing. A vibrating tuning fork is placed on the mastoid. The examiner alternately closes and opens the ear canal by pressing the tragus inwards.

Bing positive: A normal person or one with sensorineural hearing loss hears louder when the ear canal is occluded and softer when the canal is open.

Bing negative: A patient with conductive hearing loss will appreciate no change.

Solution to Question 18:

The patient diagnosed with otosclerosis will have a conductive type of hearing loss and will appreciate no change in the Bing test. Here, the Bing test is said to be negative.

Bing test is a test of bone conduction and examines the effect of occlusion of the ear canal on hearing. A vibrating tuning fork is placed on the mastoid. The examiner alternately closes and opens the ear canal by pressing on the tragus inwards.

Bing positive: A normal person or one with sensorineural hearing loss hears louder when the ear canal is occluded and softer when the canal is open.

Bing negative: A patient with conductive hearing loss will appreciate no change.

Solution to Question 19:

Gelle's test is negative in otosclerosis.

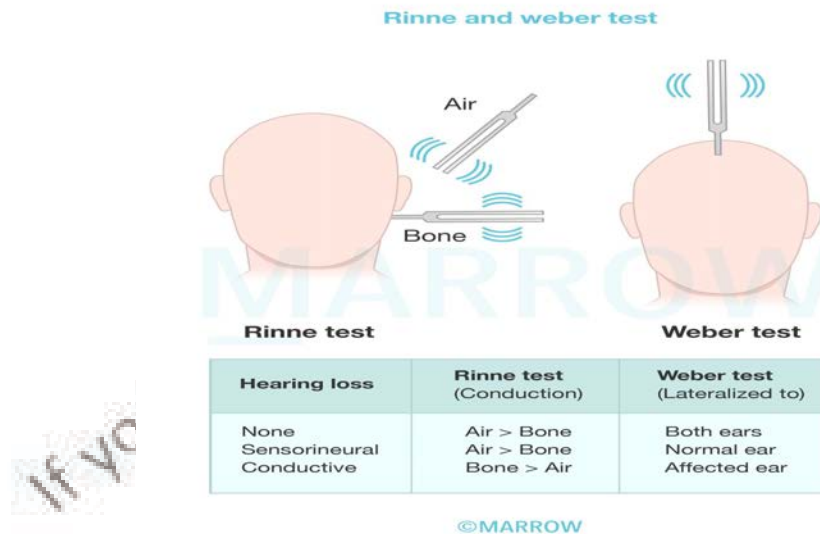
It is based on the fact that in a normal ear if pressure is increased in the external ear the stapes footplate will get medialized and will stop conducting sound (positive Gelle's test). But, in an ear with otosclerosis of the footplate, there will be no movements and the patient will continue to hear the sound (negative Gelle's test).

It was once a popular test to find out stapes fixation in otosclerosis but has now been superseded by tympanometry.

Solution to Question 20:

The patient has left-sided sensorineural deafness.

Weber's test lateralized towards the right indicates right-sided conductive deafness or left-sided sensorineural deafness. Since Rinne's test is positive on both the ears, conductive deafness can be ruled out and hence, the interpretation here is left-sided sensorineural hearing loss.



Solution to Question 21:

Noise-induced hearing loss (NIHL) damages the outer hair cells, starting in the basal turn of the cochlea. Outer hair cells are affected before the inner hair cells.

NIHL occurs after chronic exposure to less intense sounds. The damage caused by noise trauma depends on the frequency of noise (frequency of 2000 to 3000 Hz causes more damage) intensity, duration of the noise, and preexisting ear disease.

The audiogram in noise-induced hearing loss shows a typical notch at 4000Hz both for air and bone conduction known as acoustic dip/boiler's notch. Patients at this stage complain of high-pitched tinnitus and difficulty hearing in noisy surroundings. Pre-employment and annual audiograms for persons working at places where noise is above 85 dB (A) help in the early

detection of NIHL.

The human ear can hear frequencies from about 20 to 20,000 Hz. Below 20 Hz, the vibrations are infra-audible, and above 20,000 Hz, they are ultra-sonic. Pure tone audiometry can detect only up to 8000 Hz.

Extended high-frequency audiometry is recommended for early detection of hearing loss, as it can identify hearing impairments before the frequencies necessary for speech comprehension are affected. Distortion product otoacoustic emissions is a more sensitive tool than conventional audiometry, particularly in children under 5 years old for early detection of hearing loss.

A noise of 90 dB(A) SPL (acceptable noise levels, sound pressure level), 8 hours a day for 5 days per week is the maximum safe limit as recommended by the Ministry of Labour. As per WHO, 85 dB(A) is considered the highest safe exposure level up to a maximum of 8 hours.

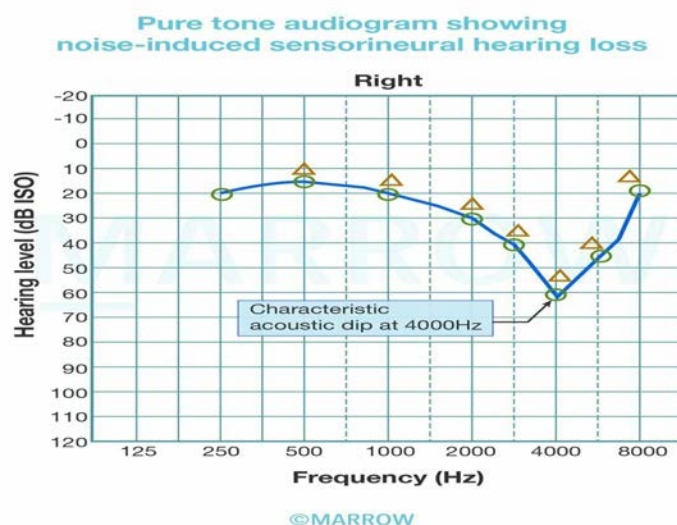
The silence zone extends to a distance of 100 meters around the premises of hospitals, nursing homes, educational institutions, and courts.

Permissible limits of noise as per the noise pollution (Regulation and Control) rules:

Noise levels are measured in dB(A), which prioritizes middle-range frequencies for assessing noise-induced hearing loss while giving less weightage to low and high frequencies.

The image below shows an acoustic dip at 4000Hz:

Zone/Area	Day(6 am to 10 pm) limits in dB(A)	Night(10 pm to 6 am) limits in dB(A)
Industrial	75	70
Commercial	65	55
Residential	55	45
Silence	50	40



Other options:

Option A: Crista ampullaris is a thickened ridge of neuroepithelium in the ampullated end of semicircular ducts.

Option B: Macula is the sensory epithelium of the utricle

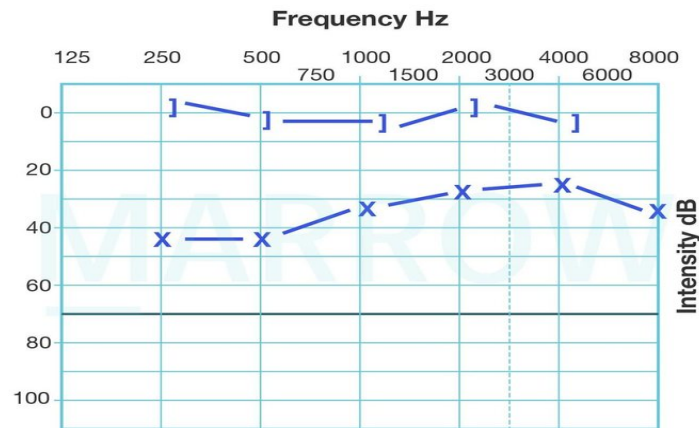
Option C: Cupula is a gelatinous mass extending from the surface of the crista (peripheral receptor located in the ampullated end of semicircular ducts) to the ceiling of the ampulla, into which project the cilia of sensory hair cells.

Sold by @Itachibot
If you purchased this from someone else,
you may have been scammed.

Audiometric tests and Special tests of hearing

Question 1:

The audiogram of a patient with difficulty in hearing is given below. What is your interpretation?

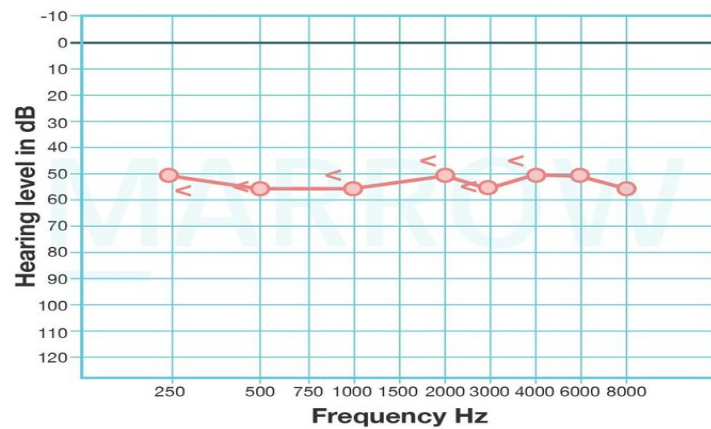


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- a) Left-sided conductive deafness
- b) Left-sided sensorineural deafness
- c) Right-sided conductive deafness
- d) Right-sided sensorineural deafness

Question 2:

A middle-aged woman presents to the ENT OPD with complaints of difficulty in hearing. A pure tone audiometry test is performed and the result is given below. What is the interpretation?

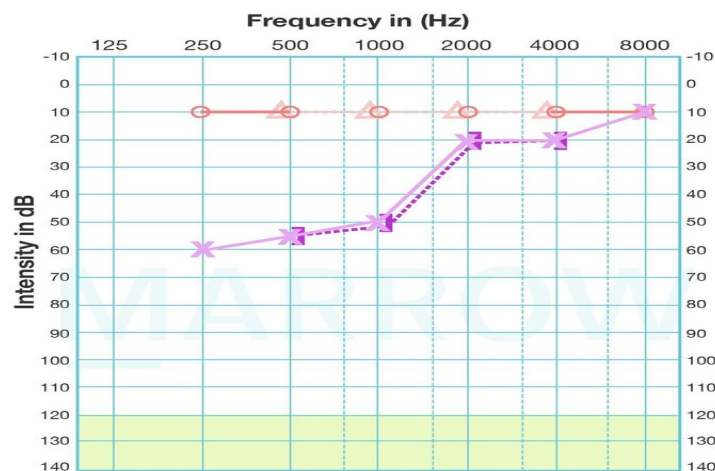


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- a) Left-sided conductive deafness
- b) Left-sided sensorineural deafness
- c) Right-sided conductive deafness
- d) Right-sided sensorineural deafness

Question 3:

The audiogram of a patient who presented with a gradually progressive hearing loss is given below. Which among the following would be the most likely diagnosis?



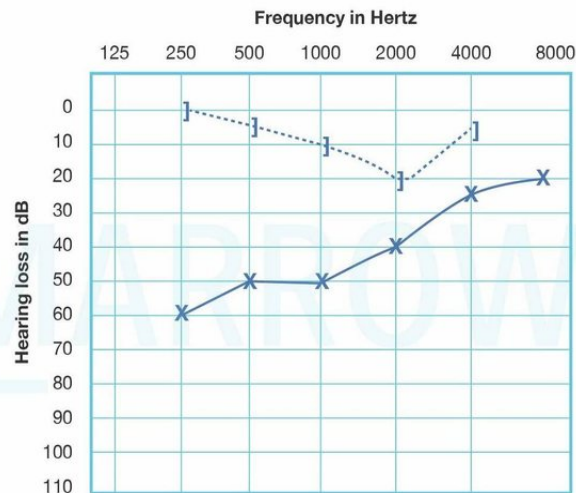
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- a) Prebycusis

- b) Ototoxicity
- c) Noise induced hearing loss
- d) Early stages of Ménière's disease

Question 4:

A 25-year-old woman presented with hearing loss that developed shortly after her first pregnancy. Her audiometry report is given below. What is the diagnosis?



- a) Ototoxicity
- b) Meniere disease
- c) Otosclerosis
- d) Noise induced hearing loss

Question 5:

A prisoner who was brought to the hospital with deafness after a brawl with another inmate is suspected of malingering. Which of the following tests can be used to assess the authenticity of his complaints?

- a) Pure-tone audiometry
- b) Speech audiometry
- c) Impedance audiometry
- d) Tone decay

Question 6:

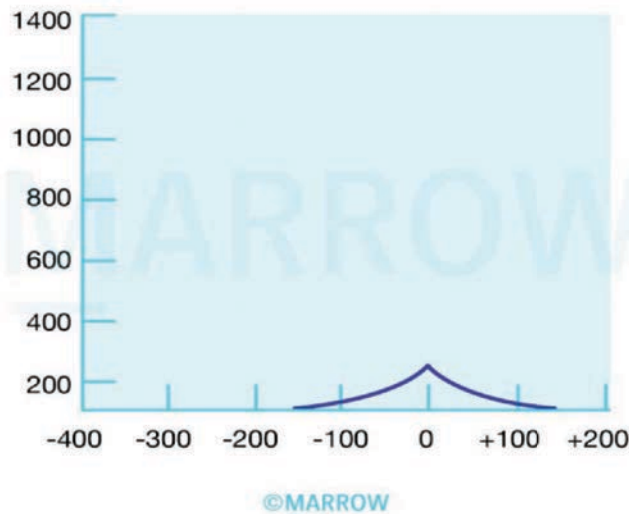
A patient with hearing loss is being evaluated with the following instrument. What is the frequency of the probe used?



- a) 220 Hz
- b) 446 Hz
- c) 512 Hz
- d) 128 Hz

Question 7:

The tympanogram of a young patient who presented with a progressive hearing loss is given below. Which of the following is the most likely diagnosis?



- a) Thin and lax tympanic membrane
- b) Otosclerosis
- c) Ossicular discontinuity
- d) Serous otitis media

Question 8:

Using impedance audiometry, stapedial reflexes are being assessed in a child with cranial nerve palsy. Which of the following cranial nerves is being tested here?

- a) 6th and 8th cranial nerves
- b) 5th and 7th cranial nerves
- c) 7th and 8th cranial nerves
- d) 6th and 7th cranial nerves

Question 9:

What is the instrument shown below?



- a) Meniette device
- b) Barany noise box
- c) Portable audiometer
- d) Higginson apparatus

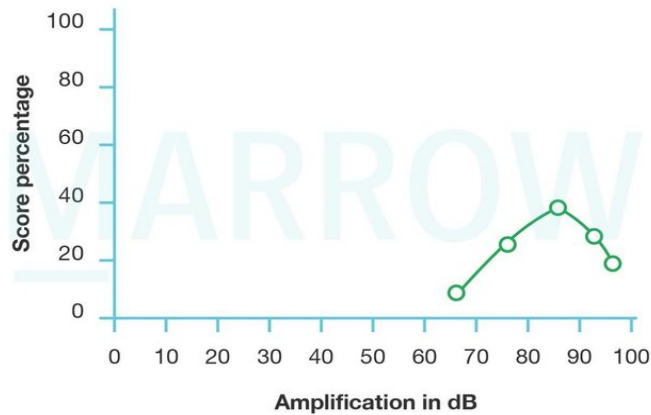
Question 10:

A woman is brought by her son to the ENT OPD due to hearing problems. He says that she always requests him to speak loud and whenever he speaks slightly louder, she tells him to stop shouting. Which of the following is the likely cause for this patient's condition?

- a) Middle ear pathology
- b) Cochlear pathology
- c) Retrocochlear disease
- d) External ear pathology

Question 11:

Patient with hearing loss undergoes speech audiometry whose report is given below. What is it suggestive of?

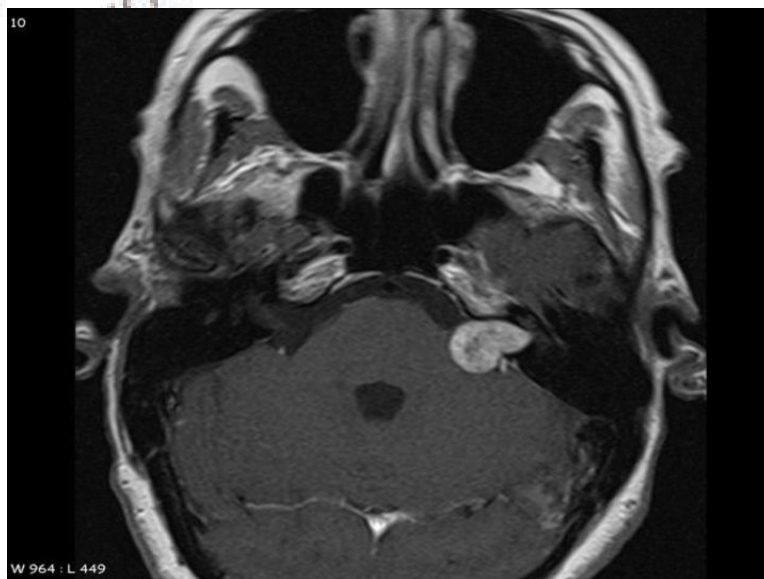


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- a) Normal finding
- b) Cochlear pathology
- c) Retrocochlear pathology
- d) Conductive deafness

Question 12:

A middle-aged woman presents to you with a progressive hearing loss accompanied by tinnitus. An MRI of the patient with gadolinium contrast is given below. Which of the following results is consistent with the diagnosis?



- a) ≤ 25 dB decay in threshold tone decay test

- b) SISI score 70 – 100%
- c) Presence of otoacoustic emissions
- d) Presence of recruitment phenomenon

Question 13:

What does summing potential in electrocochleography indicate?

- a) Electrical activity of cochlea
- b) Electrical activity of hair cells
- c) Electrical activity of distal part of auditory nerve
- d) Sum of all the above

Question 14:

A preterm neonate with an Apgar score of 4 at 5 minutes of birth undergoes otoacoustic emission study. Which of the following is false about this test?

- a) Otoacoustic emissions are generated by inner hair cells
- b) It is used as a screening test for hearing loss in infants
- c) Spontaneous type is present in 50% of normal individuals
- d) Evoked type is present in all individuals with normal hearing

Question 15:

Wave V in brainstem evoked response audiometry is produced by which of the following structures?

- a) Superior olivary complex
- b) Lateral lemniscus
- c) Inferior colliculus
- d) Distal part of cochlear nerve

Question 16:

A patient undergoes hearing assessment using cortical evoked response audiometry. Which of the following is false regarding this test?

- a) Recording is made from cortical potentials
- b) Click stimulus is used
- c) Not suitable for children
- d) Response is frequency specific

Question 17:

As a medical officer in an occupational health program, you encounter an employee with hearing loss. You are suspicious that he is faking it to claim compensation. Which of the following tests is least useful in this scenario?

- a) Stenger test
- b) Lombard test
- c) Stapedial reflex
- d) Rinne test

Question 18:

While performing a caloric test, what type of nystagmus would you expect on cold water irrigation?

- a) Horizontal nystagmus towards the same side
- b) Vertical nystagmus towards the same side
- c) Horizontal nystagmus towards the opposite side
- d) Vertical nystagmus towards the opposite side

Question 19:

Which of the following are the objective tests for the hearing?

- a) 1, 2, and 4
- b) 1,2, and 3
- c) 2,3, and 4
- d) 1,2,3, and 4

Answer Key

Question No.	Correct Option
1	a
2	d
3	d
4	c
5	c
6	a
7	b
8	c
9	b
10	b
11	c
12	c
13	b
14	a
15	b
16	b
17	d
18	c
19	a

Detailed Explanations

Solution to Question 1:

The patient has left-sided conductive hearing loss.

Interpretation of the above audiogram:

- Blue color indicates left ear.
- J indicates BC and threshold is ≤ 25 dB. BC is normal which means the sensorineural pathway is normal.
- × indicates AC and threshold is ≥ 15 dB and hence AC is abnormal.
- So, it indicates conductive deafness.
- The air-bone gap is a feature of conductive hearing loss.
- Hence, the chart is suggestive of left-sided conductive deafness.

Solution to Question 2:

The given pure tone audiometry is suggestive of right-sided sensorineural deafness.

Interpretation of the result:

- Red indicates the right ear.
- O indicates air conduction(AC). The threshold here is >15 dB indicating abnormal AC.
- $<$ indicates bone conduction(BC). The threshold is >25 dB indicating abnormal BC.
- Both AC and BC are abnormal which indicates a sensorineural hearing loss.
- There is no air-bone gap.
- Hence, it is right side sensorineural hearing loss.

Solution to Question 3:

The given audiogram is seen in Ménière's disease in its early stages. It shows an upsloping/ascending curve suggesting that the lower frequencies are affected first.

Presbycusis, ototoxicity, and sound-induced hearing loss have downsloping audiograms since the higher frequencies are affected first in these conditions.

Solution to Question 4:

The clinical scenario describes a patient with post-pregnancy hearing loss. The audiogram shows conductive deafness along with Carhart's notch (dip in bone conduction at 2000 Hz). These findings are highly suggestive of otosclerosis.

Ototoxicity, Meniere's disease, and noise-induced hearing loss cause sensory neural hearing loss (SNHL) and not conductive hearing loss.

Otosclerosis is caused by bony overgrowth involving the enchondral layer of the bony otic capsule. It presents with bilateral progressive conductive hearing loss and tinnitus. It has an autosomal dominant inheritance and can worsen with stress and hormonal changes like during pregnancy and menopause.

Stapedial otosclerosis is the most common type of otosclerosis. Here the lesion starts just in front of the oval window in an area called fistula ante fenestrae.

The presence of a reddish hue on the promontory through the tympanic membrane, known as the Schwartze sign, suggests an active focus. PTA shows loss of air conduction with air-bone gap, more for lower frequencies and Carhart's notch may be seen. Tympanometry shows an As type of tympanogram in which there is reduced compliance at or near the ambient air pressure.

Medical management with sodium fluoride is used in the active stage. Definitive treatment is stapedectomy/stapedotomy with the placement of a prosthesis. Carhart notch usually disappears

after a successful stapedectomy.

Other options:

Option A: Ototoxicity causes SNHL and shows a high-frequency hearing loss in the SNHL curve.

Option B: In Meniere's disease, pure tone audiometry shows an SNHL pattern with a rising curve in the early stages (as lower frequencies are involved first) and as the disease progresses higher frequencies are involved, the curve becomes flat or a falling type.

Option D: Noise-induced hearing loss is SNHL and PTA shows a characteristic acoustic dip at 4000 Hz both for air and bone conduction.

Solution to Question 5:

Impedance audiometry can be used to test malingering. It is an objective test and consists of tympanometry and acoustic reflex measurements.

A person who feigns total deafness and does not give any response on pure tone audiometry but shows a positive acoustic reflex is a malingerer.

Solution to Question 6:

The given image shows impedance audiometry and the frequency of the probe used here is 220 Hz.

Impedance audiometry consists of tympanometry and acoustic reflex measurements.

Tympanometry equipment consists of a probe that snugly fits into the external auditory canal and has three channels.

- To deliver a tone of 220Hz
- To absorb reflected sound
- To change the pressure in EAC

Solution to Question 7:

The tympanogram shows As type of curve suggestive of otosclerosis. It indicates that pressure is normal but compliance is reduced.

Solution to Question 8:

The 7th and 8th cranial nerves are being tested in a child by assessing the stapedial reflexes.

The afferent of stapedial reflex is the 8th cranial nerve and efferent is the 7th cranial nerve.

Solution to Question 9:

The instrument shown is Barany's noise box which is used to mask non-test ear while testing bone conduction.

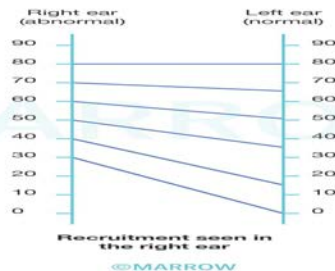
Solution to Question 10:

The given symptom is indicative of a recruitment phenomenon and is seen in cochlear hearing loss (e.g. Ménière's disease and presbycusis). It refers to abnormal growth of loudness.

It is tested by ABLB (Alternate Binaural Loudness Balance) test. Lesser intensity sounds are heard louder in the normal ear. However, a greater intensity sound is heard equally, or even better in the abnormal ear. This is referred to as abnormal growth of loudness.

However, in the conductive hearing loss, a constant difference of intensity is maintained throughout.

Alternate binaural loudness balance test



Solution to Question 11:

The given speech audiometry shows roll-over phenomenon suggestive of retrocochlear hearing loss.

With an increase in speech intensity above a particular level, the speech discrimination score falls rather than maintaining a plateau as seen in the cochlear type of sensorineural hearing loss.

Solution to Question 12:

The given clinical scenario and the MRI are suggestive of vestibular schwannoma which is a retrocochlear disease. The presence of otoacoustic emissions is a feature consistent with the diagnosis.

Vestibular schwannoma typically presents at 40-60 years of age. Progressive unilateral sensorineural hearing loss, often accompanied by tinnitus, is the presenting symptom in the majority of cases.

The MRI shows left cerebellopontine angle mass causing widening of the porous acousticus. The mass is seen extending into intracanalicular portion of the internal acoustic meatus which is suggestive of vestibular schwannoma.

Solution to Question 13:

Summating potential in electrocochleography indicates the sum of electrical activity in hair cells within the organ of Corti.

Electrocochleography (ECoG) is an objective test which records the electrical activity of the cochlea and 8th nerve in response to auditory stimuli. The recording electrode is passed through the tympanic membrane to rest over the promontory.

The three parameters measured in ECoG are:

- Cochlear microphonics – measures the electrical activity that follows the stimulus and reflects the instantaneous displacement of the basilar membrane.
- Summating potential (SP) – measures the electrical activity of the hair cells within the organ of Corti.
- Action potential (AP) – electrical activity in the distal part of the auditory nerve.

Normal SP: AP ratio is <30% but it is increased in Meniere's disease.

Solution to Question 14:

The otoacoustic emissions (OAEs) are generated by the outer hair cells of the cochlea. Presence of otoacoustic emissions is used to test hearing in a high-risk neonate as given in the clinical scenario.

Solution to Question 15:

Wave V in BERA is produced by lateral lemniscus.

The generators of ABEPs (Auditory nerve and brainstem evoked potentials) have been the subject of much research and controversy and are still not entirely agreed upon.

The literature available to us states the following:

- Scott Brown's Otorhinolaryngology- wave IV–V complex probably arises from neural activity within the superior olivary complex and nuclei of the lateral lemniscus, possibly some contribution from the inferior colliculus. So lateral lemniscus is the better option.
- Current Diagnosis and Treatment in Otorhinolaryngology- auditory brainstem response waves 2, 3, 4, and 5 represent the sequential activation of neurons as the signal is passed up the brainstem from the proximal part of the cochlear nerve, cochlear nucleus, superior olivary complex, and lateral lemniscus.
- NIH Journals- generation of wave V reflects the activity of multiple anatomic auditory structures. A sharp positive peak of wave V arises mainly from the lateral lemniscus, following a slow negative wave represents dendritic potential in the inferior colliculus.

Solution to Question 16:

In CERA or cortical evoked response audiometry, a tone stimulus is used instead of a click stimulus.

It is an objective hearing test, that records the signals from the auditory cortex (part of the brain that recognizes sound) that have been generated in response to sounds introduced to the ears.

The test gives information about the level of sound needed to produce a signal and confirms the status of the auditory cortex. This is valuable information that we can not obtain from a normal hearing test.

Difference between BERA and CERA:

BERA	CERA
The recording is made from brain stem potentials	The recording is made from cortical potentials
Click stimulus is used	Tone stimulus is used
Responses are not frequency specific	Responses are frequency specific
Can be performed in awake and restless patients	The patient must lie still throughout the process
Responses begin after 1 -10 milliseconds after stimuli	Responses begin after 50 – 300 milliseconds after stimulation
Suitable for even younger children	Unsuitable for children

Solution to Question 17:

The given scenario is suggestive of malingering and Rinne's test is least useful here.

Tests for malingering or nonorganic hearing loss:

- Stenger test
- Lombard test
- Stapedial reflex
- Inconsistent results on repeat pure tone and speech audiometry tests
- BERA

Solution to Question 18:

In caloric testing, cold water irrigation produces horizontal nystagmus towards the opposite side of irrigation. The direction of nystagmus can be remembered by the mnemonic, COWS (cold–opposite, warm–same).

The mechanism for the nystagmus is the creation of a temperature gradient by the cold water, which pulls the cupula away from the utricle, causing nystagmus towards the opposite side. As the horizontal semicircular canal is closer to the outer ear canal, most of the response comes from it and the nystagmus is horizontal.

The Fitzgerald–Hallpike test (bithermal caloric test) is done to assess this phenomenon. The patient lies in a supine position with head tilted forward at a 30° angle, ensuring the horizontal canal becomes vertical. Alternatingly, water at temperatures of 30°C and 44°C (7° below and above normal body temperature, respectively) is irrigated into the ears for a duration of 40 seconds each. The examiner closely observes the patient's eyes for the emergence of nystagmus until its endpoint is reached. The time taken from the start of irrigation to the endpoint of nystagmus is recorded and plotted on a calorigram.

If no nystagmus is evoked from either ear, the test is repeated with water at a temperature of 20°C for a duration of 4 minutes before labeling the labyrinth dead. A 5-minute interval should be allowed between testing each ear to ensure accurate results.

Solution to Question 19:

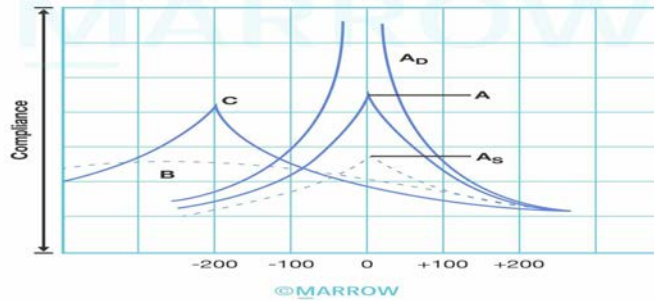
BERA, OAE, and tympanometry are the objective tests for hearing. PTA (pure tone audiometry) is a subjective test for hearing as it relies on patient responses to pure tone stimuli. Other examples of subjective tests include tuning fork tests and speech audiometry.

Tympanometry is an objective test of the middle ear and eustachian tube functions (conductive compartment of hearing). Here, the compliance of the TM, i.e., how much it moves to the pressure differentials applied from the external ear is measured.

The image below shows the types of tympanogram:

Types of tympanogram

- Type A** Normal tympanogram.
- Type A_S** Compliance is lower at or near ambient air pressure. Seen in fixation of ossicles, e.g. otosclerosis or malleus fixation.
- Type A_D** High compliance at or near ambient pressure. Seen in ossicular discontinuity or thin and lax tympanic membrane.
- Type B** A flat or dome-shaped graph. No change in compliance with pressure changes. Seen in middle ear fluid or thick tympanic membrane.
- Type C** Maximum compliance occurs with negative pressure in excess of 100 mm H₂O. Seen in retracted tympanic membrane.



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Anatomy of Nose

Question 1:

Which of the following is correct about the framework of the external nose?

- a) Upper 1/3rd bony and lower 2/3rd cartilaginous
- b) Upper ½ is bony and lower ½ is cartilaginous
- c) Upper 2/3rd is bony and lower 1/3rd is cartilaginous
- d) Upper 2/3rd cartilaginous and lower 1/3rd bony

Question 2:

Which of the following is true about the cartilaginous part of the external nose?

- a) 3 paired and 3 unpaired cartilages
- b) 3 paired and 1 unpaired cartilages
- c) 1 paired and 1 unpaired cartilages
- d) 2 paired and 2 unpaired cartilages

Question 3:

What is the shape of the septal cartilage?

- a) Quadrilateral
- b) Triangular
- c) Hexagonal
- d) Circular

Question 4:

A boxer suffered a blow to the nose. You suspect a fracture of his external nasal bones. Which of the following will you not expect to contribute to the fracture?

- a) Nasal bone
- b) Frontal bone

- c) Sphenoid bone
- d) Maxilla

Question 5:

Which of the following structures does not contribute to the formation of the nasal septum?

- a) Septal cartilage
- b) Vomer
- c) Perpendicular plate of ethmoid
- d) Perpendicular plate of sphenoid

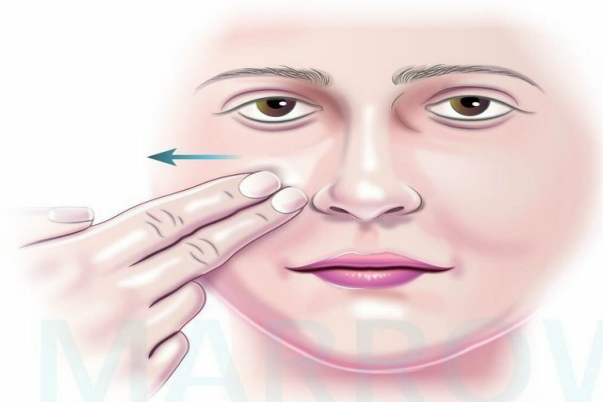
Question 6:

What are the choanae also known as?

- a) Anterior nares
- b) Posterior nares
- c) Limen nasi
- d) Agger nasi

Question 7:

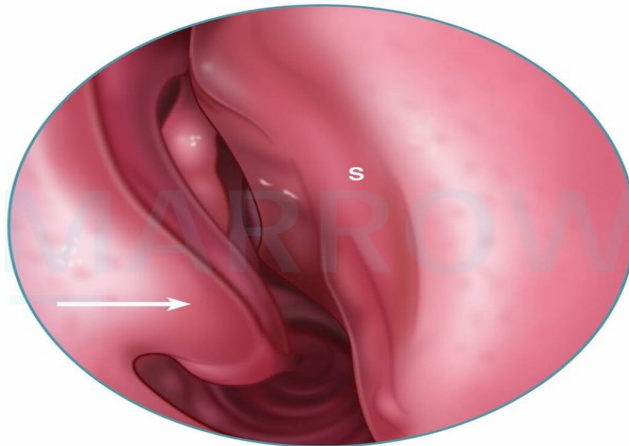
A patient presents with nasal congestion and the following test is performed. Which of the following does not contribute to the boundaries of the involved structure?



- a) Upper lateral cartilage
- b) Inferior turbinate
- c) Middle turbinate
- d) Septum

Question 8:

Given below is the endoscopic view of the right nasal cavity. Identify the marked structure?



- a) Inferior Turbinate
- b) Middle turbinate

- c) Superior Turbinate
- d) Uncinate process

Question 9:

Which of the following articulates with the maxilla?

- a) Superior turbinate
- b) Middle turbinate
- c) Inferior turbinate
- d) Supreme turbinate

Question 10:

A middle-aged woman was admitted with complaints of longstanding breathing difficulties and headaches. On evaluation, a deviated nasal septum was noted. CT image is given below. Which of the following structures is involved in this case?



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- a) Inferior turbinate
- b) Middle turbinate
- c) Superior turbinate
- d) Ethmoid sinus

Question 11:

In a patient with refractory posterior epistaxis, a transnasal endoscopic sphenopalatine artery ligation is planned. During the operation, the surgeon asks you to clip the artery at the sphenopalatine foramen. Where will you locate the site of ligation?

- a) Anterior end of middle turbinate
- b) Posterior end of middle turbinate
- c) Anterior end of superior turbinate
- d) Posterior end of superior turbinate

Question 12:

Where is hiatus semilunaris present?

- a) Superior meatus
- b) Middle meatus
- c) Inferior meatus
- d) Sphenoethmoidal recess

Question 13:

Which of the following sinuses does not open into the middle meatus?

- a) Frontal sinus
- b) Maxillary sinus
- c) Anterior ethmoidal sinus
- d) Posterior ethmoidal sinus

Question 14:

Where is the ethmoidal bulla present?

- a) Superior meatus
- b) Middle meatus
- c) Inferior meatus
- d) Sphenoethmoidal recess

Question 15:

A 10-month-old child presents with congenital dacryocystitis and requires probing of the nasolacrimal duct to relieve the obstruction. Where will the probe emerge in the nasal cavity?

- a) Superior meatus
- b) Middle meatus
- c) Inferior meatus
- d) Sphenoethmoidal recess

Question 16:

While performing diagnostic nasal endoscopy, which of the following structures would you not see during the third pass?

- a) Agger nasi
- b) Atrium of middle meatus
- c) Fronto-nasal duct
- d) Nasolacrimal duct

Question 17:

Which of the following statements is false about the olfactory area of the nose?

- a) The olfactory area is known as the dangerous area of the nose
- b) Bounded laterally by superior turbinate, medially by corresponding part of septum, above by cribriform plate of ethmoid bone
- c) Trauma can lead to spread of infection intracranially and can cause CSF rhinorrhea
- d) The olfactory area is very well ventilated and it aids in olfaction

Question 18:

A 32-year-old woman presented with nasal obstruction and rhinorrhea refractory to medical treatment. On examination, turbinates were hypertrophied. She was scheduled for vidian neurectomy. Which of the following statements is incorrect about the nerve supply of the region?

- a) Parasympathetic supply is from greater superficial petrosal nerve
- b) Sympathetic supply is by upper two thoracic segments of spinal cord
- c) Parasympathetic stimulation causes vasodilatation and sympathetic stimulation causes vasoconstriction

d) Sympathetic fibres of the nerve of pterygoid canal relay in the sphenopalatine ganglion

Question 19:

At what site does the maxillary sinus open into the middle meatus?

- a) Hiatus semilunaris
- b) Bulla ethmoidalis
- c) Infundibulum
- d) Uncinate process

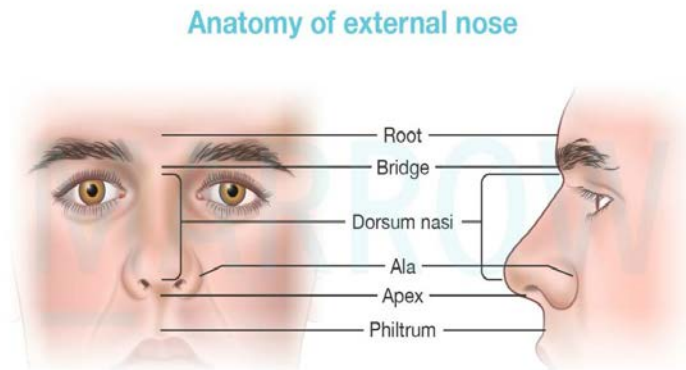
Answer Key

Question No.	Correct Option
1	a
2	b
3	a
4	c
5	d
6	b
7	c
8	a
9	c
10	b
11	b
12	b
13	d
14	b
15	c
16	d
17	d
18	d
19	c

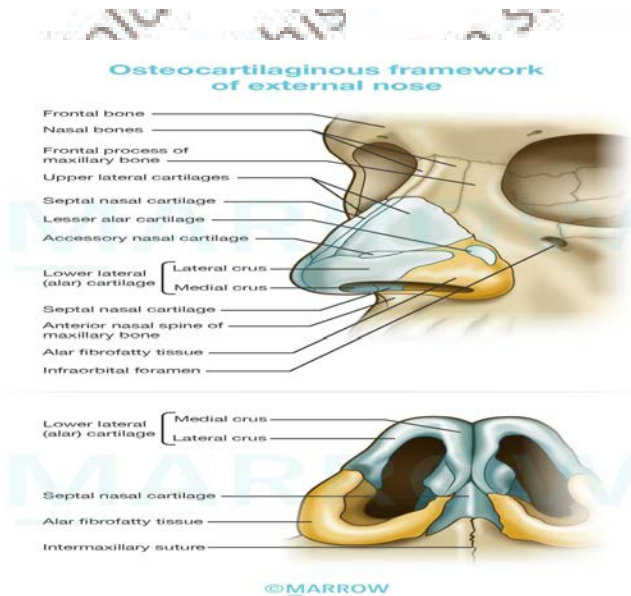
Detailed Explanations

Solution to Question 1:

Upper 1/3rd of the external nose is bony, whereas lower 2/3rd is cartilaginous.



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Solution to Question 2:

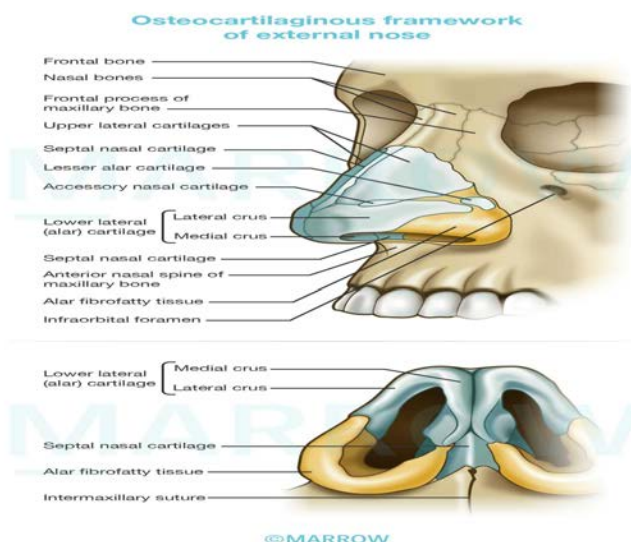
The external nose is made up of 3 paired and 1 unpaired cartilages.

Paired cartilages:

- Upper lateral cartilage - From nasal bone above to alar cartilage below.

- Lower lateral (alar) cartilage - U-shaped. Lateral crus forms the ala and medial crus runs in the columella. Limen nasi or limen vestibuli is the junction of the upper and lower lateral cartilages.
- Lesser alar (sesamoid) cartilage - multiple, above, and lateral to the alar cartilage.

The septal cartilage is the only unpaired cartilage. It forms the midline of the lower 2/3rd of the dorsum of nose and supports lower 2/3rd of the dorsum of external nose.



Solution to Question 3:

Septal cartilage is a quadrilateral cartilage.

Solution to Question 4:

Sphenoid bone does not contribute to the bony part of the external nose.

The bony part of the external nose consists of

- Two nasal bones
- Frontal process of maxilla lateral to nasal bones on each side.
- Nasal process of frontal bone in the midline between 2 nasal bones.
- Nasion is the site of the junction of nasal bone with the nasal process of the frontal bone.

Solution to Question 5:

Perpendicular plate of sphenoid does not contribute to the formation of the nasal septum.

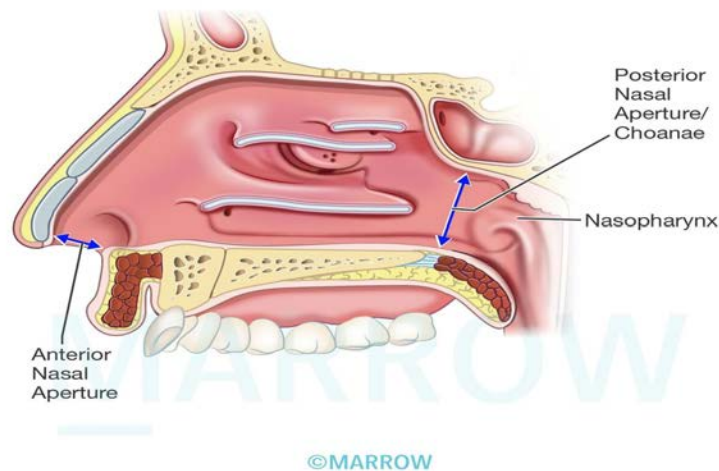
Solution to Question 6:

The choanae are also known as posterior nares.

Each nasal cavity communicates with the exterior through the anterior nares or nostril and with the nasopharynx through the posterior nares, posterior nasal aperture or the choanae.

Limen nasi or the nasal valve is the upper limit of nasal vestibule on the lateral wall. It demarcates the skin-lined vestibule with the mucosa-lined nasal cavity proper.

Agger nasi is an elevation just anterior to the attachment of middle turbinate.



Solution to Question 7:

The given image shows Cottle's test being performed and the structure involved is the nasal valve.

Middle turbinate does not contribute to the boundaries of the nasal valve.

In Cottle's test, the cheek of the patient is drawn laterally during quiet breathing. If this improves the airway on the test side, the test is considered to be positive. This suggests an abnormality of the vestibular component of the nasal valve causing nasal obstruction.

Solution to Question 8:

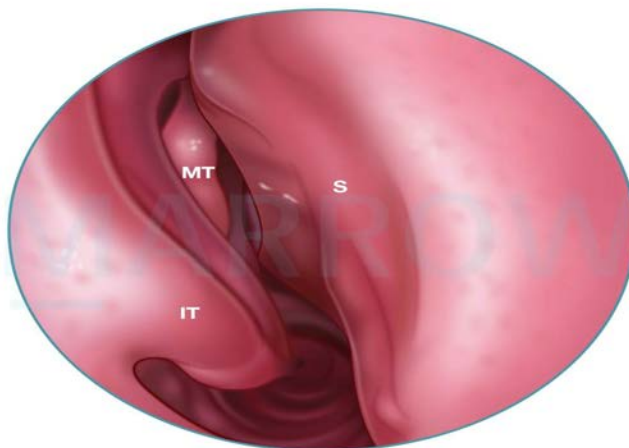
The marked structure is the lowermost projection in the lateral wall of the nose, the inferior turbinate. The middle turbinate (Option B) is seen as superior to it.

The lateral wall of the nose contains three or sometimes four projections of variable size namely superior, middle, and inferior turbinates along with a corresponding meatus under each turbinate.

The inferior turbinate is the largest turbinate and occupies the lower third of the lateral nasal wall. It is a separate bone. Middle and superior turbinates (Option C) are ethmoturbinal, part of the ethmoid bone.

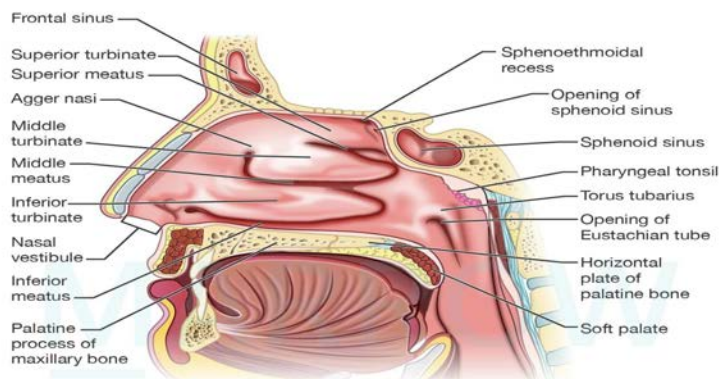
Other important structures which could be seen in the above image are:

- IT: Inferior turbinate
- S: Septum,
- MT: middle turbinate



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Lateral wall of nasal cavity

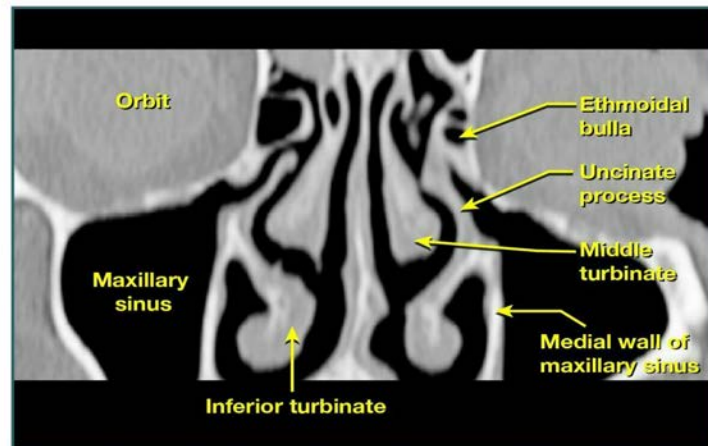


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Uncinate process (Option D) is a hook-like structure running parallel to the anterior border of bulla ethmoidalis. The posteroinferior end of the uncinate process is attached to the inferior turbinate.

The image below shows a CT image of the uncinate process.

CT scan of ostiomeatal complex

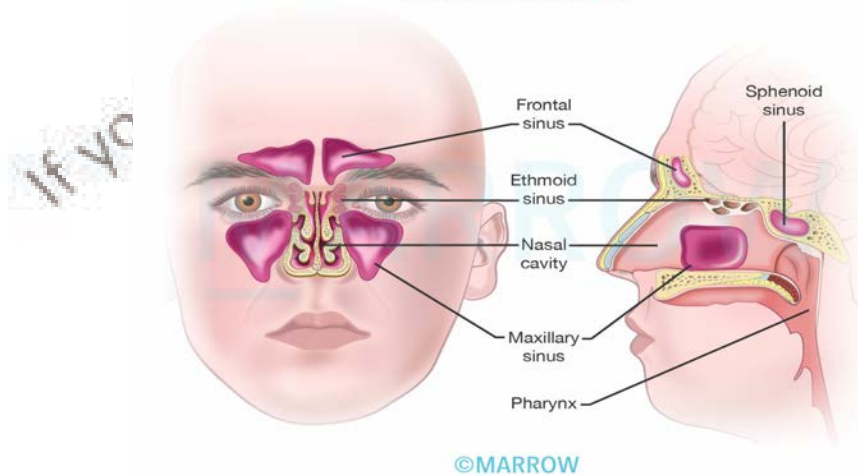


Solution to Question 9:

The inferior turbinate articulates with the nasal surface of the maxilla and the perpendicular plate of the palatine bone.

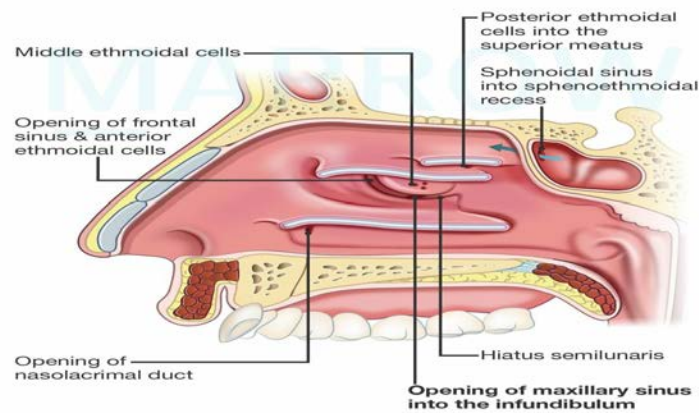
The superior, middle, and supreme turbinates are present in the labyrinth of the ethmoid bone.

Paranasal sinuses



The image given below shows the nasal cavity, inferior turbinate and opening of maxillary sinus.

Drainage of paranasal sinuses



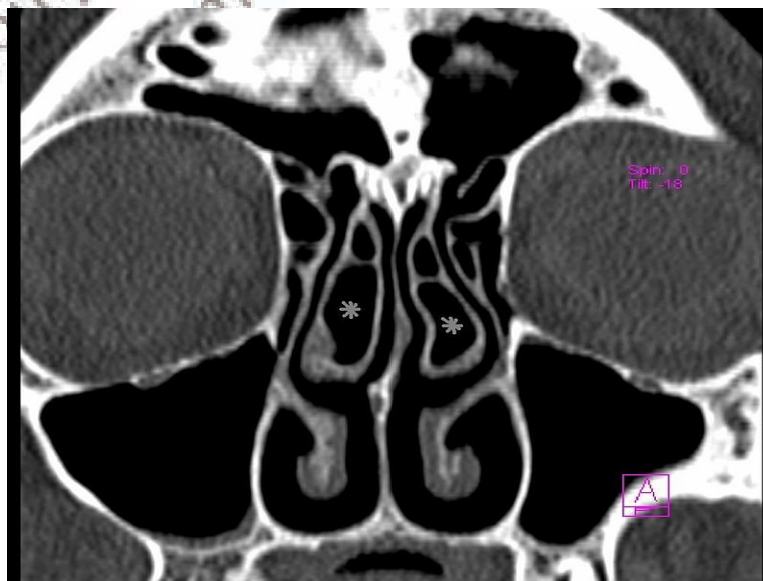
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Solution to Question 10:

The given clinical scenario and CT findings are suggestive of concha bullosa. It is the pneumatization of the middle turbinate and is one of the anatomic variations of the paranasal region.

Concha bullosa can be either unilateral or bilateral and generally occurs together with a septal deviation to the contralateral side.

In the image given below, the star-marked area is the enlarged, ballooned out middle turbinate called concha bullosa.

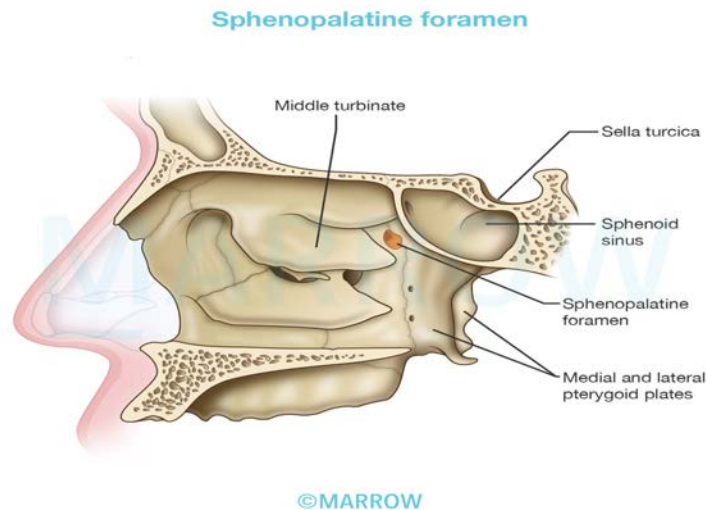


Solution to Question 11:

Transnasal endoscopic sphenopalatine artery ligation (TESPAL) is done to control refractory posterior epistaxis by ligating the sphenopalatine artery. It emerges from the sphenopalatine foramen which is situated at the posterior end of the middle turbinate.

Branches of sphenopalatine ganglion also emerge from the sphenopalatine foramen. It supplies most of the posterior 2/3rd of the nasal cavity. It can be anaesthetized by placing a pledget of cotton soaked in anaesthetic solution at the foramen.

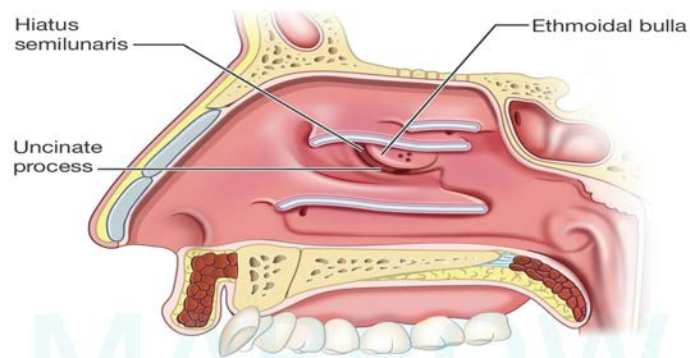
The sphenopalatine foramen is also the most common site for the origin of angiofibroma.



Solution to Question 12:

Hiatus semilunaris is present in the middle meatus. It is a two-dimensional space bounded by bulla ethmoidalis above and uncinate process below.

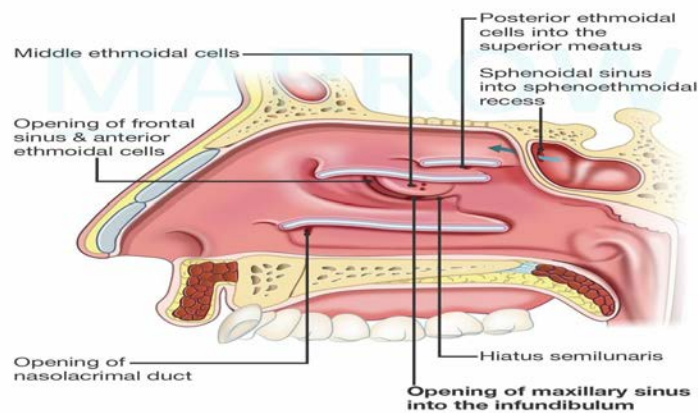
It opens into a three-dimensional area called infundibulum bounded by uncinate process, frontal process of maxilla medially, and lamina papyracea laterally. The maxillary sinus opens into lower part of infundibulum.



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The image given below shows the opening of maxillary sinus

Drainage of paranasal sinuses



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Solution to Question 13:

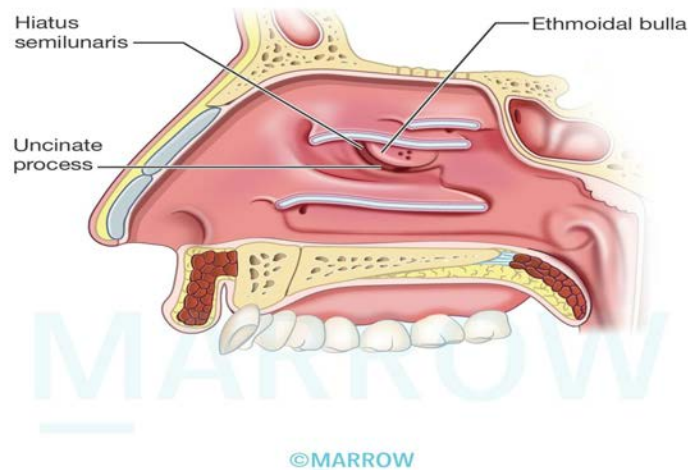
The posterior ethmoidal sinus does not open into the middle meatus. It opens into the superior meatus.

Frontal, maxillary, and anterior ethmoidal sinuses open into middle meatus.

Solution to Question 14:

Ethmoidal bulla (Bulla ethmoidalis) is an ethmoidal cell situated behind the uncinete process.

It forms the posterior boundary of the hiatus semilunaris.



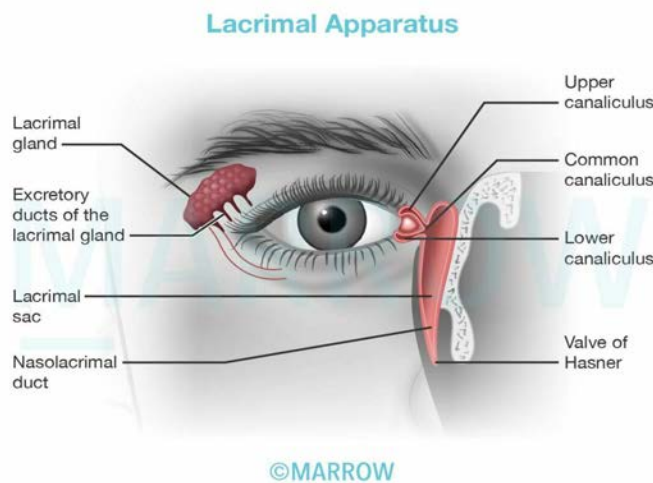
Solution to Question 15:

The probe emerges in the inferior meatus because the opening of the nasolacrimal duct is present in the inferior meatus.

The nasolacrimal duct runs downwards, backward, and laterally from the lacrimal sac and opens into the inferior meatus.

It is bounded by a mucosal valve called the valve of Hasner in the inferior meatus.

Note: In congenital dacryocystitis, probing of the nasolacrimal duct is done at 9-12 months of age if symptoms persist past 6 months of age and aren't relieved by local massage and topical antibiotics.



Solution to Question 16:

The third pass in diagnostic nasal endoscopy involves the examination of the middle meatus in detail. The nasolacrimal duct opens into the inferior meatus, hence it will not be visualized here.

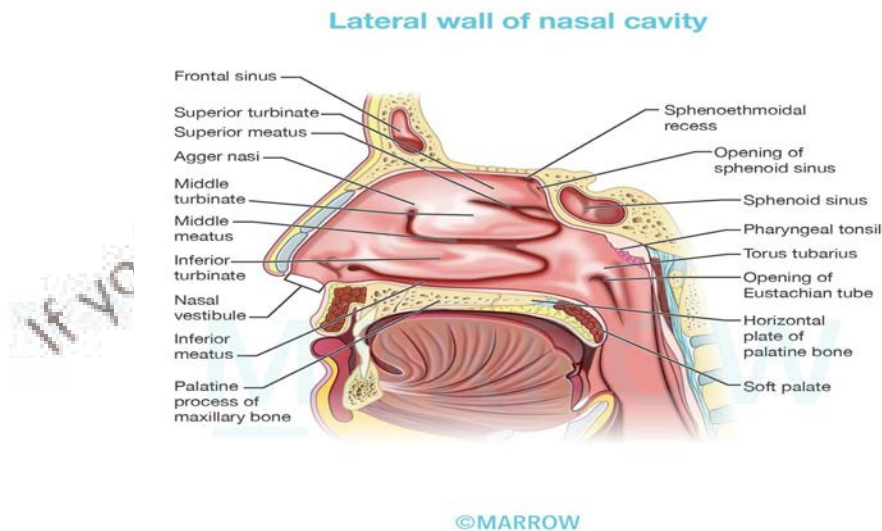
Option A: Agger nasi is an elevation just anterior to the attachment of the middle turbinate. When pneumatized and enlarged it encroaches on the frontal recess area and causes mechanical obstruction to drainage of the frontal sinus leading to sinusitis.

Option B: Atrium of the middle meatus is a shallow depression lying in front of the middle turbinate and above the nasal vestibule.

Option C: Fronto-nasal duct/frontal recess is the opening of the frontal sinus into the infundibulum.

Note: In diagnostic nasal endoscopy three passes are performed -

- First pass is the examination of the nasopharynx and inferior meatus.
- Second pass is the examination of sphenothmoidal recess, superior meatus, and openings of sphenoid sinus and posterior ethmoidal cells.
- Third pass is the examination of the middle meatus in detail.



Solution to Question 17:

The olfactory area is not well ventilated, and very little air passes through the olfactory region during quiet inspiration. Weak odors need to be sniffed, due to the same.

The upper 1/3rd of the nasal cavity forms the olfactory area which is also known as the dangerous area of the nose. Olfactory nerve fibers pass through the cribriform plate of the ethmoid. Spread of infection to the intracranial space can occur through this route. Trauma to the cribriform plate

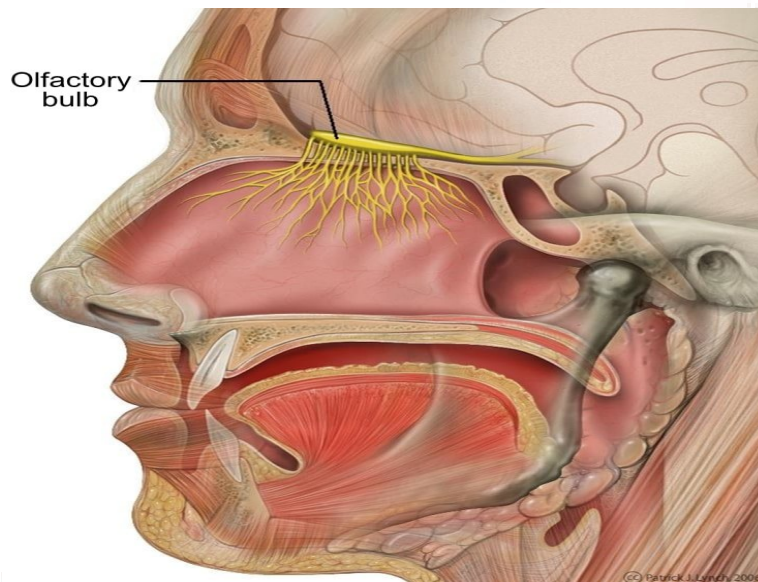
can lead to CSF rhinorrhea.

It is bounded by the following structures:

- Laterally by superior turbinate
- Medially by corresponding part of the septum
- Superiorly by cribriform plate of the ethmoid

It is lined by olfactory epithelium.

Note: The danger triangle of the face consists of the area from the corners of the mouth to the bridge of the nose. This is because retrograde infection from these areas can spread to the brain, through the venous communication (via the ophthalmic veins) between the facial vein and the cavernous sinus.

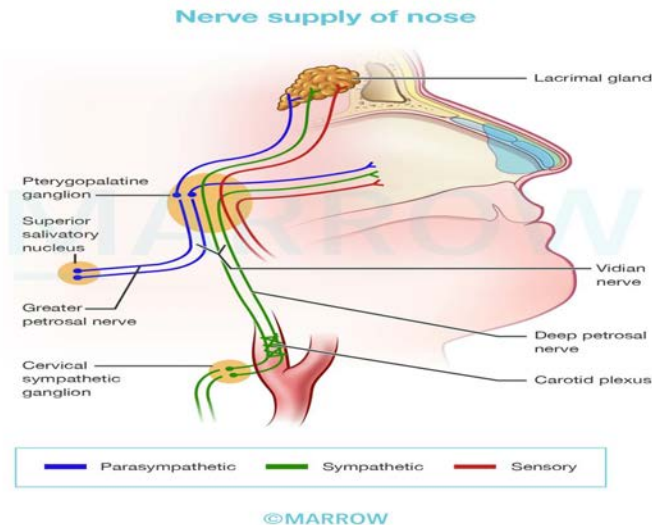


Solution to Question 18:

The given case scenario with rhinorrhea, nasal obstruction, and turbinate hypertrophy is suggestive of vasomotor rhinitis (VMR). Vidian neurectomy is the treatment of choice in VMR refractory to medical treatment. The autonomic nerve supply of the nose is unstable in cases of VMR. Sympathetic fibres of the nerve of the pterygoid canal do not relay in the sphenopalatine ganglion.

	Parasympathetic supply	Sympathetic supply
Nerve fibres arise from	Greater superficial petrosal nerve	Upper two thoracic segments of spinal cord
Course	Travel in the nerve of pterygoid canal (Vidian nerve) to the sphenopalatine ganglion	Superior cervical ganglion to deep petrosal nerve to parasympathetic fibres of greater petrosal nerve - nerve of pterygoid canal

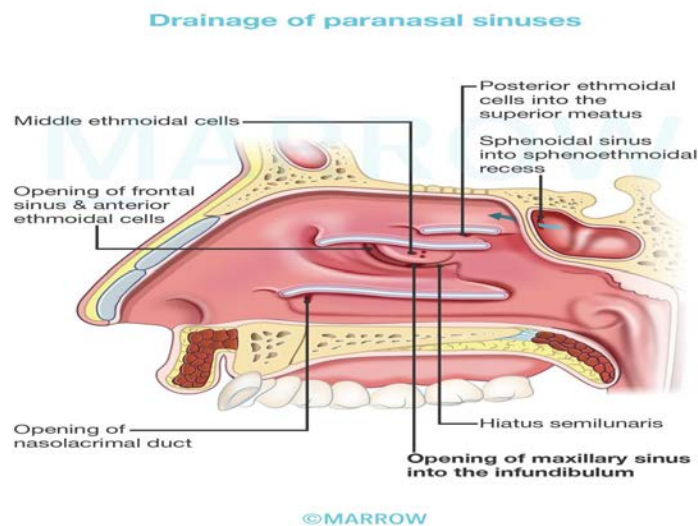
	Parasympathetic supply	Sympathetic supply
Relay	Relays in the sphenopalatine ganglion before reaching nasal cavity	Doesn't relay in sphenopalatine ganglion
Function	Vasodilation	Vasoconstriction



Solution to Question 19:

Maxillary sinus opens into middle meatus at the lower part of infundibulum, in its medial wall.

Infundibulum is the space limited medially by the uncinate process and frontal process of maxilla, and laterally by the lamina papyracea.



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Anatomy of Paranasal Sinuses

Question 1:

Which of the following is not a paranasal sinus?

- a) Frontal sinus
- b) Pyriform sinus
- c) Ethmoid sinus
- d) Maxillary sinus

Question 2:

Which is the largest paranasal sinus?

- a) Maxillary sinus
- b) Frontal sinus
- c) Ethmoid sinus
- d) Sphenoid sinus

Question 3:

Which of the following is also known as antrum of Highmore?

- a) Frontal sinus
- b) Sphenoid sinus
- c) Maxillary sinus
- d) Ethmoid sinus

Question 4:

Which of the following statements is correct about the maxillary sinus?

- a) Pyramidal in shape with base towards lateral wall of nose and apex towards zygomatic arch
- b) Pyramidal in shape with base towards zygomatic arch and apex towards lateral wall of nose.
- c) Triangular in shape

- d) Rectangular in shape

Question 5:

A pathologist is examining the specimen of a maxillectomy. Which of the following epithelium do you expect to find lining the maxillary sinus?

- a) Keratinising stratified squamous epithelium
- b) Non keratinizing stratified squamous epithelium
- c) Ciliated columnar epithelium
- d) Simple columnar epithelium

Question 6:

Which of the following is most superior in location?

- a) Frontal sinus
- b) Maxillary sinus
- c) Ethmoid sinus
- d) Sphenoid sinus

Question 7:

An anxious father brings his newborn baby to the pediatrician after dropping his baby when shifting from one arm to another. The pediatrician orders a CT head, which is normal. Which of the following structures would you not expect to find in the CT?

- a) Maxillary sinus
- b) Frontal sinus
- c) Ethmoid sinus
- d) None of the above

Question 8:

Where does the posterior ethmoidal sinus open?

- a) Superior meatus
- b) Middle meatus

- c) Inferior meatus
- d) Sphenothmoidal recess

Question 9:

Which of the following is true about sphenoid sinus?

- a) Present in greater wing of sphenoid
- b) Lined by stratified squamous epithelium
- c) Opens into sphenothmoidal recess
- d) Lateral wall is related to the pituitary gland

Question 10:

You are an ENT resident performing endoscopic biopsy from the sphenoid sinus in a COVID patient with suspected mucormycosis. Through which of the following areas will you access the sinus?

- a) Superior meatus
- b) Middle meatus
- c) Inferior meatus
- d) Sphenothmoidal recess

Question 11:

Optic nerve injury following sinus surgery (FESS) is due to removal of which of the following ethmoidal cells?

- a) Haller
- b) Agger nasi
- c) Onodi cells
- d) Bullae cells

Question 12:

While performing functional endoscopic sinus surgery, the surgeon points out the Haller cells and their clinical significance. Which of the following is correct about these cells?

- a) They are related to the maxillary sinus
- b) They are related to the optic nerve
- c) They are related to the sphenoid sinus
- d) They are related to the facial nerve

Question 13:

What is occipitomenal view with open mouth also known as?

- a) Water's view
- b) Towne's view
- c) Law's view
- d) Stenver's view

Answer Key

Question No.	Correct Option
1	b
2	a
3	c
4	a
5	c
6	a
7	b
8	a
9	c
10	d
11	c
12	a
13	a

Detailed Explanations

Solution to Question 1:

Pyriform sinus (fossa) is a part of hypopharynx. It lies on either side of the larynx and extends from pharyngoepiglottic fold to the upper end of oesophagus.

Sinuses are air-containing cavities in certain bones of skull.

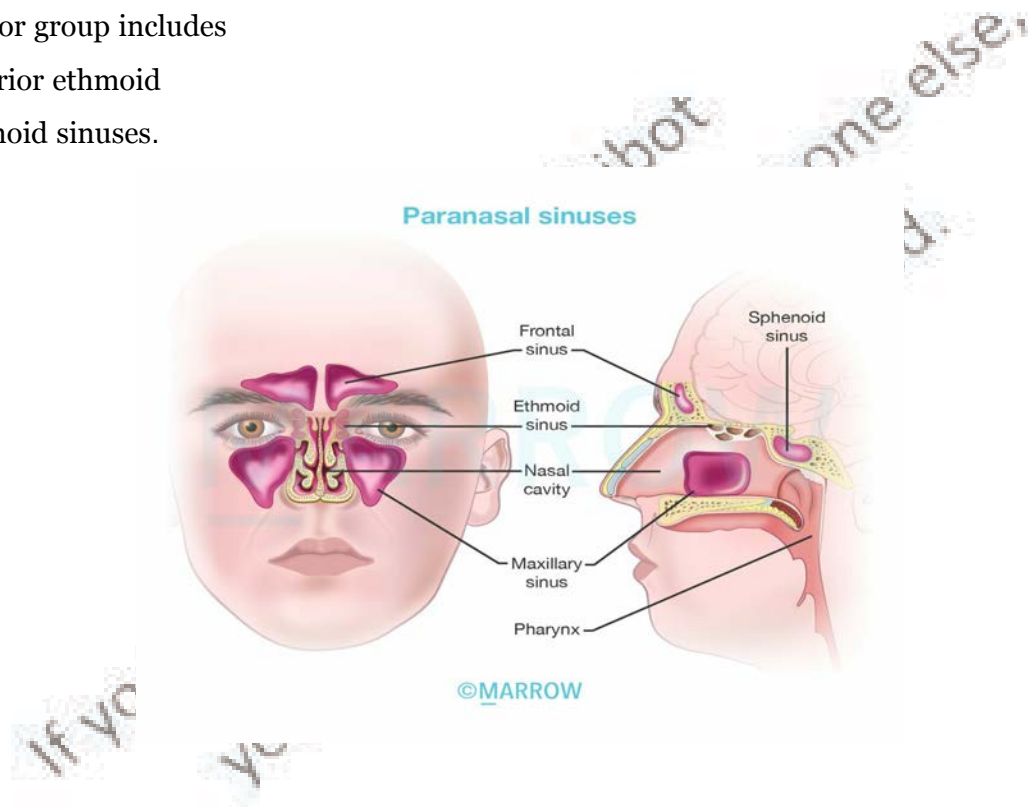
It is divided into 2 groups.

Anterior group includes

- Maxillary
- Frontal
- Anterior ethmoidal sinuses.

Posterior group includes

- Posterior ethmoid
- Sphenoid sinuses.



Solution to Question 2:

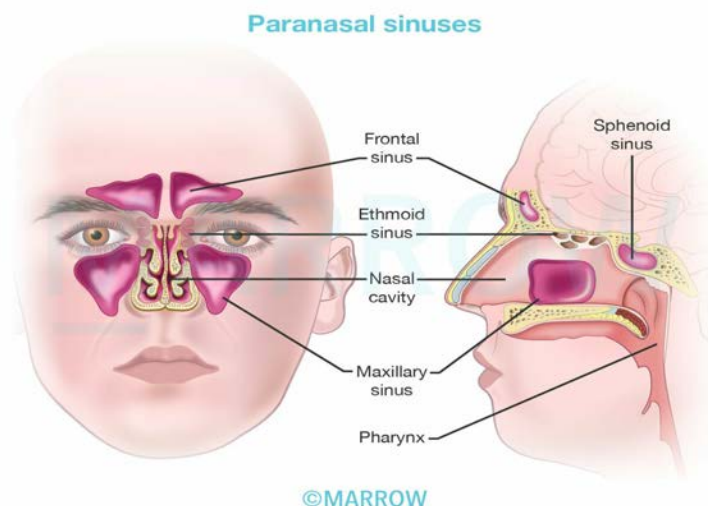
Maxillary sinus is the largest of paranasal sinuses and occupies the body of maxilla.

Solution to Question 3:

Maxillary sinus is also known as antrum of Highmore.

Solution to Question 4:

Maxillary sinus is pyramidal in shape, with base towards lateral wall of nose and apex directed laterally into the zygomatic process of maxilla and sometimes in the zygomatic bone itself.



Solution to Question 5:

Paranasal sinuses are lined by ciliated columnar epithelium with goblet cells which secrete mucus. They're lined by the mucous membrane, which is continuous with that of the nasal cavity through the ostia of sinuses.

Solution to Question 6:

Frontal sinus is the most superiorly located sinus.

It is situated between the inner and outer tables of frontal bone, above and deep to the supraorbital margin.

Solution to Question 7:

The frontal sinus is absent or rudimentary at birth. Radiologically, the frontal sinus is able to be identified at 6 years and reaches full size only after puberty.

At birth, only the maxillary and ethmoidal sinuses are present and are large enough to be clinically significant. Sphenoid sinus, though may be present, is usually undeveloped and non-aerated.

Sinuses	At birth	First radiological evidence	Reaches adult size by
Maxillary	Present	4-5 months after birth	15 years
Ethmoidal	Present	1 year	12 years
Frontal	Absent/rudimentary	6 years	Post-puberty
Sphenoid	Present	4 years	14 years

Solution to Question 8:

The posterior ethmoidal sinus opens into the superior meatus.

Solution to Question 9:

Sphenoid sinus opens into sphenoidal recess.

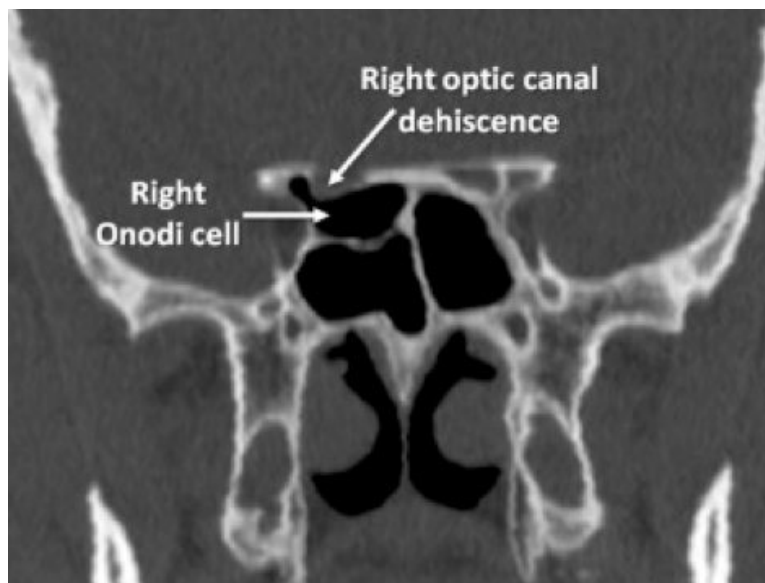
Solution to Question 10:

The sphenoid sinus is accessed through the sphenoidal recess which is situated above the superior turbinate.

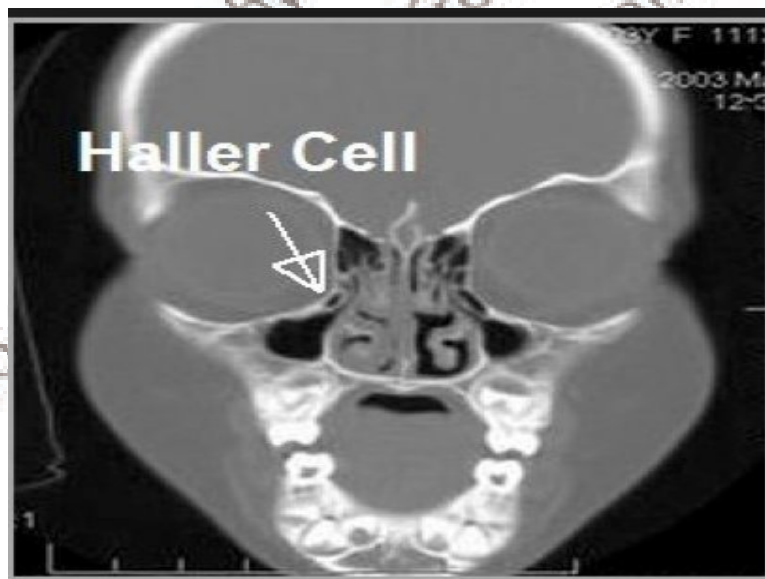
The ostium of the sphenoid sinus can be located endoscopically 1cm above the upper margin of the posterior choana near the posterior border of the septum.

Solution to Question 11:

Onodi cell is the posterior ethmoidal cell, related superiorly or laterally to the sphenoid sinus. The optic nerve lies in its lateral wall and can be injured while operating in the cells.



Option A: Haller cells are a type of ethmoidal cells that are situated on the roof of the maxillary sinus. Sometimes they may enlarge and block the opening of the maxillary sinus leading to maxillary sinusitis.



Option B: Agger nasi is an elevation just anterior to the attachment of middle turbinates.

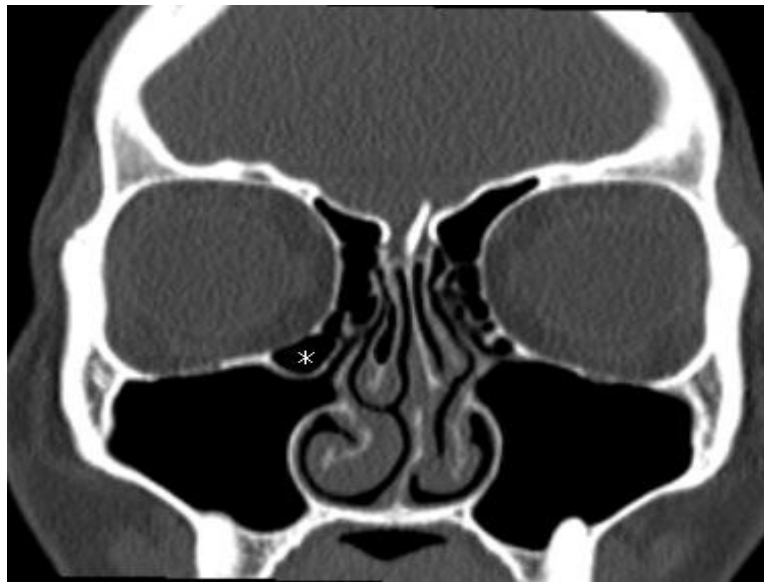
Option D: Bulla ethmoidalis is the ethmoidal cell situated behind the uncinate process and forms the posterior boundary of hiatus semilunaris through which the maxillary, frontal and anterior ethmoidal sinuses drain.

Solution to Question 12:

Haller cells (infraorbital ethmoidal air cells) are a type of ethmoidal cells that are situated in the roof of the maxillary sinus. Sometimes they may enlarge and block the opening of the maxillary

sinus leading to maxillary sinusitis.

The star-marked area in the given CT scan shows the Haller cell



Solution to Question 13:

Occipitomental view is called Water's view. Water's view with an open mouth is known as Pierre's view. Among the given options, Water's view is the best answer.

In Water's view, the patient's head is positioned such that the nose and the chin touch the film while the X-rays are projected from backward. The mouth is usually kept open, which allows visualizing the sphenoid sinus.

Structures seen in Water's view are:

- Maxillary sinuses (seen best)
- Frontal sinuses
- Sphenoid sinus (if the film is taken with an open mouth)
- Zygoma
- Zygomatic arch
- Nasal bone
- Frontal process of the maxilla
- Superior orbital fissure
- Infratemporal fossa

Other options:

Option B: Towne's view is an angled anteroposterior radiograph of the skull. Structures seen are the petrous part of the pyramids, arcuate eminence, superior semicircular canal, mastoid antrum, internal auditory canal, tympanic cavity, cochlea, and external auditory canal. This view is taken

for acoustic neuroma and apical petrositis.

Option C: Law's view is a lateral oblique X-ray view of the mastoid. Structures seen are the external auditory canal, mastoid air cells, tegmen, lateral sinus plate, and temporomandibular joint.

Option D: Stenver's view is an X-ray view of the skull where the long axis of the petrous bone is parallel to the film. Structures seen are the entire petrous pyramid, arcuate eminence, internal auditory meatus, labyrinth with its vestibule, cochlea, and mastoid antrum.

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Physiology of Nose and Paranasal Sinuses

Question 1:

During inspiration, through which part of the nasal cavity does the main current of airflow take place?

- a) Olfactory region
- b) Upper part of nasal cavity in a parabolic curve
- c) Middle part of nasal cavity in a parabolic curve
- d) Lower part of nasal cavity in a parabolic curve

Question 2:

What percentage of total airflow through nose reaches the olfactory region?

- a) 10 - 15%
- b) 30 - 40%
- c) 40 - 50%
- d) 60 - 70%

Question 3:

How frequently does the rhythmic congestion and decongestion of nasal mucosa occur?

- a) 2 - 4 hours
- b) 4 - 6 hours
- c) 6 - 8 hours
- d) 1 - 2 hours

Question 4:

Which of the following statements is false about nasal secretions?

- a) They contain immunoglobulins and lysozymes.
- b) pH of nasal secretions is around 7.

- c) They increase in response to noxious stimuli.
- d) Lactoferrin is not a component of nasal secretions.

Question 5:

Which of the following causes movement of the mucus blanket of the nose?

- a) Inspiration
- b) Expiration
- c) Nasal cycle
- d) Ciliary action

Question 6:

Which of the following does not affect movements of the cilia?

- a) Drying
- b) Smoking
- c) Adrenaline
- d) Nasal valve

Question 7:

A patient presents to you with bronchiectasis and chronic sinusitis. Chest X-Ray shows the following finding. Which of the following is the method of protection of the lower airway that's compromised in this patient?



- a) Temperature regulation
- b) Humidification
- c) Mucociliary mechanism
- d) Mucosal defense with immunoglobulins

Question 8:

As a first-year ENT resident, you receive a patient complaining of a complete lack of smell for the past week. Which of the following will not be part of your differential diagnosis?

- a) COVID-19 infection
- b) Atrophic rhinitis
- c) Olfactory nerve injury
- d) Allergic rhinitis

Question 9:

A 17-year-old girl was evaluated for delay in the onset of puberty. On further enquiry, she revealed that she has always had a very poor sense of smell. Which of the following is the most likely diagnosis?

- a) Kallmann syndrome
- b) Turner's syndrome
- c) Down's syndrome

- d) Klinefelter's syndrome

Question 10:

A 32-year-old female patient was found to be positive for SARS-CoV-2 and had a loss of taste and smell. After regaining her ability to smell after 2 months, she reported to the OPD with the complaint that all the smells she perceived were distorted, similar to the smell of burnt rubber. What is this called?

- a) Anosmia
- b) Phantosmia
- c) Parosmia
- d) Olfactory agnosia

Question 11:

Which of the following statements is true about ventilation of the nose and paranasal sinuses?

- a) Paranasal sinuses are filled with air during inspiration
- b) Paranasal sinuses are filled with air during expiration
- c) During inspiration, positive pressure is created in the nose
- d) During expiration, negative pressure is created in the nose

Question 12:

Which of the following is not a function of the paranasal sinuses?

- a) Air conditioning of inspired air
- b) Adds resonance to voice
- c) Lighten skull bones
- d) Maintain airway resistance

Answer Key

Question No.	Correct Option
1	c
2	a
3	a
4	d
5	d
6	d
7	c
8	d
9	a
10	c
11	b
12	d

Detailed Explanations

Solution to Question 1:

The main current of airflow in the nasal cavity flows through the middle part of nose between the turbinates and nasal septum. Very little air passes through the inferior meatus or olfactory region of nose.

Solution to Question 2:

About 15% of inspired air reaches the olfactory area within the nose, where it interacts with the mucus secreted by the Bowman's glands in the lamina propria and respiratory epithelium.

Solution to Question 3:

Rhythmic cyclical congestion and decongestion of the nasal mucosa is called nasal cycle. It occurs every 2.5-4 hours and is usually characteristic of an individual.

Solution to Question 4:

Nasal secretions contain muramidase (lysozyme), Lactoferrin and immunoglobulins IgA and IgE, and interferon and thus have antibacterial properties.

Their pH is around 7.

Copious flow of nasal secretions that follows irritation by noxious substance helps to wash them out.

Solution to Question 5:

The movement of the mucus blanket of the nose is by ciliary action.

The mucus blanket consists of a superficial mucus layer and a deeper serous layer, floating on the top of cilia, which are constantly beating to carry it like a conveyor belt towards the nasopharynx.

The mucus blanket moves at a speed of 5–10 mm/min and the complete sheet of mucus is cleared into the pharynx every 10–20 min. In mammals, cilia beat 10–20 times per second at room temperature.

Solution to Question 6:

Nasal valve does not affect the movements of the nasal cilia.

Movements of cilia are affected by:

- Drying
- Drugs (adrenaline)
- Excessive heat or cold
- Smoking
- Infections
- Noxious fumes like sulfur dioxide and carbon dioxide

Solution to Question 7:

The given clinical scenario of chronic sinusitis, bronchiectasis and the X-ray showing dextrocardia (situs inversus) is suggestive of Kartagener syndrome. The mucociliary mechanism is compromised in this case, due to immotile cilia.

The lower airway is protected by the following mechanisms:

- Mucociliary mechanism - The inspired pathogens and dust are trapped on the viscous mucous blanket. By beating of the cilia, the mucus is carried to the nasopharynx to be swallowed.
- Chemical mechanisms - Immunoglobulins such as IgA, IgE, and Lactoferrin, lysozyme provide innate immunity.
- Sneezing - It is a protective reflex due to irritation of the nasal mucosa.

Temperature regulation and humidification are due to the large surface area of mucosa and highly vascular cavernous spaces. This is achieved by an increase or decrease in blood flow which in turn increase or decrease the size of turbinates.

Solution to Question 8:

Anosmia refers to a complete loss of the sense of smell. Allergic rhinitis is more commonly associated with hyposmia, hence it'll not be part of differential diagnosis. Anosmia only occurs in case it leads to nasal polyposis

Causes of anosmia:

- Nasal obstruction due to nasal polyps, enlarged turbinates, or mucosal edema due to infections.
- Atrophic rhinitis, a degenerative disorder of nasal mucosa
- Peripheral neuritis (toxic or influenzal)
- Injury to olfactory nerves or olfactory bulb in fractures of anterior cranial fossa
- Intracranial lesions like abscesses, tumors, or meningitis, which cause pressure on olfactory tracts.
- COVID-19 is associated with a complete lack of taste and smell and is a common presenting symptom

Solution to Question 9:

The given clinical scenario of delay in puberty along with an impaired sense of smell is suggestive of Kallmann syndrome.

Kallmann syndrome and familial causes like primary congenital anosmia are congenital causes of anosmia.

Solution to Question 10:

Parosmia is the perversion of smell. It is seen classically in the recovery phase of postinfluenzal anosmia, and has been recently reported as a rare complication in post-COVID patients.

The probable explanation is misdirected regeneration of nerve fibers. Intracranial tumors should be ruled out in all cases of parosmia.

Option A: Anosmia is the complete lack of smell

Option B: Phantosmia is the sensation of olfaction in the absence of a stimulus

Option D: Olfactory agnosia is the inability to recognize an odour sensation

Solution to Question 11:

Paranasal sinuses are ventilated during expiration. Ventilation of sinuses is paradoxical, as they are emptied of air during inspiration and filled with air during expiration.

During inspiration, negative pressure is created in the nose by the air current, whereas during expiration, positive pressure is created in the nose, and this sets up eddy currents, which ventilate the sinuses.

Solution to Question 12:

Airway resistance is not a function of paranasal sinuses.

Resistance to the flow of air in the nose maintains the positive pressure needed to keep the alveoli open. It is due to two factors:

- Bones, cartilages, and attached muscles - fixed
- Mucosa - variable

Epistaxis

Question 1:

A 15-year-old boy was brought to the ER with a nose bleed secondary to trauma. The bleeding started an hour ago and has not stopped. Where is the bleed most likely to be from?

- a) Little's area
- b) Bony septum
- c) Superior tubinate
- d) Lateral wall of nose

Question 2:

Where is Little's area situated?

- a) Anterosuperior part of septum
- b) Anteroinferior part of septum
- c) Posterosuperior part of septum
- d) Posteroinferior part of septum

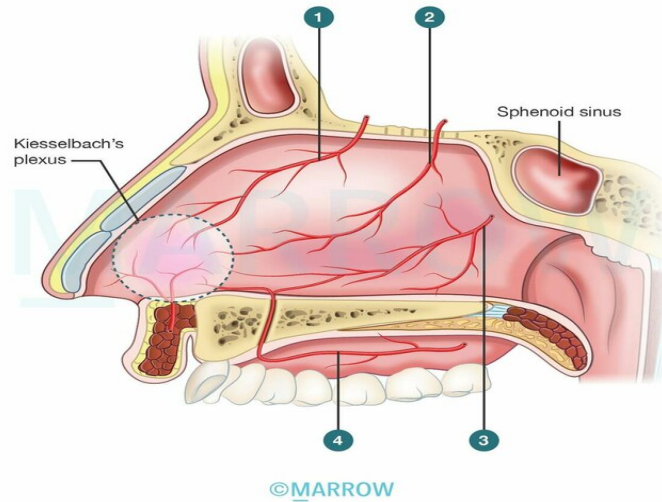
Question 3:

A worried mother brought her 4-year-old son to the ER with complaints of a nose bleed. On examination, the bleed was found to be from the Kiesselbach's plexus. Which of the following arteries does not contribute to its formation?

- a) Sphenopalatine artery
- b) Greater palatine artery
- c) Anterior ethmoidal artery
- d) Posterior ethmoidal artery

Question 4:

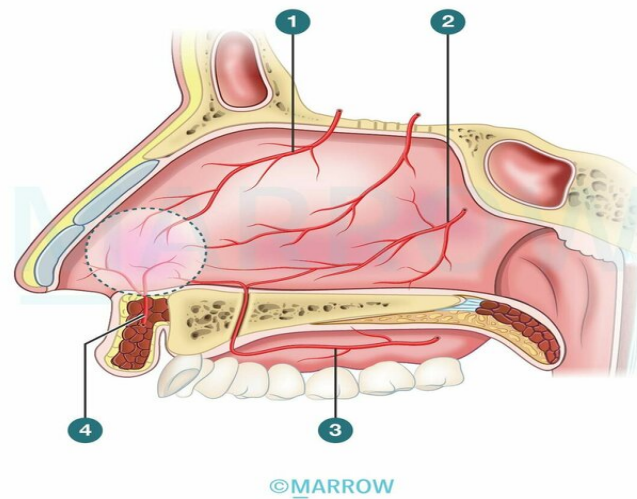
Identify the artery of epistaxis in the given image.



- a) 1
- b) 2
- c) 3
- d) 4

Question 5:

Which of the following marked structures is a branch of the internal carotid artery?



- a) 1
- b) 2

- c) 3
- d) 4

Question 6:

A 40-year-old male patient with a history of uncontrolled, chronic hypertension was rushed to the casualty after he had an episode of coughing out frank blood. On further evaluation, a diagnosis of posterior epistaxis, secondary to hypertension, was made. What is the site of the blood vessel most likely to be involved in this case?

- a) Posterior end of superior turbinate
- b) Posterior end of inferior turbinate
- c) Posterior end of middle turbinate
- d) Anterior end of septum

Question 7:

Which of the following is the most common cause of epistaxis in children?

- a) Trauma
- b) Bleeding disorders
- c) Cold weather
- d) Sinonasal infections

Question 8:

Which of the following is the most common cause of epistaxis in elderly patients?

- a) Medication
- b) Carcinoma
- c) Idiopathic
- d) Hypertension

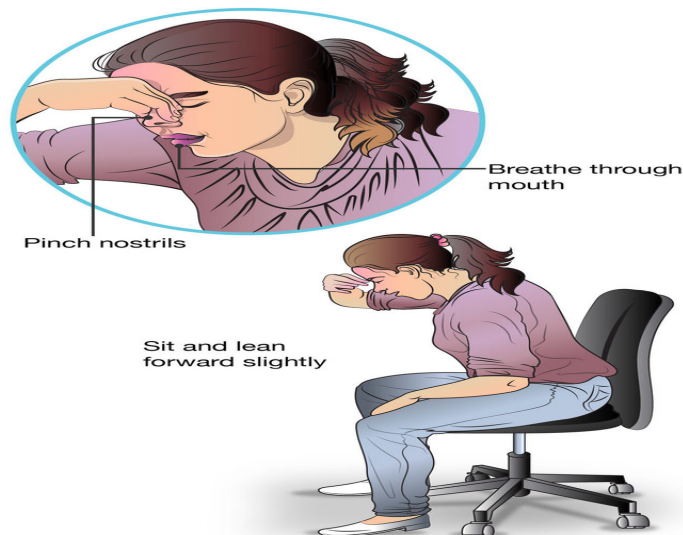
Question 9:

A 4-year-old child presents with persistent unilateral foul-smelling nasal discharge which is blood-stained. Which of the following is the most likely cause?

- a) Foreign body
- b) Atrophic rhinitis
- c) Nasal polyp
- d) Nasal angiofibroma

Question 10:

What is the maneuver depicted in the image given below?



- a) Epley's manoeuvre
- b) Trotter's method
- c) McGovern's technique
- d) Valsalva manoeuvre

Question 11:

A 60-year-old woman came to the casualty with a history of an episode of spontaneous epistaxis yesterday night. On examination, her BP was 200/110, and there was no active bleeding from the nose. Which of the following is the most appropriate next step in managing this case?

- a) Observation with BP management
- b) Internal maxillary artery ligation
- c) Nasal packing

- d) Cauterization

Question 12:

A 65-year-old hypertensive patient presented with a history of recurrent episodes of spontaneous nose bleeds over the past two weeks, each requiring medical attention. After evaluating the patient, the ENT surgeon recommended endoscopic artery ligation to prevent further episodes. Which of the following arteries is most commonly ligated in this procedure?

- a) Sphenopalatine artery
- b) Greater palatine artery
- c) Maxillary artery
- d) Internal carotid artery

Question 13:

As a first-year resident, you have scrubbed in to assist for a maxillary artery ligation procedure on a patient with intractable epistaxis. After opening the posterior wall of the maxillary sinus, the surgeon asks you to ligate the maxillary artery. Where would you ligate the artery at?

- a) Maxillary antrum
- b) Pterygopalatine fossa
- c) Neck
- d) Medial wall of orbit

Question 14:

Which artery among the following could be the source of bleeding in a case of epistaxis not controlled by external carotid artery ligation?

- a) Sphenopalatine artery
- b) Greater palatine artery
- c) Facial artery
- d) Ethmoidal artery

Question 15:

Which of the following vessels would you never ligate for the control of recurrent spontaneous epistaxis?

- a) Internal carotid artery
- b) Anterior ethmoidal artery
- c) Maxillary artery
- d) External carotid artery

Answer Key

Question No.	Correct Option
1	a
2	b
3	d
4	c
5	a
6	c
7	a
8	c
9	a
10	b
11	a
12	a
13	b
14	d
15	a

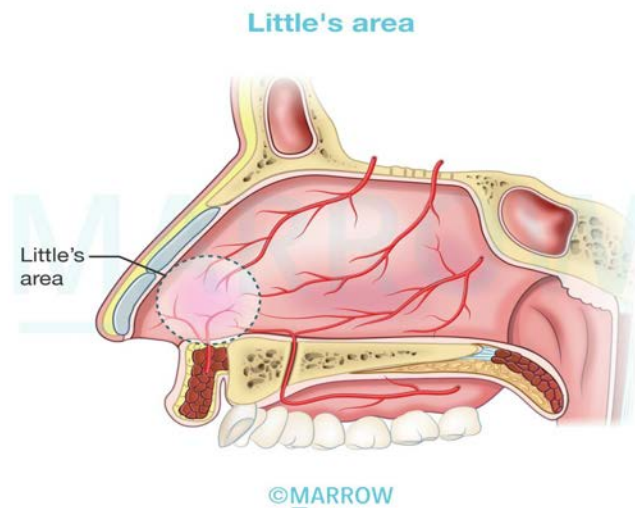
Detailed Explanations

Solution to Question 1:

The bleeding is most likely from the Little's area.

Little's area, situated at the anteroinferior part of the nasal septum is the most common site of epistaxis in children and young adults.

Factors that increase the risk of epistaxis in children when compared to adults, include the habit of nose picking and increased friability of mucosa due to frequent upper respiratory tract infections.



Solution to Question 2:

Little's area is situated in the anterior inferior part of the nasal septum, just above the vestibule.

Solution to Question 3:

The posterior ethmoidal artery does not contribute to the formation of Kiesselbach's plexus.

Kiesselbach's plexus is the vascular anastomosis situated at Little's area (anteroinferior part of the septum).

The contributing arteries are as follows:

- Sphenopalatine and greater palatine branches of the maxillary artery, which in turn is a branch of the external carotid artery.
- Septal branch of the superior labial artery which in turn is a branch of the facial artery. The facial artery is a branch of the external carotid artery.
- Anterior ethmoidal branch of the ophthalmic division of internal carotid artery.

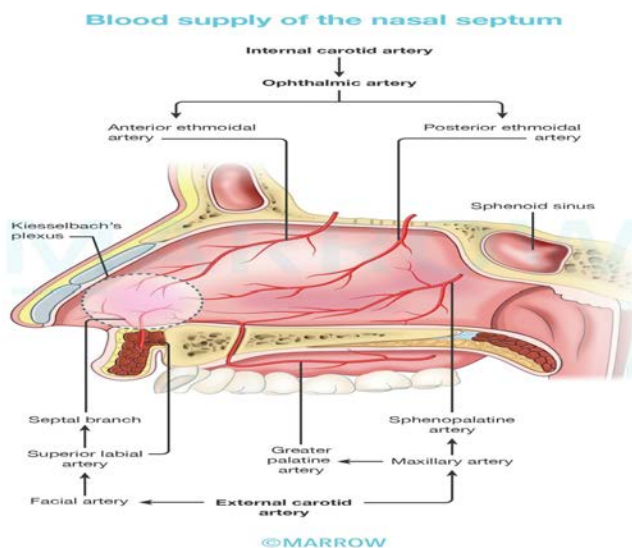
Solution to Question 4:

The structure marked 3 is the sphenopalatine artery which is also known as the artery of epistaxis.

It is called the artery of epistaxis because the majority of the blood supply to the nasal cavity is derived from it. It is also the most commonly involved artery in epistaxis.

Other structures marked in the diagram are as follows:

- 1 - Anterior ethmoidal artery
- 2 - Posterior ethmoidal artery
- 4 - Greater palatine artery



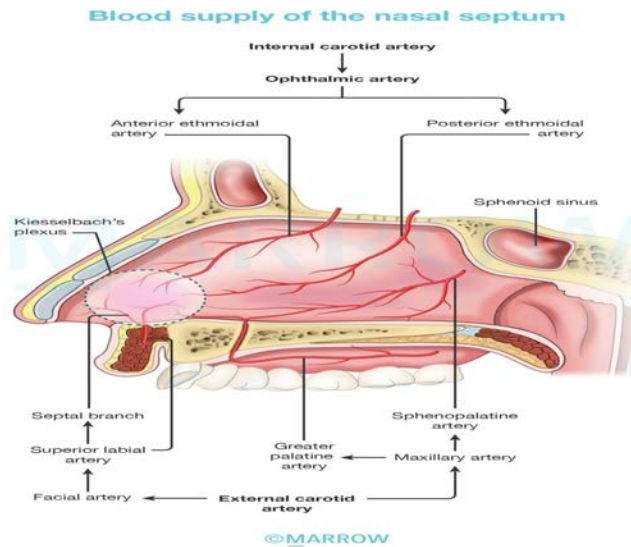
Solution to Question 5:

The structure marked 1 is the anterior ethmoidal artery, which is a branch of the ophthalmic division of the internal carotid artery.

Other structures marked in the diagram are as follows:

- 2 – Sphenopalatine artery
 - 3 – Greater palatine artery
 - 4 – Superior labial artery
- 2, 3 and 4 are derived from branches of the external carotid artery.

The nasal septum is supplied by branches of both internal and external carotid arteries.



Solution to Question 6:

The most common cause for posterior epistaxis is arterial and the vessel involved is the sphenopalatine artery. It is located in the posterior end of the middle turbinate.

Woodruff's plexus is a venous plexus located in the posterior end of the inferior turbinate and is also sometimes implicated in epistaxis associated with systemic conditions such as hypertension and coagulopathy. However, it only accounts for about 5-10% of all nose bleeds.

Note: Most common vessel involved in both anterior and posterior epistaxis is the sphenopalatine artery and not Woodruff's plexus.

Solution to Question 7:

Microtrauma due to nose picking is the most common cause of epistaxis in children.

Other common aetiological associations include upper respiratory tract infections.

Note: In children presenting with recurrent epistaxis, bleeding disorders, hereditary hemorrhagic telangiectasia, and tumors, etc. have to be ruled out.

Solution to Question 8:

The most common cause of epistaxis in elderly patients is idiopathic.

Other causes are:

- Trauma
- Coagulopathy

- Tumor - high suspicion in elderly patients presenting with epistaxis
- Infection
- Medication

Note: Due to the failure of studies to establish a causal relationship between hypertension and epistaxis, hypertension is now considered an associated risk factor and not a cause.

Risk factors associated with epistaxis are weather changes and alcohol consumption.

Solution to Question 9:

In a child presenting with persistent unilateral, foul-smelling nasal discharge, which is blood-stained, the most likely cause is a foreign body.

Foreign body can cause unilateral nasal obstruction and subsequent superimposed infection which causes malodorous discharge. It may often be blood-stained.

Option B: Atrophic rhinitis usually starts at puberty and mainly presents with nasal obstruction and foul smell from the nose. Epistaxis occurs only after the removal of crusts.

Option C: Epistaxis is a rare manifestation of polyps.

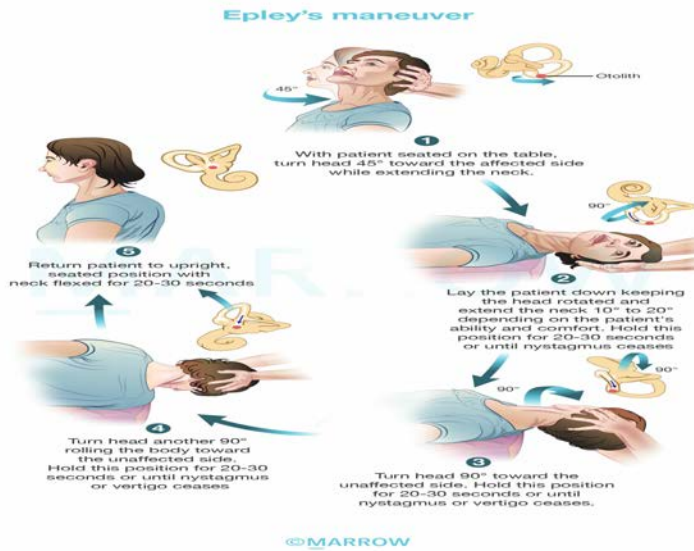
Option D: Unilateral nasal angiofibroma also presents as painless, progressive unilateral nasal obstruction, often associated with epistaxis and rhinorrhea. But it is more common in the adolescent age group.

Solution to Question 10:

The given image shows Trotter's method in which the patient is made to sit, leaning a little forward over a basin to spit any blood and breathe quietly from the mouth. It is the first step to control epistaxis.

Other options:

Option A: Epley's maneuver is used in the treatment of BPPV (Benign Paroxysmal Positional Vertigo).



Option C: McGovern's technique is used in choanal atresia. A feeding nipple with a large hole provides a good oral airway.



Option D: Valsalva maneuver is performed by forceful attempted exhalation against a closed nose and mouth. It is used as a test of cardiac function and autonomic nervous control of the heart, or to clear the ears and sinuses.

Solution to Question 11:

The most appropriate next step is observation and blood pressure management as there is no active bleeding from the nose.

Solution to Question 12:

The given clinical scenario points to a diagnosis of refractory posterior epistaxis. The artery which is most commonly ligated in a case of refractory posterior epistaxis is the sphenopalatine artery.

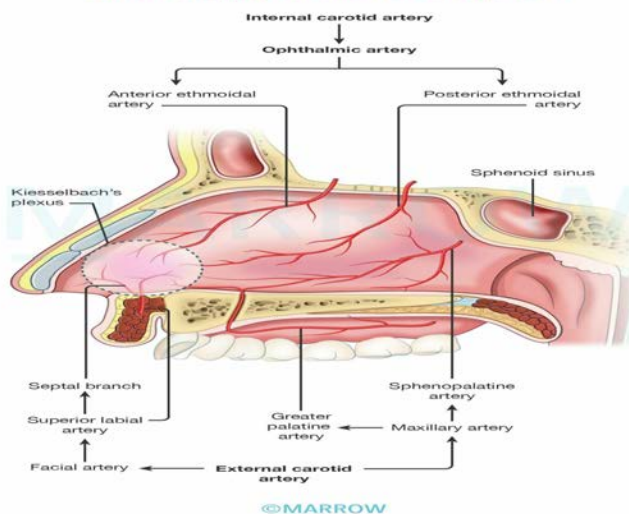
Posterior epistaxis is most commonly seen in older age groups. The bleeding is spontaneous, often due to hypertension, and is severe, requiring hospitalization. Refractory epistaxis is defined as recurrent or persistent bleeding after appropriate packing or multiple episodes of epistaxis during a short period, each requiring medical attention.

Sphenopalatine artery ligation gives high success in the control of refractory posterior bleed. Therefore, transnasal endoscopic sphenopalatine artery ligation (TESPAL) is the preferred method.

It is ligated at sphenopalatine foramen situated 1 cm behind the posterior end of middle turbinate.

	Anterior epistaxis	Posterior epistaxis
Age	Children & young adults	>40 years
Incidence	Most common type	Less common
Cause	Trauma	Spontaneous; often due to Hypertension or arteriosclerosis
Site	Little's area - anteroinferior part of the nasal septum	Posterosuperior part of the nasal cavity; difficult to localize
Bleeding	Mild- can be controlled by local pressure or nasal packing.	Severe-often requires hospitalization and posterior nasal packing.

Blood supply of the nasal septum

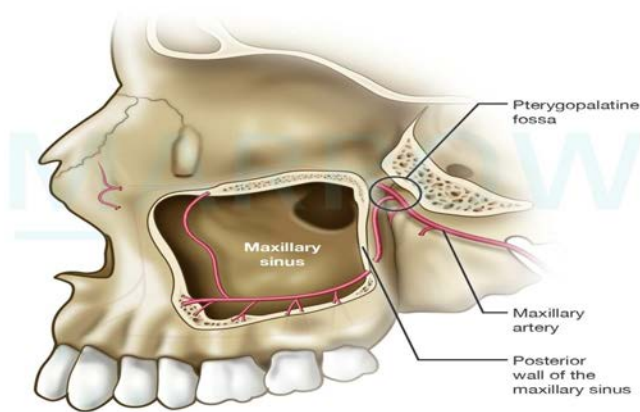


Solution to Question 13:

The maxillary artery is ligated at the pterygopalatine fossa (sphenopalatine fossa) situated behind the maxillary sinus's posterior wall.

The internal maxillary artery is accessed transantrally via a Caldwell-Luc approach. After the posterior sinus wall is removed, the internal maxillary artery and its terminal branches are identified and clipped. The posterior sinus wall is then packed with Gelfoam, and the gingivobuccal incision is closed. More recently, transoral and transnasal endoscopic approaches have also been described.

Maxillary artery within the maxillary sinus



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Solution to Question 14:

Ethmoidal artery can cause epistaxis not controlled by external carotid artery ligation.

The ethmoidal artery is a branch of internal carotid artery, whereas sphenopalatine, greater palatine and facial artery are branches of the external carotid artery. Hence if the bleeding is not controlled by external carotid artery ligation, the subsequent step would be to ligate the ethmoidal artery.

Epistaxis may be classified as anterior or posterior epistaxis.

Little's area is situated in the anterior inferior part of the nasal septum, just above the vestibule. It is the site of epistaxis in 90% of the cases.

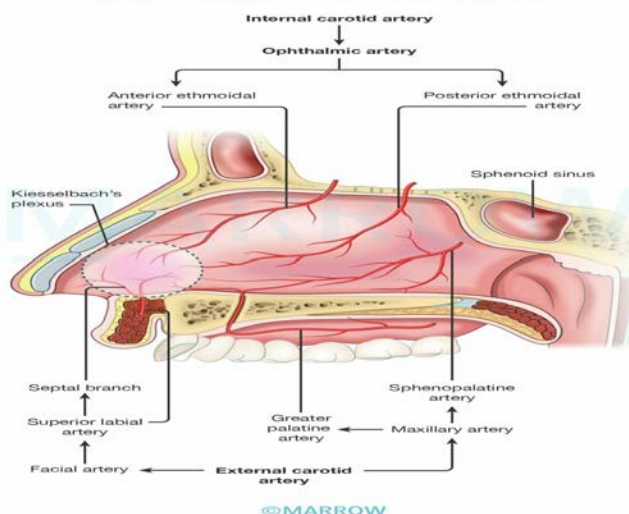
Four arteries anastomose here to form a vascular plexus called the Kiesselbach's plexus. This area is exposed to the drying effect of inspiratory current and to fingernail trauma and is the usual site for epistaxis in children and young adults.

The four contributing arteries are as follows:

- Septal branch of sphenopalatine and greater palatine branches of the maxillary artery, which in turn is a branch of the external carotid artery.
- Septal branch of the superior labial artery which in turn is a branch of the facial artery. The facial artery is a branch of the external carotid artery.
- Anterior ethmoidal branch of the ophthalmic division of internal carotid artery.

	Anterior epistaxis	Posterior epistaxis
Site	Little's area or anterior part of the lateral wall	Posterosuperior part of the nasal cavity, or difficult to localize
Age	Children, young adults	>40 years of age
Cause	Usually trauma	Spontaneous, due to hypertension or IHD
Bleeding	Mild, easily controlled with conservative measures (pressure, anterior nasal pack)	Severe requires hospitalisation and posterior nasal pack

Blood supply of the nasal septum



The stepwise management of epistaxis:

- Pinching the nose
- Hippocratic or Trotter's method
- Cauterisation
- Anterior nasal packing
- Posterior nasal packing
- Endoscopic Cauterisation
- Ligation
- Embolization

Ligation should be performed as close as possible to the likely bleeding point. The recommended hierarchy of ligation to be followed is Sphenopalatine artery ° Internal maxillary artery ° External carotid artery ° Anterior /posterior ethmoidal artery.

Ethmoidal vessels are exposed in the medial wall of the orbit by an external ethmoid (Lynch) incision.



Solution to Question 15:

Internal carotid artery is not ligated for controlling epistaxis due to the high percentage of cerebral complications and mortality.

Ligation of the following arteries is done in epistaxis control :

- Trans nasal Endoscopic Sphenopalatine artery ligation (TESPAL)
- If not controlled, maxillary artery ligation
- If still not controlled, external carotid artery ligation

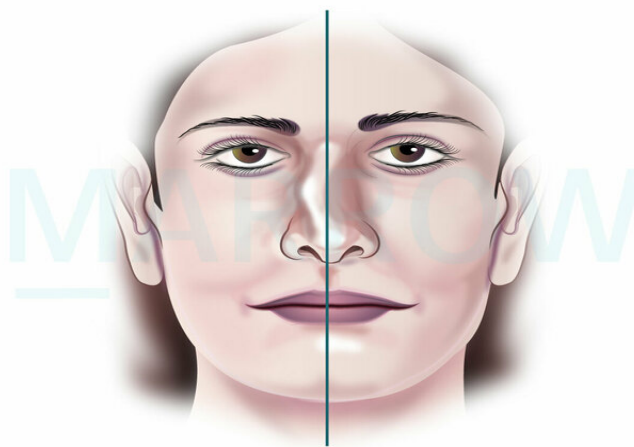
- In case, bleeding site is superior to the level of middle turbinate, then anterior and posterior ethmoidal vessels should be ligated.

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Congenital Anomalies of Nose and Disease of Nose

Question 1:

A patient presented with a deformity as shown in the image below. What is this deformity known as?



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- a) Crooked nose
- b) Deviated nose
- c) Hump nose
- d) Saddle nose

Question 2:

In which of the following patients will you not expect a saddle nose deformity?

- a) A VDRL positive patient with gumma of the nasal septum
- b) A patient with lateral madarosis and leonine facies
- c) A boxer who developed bilateral nasal obstruction after a trauma to the nose
- d) A patient with growth hormone excess and a large nose.

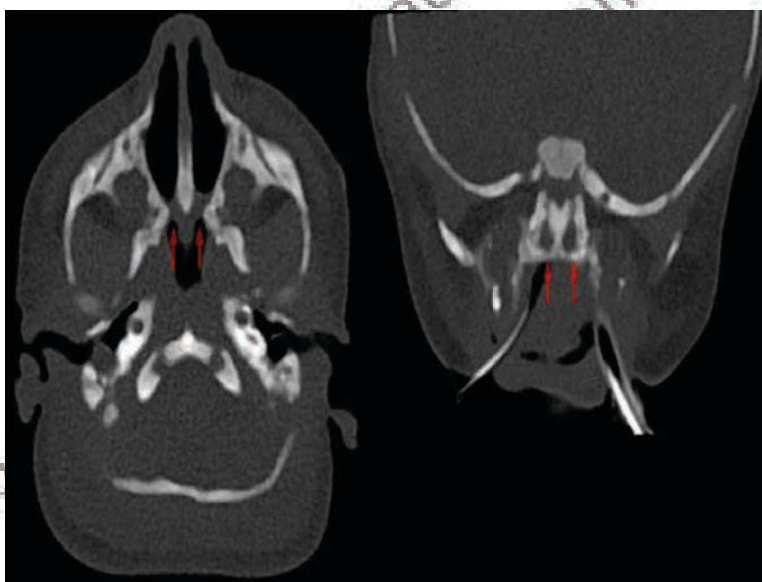
Question 3:

A newborn developed respiratory distress immediately after delivery. The paediatrician was unable to pass the nasogastric tube through both the nares. Which of the following statements is false about the above diagnosis?

- a) It is due to persistence of buconasal membrane
- b) Can be associated with CHARGE syndrome.
- c) Bony obstruction is more common than membranous obstruction
- d) Immediate tracheostomy to be done to maintain the airway.

Question 4:

What is the most common presentation of a patient with the pathology shown in the CT below?



- a) Respiratory distress
- b) Aspiration
- c) Regurgitation of feeds
- d) Post nasal drip

Question 5:

Choanal atresia is not associated with which of the following defects?

- a) Coloboma of eye
- b) Heart defects

- c) Retarded growth
- d) Renal angiomyolipoma

Question 6:

A 2-year-old child was brought with a soft, compressible swelling at the root of the nose. What is the most likely diagnosis?

- a) Meningoencephalocele
- b) AV malformation
- c) Lacrimal sac cyst
- d) Ethmoid cyst

Question 7:

Which of the following statements about nasal glioma is false?

- a) They are predominantly extranasal
- b) It presents with CSF leak
- c) CT Scan provides bony details
- d) Furstenberg test is negative

Question 8:

A surgeon suspects a rodent ulcer in a patient who presented with a finding as shown in the image. What is this condition also called?



- a) Rhinophyma
- b) Squamous cell carcinoma
- c) Basal cell carcinoma
- d) Cutaneous anthrax

Question 9:

In a patient with findings as shown in the image, which of the following statements is false?



- a) It is known as elephantiasis of the nose
- b) It is seen in long standing cases of acne rosacea

- c) It is commonly seen in middle aged women
- d) It is due to hypertrophy of sebaceous glands

Question 10:

A patient presented with atrophic dry nasal mucosa, extensive crustations, and a woody hard external nose. What is the most likely diagnosis?

- a) Atrophic rhinitis
- b) Rhinophyma
- c) Rhinosporidiosis
- d) Rhinoscleroma

Question 11:

A pathologist observed Mikulicz cells and Russell bodies on microscopy in a biopsy specimen taken from the nose. Which of the following is a false statement regarding the diagnosis?

- a) Seen in endemic regions
- b) Medical management include streptomycin and tetracycline.
- c) Surgery is the treatment of choice for this disease
- d) Causative organism can be cultured

Question 12:

Snuffles are seen in which stage of syphilis?

- a) Early congenital syphilis
- b) Late congenital syphilis
- c) Primary syphilis
- d) Secondary syphilis

Question 13:

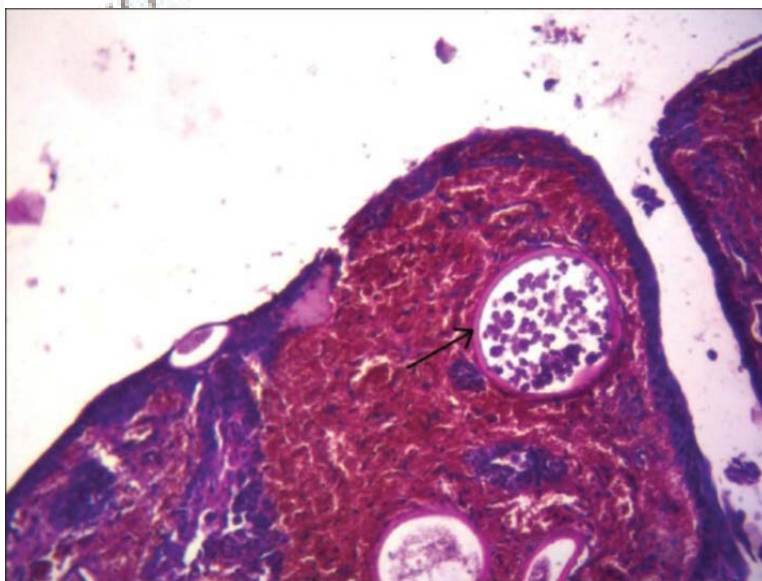
An eighteen-year-old boy presented with a lesion over the nose as shown in the image. What is your diagnosis?



- a) Rhinoscleroma
- b) Rhinosporidiosis
- c) Lupus vulgaris
- d) Syphilis

Question 14:

Given below is a histological section of a biopsy taken from a farmer who presented with a vascular polypoidal mass protruding from his left nostril. What is the causative organism for this condition?



- a) Klebsiella rhinoscleromatis

- b) Rhinosporidium seeberi
- c) Mycobacterium ulcerans
- d) Rhizopus Oryza

Question 15:

Rhinosporidium seeberi belongs to which group of organisms?

- a) Fungus
- b) Aquatic protista
- c) Bacteria
- d) Protophyta

Question 16:

A farmer presented with a mass protruding from his right nostril as shown in the figure below. On examination, mass bleeds easily on touch. Which of the statements is false regarding the diagnosis?



- a) Water borne is the most common route of transmission
- b) Endemic in southern states of India
- c) Affects immunocompromised individuals
- d) Bleeding polypoid mass

Question 17:

A man who works in a cattle ranch presented with complaints of a mass protruding from the left nostril and blood-stained nasal discharge. The mass is pink, leafy, and polypoidal and is attached to the nasal septum. Biopsy from the lesion showed several sporangia. What is the treatment of choice for the above condition?

- a) Streptomycin and tetracycline
- b) Radiotherapy
- c) Excision with cauterisation
- d) Dapsone

Question 18:

Which of the following statements is false about nasal T-cell lymphoma?

- a) It is also known as Stewart's granulomas
- b) It is associated with EBV
- c) It is a locally invasive tumor.
- d) Surgery is the treatment of choice

Answer Key

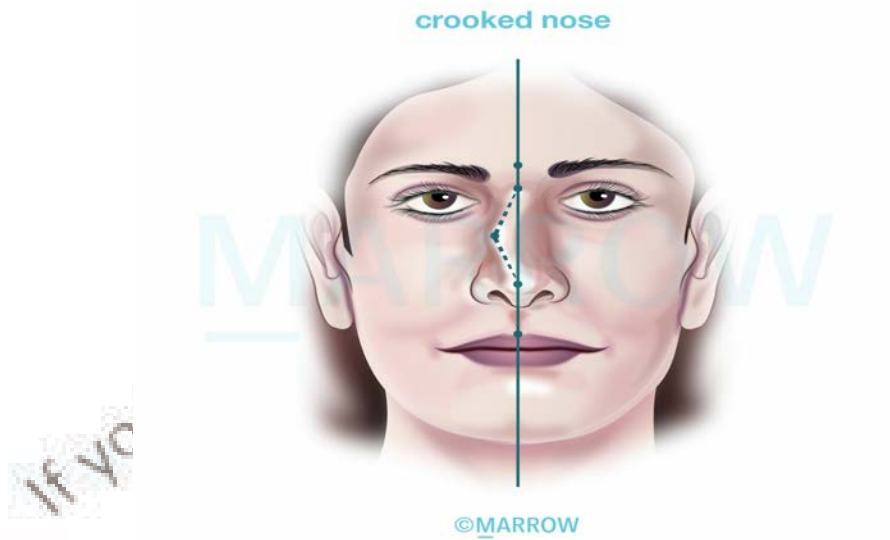
Question No.	Correct Option
1	a
2	d
3	d
4	a
5	d
6	a
7	b
8	c
9	c
10	d
11	c
12	a

13	c
14	b
15	b
16	c
17	c
18	d

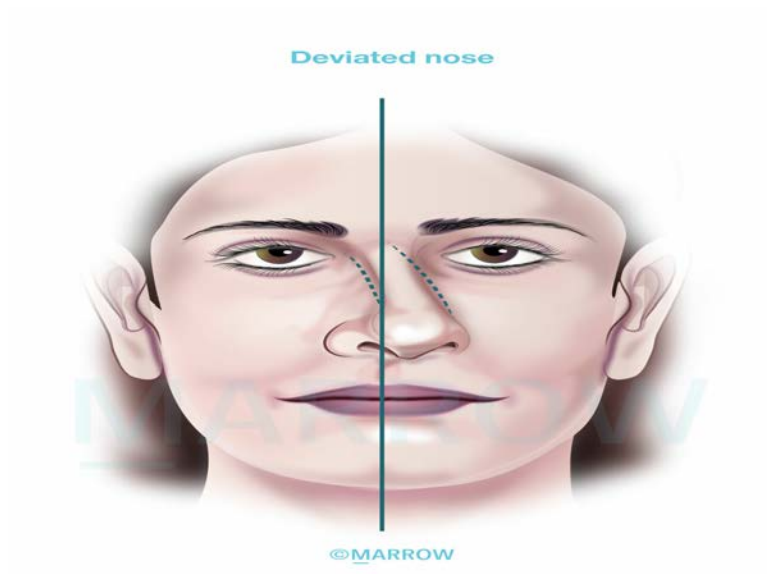
Detailed Explanations

Solution to Question 1:

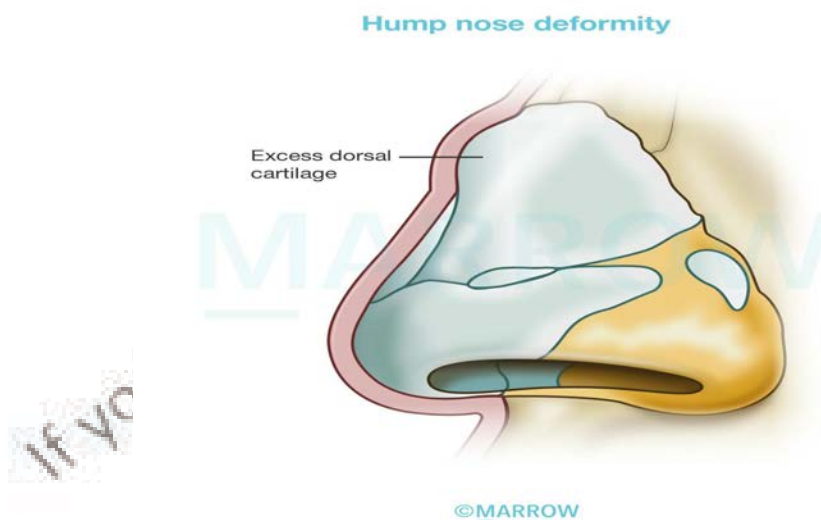
In the given image, the midline of the dorsum of the nose from the frontonasal angle to the tip is curved in a C- or S-shaped manner which is known as crooked nose deformity.



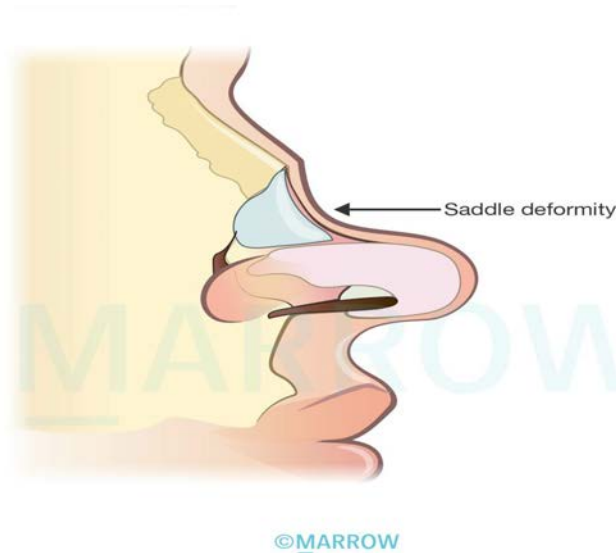
Option B: Deviated nose - midline is straight but deviated to one side.



Option C: Hump nose is due to excess of bony or cartilaginous dorsum.



Option D: Saddle nose is depressed nasal dorsum.



Solution to Question 2:

Option D is a case of acromegaly, in which there may be enlargement of the nose. Saddle nose deformity only occurs when the bridge of the nose collapses.

Causes for saddle nose/depressed dorsum of nose:

- Nasal trauma causing depressed fractures - most common etiology.
- Excessive removal of septum in submucous resection
- Destruction of septal cartilage by haematoma or abscess
- Leprosy
- Syphilis
- Tuberculosis.

Option A is a case of tertiary syphilis.

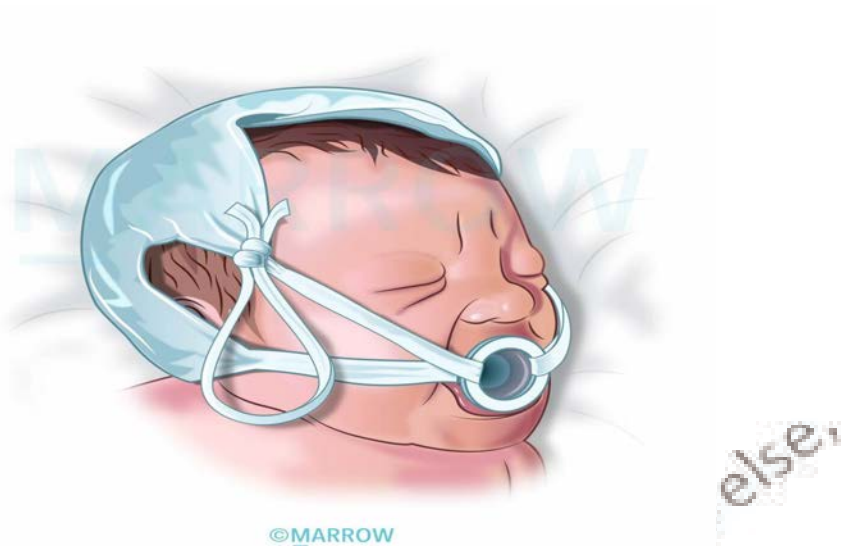
Option B is a case of lepromatous leprosy.

Option C is a case of septal hematoma.

Solution to Question 3:

The above clinical scenario is suggestive of bilateral choanal atresia and immediate tracheostomy is not indicated.

Simply securing a feeding nipple with a large hole provides a good oral airway and forces the infant to breathe through the mouth. This is known as McGovern's technique and obviates the need for tracheostomy.



Solution to Question 4:

Neonates with bilateral choanal atresia (image) present with respiratory distress at birth. Neonates are natural nose breathers, and do not breathe from their mouths.

Solution to Question 5:

Choanal atresia may be associated with CHARGE syndrome. Renal angiomyolipoma is not a part of CHARGE syndrome.

- C- Coloboma of the eye
- H- Heart defects
- A- Atresia of the choana
- R- Retarded growth and development
- G- Genital anomalies
- E- Ear anomalies and hearing loss.

Renal angiomyolipoma - It is one of the major features of tuberous sclerosis complex.

Solution to Question 6:

The above clinical scenario of isolated solitary mass arising from the root of the nose should raise the index of suspicion of nasal meningocele or meningoencephalocele. It is the herniation of brain tissue along with its meninges through a congenital bony defect.

Nasal obstruction is the most common symptom. It presents as a cystic, bluish, compressible, translucent mass at the root of the nose (nasofrontal variety), side of the nose (nasoethmoid variety), or at the anteromedial aspect of the orbit (nasoorbital variety).

Furstenberg's test: compression of the ipsilateral internal jugular vein usually causes an encephalocele to enlarge.

Investigation: CT or MRI.

Treatment: Transnasal endoscopic excision with the closure of the defect using bone or cartilage graft along with temporalis fascia and fat.



Solution to Question 7:

There is no CSF leakage in nasal glioma. It is a nipped-off portion of an encephalocele with no communication with the cranial cavity.

They usually present as a firm, subcutaneous swelling causing nasal obstruction.

Furstenberg's test - compression of the ipsilateral internal jugular vein usually causes an encephalocele to enlarge, but not a glioma. So, this test is negative in nasal glioma.

60% of nasal gliomas are extra-nasal, 30% are intranasal, and 10% are both intranasal and extra-nasal.

CT scan reveals bony details while MRI reveals soft tissue details of a nasal glioma.

Solution to Question 8:

Basal cell carcinoma also known as rodent ulcer is the most common malignant tumour involving skin of nose.

Solution to Question 9:

The given image is diagnostic of rhinophyma. It is commonly seen in men past middle age.

Rhinophyma or potato tumour or elephantiasis of nose is due to hypertrophy of sebaceous glands. It is usually associated with long-standing cases of acne rosacea. Mostly seen at the tip of the nose and causes unsightly appearance and obstruction to breathing and vision.

Treatment: Debulk of the tumor with carbon dioxide laser and the area is allowed to re-epithelialize. Sometimes the tumor is completely excised and skin is grafted over the raw area.

Solution to Question 10:

The given clinical condition is suggestive of rhinoscleroma.

Rhinoscleroma is caused by *Klebsiella rhinoscleromatis* (Frisch bacillus).

It has 3 stages:

- Atrophic stage– Crusting and foul smelling nasal discharge similar to atrophic rhinitis.
- Granulomatous stage– Painless, non ulcerative Granulomatous nodules with subdermal infiltration giving feel of woody/hebra nose.
- Cicatricial stage– Fibrosis leading to stenosis of vestibule, pharynx, larynx. Deformity is called tapir nose.



Option A: Atrophic rhinitis doesn't cause woody nose.

Option B: Rhinophyma or potato tumor is due to hypertrophy of sebaceous gland and presents as swelling over tip of nose.

Option C: Rhinosporidiosis is a granulomatous condition caused by *Rhinosporidium seeberi* and presents with nasal discharge, epistaxis and nasal obstruction by a polypoidal mass.

Solution to Question 11:

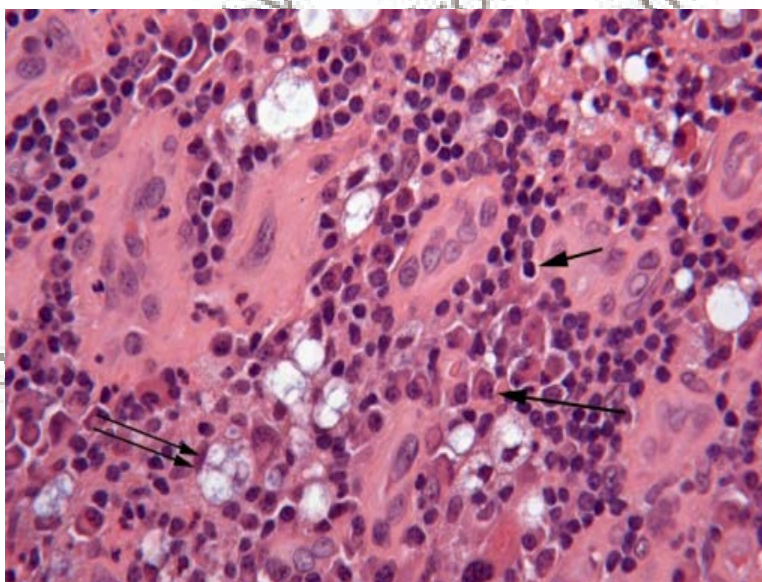
Biopsy findings of Mikulicz cells and Russell's bodies are characteristic findings in rhinoscleroma. Medical management is the mainstay treatment for rhinoscleroma. Streptomycin (1 g/day) and tetracycline (2 g/day) are given for a minimum of 4-6 weeks and continued till 2 consecutive biopsies are negative. Surgery is reserved to establish airway or correct deformity.

Biopsy in the case of rhinoscleroma shows infiltration of submucosa with plasma cells, lymphocytes, eosinophils, Mikulicz cells, and Russell bodies.

Mikulicz cells are large foam cells with a central nucleus and vacuolated cytoplasm containing causative bacilli.

Russell bodies are homogenous eosinophilic inclusion bodies found in the plasma cells. They occur due to the accumulation of immunoglobulins secreted by the plasma cells.

The following image shows rhinoscleroma with Mikulicz cells.



H&E showing a mixture of plasma cells (arrow), lymphocytes (short arrow), Mikulicz cells which are vacuolated macrophages containing causative bacilli (double arrow).

Option A: Rhinoscleroma is endemic in several parts of the world. In India, it is seen more often in the northern than in the southern parts.

Option D: Organism can be cultured from biopsy material.

Solution to Question 12:

Snuffles is seen in early congenital syphilis.

Nasal features in different stages of syphilis:

Stage	Feature
Early congenital syphilis	Snuffles
Late congenital syphilis	Gumma on the nasal septum
Primary syphilis	Primary chancre of the vestibule of nose
Secondary syphilis	Simple rhinitis with crusting and fissuring in the nasal vestibule
Tertiary form of acquired syphilis	Gumma on the nasal septum

Solution to Question 13:

The image shows apple jelly nodules seen in lupus vulgaris which is a low-grade tuberculous infection.

These are brown, gelatinous nodules commonly affecting nasal vestibule or the skin of nose and face.

Solution to Question 14:

The above clinical scenario along with the histology (showing sporangia) is suggestive of rhinosporidiosis caused by *Rhinosporidium seeberi*.

Rhinoscleroma is caused by *Klebsiella rhinoscleromatis*.

Solution to Question 15:

Rhinosporidium seeberi is an aquatic protista.

Molecular biological techniques have demonstrated that this organism is an aquatic protistan parasite and it is currently included in a new class, the Mesomycetozoea, along with organisms causing similar infections in amphibians and fish.

Solution to Question 16:

Rhinosporidiosis affects immunocompetent individuals.

Solution to Question 17:

Surgical management (excision of the lesion with cauterisation of the base), is always the treatment of choice in rhinosporidiosis.

No drug treatment has proved effective, although dapsone is tried in some cases. Followup of the patient after surgical management is necessary, as recurrence is common.

Solution to Question 18:

T-cell lymphoma is treated by radiation when it is localised and by chemotherapy when it is disseminated.

The condition is also called as:

- Non-healing midline granuloma
- Midline granulomas reticulosis
- Stewart's granuloma
- Nasal T-cell lymphoma
- Lethal midline granuloma

It is associated with EBV. Cells express T cell markers like CD56.

Nasal obstruction and epistaxis are the presenting features. Destruction of mid-facial structures may be caused by the local invasion.

Histologically, polymorphic lymphoid tissue with angiocentric and angioinvasive features are seen.

Rhinitis

Question 1:

Which of the following is the most common cause of common cold?

- a) Adenovirus
- b) Rhinovirus
- c) Influenza virus
- d) Parainfluenza virus

Question 2:

You are an intern examining a patient with nasal obstruction and nasal discharge. The anterior rhinoscopy shows mulberry appearance of the inferior turbinate. What is the most likely diagnosis?

- a) Atrophic rhinitis
- b) Coryza
- c) Hypertrophic rhinitis
- d) Vasomotor rhinitis

Question 3:

Which of the following disease is also called ozaena?

- a) Atrophic rhinitis
- b) Allergic rhinitis
- c) Vasomotor rhinitis
- d) Hypertrophic rhinitis

Question 4:

A middle-aged woman with nasal discharge and crusting is diagnosed to have primary atrophic rhinitis. Which of the following organisms is most commonly associated with this condition?

- a) *Coccobacillus foetidus ozaenae*
- b) *Streptococcus pneumonia*
- c) *Klebsiella ozaena*
- d) *Klebsiella rhinoscleroma*

Question 5:

Which of the following is false about atrophic rhinitis?

- a) More common in females
- b) Starts at puberty
- c) Unilateral condition
- d) Merciful anosmia

Question 6:

A 25-year-old woman presented to the OPD with foul-smelling nasal discharge and obstruction. On examination, crusts are present bilaterally and on the removal of the crust, bleeding is present. Which of the following is least likely to cause this?

- a) Syphilis
- b) Wegener's granulomatosis
- c) Deviated nasal septum
- d) Rhinoscleroma

Question 7:

In a patient with atrophic rhinitis, what is the main reason for nasal obstruction?

- a) Hypertrophy of turbinates
- b) Crust formation
- c) Excessive synechiae
- d) Polyps

Question 8:

You are an intern in the ENT department preparing the alkaline nasal solution for a patient with atrophic rhinitis. Which of the following will you not use for the preparation?

- a) Sodium chloride
- b) Potassium chloride
- c) Sodium bicarbonate
- d) Sodium baborate

Question 9:

An adolescent girl with atrophic rhinitis is using kemicetine antiozaena solution. Which of the following is not a component of this solution?

- a) Chloramphenicol
- b) Estradiol
- c) Vitamin D2
- d) Sodium bicarbonate

Question 10:

A housewife presents with complaints of a blocked nose, loss of smell and dry, greenish crusty nasal discharge. On examination, nasal cavities appear roomy and turbinates are atrophied. Which of the following surgeries can be done in this patient?

- a) Submucous resection
- b) Young's operation
- c) Septoplasty
- d) Vidian neurectomy

Question 11:

A 40-year-old ironsmith came to the ENT clinic with complaints of dry nose, nasal crusting and bleeding on removal of the crusts. A diagnosis of rhinitis sicca was made. Which of the following is false regarding this condition?

- a) More common in people working in hot, dry, dusty surroundings
- b) Mainly affects posterior 2/3rd of nose
- c) Squamous metaplasia is seen

- d) Antibiotic & steroid ointment is used for treatment

Question 12:

Which of the following is false about rhinitis caseosa?

- a) Associated with suppurative sinusitis
- b) More common in males
- c) Also known as nasal cholesteatoma
- d) It is a bilateral condition

Question 13:

Allergic rhinitis is mediated by which of the following?

- a) IgA
- b) IgE
- c) IgD
- d) IgM

Question 14:

A mother brought her a 6-year-old child with watery nasal discharge, nasal obstruction, frequent sneezing and itching in the nose. She also adds that the symptoms occur during the winter season and there is a history of asthma in the family. Which of the following is not associated with this condition?

- a) Dennie-Morgan lines
- b) Allergic salute
- c) Atrophy of turbinates
- d) Serous otitis media

Question 15:

An ENT resident is evaluating a patient with suspected allergic rhinitis. On anterior rhinoscopy, which of the following is most likely to be seen?

- a) Pale and edematous mucosa

- b) Crusts that bleed on removal
- c) Atrophied turbinates
- d) Congested and edematous mucosa

Question 16:

A 30-year-old man came with complaints of excessive nasal discharge, sneezing, and nasal obstruction. There is no history of allergen exposure or asthma in the family. You are suspecting vasomotor rhinitis. Which of the following statements is true regarding this condition?

- a) It is an allergic condition.
- b) It occurs due to sympathetic overactivity.
- c) Vidian neurectomy is a treatment option.
- d) Pale and atrophied turbinates are seen.

Question 17:

Prolonged use of which of the following drugs can cause rhinitis medicamentosa?

- a) Steroid nasal spray
- b) Nasal decongestants
- c) Antihistamines
- d) Leukotriene receptor antagonists

Question 18:

Surgery where nostrils are partially or completely occluded is done for which condition?

- a) Vasomotor rhinitis
- b) Atrophic rhinitis
- c) Invasive aspergillosis
- d) Allergic rhinitis

Answer Key

Question No.	Correct Option
1	b
2	c
3	a
4	c
5	c
6	c
7	b
8	b
9	d
10	b
11	b
12	d
13	b
14	c
15	a
16	c
17	b
18	b

Detailed Explanations

Solution to Question 1:

The most common cause of the common cold is Rhinovirus.

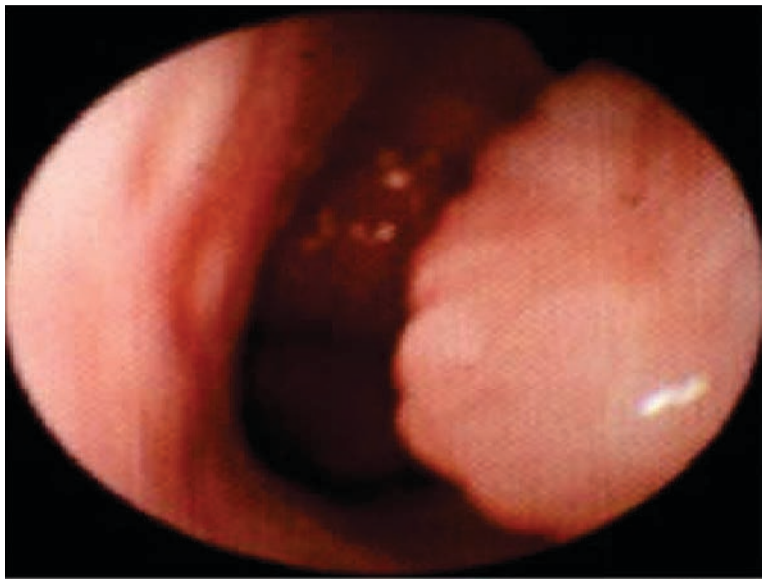
Adenovirus, picornavirus and its subgroups such as Coxsackievirus and enteric cytopathic human orphan (ECHO) virus are also usually implicated. Influenza and parainfluenza virus causes the flu.

Solution to Question 2:

The mulberry appearance of turbinate is seen in patients with hypertrophic rhinitis.

It is characterized by thickening of mucosa, submucosa, seromucinous glands, periosteum and bone.

Maximum changes are seen in the inferior turbinate.



Solution to Question 3:

Atrophic rhinitis is also known as ozaena.

Solution to Question 4:

The most common organism causing primary atrophic rhinitis is *Klebsiella ozaena*.

Various organisms have been cultured from cases of atrophic rhinitis such as *Coccobacillus foetidus ozaenae*, *Proteus vulgaris*, *Escherichia coli*, staphylococci, diphtheroids, and streptococci but they are all considered to be secondary invaders responsible for foul smell rather than the primary causative organisms of the disease.

Etiology of atrophic rhinitis:

Primary atrophic rhinitis:

- Hereditary- Involves more than one member of a family
- Endocrinal- More common in females, starts at puberty and regresses after menopause
- Racial- More common in mongoloid and caucasian races
- Nutritional deficiency- Vitamin A, D, iron deficiencies
- Autoimmune- Antigens of the nasal mucosa
- Organism- *Klebsiella ozaenae*.

Secondary atrophic rhinitis:

- Specific infections like syphilis, lupus, leprosy and rhinoscleroma
- Long-standing purulent sinusitis

- Radiotherapy to nose
- Excessive surgical removal of turbinates
- Diphtheroids, Proteus vulgaris, Escherichia coli, staphylococci and streptococci.

Solution to Question 5:

Atrophic rhinitis is mostly a bilateral condition.

Solution to Question 6:

The given clinical scenario is suggestive of bilateral atrophic rhinitis. Only deviated nasal septum causes unilateral atrophic rhinitis. The others cause bilateral atrophic rhinitis.

Etiology of atrophic rhinitis:

Primary atrophic rhinitis:

- Hereditary- Involves more than one member of a family
- Endocrinal- More common in females, starts at puberty and regresses after menopause
- Racial- More common in mongoloid and caucasian races
- Nutritional deficiency- Vitamin A, D, iron deficiencies
- Autoimmune- Antigens of the nasal mucosa
- Organism- Klebsiella ozaenae.

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- Specific infections like syphilis, lupus, leprosy and rhinoscleroma
- Long-standing purulent sinusitis
- Radiotherapy to nose
- Excessive surgical removal of turbinates
- Diphtheroids, Proteus vulgaris, Escherichia coli, staphylococci and streptococci.

Solution to Question 7:

Nasal obstruction in a patient with atrophic rhinitis is due to excessive crust formation.

Turbinates are atrophied here.

Solution to Question 8:

Potassium chloride is not a part of the alkaline nasal solution used in the treatment of atrophic rhinitis.

The composition of the alkaline solution is:

- Sodium chloride - 2 parts
- Sodium bicarbonate - 1 part
- Sodium biborate - 1 part

It is dissolved in 280 mL of water and the nasal cavities are irrigated to loosen the crusts and ease its removal.

Other treatment options:

- 25% glucose in glycerine
- Estradiol spray
- Submucosal placental extract
- Local antibiotics
- Potassium iodide
- Systemic streptomycin

Solution to Question 9:

Sodium bicarbonate is not a component of kemacetine antiozaena solution. It is used in alkaline nasal solution.

The components of kemacetine antiozaena solution are:

- Chloramphenicol
- Oestradiol
- Vitamin D2.

It is used in the medical management of atrophic rhinitis.

Solution to Question 10:

The given clinical scenario is suggestive of atrophic rhinitis. Young's and modified Young's operations are done for the treatment of atrophic rhinitis.

In Young's operation, both the nostrils are closed completely just within the nasal vestibule by raising flaps. They are opened after 6 months or later where the mucosa may revert back to normal and crusting will be reduced.

Modified Young's operation aims at partially closing the nostrils.

Options A and C: Submucous resection and septoplasty are done for deviated nasal septum.

Option D: Vidian neurectomy is done for vasomotor rhinitis.

Solution to Question 11:

Rhinitis sicca affects the anterior 1/3rd of the nose since this part is exposed to the effect of the dry, dusty air.

Rhinitis sicca is a crust-forming disease seen in patients who work in hot, dry, and dusty surroundings, e.g. bakers, iron- and goldsmiths. It is usually confined to the anterior third of the nose particularly of the nasal septum. The ciliated columnar epithelium undergoes squamous metaplasia with atrophy of the seromucinous glands.

Treatment consists of correcting occupational surroundings and application of bland ointment with an antibiotic and steroid combination to the affected part.

Solution to Question 12:

Rhinitis caseosa is a unilateral condition and not bilateral.

It is also known as nasal cholesteatoma and is more common in males. There will be an accumulation of secretions in the nasal cavity which undergoes putrefaction resulting in the formation of caseous material. It is usually associated with suppurative sinusitis.

Treatment includes the removal of debris and granulation tissue and free drainage of the affected sinus.

Solution to Question 13:

Allergic rhinitis is an IgE-mediated immunologic response of nasal mucosa to airborne allergens.

Patients typically present with recurrent bouts of sneezing, nasal obstruction, watery nasal discharge, and itching and clinical signs include allergic shiners (dark circles around the eyes) and allergic salute (black transverse nasal crease on the dorsum of the nose due to repeated upward rubbing).

Treatment options include:

- Avoidance of the allergen
- Treatment with drugs like oral antihistamines, intranasal corticosteroids, intranasal cromolyn sodium
- Immunotherapy is through the delivery of allergen in gradual increasing doses till the maintenance dose is reached. It suppresses the development of IgE. It has to be given for a year to see significant improvement in symptoms.

Intranasal corticosteroids (INCS) remain the preferred monotherapy for persistent allergic rhinitis (AR). However, additional studies support the additive benefit of combination treatment with INCS and intranasal antihistamines in AR. For patients receiving INCS, an intranasal decongestant may be added to the treatment for up to 4 weeks.

Solution to Question 14:

The given clinical scenario of a child with watery nasal discharge, frequent sneezing, itching of the nose, and seasonal variation is suggestive of allergic rhinitis, and hypertrophy of turbinates is seen, not atrophy.

Other features of allergic rhinitis:

Option A: Dennie-Morgan lines - transverse skin creases in the infraorbital area.



Option B: Allergic salute - is a transverse nasal crease that appears as a black line across the middle of the dorsum of the nose due to constant upward rubbing of nose simulating a salute.



Option D: Serous otitis media occurs as a result of eustachian tube blockage.

Solution to Question 15:

A patient with perennial allergic rhinitis will have a pale and edematous nasal mucosa with swollen turbinates.

Crusts that bleed on removal and atrophied turbinates are features seen in atrophic rhinitis.

Note: Typical allergic mucosa appears pale and swollen, while the mucosa is reddened in acute infections and after the overuse of topical medications.

Solution to Question 16:

Vidian neurectomy is a treatment option in vasomotor rhinitis.

Vasomotor rhinitis is a non-allergic condition that occurs due to parasympathetic overactivity, causing vasodilatation and congestion of nasal mucosa. The vidian nerve carries the parasympathetic supply to the nose.

The usual presentation is excessive rhinorrhea, sneezing, and nasal obstruction which is similar to allergic rhinitis. But the turbinates will be congested and edematous.

Treatment includes antihistamines, oral nasal decongestants, and topical steroids. In case of failed medical treatment, vidian neurectomy is done.

Solution to Question 17:

Rhinitis medicamentosa is caused by the prolonged use of nasal decongestants.

The treatment includes the withdrawal of nasal decongestant drops, a short course of systemic steroid therapy, and in some cases, surgical reduction of turbinates if they have become hypertrophied.

Solution to Question 18:

Complete closure of the nostrils (Young's operation) or partial closure of the nostrils (Modified Young's operation) is done for the treatment of atrophic rhinitis.

In Young's operation, both nostrils are closed completely by raising flaps within the nasal vestibule. They are opened after 6 months. During this time, the mucosa becomes normal, and the crusting is reduced.

In Modified Young's operation, the procedure has been modified to avoid discomfort from complete closure of the nostrils. Here the nostrils are only partially closed. The results are the same when compared to Young's operation.

Atrophic rhinitis (ozaena) is a chronic inflammation of the nose with atrophy of nasal mucosa and turbinate bones. Roomy nasal cavities with foul-smelling crusts are seen. The patient presents with a foul smell from the nose, marked anosmia (merciful anosmia), and paradoxical nasal obstruction (nasal obstruction despite wide nasal passages). Paradoxical nasal obstruction occurs due to degenerative changes in the sensory nerves of the nasal mucosa, leading to an inability to sense airflow.

Medical management includes nasal irrigation and removal of crusts. 25% glucose in glycerine, antibiotics (quinolones or metronidazole), estradiol nasal spray, and mucolytics can be used.

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Disorder of Nasal Septum and Nasal Polyposis

Question 1:

Which of the following is not a cause for deviated nasal septum?

- a) Birth trauma
- b) Developmental error
- c) Fractures of nasal septum
- d) Allergic rhinitis

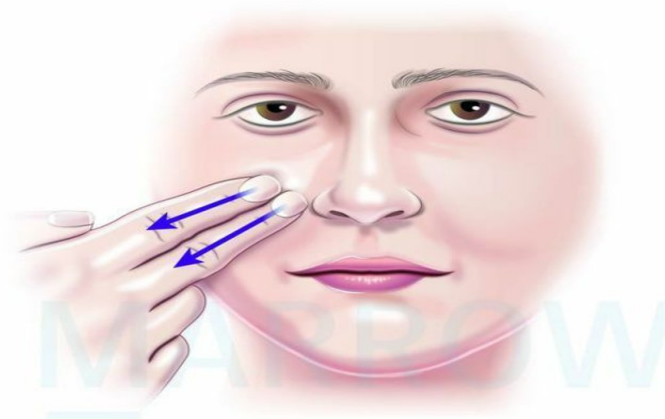
Question 2:

A 3rd-year MBBS student is examining a case of deviated nasal septum. Which of the following findings is unlikely to be seen in this patient?

- a) Hypertrophy of turbinates
- b) Nasal polyp
- c) Epistaxis
- d) Maxillary sinus tenderness

Question 3:

What clinical test is depicted in the image?



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- a) Epley's maneuver
- b) Heimlich maneuver
- c) Cottle's test
- d) Trotter's method

Question 4:

An ENT surgeon is performing septoplasty in a patient with deviated nasal septum. Which of the following is not an indication for the surgery?

- a) Persistent nasal obstruction
- b) Recurrent sinusitis
- c) Sleep apnoea
- d) Nasal discharge

Question 5:

A patient diagnosed with deviated nasal septum is undergoing submucous resection. What is the key incision is used in this procedure?

- a) Killian's incision
- b) Shobinger incision
- c) Freer's incision
- d) Weber Ferguson incision

Question 6:

A 20-year-old college student presented with complaints of facial pain, nasal discharge, headache, and nasal bleeding. He has a history of recurrent sinusitis in the past. On anterior rhinoscopy examination, the nasal septum is deviated. Which of the following is the treatment of choice?

- a) Submucous resection
- b) Septoplasty
- c) Functional endoscopic sinus surgery
- d) Rhinoplasty

Question 7:

Which of the following is a false statement regarding septoplasty?

- a) It is done in cases of symptomatic nasal obstruction.
- b) Freer's incision is used.
- c) Mucoperiosteal flap is raised on both sides.
- d) It is a conservative procedure.

Question 8:

A mother brought her 5-year-old child with complaints of pain and swelling in the nose and difficulty in breathing after an alleged history of fall while playing. On examination, there is bilateral swelling of the nasal septum which is tender and erythematous. Which of the following is false about the given condition?

- a) Trauma is an important cause
- b) Can result in hump nose deformity
- c) Can lead to abscess formation
- d) Accumulation of blood under the perichondrium

Question 9:

A 6-year-old boy fell from the bed and developed swelling in the nose and difficulty in breathing. Which of the following is the best management for this child?

- a) Observation
- b) Systemic antibiotics
- c) Surgical drainage
- d) Nasal packing

Question 10:

Which of the following is the most common cause of septal perforation?

- a) Septal abscess
- b) Trauma

- c) Syphilis
- d) Tuberculosis

Question 11:

A middle-aged man with tertiary syphilis presented with complaints of epistaxis, nasal crusting, and nasal twang on speaking. On examination, nasal septal perforation was noted. What is the most likely site for the perforation?

- a) Bony septum
- b) Cartilaginous septum
- c) Both bony and cartilaginous septum
- d) Junction of the bony and cartilaginous septum

Question 12:

A 30-year-old woman from a rural area presented with foul-smelling nasal discharge, nasal crusts, and epistaxis. She was diagnosed with atrophic rhinitis. On examination, maggots were seen in the nasal cavity. Which of the following is used in the treatment for this patient?

- a) Chloroform diluted with water
- b) Liquid paraffin
- c) Lignocaine spray
- d) Systemic antibiotics

Question 13:

Bernoulli's phenomenon explains the pathogenesis of which of the following conditions?

- a) Zenker's diverticulum
- b) Nasal polyp
- c) Deviated nasal septum
- d) Laryngomalacia

Question 14:

A 45-year-old man was prescribed aspirin for post-MI prophylaxis. But whenever he takes the medication, he develops difficulty in breathing, sneezing, and lacrimation. He has a known family history of bronchial asthma. Which of the following is related to his condition?

- a) Ethmoidal polyp
- b) Angiofibroma
- c) Nasopharyngeal carcinoma
- d) Nasal glioma

Question 15:

Which of the following is not part of Kartagener's syndrome?

- a) Bronchiectasis
- b) Chronic sinusitis
- c) Azoospermia
- d) Situs inversus

Question 16:

What is Young's syndrome?

- a) Neutrophilia with red papular rash
- b) Infertility with bronchiectasis
- c) Mitral stenosis with ASD
- d) Asthma, aspirin hypersensitivity, nasal polyp

Question 17:

A 40-year-old woman came to OPD with complaints of bilateral nasal obstruction, headache, sneezing and nasal discharge. On examination, multiple smooth pale glistening masses are present bilaterally that don't bleed on touch. Which of the following is false about the diagnosis?

- a) Allergy is an etiological factor
- b) The site of origin is lateral wall of the nose
- c) FESS is the treatment of choice
- d) The mass can grow posteriorly and occupy the choana

Question 18:

You are an intern examining a child who presented with nasal obstruction and discharge. On anterior rhinoscopy, there is a single smooth glistening mass in the right nasal cavity which can be moved up and down with a probe and doesn't bleed on touch. The mass is extending posteriorly and occupying the choana. Which of the following is false regarding this condition?

- a) Infection and allergy are the main etiological factors
- b) The site of origin is maxillary antrum
- c) FESS is the treatment of choice
- d) Can have a hypernasal voice

Question 19:

A middle-aged man with recurrent diffuse ethmoidal polyps is scheduled for a laser-assisted FESS procedure. Which of the following lasers is ideal in this scenario?

- a) ND-YAG laser
- b) KTP laser
- c) CO₂ laser
- d) Argon laser

Answer Key

Question No.	Correct Option
1	d
2	b
3	c
4	d
5	a
6	b
7	c
8	b
9	c

10	b
11	a
12	a
13	b
14	a
15	c
16	b
17	d
18	d
19	b

Detailed Explanations

Solution to Question 1:

Allergic rhinitis does not cause deviated nasal septum (DNS).

Etiology of DNS:

- Trauma - during passage through the birth canal or trauma after birth leading to fracture of the nasal septum
- Developmental error - may be associated with cleft lip, cleft palate
- Racial factor - more common in Caucasians
- Hereditary - affects more than one member of the same family

Solution to Question 2:

Nasal polyp will not be seen in a patient with deviated nasal septum.

Clinical features of DNS:

- Nasal obstruction - unilateral in C-shaped deviation and bilateral in S-shaped deviation
- Headache - may be due to associated sinusitis or due to spur pressing on the lateral wall of the nose
- Sinusitis - due to obstruction of ostia and poor ventilation of sinuses
- Epistaxis - from vessels over spur or due to crusting of the nasal mucosa
- Anosmia - especially in bilateral nasal obstruction
- Compensatory hypertrophy of turbinates on roomier side
- Frequent middle ear infections

- Association with external deformities

Solution to Question 3:

The clinical test depicted in the image is Cottle's test where the cheek is drawn laterally while the patient breathes quietly.

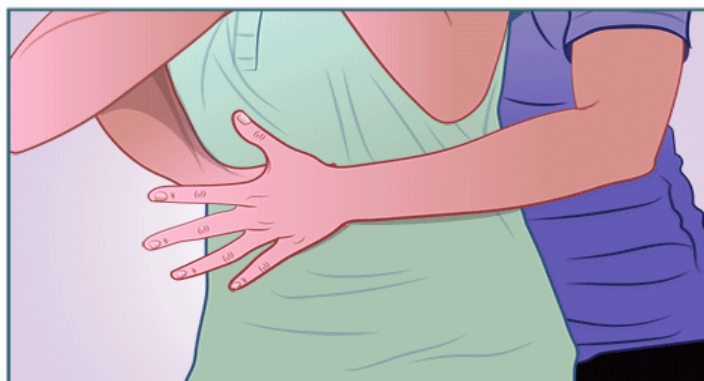
This test is used to assess the patency of the nasal valve in patients with a deviated nasal septum. If the nasal airway improves on the test side, the test is positive and indicates an abnormality of the vestibular component of the nasal valve on that side.

Option A: Epley's maneuver is used in the treatment of BPPV.



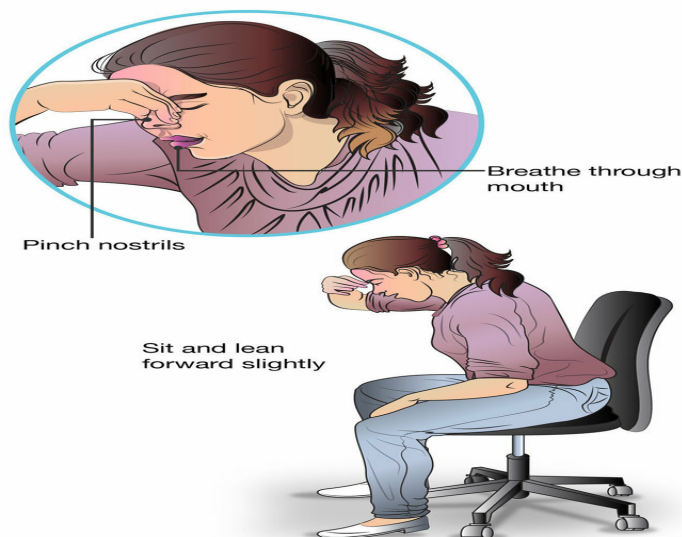
Option B: Heimlich manoeuvre is used in the treatment of accidental inhalation of foreign body. Procedure: Stand behind the person and place your arms around his lower chest and give four abdominal thrusts. The residual air in the lungs can dislodge the foreign body providing some airway.

Heimlich maneuver



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Option D: Trotter's method is used in the treatment of epistaxis. In Trotter's method, the patient is made to sit, leaning a little forward over a basin to spit any blood. He should pinch the nose firmly for 5 to 10 mins and breathe quietly from the mouth.



Solution to Question 4:

Nasal discharge is not an indication for surgery in a patient with deviated nasal septum (DNS).

Indications for surgery in DNS:

- Symptoms of nasal obstruction and recurrent headaches
- Recurrent sinusitis and otitis media
- Recurrent epistaxis from the septal spur

- Septal deviation causing sleep apnea or hypopnoea syndrome

Other indications for SMR (submucous resection) and septoplasty:

- As a part of septorhinoplasty for cosmetic correction of external nasal deformities
- As a preliminary step in hypophysectomy (trans-septal trans-sphenoidal approach) or vidian neurectomy (transseptal approach)
- For the approach to middle meatus or frontal recess in endoscopic sinus surgery when deviated septum obstructs the view and access to these areas
- To gain access for endoscopic dacryocystorhinostomy

Solution to Question 5:

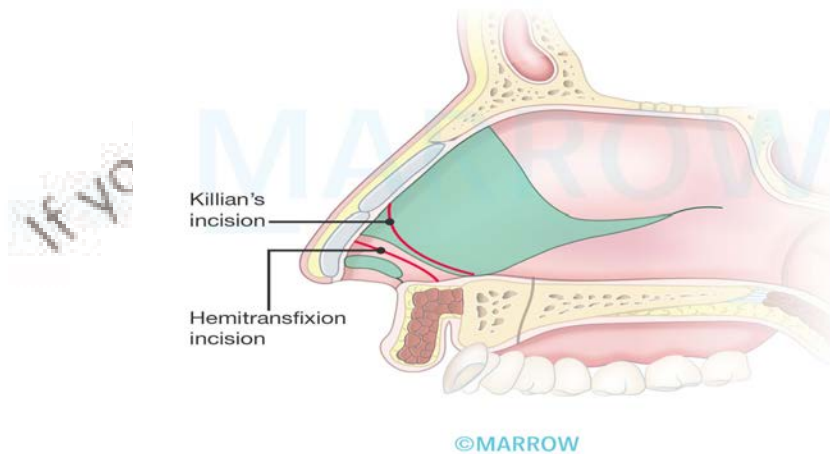
Killian's incision is used in submucosal resection. It is a slightly curvilinear incision, taken 2–3 mm above the caudal end of septal cartilage on the concave side of the deviated nasal septum.

Option B: Shobinger incision is used in modified radical neck dissection.

Option C: Freer's hemitransfixation incision is used in septoplasty. It is a septocolumellar incision taken between the caudal end of septal cartilage and columella.

Option D: Weber Ferguson incision is used in maxillectomy.

Killian's and Hemitransfixion incision



Solution to Question 6:

The given clinical scenario is a case of deviated nasal septum and septoplasty is the treatment of choice.

Both submucous resection and septoplasty can be done for DNS. Septoplasty is a conservative approach and more commonly done nowadays. Septoplasty is also the preferred surgery for DNS

in less than 18 yrs, if DNS is bilateral and severe.

Submucous resection	Septoplasty
Non-conservative procedure with extensive dissection of cartilage	Conservative procedure
Not done in < 18 yrs	Can be done adults and persons < 18 yrs if bilateral and severe
Killian's incision is used	Freer's incision is used
Mucoperichondrial/periosteal flaps raised on both sides of the septum	Flaps raised on only one side
More complications	Less complications
Reoperation is difficult	Reoperation is easier

Solution to Question 7:

Septoplasty is a conservative approach where mucoperichondrial/periosteal flap is generally raised only on one side. As much of the septal framework as possible is retained.

Submucous resection	Septoplasty
Non-conservative procedure with extensive dissection of cartilage	Conservative procedure
Not done in < 18 yrs	Done children and adults
Killian's incision is used	Freer's incision is used
Mucoperichondrial/periosteal flaps raised on both sides of the septum	Flaps raised on only one side
More complications	Fewer complications
Reoperation is difficult	Reoperation is easier

Solution to Question 8:

The given clinical scenario with bilateral septal swelling following trauma is suggestive of a septal hematoma and can result in saddle nose deformity, not hump nose deformity.

Septal hematoma is a collection of blood under the perichondrium or periosteum of the nasal septum. It is caused due to trauma, septal surgery, and bleeding disorders. Clinical features include pain, swelling, and bilateral nasal obstruction (the most common presenting feature), and

sometimes, frontal headache.

Complications include septal abscess, depression of dorsum (saddle nose), and permanently thickened septum.

Solution to Question 9:

The given clinical scenario with swelling in the nose after trauma is suggestive of septal hematoma and it needs immediate surgical drainage.

Treatment of septal hematoma:

- Aspiration of small hematomas
- Incision and drainage of large hematomas followed by nasal packing
- Systemic antibiotics to prevent abscess formation

Solution to Question 10:

Trauma is the most common cause of septal perforation.

Causes of septal perforation:

- Trauma - nasal surgery, nose picking, septal cauterization, nasal packing, septal hematoma, foreign bodies
- Surface irritants - cocaine, heroin, decongestant nasal sprays
- Infections - syphilis, typhoid, tuberculosis, lepromatous leprosy, rhinosporidiosis, rhinoscleroma, aspergillosis
- Neoplastic - melanoma, SCC, adenocarcinoma, lymphoma
- Inflammatory - sarcoidosis, Crohn's disease, RA, Granulomatosis with polyangitis

Small anterior perforations may present with a whistling sound during respiration. Large perforations may obstruct the nose or even bleed due to the formation of crusts.

Treatment depends on the cause. Small asymptomatic perforations can be closed by plastic flaps. In larger perforations treatment is aimed at keeping the nose crust-free using nasal douches and bland ointments.

Solution to Question 11:

The given patient is having nasal septal perforation due to syphilis. The most common site of involvement in syphilis is the bony nasal septum.

Other chronic granulomatous conditions like lupus, tuberculosis, and leprosy can cause perforation in the cartilaginous part. Wegener's granulomatosis causes perforation of both bony and cartilaginous parts.

Solution to Question 12:

Myiasis (maggots in the nasal cavity) is a complication of atrophic rhinitis. It is treated with chloroform diluted with water.

The maggots are larval forms of flies especially genus *Chrysomia*. The flies are attracted by the foul-smelling nasal discharge in conditions like atrophic rhinitis, syphilis, leprosy, and lay eggs in the nasal cavity. These eggs hatch into larvae within 24 hrs and cause intense irritation, lacrimation, sneezing, and blood-tinged nasal discharge.

Treatment:

- All visible maggots should be picked up with forceps
- Instillation of chloroform water and oil kills them
- Nasal douche with warm saline is used to remove slough, crusts, and dead maggots.

Solution to Question 13:

Bernoulli's phenomenon explains the pathogenesis of nasal polyp formation.

Bernoulli's principle states that when the velocity of air increases, the lateral pressure will decrease. When air passes through the nasal valve area which is the narrowest region of the nasal cavity, velocity increases. This creates a negative pressure that sucks mucosa of sinuses into the nasal cavity leading to polyp formation.

Solution to Question 14:

The given scenario is suggestive of Samter's triad where the patient will have aspirin intolerance, asthma, and ethmoidal polyps.

Other etiological factors of ethmoidal polyp:

- Chronic rhinosinusitis especially non-allergic rhinitis with eosinophilia syndrome
- Allergic fungal sinusitis
- Asthma
- Cystic fibrosis
- Kartagener syndrome - This consists of bronchiectasis, sinusitis, situs inversus and ciliary dyskinesia
- Young syndrome - It consists of sinopulmonary disease and azoospermia
- Churg–Strauss syndrome - Consists of asthma, fever, eosinophilia, vasculitis and granuloma

Solution to Question 15:

Azoospermia is not a component of Kartagener's syndrome.

Kartagener syndrome is characterized by the triad of:

- Bronchiectasis
- Sinusitis
- Situs inversus

Kartagener syndrome is one of the etiology for ethmoidal polyps.

Solution to Question 16:

Young's syndrome consists of sinopulmonary disease and azoospermia.

Option A: Neutrophilia with red papular rash is seen in Sweet's syndrome/acute febrile neutrophilic dermatosis.

Option C: Mitral stenosis with ASD forms part of Lutembacher syndrome.

Option D: Nasal polyp, asthma, aspirin hypersensitivity form Samter's triad.

Solution to Question 17:

The given case scenario with multiple bilateral nasal masses in a middle-aged woman is suggestive of ethmoidal polyp and it doesn't grow posteriorly to occupy choana.

Solution to Question 18:

The given clinical scenario with findings of a single, unilateral polyp that extends posteriorly to occupy the choana in a child is suggestive of antrochoanal polyp. It can present with hyponasal voice and not hypernasal voice.

Solution to Question 19:

KTP laser (potassium titanyl phosphate) is the ideal laser for the treatment of nasal polyps.

Option A: Nd-YAG laser due to its greater depth of penetration has a higher risk of complications within the nasal cavity.

Option C: CO₂ lasers are primarily used for laryngeal interventions. Its use in nasal sinus surgery is restricted as it causes extensive scarring and it cannot be delivered through a flexible fiber-optical system.

Option D: The use of argon lasers is limited in sinus surgery due to the high cost of the laser tube and the need for a high current power supply and water flow for cooling.

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Trauma of Nose and Face

Question 1:

A biker met with an accident and sustained a maxillofacial injury. Which of the following is the first step in the primary care of this patient?

- a) Assessment of level of consciousness
- b) Ensure adequate airway
- c) Control of hemorrhage
- d) Fracture reduction

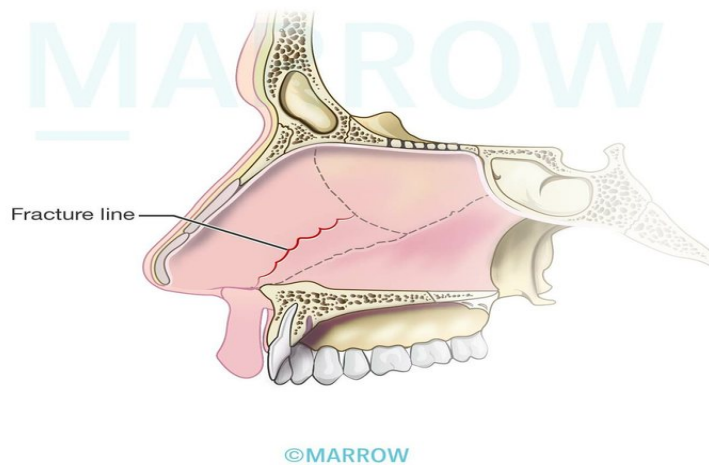
Question 2:

Which of the following facial bones is most commonly fractured?

- a) Zygomatic bone
- b) Nasal bone
- c) Maxilla
- d) Mandible

Question 3:

A 30-year-old man presented with a nasal bleed after sustaining a blow to the nose in a fistfight. On evaluation, a nasal septal fracture similar to the one in the image below was identified. What is your diagnosis?



- a) Jarjaway fracture
- b) Citelli fracture
- c) Chevallet fracture
- d) Tripod fracture

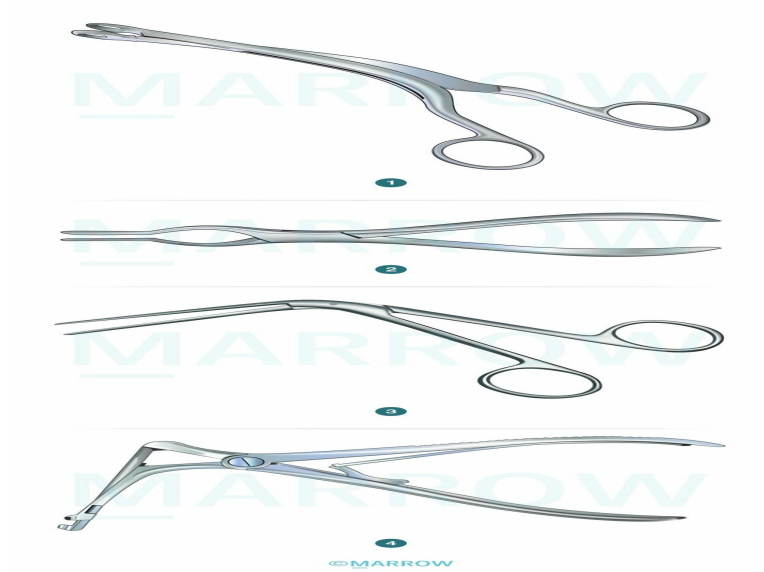
Question 4:

A young adult presents with a deformed nose after falling from a ladder. The CT scan shows a displaced nasal fracture. What is the ideal time to reduce the fracture in this patient?

- a) Within 12 hours following the injury.
- b) 5-7 days following injury
- c) After 2 weeks of injury
- d) After 4 weeks of injury

Question 5:

As an ENT resident, you are assisting in the reduction of a nasal bone fracture. Which of the following instruments would you use?



- a) 1
- b) 2
- c) 3
- d) 4

Question 6:

A boxer is wheeled into the ER with complaints of seeing double after having suffered a severe blow to his face during a match. On examination, there is a flattening of malar prominence, trismus and a step deformity of the infraorbital margin. What is the likely diagnosis?

- a) Fracture of nasal bone
- b) Fracture of maxilla
- c) Fracture of orbital floor
- d) Fracture of zygoma

Question 7:

You are an ENT resident evaluating a patient with a fracture of the zygoma. Which of the following findings would you not expect to see in this patient?

- a) Trismus
- b) Diplopia
- c) CSF rhinorrhea

d) Infraorbital anesthesia

Question 8:

A patient with a tripod fracture is having decreased sensation over the cheek. Which of the following nerve injuries can result in this?

- a) Supraorbital nerve
- b) Infraorbital nerve
- c) Lingual nerve
- d) Facial nerve

Question 9:

Which of the following is called blow-out fractures?

- a) Fracture of floor of orbit
- b) Fracture of lateral wall of orbit
- c) Fracture of medial wall of orbit
- d) Fracture of roof of orbit

Question 10:

A teenage boy suffered trauma to his left eye while playing cricket. He is rushed to the ER in severe pain. His X-ray appears as below. What is the characteristic sign noted?



- a) Tear drop sign
- b) Phelp's sign
- c) Holman Miller sign
- d) Antral sign

Question 11:

A 19-year-old college student presented with complaints of seeing double after sustaining a blow to his face in a scuffle with his seniors. On examination, there is a step deformity and periorbital emphysema. Entrapment of which of the following muscles is likely to be responsible for his double vision?

- a) Superior oblique
- b) Inferior rectus
- c) Superior rectus
- d) Medial rectus

Question 12:

In which of the following fractures Le-Fort classification is used?

- a) Nasal bone
- b) Maxilla
- c) Mandible
- d) Zygoma

Question 13:

Which of the following bones is not involved in Le-Fort fractures?

- a) Maxilla
- b) Mandible
- c) Nasal bone
- d) Lacrimal bone

Question 14:

The CT scan of a patient who met with a motorcycle accident shows a fracture of the maxilla. The fracture line passes through the floor of the nose separating the palate from the rest of the face. How will you classify the fracture?

- a) Le-Fort I
- b) Le-Fort II
- c) Le-Fort III
- d) Le-Fort IV

Question 15:

Which of the following is the most common site for fracture of the mandible in children?

- a) Condyle
- b) Angle
- c) Body
- d) Symphysis

Question 16:

A 23-year-old college student presented with jaw pain and trismus after falling from a bike. A diagnosis of mandibular fracture was made. Which of the following classification is used for this?

- a) Le-Fort classification
- b) Dingman's classification
- c) Gartland classification
- d) Pauwell classification

Question 17:

Following trauma, a patient presented with clear watery nasal discharge that cannot be sniffed back. Fractures of which of the following bones is the most likely cause for this?

- a) Cribriform plate
- b) Tegmen tympani

- c) Frontal bone
- d) Sphenoid bone

Question 18:

Which of the following fractures is not associated with CSF rhinorrhea?

- a) Le-Fort I
- b) Le-Fort II
- c) Le-Fort III
- d) Naso-orbito-ethmoid fracture

Question 19:

A middle-aged woman, who had undergone a polypectomy complains of watery nasal discharge on bending forward. Which of the following is false regarding this condition?

- a) Reservoir sign seen
- b) Halo sign seen
- c) Discharge doesn't stiffen handkerchief
- d) Discharge can be sniffed back

Question 20:

In a patient with suspected CSF rhinorrhea, the presence of which of the following in the nasal discharge will confirm the diagnosis?

- a) β 2 microglobulin
- b) β 2 transferrin
- c) Transthyretin
- d) α 2 microglobulin

Question 21:

Which of the following is false about CSF rhinorrhea?

- a) Ethmoid is the most common site of leak

- b) Glucose is present
- c) Intrathecal fluorescein study can be used to detect site of leak
- d) Double ring sign is absent

Question 22:

What is the preferred line of management in a patient with CSF rhinorrhea who is otherwise stable?

- a) Immediate surgical repair
- b) Conservative measure
- c) Nasal packing
- d) Trotter's method

Answer Key

Question No.	Correct Option
1	b
2	b
3	a
4	b
5	b
6	d
7	c
8	b
9	a
10	a
11	b
12	b
13	b
14	a
15	a
16	b
17	a
18	a

19	d
20	b
21	d
22	b

Detailed Explanations

Solution to Question 1:

The first step in primary care of a patient with a maxillofacial injury is securing an adequate airway.

Evaluation of facial fractures:

- A & B – Airway & Breathing: the first objective is to ensure an adequate airway by intubation or tracheostomy.
- C – Circulation: control of haemorrhage by pressure or ligation of bleeding vessels.
- Rule out head and spinal injuries.
- Neurological evaluation.
- Examine for injuries in other parts of the body.
- Care of the wound.
- Documentation of the injuries for medico-legal purposes.
- Examination of the facial skeleton – mandible, maxilla, zygomatic and orbital fracture.

Solution to Question 2:

The most commonly fractured bone of the face is the nasal bone because of the projection of the nose on the face.

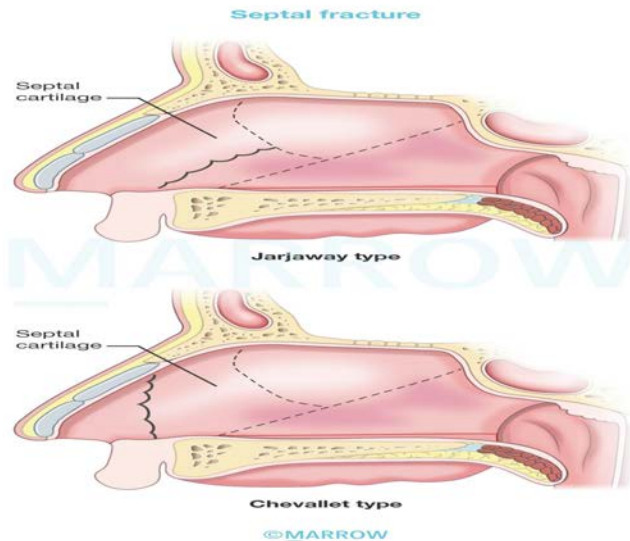
Solution to Question 3:

The given clinical scenario of nasal bleed following trauma and the image showing a horizontal fracture line is suggestive of Jarjaway fracture.

It is a fracture of the nasal septum that results from direct blows from the front. The fracture line runs horizontally from just above the anterior nasal spine to just above the junction of septal cartilage with vomer.

Option C: Chevallet fracture results from blows from below. The fracture line runs vertically from the anterior nasal spine upwards to the junction of the bony and cartilaginous dorsum of the nose.

Option D: Tripod fracture is another name for zygoma fracture.



Solution to Question 4:

The ideal time for reduction of fracture of the nasal bone is 5-7 days following injury, once the edema has subsided.

The presence of edema interferes with the accurate reduction of the fracture. In most cases, the deformity is due to edema. So, ice packs and head elevation are advised initially after injury. After 3-5 days, the edema would subside and the underlying deformity will be apparent on clinical assessment.

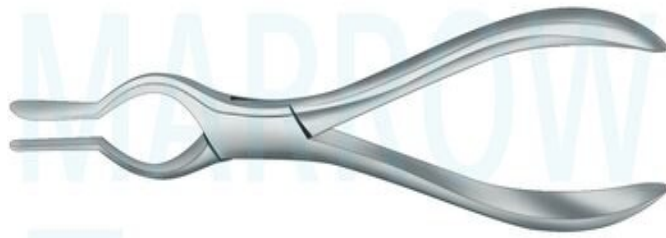
After 2 weeks it is difficult to reduce since the process of callus formation and the bony and fibrous union would have started.

Solution to Question 5:

Walsham forceps (Image 2) is used to reduce a nasal bone fracture.

Asch forceps is mainly used to reduce nasal septum fracture.

Walsham forceps



©Marrow

Asch's forceps



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Option A: Luc's forceps is used in Caldwell Luc operation, submucous resection, and polypectomy.

Luc's forceps



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Option C: Tilley's forceps is used in nasal packing in the epistaxis.

Tilley's forceps



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Option D: Citelli's forceps is used for bone punch.

Citelli's forceps



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Solution to Question 6:

The given clinical scenario of diplopia, flat malar prominence, trismus, and step deformity in the infraorbital margin is suggestive of fracture of zygoma or tripod fracture.

Zygoma is the second most frequently fractured bone in the face. The zygoma fracture is separated at its three processes - the zygomaticofrontal suture, infraorbital margin, and the zygomaticotemporal suture. It has the following clinical features:

- Flattening of malar prominence.
- Trismus, due to depression of zygoma on the underlying coronoid process.
- Anesthesia in the distribution of the infraorbital nerve.
- Step deformity of infraorbital margin.
- Oblique palpebral fissure, due to the displacement of lateral palpebral ligament.
- Restricted ocular movements, due to entrapment of inferior rectus muscle. It may cause diplopia.
- Periorbital emphysema, due to escape of air from the maxillary sinus on nose blowing.

The following x-ray shows a right-sided zygoma fracture:



Solution to Question 7:

CSF rhinorrhea is not seen in patients with fractures of the zygoma.

It has the following clinical features:

- Flattening of malar prominence.
- Step deformity of infraorbital margin.
- Anesthesia in the distribution of the infraorbital nerve.
- Trismus, due to depression of zygoma on the underlying coronoid process.
- Oblique palpebral fissure, due to the displacement of lateral palpebral ligament.
- Restricted ocular movements, due to entrapment of inferior rectus muscle. It may cause diplopia.
- Periorbital emphysema, due to escape of air from the maxillary sinus on nose blowing.

The following x-ray shows a right-sided zygoma fracture:



Solution to Question 8:

Infraorbital nerve injury results in decreased sensation over the cheek in tripod(zygomatic) fracture.

Solution to Question 9:

Fracture of the floor of the orbit is called blow-out fracture.

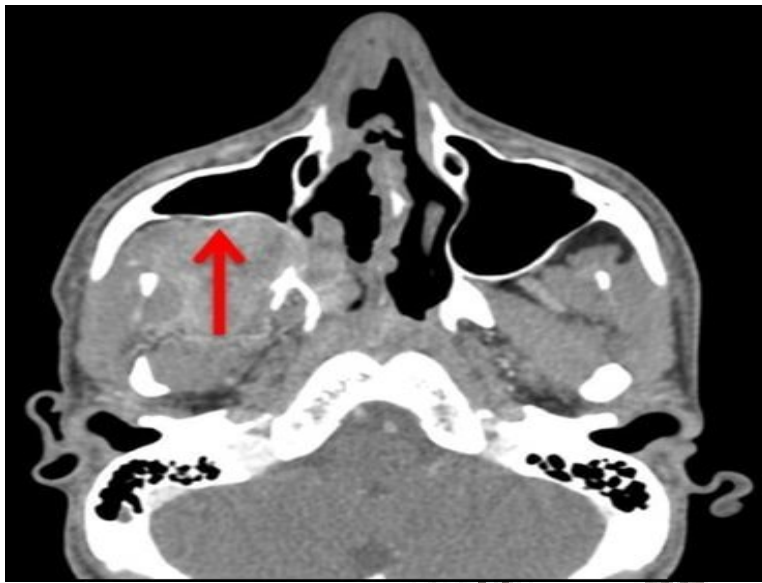
It occurs when a large blunt object strikes the globe.

Solution to Question 10:

The given X-ray shows the tear-drop sign. It is seen in orbital floor or blow-out fractures when the orbital contents herniate into the maxillary antrum.

Option B: Phelps' sign is seen in CT scan in glomus jugulare tumor. It is the erosion of the normal crest of bone between the carotid canal and jugular foramen.

Options C and D: Antral and Holman Miller signs are the same. It is the anterior bowing of the posterior wall of the maxillary sinus and is pathognomonic of juvenile nasopharyngeal angiofibroma.



Solution to Question 11:

The given clinical scenario with diplopia, periorbital emphysema, and a step deformity can be seen in both orbital floor fracture as well as tripod fracture. In both conditions diplopia is due to:

- Entrapment of inferior rectus and inferior oblique muscles
- Displacement of the eyeball.

The common features seen in both orbital floor fracture and tripod fracture are:

- Periorbital emphysema
- Step deformity of infraorbital margin
- Diplopia
- Enophthalmos.

Solution to Question 12:

Le-Fort classification is used in the fracture of the maxilla.

The 3 types are:

- Le Fort I (transverse fracture)
- Le Fort II (pyramidal fracture)
- Le Fort III (craniofacial disjunction).

Solution to Question 13:

The mandible is not involved in Le-Fort's fractures.

Solution to Question 14:

The CT findings describe a Le-Fort I fracture of the maxilla. It is also known as Guerin's fracture.

Solution to Question 15:

The most common site of fracture of the mandible in children is the condyle. It is because the condylar head is connected to the vertical ramus of the bone by a thin neck which gives away easily to an impact.

Mandibular fractures:

Children:

- Due to falls and sports injuries.
- Most common site - condyle.

Adults > 50 years:

- Most commonly due to falls.
- Most common site - body (the molar area is the weakest area in an edentulous mandible).

Young Adults:

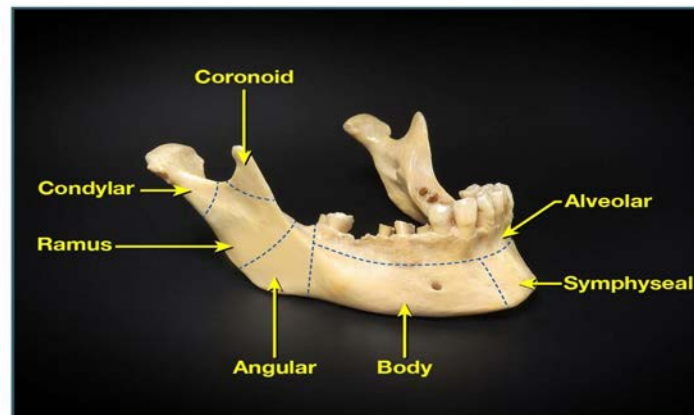
- Due to falls, assaults, and accidents.
- The site of fracture depends on the nature of impact and the direction of force applied.

Solution to Question 16:

Dingman's classification is used in the classification of fractures of the mandible.

Dingman's classification is based on the region in which the fracture is located. The seven regions used in this classification are as shown below:

Dingman and Natvig classification of mandibular fractures



Option A: Le-Fort classification is used in maxillary fractures.

Option C: Gartland classification is used in supracondylar fractures of the humerus.

Option D: Pauwell classification is used in the neck of femur fractures.

Solution to Question 17:

The given clinical scenario is suggestive of CSF rhinorrhea and fracture of the cribriform plate of the ethmoid is the most common cause for this.

Causes of CSF rhinorrhea:

- Trauma (accidental or surgical) - most of the cases follow trauma. Surgical trauma includes endoscopic sinus surgery, trans-sphenoidal hypophysectomy, nasal polypectomy, or skull base surgery.
- Inflammations - mucocoeles of sinuses, sinonasal polyposis, the fungal infection of sinuses and osteomyelitis, can all erode the bone and dura.
- Neoplasms - tumours, both benign and malignant, invading the skull base.
- Congenital lesions - meningocele, meningoencephaloceles, and gliomas can have associated skull base defects.
- Idiopathic - where the cause is unknown and the patient has a spontaneous leak.

Solution to Question 18:

Le-Fort I fracture is not associated with CSF rhinorrhea. Whereas, fractures like Le-Fort II, III and naso-orbito-ethmoid fracture are associated with CSF rhinorrhea.

Solution to Question 19:

The given clinical scenario is suggestive of CSF rhinorrhea secondary to surgery.

Identifying features of CSF Rhinorrhea:

1. The CSF discharge is thin, watery, and clear in appearance.
2. It cannot be sniffed back. (Option D)
3. It does not stiffen a handkerchief.
4. It shows positional variation: The watery CSF discharge occurs suddenly and gushes in drops only on bending the head forward/straining.
5. The Halo sign is a marker of CSF rhinorrhea post-head trauma. It is considered positive when a clear ring surrounds a central bloody spot after the discharge is dropped on a handkerchief/paper towel.
6. Reservoir sign: The discharge is seen on rising in the morning and bending the head, as fluid accumulated in the sinuses empties into the nose.

Solution to Question 20:

Presence of β 2 transferrin protein in the nasal discharge confirms the presence of CSF in the specimen.

β 2 transferrin protein detection by electrophoresis is a sensitive and specific test for CSF. It is secreted by meninges and choroid plexus.

Since it is absent in nasal secretions it helps to differentiate CSF from nasal secretions. It is also present in aqueous humor and perilymph.

Another method is glucose testing by oxidase peroxidase or biochemical estimation. CSF glucose is 60% of blood glucose whereas it is $<10\text{mg/dl}$ in nasal secretions. It is not reliable in associated meningitis (reduced levels giving false-negative results) and when CSF is mixed with blood.

Solution to Question 21:

Double ring sign is present in a case of CSF rhinorrhea.

Double ring sign or Halo sign or Target sign - when CSF mixed with blood is collected on a piece of filter paper, it will show a central red spot (blood) and a peripheral lighter halo.

Option A: Fracture cribriform plate of ethmoid is the most common cause.

Option B: CSF glucose is 60% of blood glucose.

Option C: Intrathecal fluorescein study is used to detect the site of leak of CSF.

Solution to Question 22:

Patients with CSF rhinorrhea who are stable, are managed conservatively.

The conservative measures are:

- Bed rest
- Elevation of the head end
- Stool softeners
- Avoidance of nose-blowing, sneezing, and straining.

Prophylactic antibiotics can be used to prevent meningitis and acetazolamide to decrease CSF formation. Surgery is done if the conservative management fails.

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If you purchased this from someone else,
you may have been scammed.

Tumors of Nose and PNS

Question 1:

Which of the following statements about inverted papilloma is false?

- a) Also known as Schneiderian papilloma
- b) Arises from the lateral wall of nose
- c) More common in women
- d) Presents with unilateral nasal obstruction and epistaxis

Question 2:

A 46-year-old man presented with left-sided nasal blockage and discharge for 3 years. On anterior rhinoscopy, a pale, red polypoidal mass is seen in the left nostril. He was diagnosed with inverted papilloma. Which of the following statements is false about the management of his condition?

- a) Endoscopic surgery is performed.
- b) Postoperative radiotherapy is usually given
- c) Recurrence after removal is common
- d) MRI is the first imaging modality in follow-up

Question 3:

Which of the following is the most common malignant tumor of the nasal cavity?

- a) Squamous cell carcinoma
- b) Basal cell carcinoma
- c) Malignant melanoma
- d) Neuroblastoma

Question 4:

Which of the following is also known as nose picker's carcinoma?

- a) Inverted papilloma

- b) Basal cell carcinoma
- c) Squamous cell carcinoma
- d) Esthesioneuroblastoma

Question 5:

A 15-year-old presented with a cherry red, polypoidal mass detected on anterior rhinoscopy. Biopsy of the mass suggested that it originated from the olfactory neural epithelium. Which of the following is the diagnosis?

- a) Glioma
- b) Esthesioneuroblastoma
- c) Nasopharyngeal carcinoma
- d) Malignant melanoma

Question 6:

A patient is planned for surgical resection of esthesioneuroblastoma. Which of the following statements is false about the same?

- a) Also known as olfactory neuroblastoma
- b) Bimodal age distribution
- c) Highly vascular tumor
- d) Lymph node metastasis does not occur

Question 7:

Which of the following is the most common site for osteomas?

- a) Maxillary sinus
- b) Frontal sinus
- c) Ethmoid sinus
- d) Sphenoid sinus

Question 8:

The most common site of occurrence of fibrous dysplasia is

- a) Maxillary sinus
- b) Mandible
- c) Frontal sinus
- d) Ethmoidal sinus

Question 9:

Which of the following is the most common site for ossifying fibroma?

- a) Frontal sinus
- b) Maxilla
- c) Mandible
- d) Maxillary sinus

Question 10:

A patient is referred to the ENT OPD with left-sided nasal obstruction, disfigurement of the face, and displacement of his left eye. Imaging of paranasal sinuses shows an ill-defined expanding lesion with ground glass appearance in the maxillary sinus. What is the diagnosis?

- a) Osteoma
- b) Fibrous dysplasia
- c) Maxillary carcinoma
- d) Maxillary polyp

Question 11:

Which of the following histological varieties is the most common among paranasal sinus carcinomas?

- a) Squamous cell carcinoma
- b) Basal cell carcinoma
- c) Adenocarcinoma
- d) Adenoid cystic carcinoma

Question 12:

Where is squamous cell carcinoma most commonly found in the paranasal sinuses?

- a) Maxillary sinus
- b) Frontal sinus
- c) Sphenoid sinus
- d) Ethmoid sinus

Question 13:

A carpenter was diagnosed with sinonasal carcinoma and planned for surgical resection. Which of the following histological varieties would be most likely in this case?

- a) Squamous cell carcinoma
- b) Adenocarcinoma
- c) Melanoma
- d) Anaplastic carcinoma

Question 14:

Which of the following is an early feature of maxillary carcinoma?

- a) Blood tinged nasal discharge
- b) Enlarged cervical lymph node
- c) Proptosis
- d) Dental pain

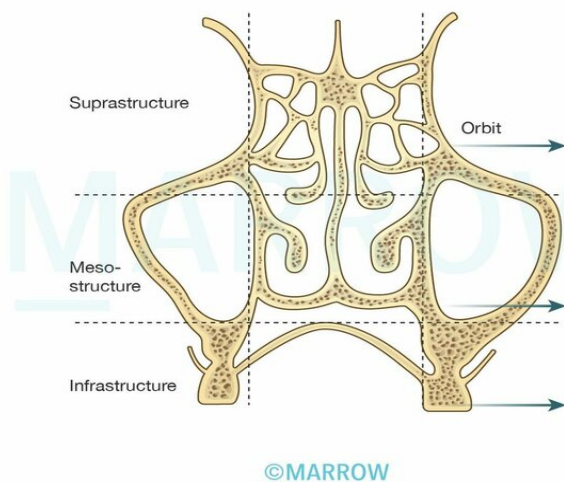
Question 15:

Which of the following conditions is classified using Ohngren's classification?

- a) Maxillary sinusitis
- b) Frontal sinusitis
- c) Maxillary carcinoma
- d) Frontal carcinoma

Question 16:

A 43-year-old factory worker presents to you with nasal obstruction, cheek swelling, and diplopia. The ENT resident suspects maxillary carcinoma and uses the following image to classify the same. Which one of the following classification systems are used?



- a) Ohngren's classification
- b) Lederman's classification
- c) AJCC classification
- d) TNM classification

Question 17:

You have a patient with maxillary sinus carcinoma. Which of the following lymph nodes will be most commonly involved?

- a) Submandibular
- b) Submental
- c) Clavicular
- d) Preauricular

Question 18:

In a known case of maxillary carcinoma, histopathology of the surgically resected region shows that the tumor has eroded into the floor and medial wall of the orbit and the frontal sinus. Which of the following is the best management strategy?

- a) Surgery
- b) Radiotherapy only
- c) Surgery + chemotherapy
- d) Surgery + radiotherapy

Answer Key

Question No.	Correct Option
1	c
2	b
3	a
4	c
5	b
6	d
7	b
8	a
9	c
10	b
11	a
12	a
13	b
14	a
15	c
16	b
17	a
18	b

Detailed Explanations

Solution to Question 1:

Inverted papilloma is more common in males.

Solution to Question 2:

Radiotherapy is not advised in inverted papilloma since it may induce malignancy.

Option A: Endoscopic approach is preferred to surgical resection.

Option C: There is a high rate of recurrence of inverted papillomas.

Option D: MRI is the first imaging modality to perform during follow up.

Solution to Question 3:

Squamous cell carcinoma is the most common malignant tumor of the nasal cavity, seen in about 80% of cases.

Solution to Question 4:

Squamous cell carcinoma of the nasal cavity arising from the nasal septum is called nose picker's cancer.

Option A: Inverted papilloma is also known as transitional cell papilloma or Ringertz tumor or Schneiderian papilloma.

Option B: Basal cell carcinoma is also known as rodent ulcer.

Option D: Esthesioneuroblastoma is also known as olfactory neuroblastoma.

Solution to Question 5:

The given clinical scenario suggests the diagnosis of esthesioneuroblastoma. It arises from the olfactory epithelium of the upper third of the nose.

Solution to Question 6:

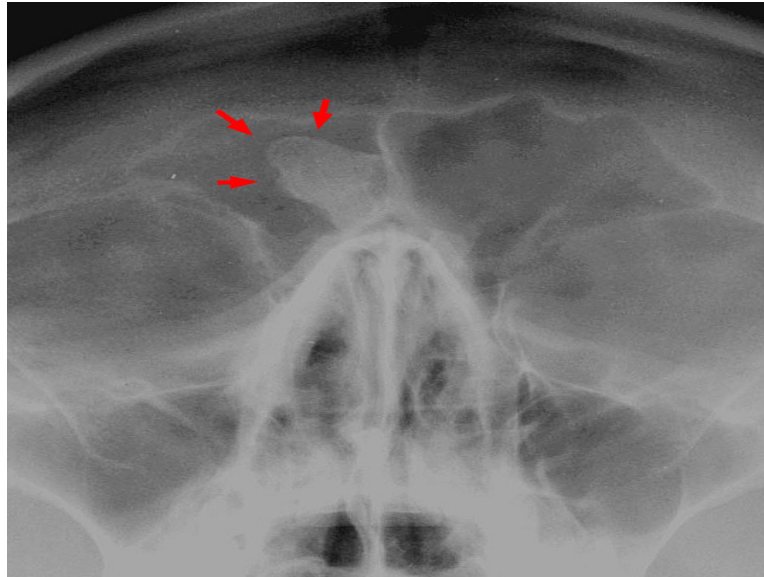
Lymph node metastasis occurs in 10-15% of the cases.

Solution to Question 7:

Osteomas are most commonly seen in the frontal sinus followed by ethmoid and maxillary sinuses.

They remain asymptomatic and are incidentally found on x-rays. Treatment is required only when they cause obstructive symptoms or pressure effects.

The image given shows the x-ray image of an osteoma in the frontal sinus.



Solution to Question 8:

The maxilla is the most common site for fibrous dysplasia.

Fibrous dysplasias and ossifying fibromas are both skeletal lesions where normal architecture is replaced by vascularized fibrous tissue.

However, the difference between the two is as follows -

	Fibrous dysplasia	Ossifying fibroma
Site	More common in the maxilla.	More common in the mandible
Radiology	No well-defined edge, lesion merges with surroundings	Well demarcated lesion
Pathology	Strongly reactive for osteocalcin	Weakly reactive for osteocalcin

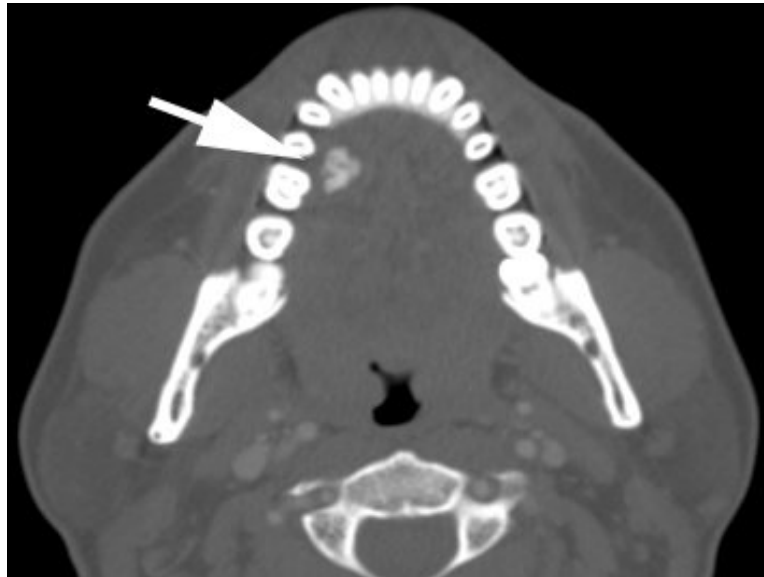
Solution to Question 9:

The mandible is the most common site for the occurrence of ossifying fibroma.

Fibrous dysplasias and ossifying fibromas are both skeletal lesions where normal architecture is replaced by vascularized fibrous tissue.

However, the difference between the two is as follows -

	Fibrous dysplasia	Ossifying fibroma
Site	More common in the maxilla.	More common in the mandible
Radiology	No well-defined edge, lesion merges with surroundings	Well demarcated lesion
Pathology	Strongly reactive for osteocalcin	Weakly reactive for osteocalcin



Solution to Question 10:

The diagnosis is fibrous dysplasia. A ground-glass appearance in CT or x-ray is characteristic of fibrous dysplasia.

In fibrous dysplasia, bone is replaced by fibrous tissue. The most common site is the maxillary sinus, followed by the ethmoid and frontal sinuses.

It can present with disfigurement of the face, nasal obstruction, and displacement of the eye.

The x-ray image below shows a ground-glass appearance present in the left maxillary sinus due to fibrous dysplasia. The lesion is not well demarcated and merges with the surrounding area. There is orbital displacement on the left side.



Solution to Question 11:

Squamous cell carcinoma is the most common malignant tumor of the paranasal sinuses.

Solution to Question 12:

Squamous cell carcinoma is the most common in the maxillary sinuses followed in turn by ethmoids, frontal and sphenoid sinuses.

Solution to Question 13:

Adenocarcinoma is the most common variety in this case. Workers of the furniture industry or woodworkers develop adenocarcinoma.

It is most commonly found in the ethmoid sinuses and upper nasal cavity.

Workers engaged in nickel refining may develop squamous cell carcinoma and anaplastic carcinoma.

Solution to Question 14:

Blood-tinged nasal discharge is an early feature of maxillary carcinoma.

Other early features include:

- Nasal stuffiness

- Facial paraesthesias
- Pain
- Epiphora.

These symptoms may be missed or simply treated as sinusitis.

Option B: Lymphatic spread occurs in the late stages of the disease. Among the cervical lymph nodes, submandibular and upper jugular lymph nodes are enlarged.

Option C: Proptosis is a late feature that occurs after the invasion of the orbit.

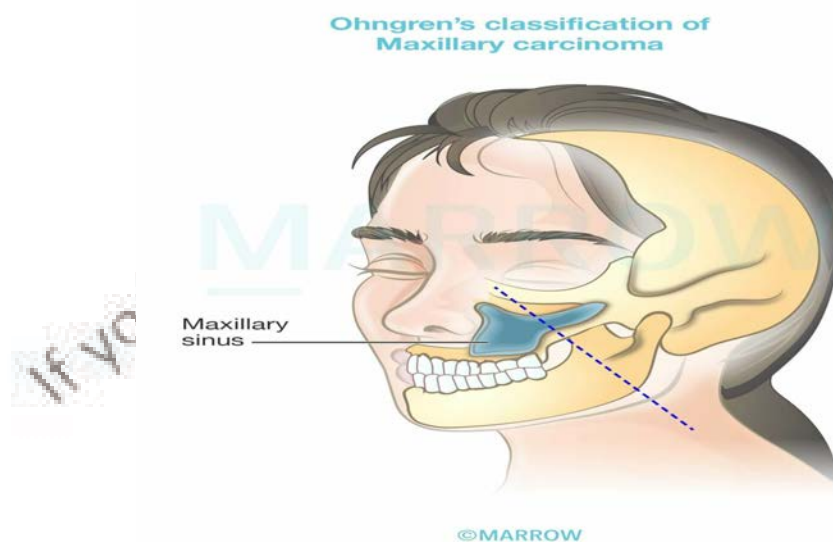
Option D: Dental pain is a late feature that occurs due to expansion of the alveolus after the tumor spreads inferiorly.

Solution to Question 15:

Ohngren's classification is used for maxillary carcinoma.

An imaginary plane is drawn, extending between medial canthus of eye and the angle of mandible.

Ohngren's line(below image):



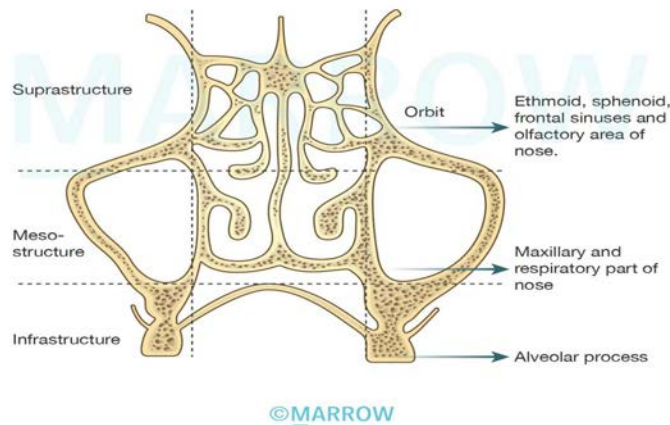
Growths situated above this plane (suprastructural) have a poorer prognosis than those below it (infrastructural).

Solution to Question 16:

Ledermann's classification is used to classify maxillary carcinoma in this case.

It uses two horizontal lines of Seibileau; one passing through the floors of orbits and the other through floors of antra.

Ledermann's classification of maxillary carcinoma



Solution to Question 17:

The submandibular and jugulodigastric nodes are the most commonly involved in maxillary sinus carcinoma.

Maxillary and ethmoid sinus drain primarily into retropharyngeal nodes but, these nodes are inaccessible for palpation.

Solution to Question 18:

Radiotherapy is preferred here, as in this case surgery has already been performed. The tumor staging is T4 by AJCC classification.

T3 and T4 lesions are treated by combined modalities of radiation and surgery. Radiation in such cases may be given preoperatively or postoperatively.

T1 and T2 lesions without nodal or distant metastasis are treated with surgery or radiation with equal results.

Stage	Description
T1	Tumour restricted to maxillary sinus mucosa
T2	Bony erosion- extension into hard palate or middle meatus, except posterior wall and pterygoid plates

Stage	Description
T3	Invasion into suprastructures such as: Posterior wall of maxillary sinus Subcutaneous tissues Floor or medial wall of orbit Pterygoid fossa Ethmoid sinus
T4a	Tumour invades any of the following: Anterior orbital contents The skin of the nose or cheek Infratemporal fossa Pterygoid plates Cribriform plate Sphenoid or frontal sinuses
T4b	Tumour invades any of the following: Orbital apex Brain/dura Middle cranial fossa Nasopharynx

Sinusitis and its Complication

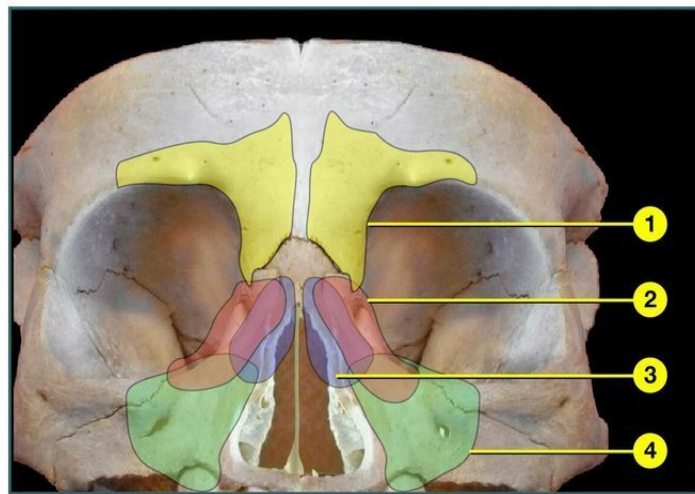
Question 1:

Which of the following is the most common cause of sinusitis?

- a) Nasal infections
- b) Dental infections
- c) Trauma
- d) Swimming and diving

Question 2:

Identify the most common sinus involved in sinusitis in adults.



- a) 4
- b) 3
- c) 2
- d) 1

Question 3:

A patient with sinusitis is least likely to present with which of the following symptoms?

- a) Occipital headache
- b) Eyelid edema
- c) Diplopia
- d) Postnasal drip

Question 4:

A 28-year-old IT professional with a recent history of upper respiratory tract infection has a headache that worsens during office hours. The headache comes upon waking, tends to be at its worst by about midday and improves as the day progresses. This characteristic periodicity is seen in?

- a) Maxillary sinusitis
- b) Frontal sinusitis
- c) Ethmoid sinusitis
- d) Sphenoid sinusitis

Question 5:

Which of the following is the most common sinusitis in children?

- a) Maxillary
- b) Frontal
- c) Sphenoid
- d) Ethmoid

Question 6:

A 12-year-old child diagnosed with sphenoid sinusitis was brought to the hospital with complaints of a fever and headache. The headache would most likely be localised to?

- a) Frontal
- b) Occiput
- c) Temporal region
- d) Root of nose.

Question 7:

A middle-aged man with a history of recurrent episodes of sinusitis for two years presented with complaints of a dull aching pain over his cheek, persistent nasal obstruction which worsens at night and an urge to constantly clear his throat. Which of the following anterior rhinoscopy findings is pathognomonic of the condition?

- a) Muco-pus in middle meatus
- b) Inferior turbinate atrophy
- c) Inferior turbinate hypertrophy
- d) Crusting on the turbinates

Question 8:

A 21-year-old female patient presented with fever, headache and, hawking sensation. Suspecting severe sinusitis, you order an X-ray of the PNS. Which view would be best in capturing all the paranasal sinuses in a single film?

- a) Lateral View
- b) Caldwell's view
- c) Water's view
- d) Submento-vertical view

Question 9:

A patient with deviated nasal septum presents with a headache that starts upon waking and reaches its peak by about midday. On examination, she has upper eyelid oedema and tenderness on pressing just above the medial canthus. Which of the following X-ray views is preferred to best evaluate the sinus involved?

- a) Water's view
- b) Caldwell view
- c) Schuller's view
- d) Water's view with mouth open

Question 10:

Which of the following is used to make a definitive diagnosis of sinusitis?

- a) X Ray PNS
- b) Nasal endoscopy

- c) Proof puncture
- d) Transillumination test

Question 11:

Which of the following complications would you not expect to see in a patient diagnosed with sinusitis?

- a) Orbital cellulitis
- b) Brain abscess
- c) Eye lid edema
- d) Soft palate paralysis

Question 12:

Which of the following is the most common complication following acute rhinosinusitis?

- a) Orbital cellulitis
- b) Meningitis
- c) Osteomyelitis
- d) Septicaemia

Question 13:

A 60-year-old lady presents with pain and swelling around the left upper eyelid region, diplopia and a headache. She gives a history of recurrent episodes of rhinosinusitis in the past. On examination, there is a downward, and outward deviation of the eyeball & the most prominent part of the swelling has a punctum with a mucopurulent discharge, just above the upper eyelid. Which of the following sinuses is most commonly involved in this complication?

- a) Frontal sinus
- b) Maxillary sinus
- c) Ethmoid sinus
- d) Sphenoid sinus

Question 14:

Which of the following statements is false about mucocoeles?

- a) Frontal sinus is most commonly involved
- b) Presents as a cystic swelling in the superomedial quadrant
- c) Eyeball is pushed downwards, forwards and medially
- d) Treatment is frontoethmoidectomy

Question 15:

A young man was referred to the maxillofacial surgeon with complaints of pain and tenderness over the left cheek, inability to move the left eye, and numbness over the left forehead. He gave a history of chronic sinusitis for two years. Superior orbital fissure syndrome was suspected. Which of the following nerves is least likely to be involved in this condition?

- a) Oculomotor nerve
- b) Ophthalmic division of trigeminal nerve
- c) Maxillary division of trigeminal nerve
- d) Trochlear nerve

Question 16:

Which of the following statements is not true?

- a) Osteomyelitis of maxilla is more common in children
- b) Osteomyelitis of frontal bone is more common in adults
- c) Ethmoidal sinusitis spreads through lamina papyracea
- d) Pott's puffy tumor is related to osteomyelitis of maxilla

Question 17:

A middle-aged patient with a history of recurrent upper respiratory tract infections presents with diplopia and protrusion of the left eyeball. Inflammation of which of the following paranasal sinuses would most likely lead to this complication?

- a) Maxillary sinus
- b) Frontal sinus
- c) Ethmoid sinus

- d) Sphenoid sinus

Question 18:

A procedure that involves puncturing and irrigating the sinus is planned for a patient with a history of chronic sinusitis. For which of the following paranasal sinuses is this procedure performed?

- a) Maxillary sinus
- b) Frontal sinus
- c) Ethmoid sinus
- d) Sphenoid sinus

Question 19:

In which of the following conditions is antral lavage not indicated?

- a) Treatment of acute maxillary sinusitis
- b) Diagnosis of chronic maxillary sinusitis
- c) Treatment of chronic maxillary sinusitis
- d) Collect specimen from maxillary antrum

Question 20:

While performing a therapeutic antral lavage of the maxillary sinus, the patient developed a fatal complication. He stopped breathing, the colour of his face became ashen grey, and the pupils became dilated. Attempts to resuscitate the patient were futile, and he was declared dead after 20 minutes. Which of the following could be the most likely cause?

- a) Maxillary artery thrombus
- b) Air embolism
- c) Bleeding
- d) Meningitis

Question 21:

Which of the following sinuses is accessed via the Caldwell-Luc approach?

- a) Maxillary sinus
- b) Frontal sinus
- c) Ethmoid sinus
- d) Sphenoid sinus

Question 22:

Which of the following structures is pierced while creating a nasoantral window in Caldwell Luc operation?

- a) Superior meatus
- b) Middle meatus
- c) Inferior meatus
- d) Uncinate process

Question 23:

Which of the following is the most common complication of Caldwell Luc operation?

- a) Bleeding
- b) Oroantral fistula
- c) Infra orbital nerve injury
- d) Orbital cellulitis

Question 24:

What does FESS stand for?

- a) Functional endonasal sinus surgery
- b) Functional endoscopic sinus surgery
- c) Factual endoscopic sinus surgery
- d) Factual endonasal sinus surgery

Question 25:

Which of the following conditions is not an indication for functional endoscopic sinus surgery?

- a) Chronic sinusitis
- b) Epistaxis
- c) Nasal polyp
- d) Cavernous sinus thrombosis

Question 26:

Which of the following conditions is a contraindication for endoscopic sinus surgery?

- a) Chronic bacterial rhinosinusitis
- b) Allergic fungal sinusitis with mycetoma
- c) Acute rhinosinusitis with epidural abscess
- d) Repair of cerebrospinal fluid leak

Question 27:

Which of the following structures cannot be accessed by performing an endoscopic nasal surgery?

- a) Pituitary
- b) Optic nerve
- c) Cerebellum
- d) Lacrimal sac

Question 28:

A 45-year-old-man with a history of chronic sinusitis unresponsive to medications is planned for functional endoscopic sinus surgery. Which of the following is the first step in the procedure?

- a) Middle turbinectomy
- b) Uncinectomy
- c) Bullectomy
- d) Sphenoidotomy

Question 29:

You are a third-year ENT resident performing a functional endoscopic sinus surgery on a patient diagnosed with chronic maxillary sinusitis. During the procedure, the lateral wall of the infundibulum is punctured adjacent to the maxillary sinus ostium. Why is this step done?

- a) To approach the middle meatus
- b) To approach the nasolacrimal duct
- c) To approach the superior meatus
- d) To approach the ethmoidal sinus

Question 30:

Water's view is used to best visualize which of the following sinuses?

- a) Maxillary sinus
- b) Ethmoidal sinus
- c) Frontal sinus
- d) Sphenoid sinus

Question 31:

The radiograph shown below is done for better assessment of the frontal sinus. What is the common name of this view?



- a) Water's view
- b) Caldwell view

- c) Pierre's view
- d) Towne's view

Answer Key

Question No.	Correct Option
1	a
2	a
3	c
4	b
5	d
6	b
7	a
8	a
9	b
10	b
11	d
12	a
13	a
14	c
15	c
16	d
17	c
18	a
19	a
20	b
21	a
22	c
23	c
24	b
25	d
26	c
27	c
28	b
29	a

30	a
31	b

Detailed Explanations

Solution to Question 1:

Nasal infections are the most common cause of sinusitis since sinus mucosa is a continuation of nasal mucosa and infections from the nose can travel directly by continuity or by way of submucosal lymphatics.

The most common cause of acute sinusitis is viral rhinitis followed by bacterial invasion.

Other causes:

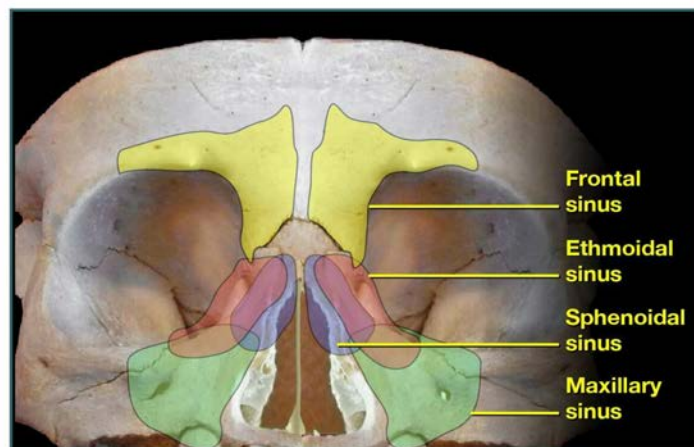
- Dental infections cause maxillary sinus infections.
- Trauma
- Swimming and diving.

Solution to Question 2:

The structure marked 4 is the maxillary sinus and is the most common sinus involved in acute sinusitis in adults.

Order of frequency of sinuses involved in acute sinusitis in adults is maxillary > ethmoid > frontal > sphenoid.

Paranasal sinuses



Solution to Question 3:

Diplopia is least likely to be seen in sinusitis.

Patients with sinusitis typically present with a dull throbbing headache which worsens on moving the head or bending forward, nasal discharge and blockage, & constitutional symptoms like fever, general malaise and bodyache.

Option A: Occipital headache can be seen in sphenoid sinusitis.

Option B: Eyelid edema can be seen in frontal and ethmoidal sinusitis.

Option D: Post-nasal drip and nasal discharge is a common finding in sinusitis.

Solution to Question 4:

Headache in frontal sinusitis shows characteristic periodicity, i.e. comes upon waking, gradually increases, reaches its peak by about midday, and then starts subsiding. It is also called office headache because of its presence only during office hours.

Solution to Question 5:

The ethmoid sinus is the most common sinus involved in pediatric sinusitis.

It usually occurs when the infection spreads through the lamina papyracea.

Solution to Question 6:

Headache in sphenoid sinusitis is localized to occiput or vertex.

Frontal headache is seen in maxillary and frontal sinusitis. Headache in frontal sinusitis has characteristic periodicity.

Solution to Question 7:

The given clinical history is suggestive of chronic maxillary sinusitis. Mucoïd or mucopurulent discharge in the middle meatus seen on anterior rhinoscopy is a pathognomonic finding in chronic maxillary sinusitis.

Other signs include mucosal edema primarily in the middle meatus as well as polyps. Clinically, patients present with nasal blockage, congestion, postnasal drip, reduced olfaction, and facial pain or pressure. On CT, mucosal changes within the osteomeatal complex and sinuses may be seen.

Solution to Question 8:

The lateral view shows all the paranasal sinuses in a single film.

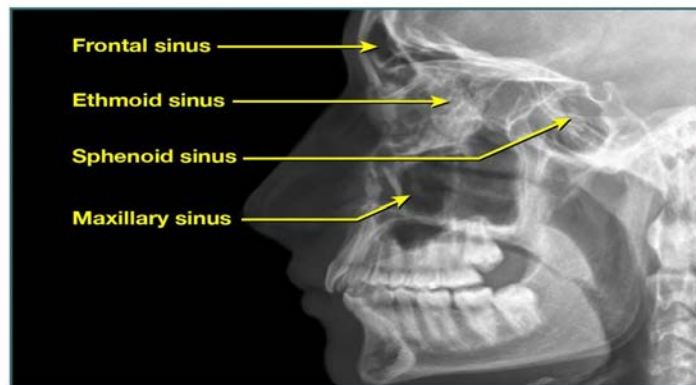
The lateral side of the skull lies against the film, and an X-ray beam is projected perpendicular to the other side.

Structures seen are:

- Anterior and posterior extent of the sphenoid, frontal sinus
- Maxillary sinuses
- Ethmoid sinuses
- Sella turcica
- Alveolar process
- Condyle and neck of the mandible.

Note: Though all the abnormalities and diagnoses of Paranasal sinuses cannot be concluded through the lateral view alone, the lateral view radiographically covers all the sinuses in the same view.

Lateral view



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Solution to Question 9:

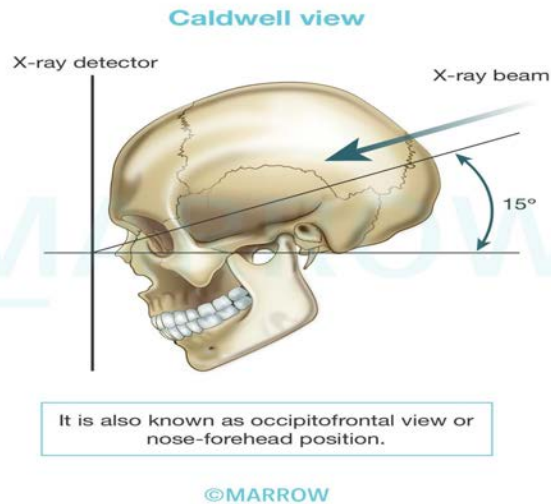
The given clinical scenario points to a diagnosis of frontal sinusitis. The best view for evaluating the frontal sinus is Caldwell's view.

It is also known as occipitofrontal view or nose-forehead position.

Structures seen are:

- Frontal sinuses
- Ethmoid sinuses

- Maxillary sinuses
- Frontal process of the zygoma and the zygomatic process of the frontal bone
- Superior margin of orbit and lamina papyracea
- Superior orbital fissure
- Foramen rotundum (inferolateral to superior orbital fissure)



Solution to Question 10:

Diagnostic nasal endoscopy findings are used to confirm the diagnosis of sinusitis.

Indications of diagnostic nasal endoscopy:

- To diagnose any disease of the nose and PNS
- To find the source of bleeding in epistaxis
- To assess response to medical or surgical treatment of the nose and PNS disease
- To take a precise biopsy from the nose and nasopharynx

Solution to Question 11:

Soft palate paralysis is not a known complication of sinusitis.

Complications of paranasal sinus infection:

- Local
- Mucocele/Mucopyocele
- Mucous retention cyst

- Osteomyelitis - Frontal bone (more common), maxilla
- Orbital
- Preseptal inflammatory oedema of lids
- Subperiosteal abscess
- Orbital cellulitis
- Orbital abscess
- Superior orbital fissure syndrome
- Orbital apex syndrome
- Intracranial
- Meningitis
- Extradural abscess
- Subdural abscess
- Brain abscess
- Cavernous sinus thrombosis
- Descending infections
- Focal infections

Solution to Question 12:

Orbital cellulitis is the most common complication following acute rhinosinusitis.

Complications of acute rhinosinusitis are typically classified as orbital (60–75 per cent), intracranial (15–20 per cent) and osseous (5–10 per cent), and local and descending infections.

Solution to Question 13:

The given clinical history and findings are suggestive of a mucocele secondary to frontal sinusitis. The frontal sinus is most commonly involved in mucocele formation.

The sinuses commonly affected by mucocele in the order of frequency are the frontal>ethmoidal>maxillary>sphenoid

The image given below shows a coronal MRI demonstrating a frontal mucocele and an orbital cystic mass.



Solution to Question 14:

Eyeballs are pushed downwards, forwards and laterally.

Option A: Frontal sinus is most commonly involved in mucocele.

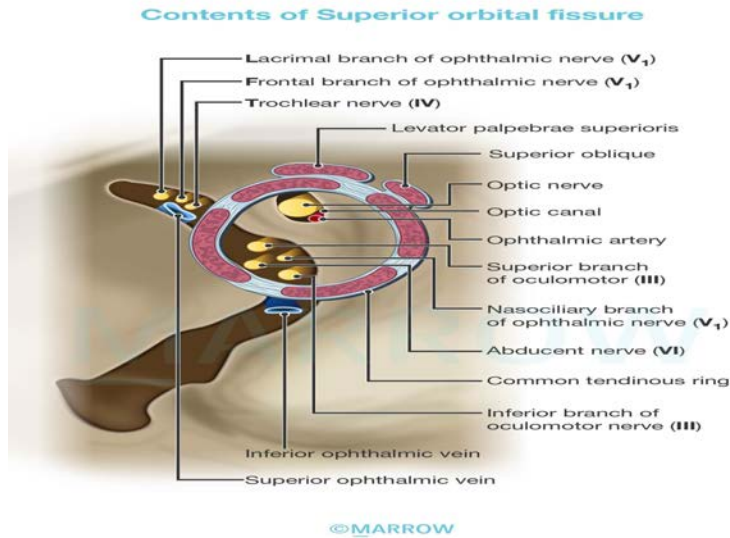
Option B: It presents as a cystic swelling in the superomedial quadrant of orbit.

Option D: Treatment is frontoethmoidectomy with free drainage of the frontal sinus into the middle meatus.

Solution to Question 15:

The maxillary division of trigeminal nerve does not pass through the superior orbital fissure and thus, is the least likely nerve to be involved in this condition.

Superior orbital fissure syndrome is characterized by the compression of cranial nerves III, IV, VI, and ophthalmic division of V. The superior orbital fissure lies between the greater and lesser wings of the sphenoid. The structures passing through include cranial nerves III, IV, VI, the ophthalmic branch of V, and the superior ophthalmic vein.



Orbital apex syndrome is superior orbital fissure syndrome with additional involvement of optic nerve and maxillary division of trigeminal nerve.

Solution to Question 16:

Pott's puffy tumor refers to osteomyelitis of the frontal bone with the development of a subperiosteal abscess manifesting as a puffy swelling on the forehead or scalp. It usually occurs as a complication of frontal sinusitis.

Option A: Osteomyelitis of the maxilla is more often seen in infants and children than adults because of the presence of spongy bone in the anterior wall of the maxilla.

Option B: Osteomyelitis of frontal bone is more often seen in adults as frontal sinus is not developed in infants and children.

Option C: Ethmoidal sinusitis spreads through lamina papyracea

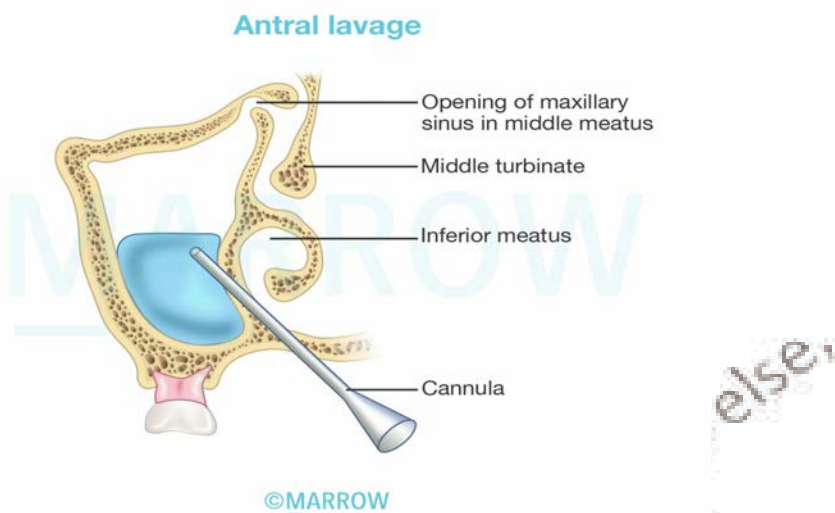
Solution to Question 17:

The given clinical vignette is suggestive of orbital complications secondary to sinusitis. The orbit and its contents are closely related to the ethmoid, frontal, and maxillary sinuses, but most of the complications, however, follow infection of ethmoids as they are separated from the orbit only by a thin lamina of bone called lamina papyracea.

Solution to Question 18:

Proof puncture or antral lavage involves puncturing the medial wall of maxillary sinus in the region of inferior meatus to allow for irrigation and drainage of the sinus.

It is called proof puncture, as the presence of an infection can be proven during the procedure.



Solution to Question 19:

In acute maxillary sinusitis, antral lavage is actually contraindicated for the risk of osteomyelitis.

Antral lavage is usually done therapeutically to remove pus and diagnostically for confirmation of chronic maxillary sinusitis. It can also be done to collect the specimen of the antral contents for culture and sensitivity, or cytological examination to exclude early malignancy.

Solution to Question 20:

Air embolism is a rare but fatal complication that can occur during antral lavage through the veins that are damaged during the puncture.

This complication can be prevented by avoiding the insufflation of air into the antrum after lavage.

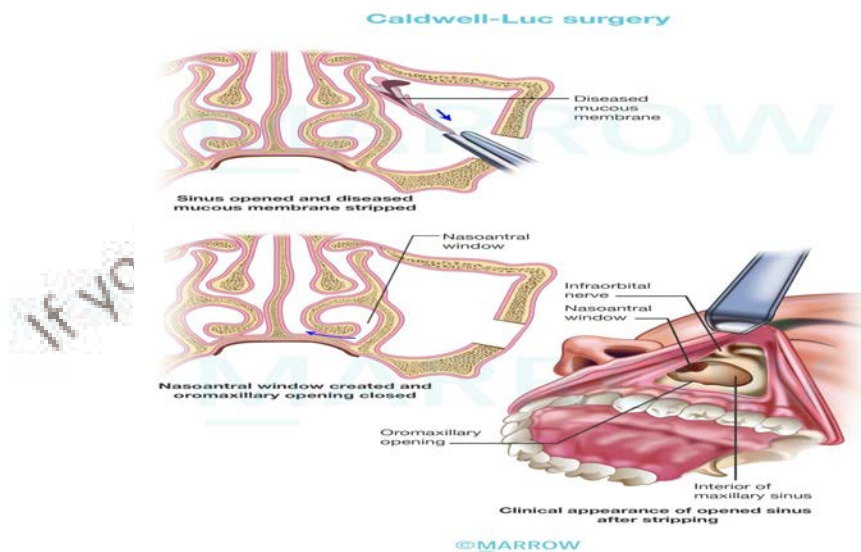
Solution to Question 21:

Caldwell–Luc operation is a surgery where the maxillary antrum is opened through canine fossa by sublabial approach and pathology inside the antrum dealt with.



Solution to Question 22:

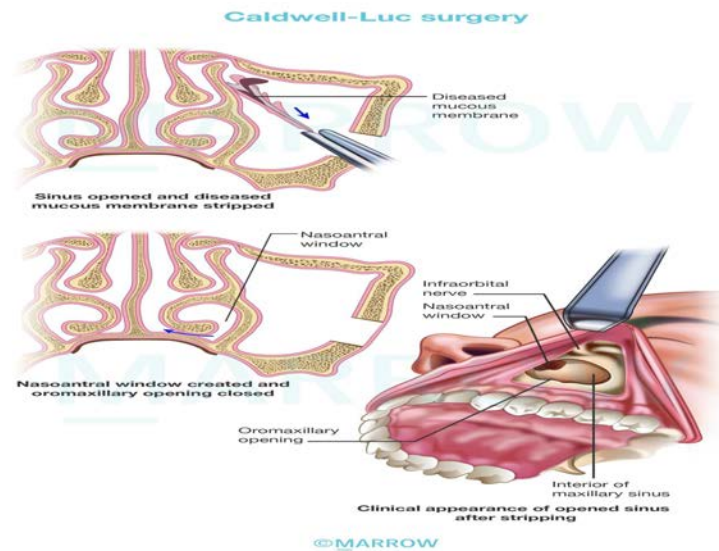
Nasoantral window of $\approx 1.5\text{cm}$ in diameter is created through the inferior meatus.



Solution to Question 23:

The most common complication of Caldwell Luc operation is infraorbital nerve injury.

Paraesthesia due to damage of the infraorbital nerve may be temporary or permanent and is reported in 21% of cases.



Solution to Question 24:

FESS stands for Functional Endoscopic Sinus Surgery.

FESS is a minimally invasive procedure that uses nasal endoscopes to enlarge the nasal drainage pathways of the paranasal sinuses to improve sinus ventilation and allow access to topical medications.

Solution to Question 25:

Cavernous sinus thrombosis is not an indication for functional endoscopic sinus surgery(FESS).

Indications of FESS:

- Chronic bacterial sinusitis unresponsive to adequate medical treatment
- Recurrent acute bacterial sinusitis
- Fungal sinusitis with fungal ball
- Nasal polyps
- Mucocele
- Control of epistaxis
- Removal of foreign body from the nose or sinus
- Barotrauma sinusitis (pain in the face during descent in aircraft or descent whilst scuba diving)

Solution to Question 26:

Acute rhinosinusitis with epidural abscess is a contraindication for functional endoscopic sinus surgery.

The endoscopic approach is contraindicated in the presence of an intracranial complication of acute infection such as:

- Meningitis
- Subperiosteal abscess
- Epidural abscess
- Cavernous sinus thrombosis
- Any visual loss

Note: Endoscopic sinus surgery is indicated in chronic bacterial rhinosinusitis, allergic fungal sinusitis with mycetoma and repair of cerebrospinal fluid leak.

Solution to Question 27:

Cerebellum cannot be accessed by performing endoscopic nasal surgery.

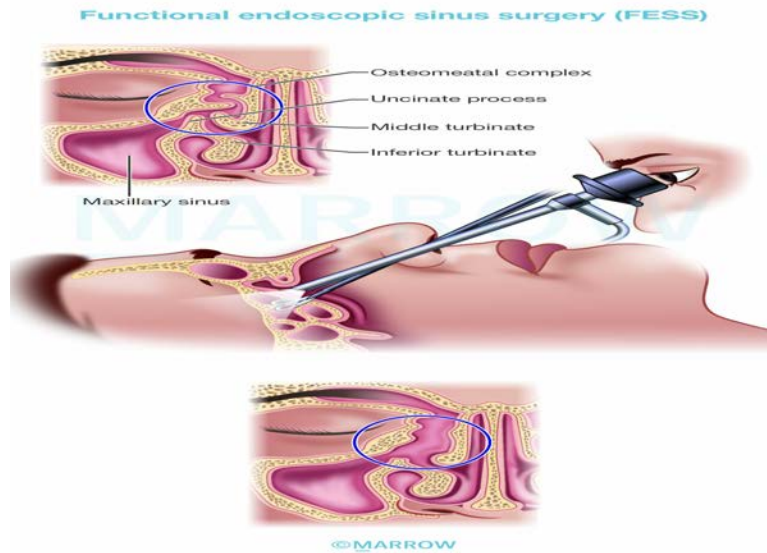
Apart from the nose and paranasal sinuses, endoscopic surgery can be used to access the orbit, optic nerve, lacrimal sac, and pituitary gland.

Indications of advanced nasal endoscopic techniques:

- Removal of benign tumors, e.g. inverted papillomas or angiofibromas
- Orbital abscess or cellulitis management.
- Dacryocystorhinostomy
- Repair of CSF leak
- Pituitary surgery
- Optic nerve decompression
- Orbital decompression for Graves' disease
- Control of posterior epistaxis (endoscopic clipping of sphenopalatine artery)
- Choanal atresia

Solution to Question 28:

Uncinectomy where uncinate process is incised with sickle knife and removed with Blakesley forceps is the first definitive surgical step in FESS.



Solution to Question 29:

Infundibulotomy, also known as uncinectomy, is the first surgical step in FESS and it is done to approach the middle meatus.

Solution to Question 30:

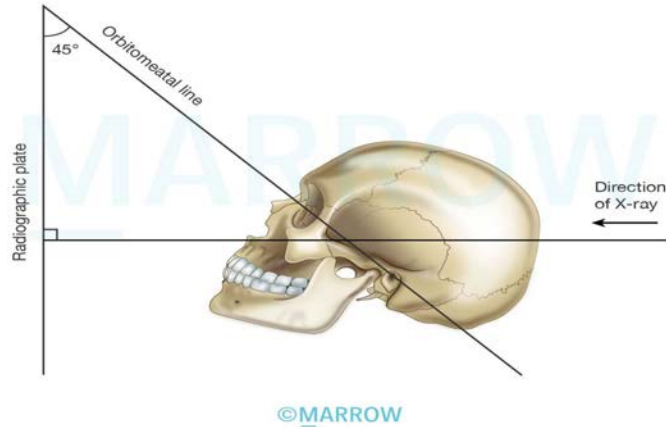
Water's view is used to best visualize the maxillary sinus. Frontal and ethmoidal sinuses can also be visualized. Sphenoid sinus can be seen in Water's view with open mouth.

It is also known as the occipitomeatal view. The X-ray beam is angled at 37° - 45° to the orbitomeatal line.

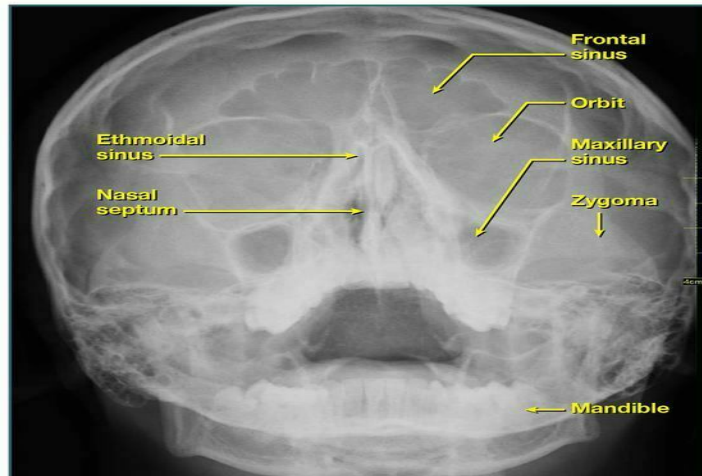
The recommended imaging views for different sinuses are as follows:

- Ethmoidal sinus: Rhese's view
- Frontal sinus: Caldwell's view
- Sphenoid sinus: Hirtz view

Water's radiographic view of skull



Water's view / Occipitomeatal view



Solution to Question 31:

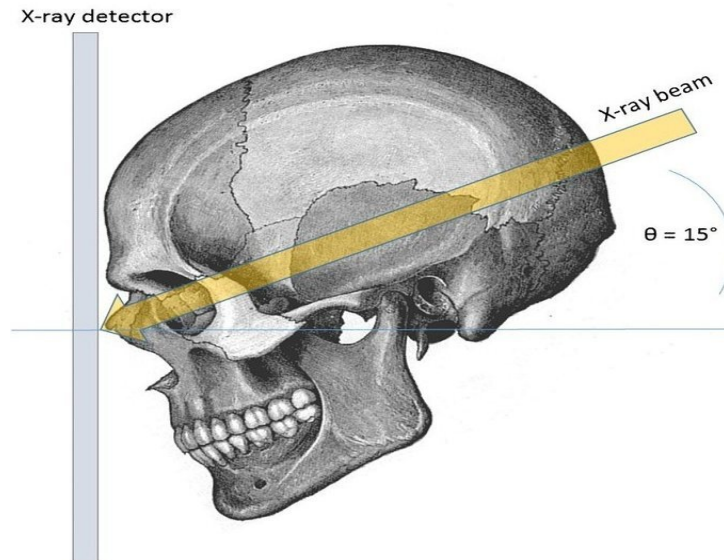
The frontal sinus is best seen in Caldwell's view (occipitofrontal view or nose-forehead position).

The structures seen in Caldwell's view include:

- Frontal sinuses (seen best)
- Ethmoid sinuses
- Maxillary sinuses
- Frontal process of the zygoma and the zygomatic process of the frontal bone
- Superior margin of the orbit and lamina papyracea
- Superior orbital fissure

- Foramen rotundum (inferolateral to superior orbital fissure)

The nose and forehead are touching the X-ray film and the X-ray beam is projected 15 degrees caudally in Caldwell's view.



Other options:

Option A: Water's view (occipitomeatal view or nose-chin position) is a radiographic view where an X-ray beam is angled at 45 degrees to the orbitomeatal line. Structures seen are the maxillary sinus (best seen), sphenoid sinus (if the film is taken with an open mouth), frontal sinus, zygoma, zygomatic arch, nasal bone, and infratemporal fossa.

Option C: Pierre's view is the occipitomeatal with an open mouth. It helps in visualizing the sphenoid sinus as well, in addition to the sinuses that can be seen in water's view.

Option D: Towne's view is an angled anteroposterior radiograph of the skull. Structures seen are the petrous part of the pyramids, arcuate eminence, superior semicircular canal, mastoid antrum, internal auditory canal, tympanic cavity, cochlea, and external auditory canal. This view is taken for acoustic neuroma and apical petrositis.

Anatomy and Physiology of Pharynx

Question 1:

What is the lower limit of the pharynx?

- a) Upper border of cricoid cartilage
- b) Lower border of cricoid cartilage
- c) Upper border of thyroid cartilage
- d) Lower border of thyroid cartilage

Question 2:

A 55-year-old man is diagnosed with carcinoma limited to the hypopharynx. Which of the following sites will not be involved?

- a) Posterior pharyngeal wall
- b) Epiglottis
- c) Pyriform Sinus
- d) Post-cricoid region

Question 3:

On oral examination of a patient with recurrent upper respiratory tract infection, you notice that the Waldeyer's group of lymph nodes is enlarged. Which of the following is not included in Waldeyer's group of lymph nodes?

- a) Tonsils
- b) Submandibular lymph node
- c) Lingual tonsils
- d) Adenoids

Question 4:

Which of the following is not a part of MALT structures?

- a) Tonsils

- b) Adenoids
- c) Retropharyngeal Lymph nodes
- d) Posterior pharyngeal nodules

Question 5:

You are the junior resident in the ENT department performing a posterior rhinoscopy procedure on a patient. Where will you locate the pharyngeal opening of the eustachian tube?

- a) 2.25cm behind posterior end of the inferior turbinate
- b) 1.25cm behind posterior end of the inferior turbinate
- c) 2.25cm behind posterior end of the middle turbinate
- d) 1.25cm behind posterior end of the middle turbinate

Question 6:

You are observing a tissue sample from the nasopharynx through an electron microscope. Which of the following epithelium will you see?

- a) Squamous epithelium
- b) Cuboidal epithelium
- c) Pseudostratified ciliated columnar epithelium
- d) Transitional epithelium

Question 7:

Which of the following is the most common site for nasopharyngeal carcinoma?

- a) Sinus of Morgagni
- b) Passavant's ridge
- c) Fossa of Rosenmüller
- d) Rathke's pouch

Question 8:

A 50-year-old man has been diagnosed with nasopharyngeal carcinoma which has spread to the sinus of Morgagni. Which of the following structures is least likely to be affected due to

the spread into this space?

- a) Eustachian tube
- b) Tensor veli palatini
- c) Levator veli palatini
- d) Descending pharyngeal artery

Question 9:

Which of the following structures is present in the oropharynx?

- a) Pharyngotympanic tube
- b) Fossa of Rosenmuller
- c) Lingual tonsil
- d) Pyriform fossa

Question 10:

A 50-year-old man has been diagnosed with oropharyngeal carcinoma. Which of the following group of lymph nodes does the sentinel node belong to?

- a) Level IA
- b) Level IB
- c) Level 2
- d) Level 3

Question 11:

Which of the following muscles form the Passavant's ridge?

- a) Stylopharyngeus
- b) Palatopharyngeus
- c) Salpingopharyngeus
- d) Levator veli palatini

Question 12:

A 30-year-old woman suffering from iron deficiency anemia presents to you with difficulty in swallowing. On examination, glossitis and cheilitis are observed. There are oesophageal webs seen on direct laryngoscopy. Which of the following is the most common site for carcinoma to develop in this patient?

- a) Posterior pharyngeal wall
- b) Post cricoid region
- c) Pyriform sinus
- d) Oropharynx

Question 13:

A 30-year-old man presents to you with a foreign body sensation in his throat. On examination, you notice the impaction of a fishbone in his pyriform sinus. Which group of lymph nodes will you find enlarged in case of its infection?

- a) Upper deep cervical nodes
- b) Prelaryngeal nodes
- c) Mediastinal nodes
- d) Posterior triangle nodes

Answer Key

Question No.	Correct Option
1	b
2	b
3	b
4	c
5	b
6	c
7	c
8	d
9	c
10	c
11	b
12	b

Detailed Explanations

Solution to Question 1:

The pharynx extends from the base of the skull (basiocciput and basisphenoid) to the lower border of the cricoid cartilage. It continues as the oesophagus.

Pharynx is a 12-14cm long conical fibromuscular tube forming the common upper part of the air and food passages.

Solution to Question 2:

Epiglottis is part of the supraglottic larynx.

The hypopharynx is made of 3 parts:

- Pyriform sinus
- Posterior Pharyngeal Wall
- Post-cricoid region

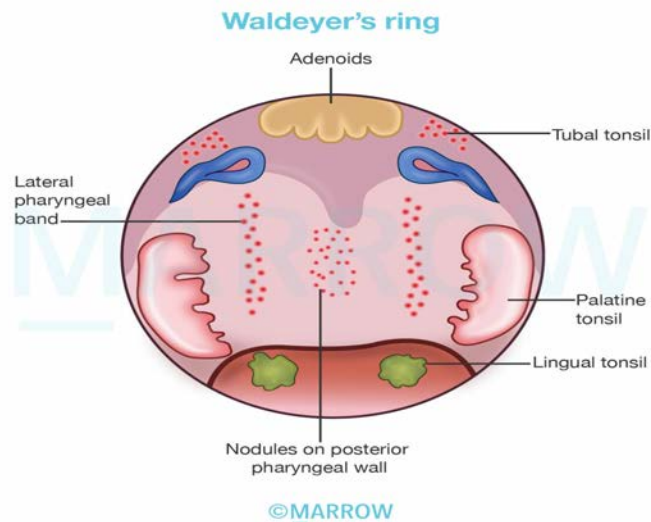
Solution to Question 3:

The submandibular group of lymph nodes is not included as a part of Waldeyer's ring.

Scattered throughout the pharynx in its subepithelial layer is the lymphoid tissue which is aggregated at places to form masses, collectively called Waldeyer's ring.

The masses are:

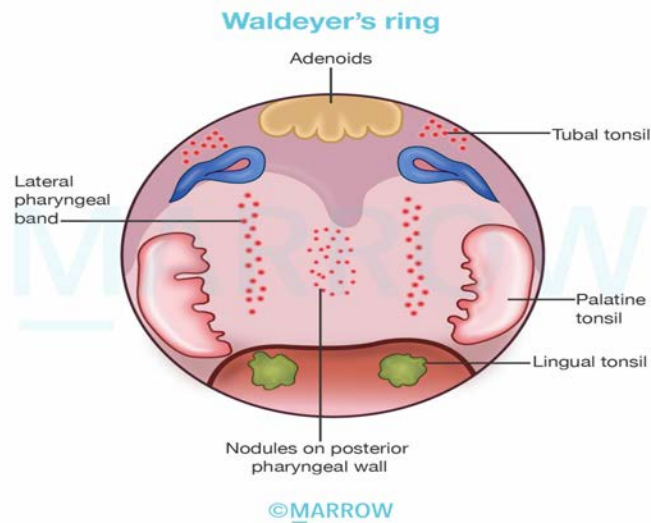
- Adenoids or nasopharyngeal tonsils
- Palatine tonsils or simply tonsils
- Lingual tonsils
- Tubal tonsils
- Lateral pharyngeal bands
- Nodules in the posterior pharyngeal wall



Solution to Question 4:

MALT stands for mucosal associated lymphoid tissue. Retropharyngeal lymph nodes are not a part of it.

The rest of the structures are a part of Waldeyer's ring.



Solution to Question 5:

The pharyngeal opening of the eustachian tube is situated in the lateral wall of the nasopharynx 1.25cm behind the posterior end of the inferior turbinate.

Solution to Question 6:

The nasopharynx is lined by pseudostratified ciliated columnar epithelium. Functionally, it is the posterior extension of the nasal cavity.

Other parts of the pharynx are lined by stratified squamous epithelium.

Solution to Question 7:

Fossa of Rosenmüller is the most common site for carcinoma in the nasopharynx.

Solution to Question 8:

Descending pharyngeal artery does not pass through the sinus of Morgagni and hence least likely to be affected. All the other options are contents of the sinus of Morgagni.

The sinus of Morgagni is a space between the base of the skull and the upper free border of the superior constrictor muscle.

The structures passing through the space are:

- Eustachian tube
- Levator veli palatini
- Tensor veli palatini
- Ascending palatine artery- branch of the facial artery

Trotter's triad or sinus of Morgagni syndrome is seen in nasopharyngeal carcinoma which spreads laterally to involve the sinus of Morgagni. It is characterized by:

- Conductive hearing loss
- Ipsilateral immobility of the palate
- Neuralgic pain

Solution to Question 9:

Lingual tonsils are lymphoid tissues seen in the base of the tongue, hence a part of the oropharynx.

Oropharynx includes:

- Base of tongue
- Palatine tonsils

- Valleculla
- Anterior and posterior pillars
- Posterior pharyngeal wall

The pharyngeal end of the pharyngotympanic tube and fossa of Rosenmuller is present in the lateral wall of the nasopharynx.

The pyriform fossa is a part of the hypopharynx.

Solution to Question 10:

The jugulodigastric node is the sentinel lymph node and belongs to the level 2 group of lymph nodes.

This is the node enlarged in both tonsillitis and base of tongue carcinoma as lymphatics from the oropharynx drain into the upper jugular chain.

Level	Lymph node
1a	Submental
1b	Submandibular
2	Upper jugular
3	Middle jugular
4	Lower jugular
5	Posterior triangle
6	Anterior group
7	Mediastinal

Solution to Question 11:

Passavant's ridge is a mucosal ridge raised by the fibers of the palatopharyngeus.

It encircles the posterior and lateral walls of nasopharyngeal isthmus. Soft palate, during its contraction makes firm contact with the ridge to cut off nasopharynx from the oropharynx during deglutition or speech.

Solution to Question 12:

The given clinical picture is suggestive of a diagnosis of Plummer-Vinson syndrome. The post-cricoid region is a common site for carcinoma in women suffering from Plummer-Vinson syndrome.

Plummer-Vinson syndrome/ Patterson Kelly Brown syndrome is a condition that can occur in people with long-term (chronic) iron deficiency anemia. The triad seen in these patients:

- Post cricoid webs
- Glossitis
- Angular cheilitis

Post cricoid region is the part of the anterior wall of the laryngopharynx between the upper and lower borders of the cricoid lamina.

Solution to Question 13:

Pyriform sinus is richly supplied by lymphatics and drains into the upper deep cervical group of lymph nodes.

The lymphatics exit through the thyrohyoid membrane to drain into the upper jugular chain.

Pyriform sinus lies on either side of the larynx and extends from the pharyngoepiglottic fold to the upper end of the esophagus. It forms the lateral channel for food. The rich lymphatic network of pyriform fossa explains the high frequency of nodal metastasis seen in carcinoma of this region.

Adenoids

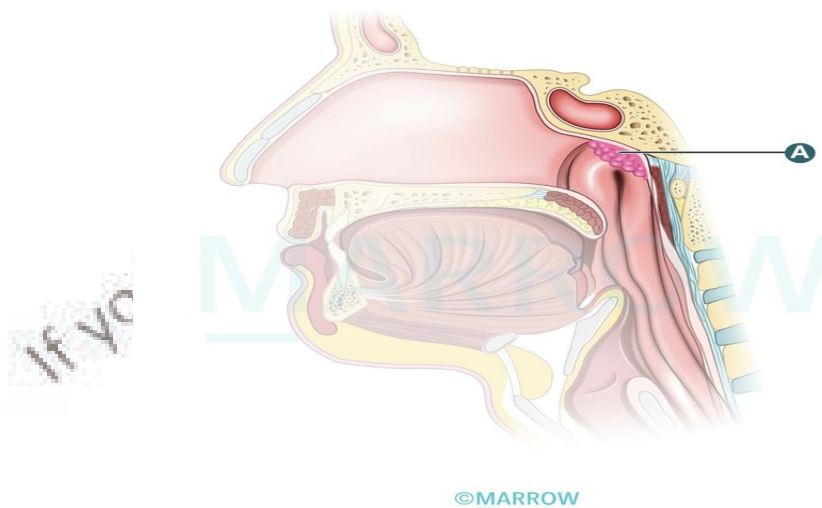
Question 1:

What are adenoids also known as?

- a) Nasopharyngeal tonsil
- b) Palatine tonsil
- c) Faucial tonsil
- d) Lingual tonsil

Question 2:

Which of the following statements is false about the structure marked as A?



- a) It is part of the Waldeyer's ring
- b) Hypertrophies up to the age of 6 years and starts to regress from puberty
- c) Hypertrophy causes rhinolalia clausa
- d) It is surrounded by a capsule

Question 3:

Which of the following is not included in the signs and symptoms of the condition represented below?



- a) Mouth breathing
- b) Failure to thrive
- c) Sleep apnoea
- d) Prominent nasolabial fold

Question 4:

A 5-year-old girl is brought to you by her mother with complaints of mouth breathing. She has a flat, expressionless long face with an open mouth. On examination, a high arched palate and crowded upper teeth are noted. What is the voice abnormality that can be noted in this child?



- a) Hot potato voice
- b) Rhinolalia clausa
- c) Rhinolalia aperta
- d) Staccato voice

Question 5:

A 5-year-old child is brought by his mother to the ENT OPD with frequent upper respiratory tract infections. His mother gives a history of loud snoring at night and posterior rhinoscopy examination shows enlarged adenoids. Which of the following will you not advise in the management of this patient?

- a) Decongestants
- b) Antihistaminics
- c) Adenoidectomy
- d) Beta 2 agonists

Question 6:

Which of the following is not an indication for an adenoidectomy procedure?

- a) Multiple adenoids
- b) Recurrent upper respiratory tract infections
- c) Recurrent middle ear infections with deafness

d) Speech abnormality

Question 7:

A 6-year-old child has been posted for an adenoidectomy procedure. Which of the following conditions if present in this child will be a contraindication to performing the surgery?

- a) Large adenoids**
- b) Large tonsils**
- c) Cleft lip**
- d) Cleft palate**

Question 8:

A school-going child presents with a history of recurrent upper respiratory tract infections, mouth breathing, and impaired hearing. On examination, a high arched palate is noted. What is the management of this condition?

- a) Tonsillectomy**
- b) Adenoidectomy**
- c) Adenoidectomy with grommet insertion**
- d) Myringotomy with grommet insertion**

Question 9:

An 8-year-old boy diagnosed with adenoid hypertrophy has undergone an adenoidectomy procedure. Two weeks following the procedure, he developed Grisel syndrome. What is the reason for this condition?

- a) Damage to upper incisors**
- b) Temporomandibular joint dislocation**
- c) Paraspinal muscle spasm**
- d) Damage to lower incisors**

Question 10:

Where is Thornwaldt cyst located?

- a) Ear
- b) Nasopharynx
- c) Oropharynx
- d) Piriform fossa

Question 11:

A 30-year-old patient has a Thornwaldt abscess that got infected from a Thornwaldt's cyst. Which of the following will you not recommend for the treatment of his condition?

- a) Antibiotics
- b) Marsupialisation
- c) Removal of cyst
- d) Anti tubercular treatment

Answer Key

Question No.	Correct Option
1	a
2	d
3	d
4	b
5	d
6	a
7	d
8	c
9	c
10	b
11	d

Detailed Explanations

Solution to Question 1:

Adenoids are commonly known as nasopharyngeal tonsil. It is situated at the junction of the roof and posterior wall of the nasopharynx.

Solution to Question 2:

The structure marked A is the adenoids. It is not surrounded by a capsule.

Adenoids are comprised of vertical ridges of lymphoid tissues separated by deep clefts. They form the superior aspect of Waldeyer's ring. They have no crypts and no capsule, unlike palatine tonsils which have both.

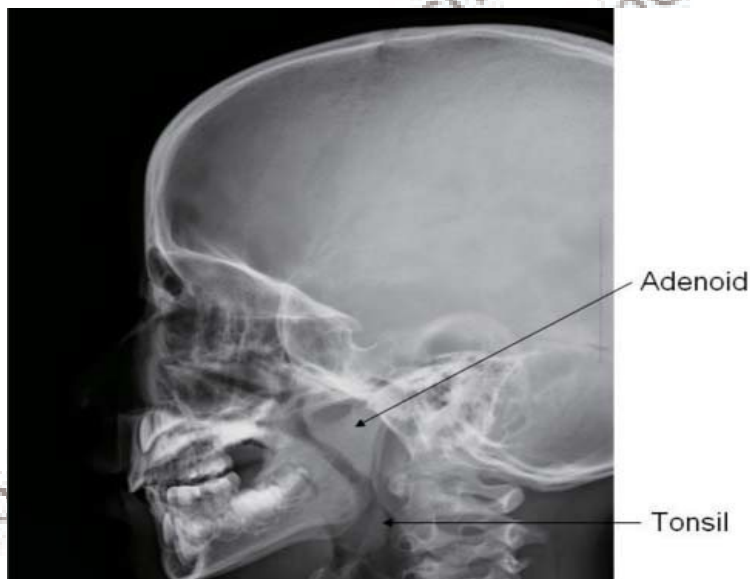
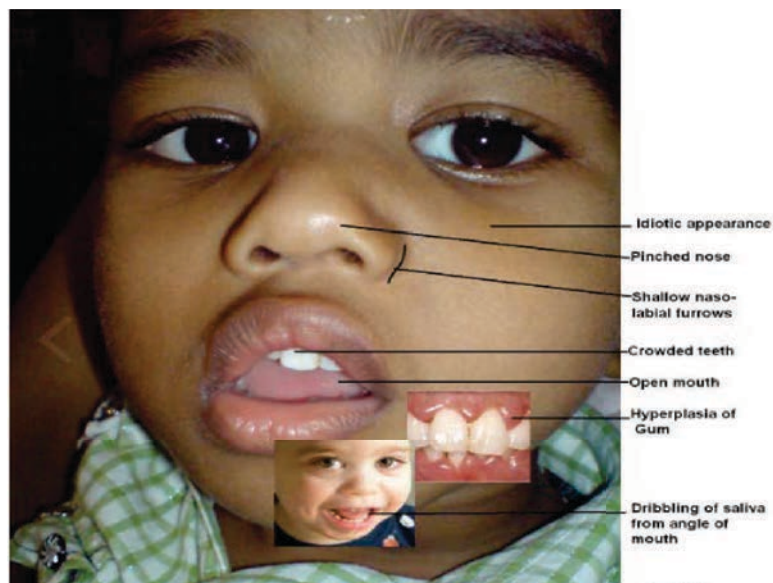
It is present at birth, enlarges up to the age of 6 years, and then starts regressing from puberty. It completely disappears by the age of 20 years.

Solution to Question 3:

The given x-ray shows enlarged adenoids and suggests the diagnosis of adenoid hypertrophy. Patients show an absent nasolabial fold.

Clinical features:

- Nasal obstruction is the most common symptom. This leads to mouth breathing. Nasal obstruction also interferes with feeding or suckling in a child. As respiration and feeding cannot take place simultaneously, a child with adenoid enlargement fails to thrive.
- Nasal discharge, sinusitis, and epistaxis.
- Bilateral conductive hearing loss and serous otitis media due to eustachian tube obstruction.
- Rhinolalia clausa - hyponasal voice.
- Adenoid facies - elongated face with a dull expression, pinched up nose, absent nasolabial crease, open mouth, prominent and crowded upper teeth.
- Sleep apnoea - chronic cases lead to hypoxemia and pulmonary vasoconstriction. This consequently leads to pulmonary hypertension and cor pulmonale.
- Aproxexia - lack of concentration.



Solution to Question 4:

The given clinical picture of a child with adenoid facies is suggestive of a diagnosis of adenoid hypertrophy. Rhinolalia clausa/ hyponasal voice can be seen in these patients.

It is also seen in conditions like common cold, nasal allergy, etc.

Option A - Hot potato voice is a muffled and thick speech often seen in a peritonsillar abscess (quinsy).

Option C - Rhinolalia aperta causes a hypernasal voice.

Option D - Staccato voice or scanning speech is a broken speech seen in cerebellar lesions as well as multiple sclerosis.

Solution to Question 5:

Beta 2 agonists have no role in conditions of upper airway obstruction like adenoid hypertrophy. They are bronchodilators used in lower airway obstructive diseases.

When symptoms of adenoid hypertrophy are not marked, breathing exercises, decongestant nasal drops, and antihistamines for any coexisting nasal allergy can cure the condition without resorting to surgery.

Marked adenoid hypertrophy with associated complications can be treated with adenoidectomy and grommet insertion.

Solution to Question 6:

Multiple adenoids is not an indication for adenoidectomy.

Indications of adenoidectomy:

- Adenoid hypertrophy causing snoring, mouth breathing, sleep apnoea syndrome or speech abnormalities, i.e. (rhinolalia clausa)
- Recurrent rhinosinusitis
- Chronic otitis media with effusion associated with adenoid hyperplasia
- Recurrent ear discharge in benign CSOM associated with adenoiditis/adenoid hyperplasia
- Dental malocclusion. Adenoidectomy does not correct dental abnormalities but will prevent its recurrence after orthodontic treatment

Solution to Question 7:

Cleft palate is a contraindication to perform adenoidectomy as removal of adenoids causes velopharyngeal insufficiency.

Velopharyngeal insufficiency is a condition where the soft palate does not close tightly against the back of the throat. This leads to air coming out the nose (characterized by hypernasality) during speech.

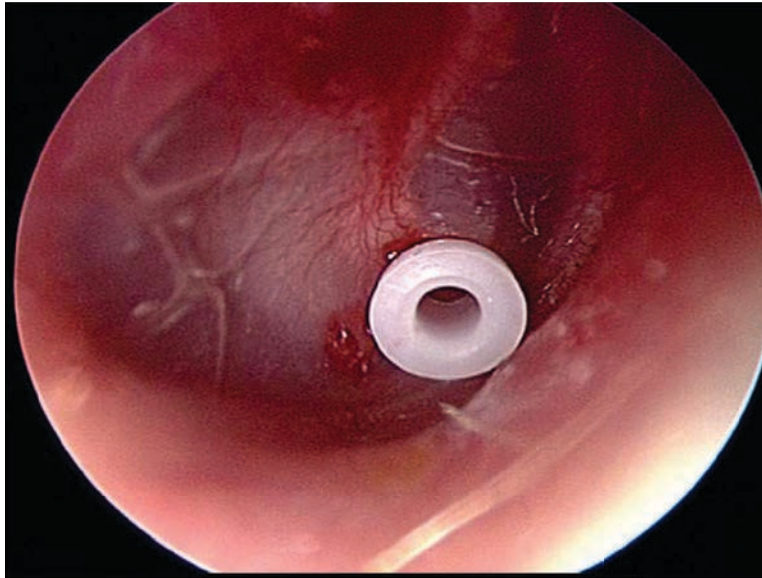
Contraindications for adenoidectomy:

- Cleft palate or submucous palate.
- Hemorrhagic diathesis.
- Acute infection of the upper respiratory tract.

Solution to Question 8:

The given clinical picture is suggestive of adenoid hypertrophy. The best line of management is adenoidectomy with grommet insertion.

A grommet is inserted to provide continued aeration of middle ear and used in cases where fluid collection recurs.



Solution to Question 9:

Grisel syndrome involves paraspinal muscle spasm that occurs post adenoidectomy.

Grisel syndrome refers to non-traumatic atlantoaxial joint subluxation. The patient complains of neck pain and develops torticollis. It is mostly due to spasm of paraspinal muscles but can also be due to atlantoaxial dislocation requiring cervical collar and even traction.

Solution to Question 10:

Thornwaldt cyst is located in the midline of the posterior wall of the nasopharynx in the adenoid mass.

It is an epithelial-lined median recess found within the adenoid mass and extends from pharyngeal mucosa to the periosteum of the basiocciput. It represents the attachment of notochord to the pharyngeal endoderm during embryonic life.

When infected, it may be the cause of persistent postnasal discharge or crusting. Sometimes an abscess can form in the bursa (Thornwaldt's disease). It is managed surgically by marsupialization of the swelling and complete removal of the lining of the cyst.



Solution to Question 11:

Antitubercular treatment has no role in the treatment of Thornwaldt's abscess.

Thornwaldt's abscess is formed due to the infection of Thornwaldt cyst. Treatment includes administration of antibiotics to treat the infection and marsupialization of the cystic swelling. There should be adequate removal of its lining membrane.

Tonsils

Question 1:

Which of the following statements is incorrect about tonsils?

- a) Bed of the tonsil is formed by the superior constrictor and styloglossus muscles.
- b) It is covered by keratinizing stratified squamous epithelium on the medial surface.
- c) Crypta magna is a remnant of the ventral part of second pharyngeal pouch.
- d) A capsule covers the lateral surface of the tonsils.

Question 2:

At what age do the tonsils attain their maximum size?

- a) 3 yr
- b) 5 yrs
- c) 8 yrs
- d) 12 yrs

Question 3:

A patient diagnosed with oral carcinoma was referred to the ENT department for evaluation. While examining the oral cavity, the consultant asked you to examine the ENT surgeon's graveyard to check for any induration or ulcers. What is the area being referred to here?

- a) Anterior tonsillar fossa
- b) Superior tonsillar recess
- c) Tonsillolinguual sulcus
- d) Lingual tonsils

Question 4:

Which of the following is the main arterial supply of palatine tonsils?

- a) Tonsillar branch of the facial artery
- b) Descending palatine branch of maxillary artery

- c) Buccal branch of maxillary artery
- d) Ascending palatine branch of facial artery

Question 5:

You are the second-year ENT resident performing tonsillectomy in a patient with chronic tonsillitis. While creating a dissection plane between the capsule and the bed of the tonsil, you accidentally deepened the incision, injuring the tonsillar bed. Which of the following structures is least likely to be injured?

- a) Glossopharyngeal nerve
- b) Styloid process
- c) Submandibular salivary gland
- d) Lateral pterygoid muscle

Question 6:

A 7-year-old child was brought to the hospital by his mother with complaints of a sore throat, earache, and pain on swallowing for three days. On examination, the tonsils were enlarged and hyperaemic. Which of the following is the most common organism implicated in this condition?

- a) Staphylococcus aureus
- b) β -haemolytic streptococcus
- c) Haemophilus influenza
- d) Streptococcus pneumoniae

Question 7:

A young lady presents to the ENT OPD with the finding shown below. Which of the following group of lymph nodes will primarily enlarge in this condition?



- a) Jugulo-digastric
- b) Jugulo-omohyoid
- c) Submandibular
- d) Retropharyngeal

Question 8:

A middle-aged man presented with a history of fever, sore throat, and dysphagia for three days. Which of the following variants is not expected to be seen on oral examination?

- a) Follicular tonsillitis
- b) Membranous tonsillitis
- c) Fibroid tonsillitis
- d) Parenchymatous tonsillitis

Question 9:

An anxious mother brought her 4-year-old-child to the hospital with complaints of sudden onset fever and refusal to eat. On examination, the given finding was noted and he was started on analgesics and antibiotics. Which among the following is not a known complication of this condition?



- a) Excessive bleeding
- b) Acute otitis media
- c) Rheumatic fever
- d) Deep neck space infections

Question 10:

Which of the following nerves is responsible for referred otalgia from tonsillitis?

- a) Glossopharyngeal nerve
- b) Facial nerve
- c) Trigeminal nerve
- d) Vagus nerve

Question 11:

A 15-year-old girl presented to the ENT surgeon with a history of loud snoring, restless sleep leading to daytime somnolence, and poor academic performance. Her medical history was unremarkable, except for recurrent episodes of acute tonsillitis. Which of the following types of tonsillitis may be implicated in this condition?

- a) Acute membranous
- b) Acute follicular
- c) Chronic fibroid

- d) Chronic parenchymatous**

Question 12:

You are an intern posted in the ENT department working up a case of chronic follicular tonsillitis. Which of the following clinical signs would you not expect to see in this patient on examining the oral cavity?

- a) Irwin-Moore sign**
- b) Enlarged jugulo-digastric nodes**
- c) Congested anterior pillars**
- d) Müller's sign**

Question 13:

An ENT junior resident observed a symptomless yellowish swelling over the tonsil when he was examining a patient for a buccal sore. The swelling was easily drained. What is the likely diagnosis?

- a) Intratonsillar abscess**
- b) Tonsillolith**
- c) Tonsillar cyst**
- d) Follicular tonsillitis**

Question 14:

On examining the oral cavity of a patient who presented with fever and a sore throat, you notice a greyish-white membrane over the tonsils. Which of the following conditions would you not consider as a differential diagnosis?

- a) Streptococcal tonsillitis**
- b) Diphtheria**
- c) Ludwig's angina**
- d) Infectious mononucleosis**

Question 15:

Which of the following organisms is implicated in Vincent's angina?

- a) β -haemolytic streptococci
- b) *Bacillus fusiformis*
- c) *Treponema pallidum*
- d) Microaerophilic streptococci

Question 16:

A worried mother brought her 6-year-old unimmunized child to the ER with complaints of fever, cough, and shortness of breath. On examination, a pseudomembrane was noted in the oral cavity along with the neck finding as shown in the image. Which of the following statements is true regarding the membrane observed?



- a) Produced by the endotoxin
- b) Firmly adherent and bleeds on removal
- c) Confined to the tonsil
- d) Pearly white in color

Question 17:

Which of the following complications would you not expect to see in a patient diagnosed with diphtheria?

- a) Myocarditis

- b) Arrhythmias
- c) Airway obstruction
- d) Hard palate paralysis

Question 18:

Following finding was noted in a patient who presented with difficulty breathing and a bull-neck appearance. Which of the following statements is true regarding the treatment of this patient?



- a) Treatment is started after the culture comes positive
- b) Treatment is with penicillin alone
- c) Antitoxin is reserved for complication
- d) Sensitivity to antitoxin should be tested prior to therapy

Question 19:

Which of the following cases cannot be posted for an elective tonsillectomy?

- a) A 7-year-old child with acute tonsillitis
- b) A young adult with chronic tonsillitis refractory to treatment
- c) A patient with unilateral enlargement of tonsil with suspected malignancy
- d) A middle-aged man with recurrent quinsy

Question 20:

Which of the following is not a contraindication to perform a tonsillectomy?

- a) Acute tonsillitis
- b) Bleeding disorders
- c) Cleft palate
- d) Diphtheria carrier

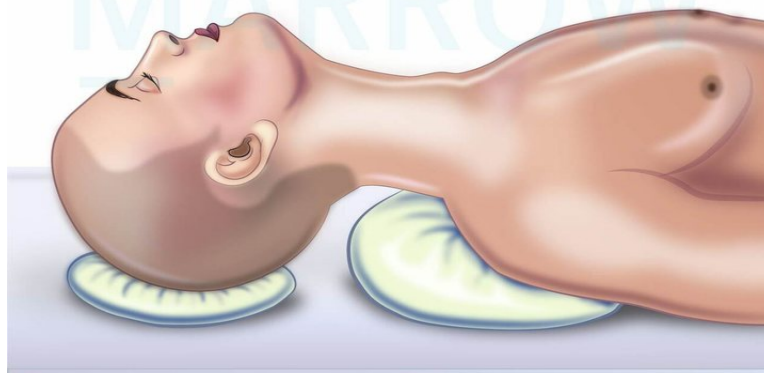
Question 21:

A patient develops bleeding from the surgical site five hours after an elective tonsillectomy. Which among the following would best describe the condition?

- a) Primary haemorrhage
- b) Secondary haemorrhage
- c) Reactionary haemorrhage
- d) Tertiary haemorrhage

Question 22:

While preparing for a procedure, the patient was kept in a specific position as depicted in the image. In which of the following procedures is this position not adopted?



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- a) Tonsillectomy

- b) Adenoidectomy
- c) Direct laryngoscopy
- d) Tracheostomy

Answer Key

Question No.	Correct Option
1	b
2	d
3	c
4	a
5	d
6	b
7	a
8	c
9	a
10	a
11	d
12	d
13	c
14	c
15	b
16	b
17	d
18	d
19	a
20	d
21	c
22	c

Detailed Explanations

Solution to Question 1:

The medial surface of the tonsil is covered by non-keratinizing stratified squamous epithelium.

Tonsils are ovoid masses of lymphoid tissue situated in the lateral wall of the oropharynx between the anterior and posterior pillars.

The medial surface of the tonsil is covered in 12-15 crypts. Crypta magna or intratonsillar cleft is the largest & deepest of the crypts. It is situated near the upper part of the tonsil and represents the ventral part of the second pharyngeal pouch.

The lateral surface of the tonsil presents a well-defined fibrous capsule.

The bed of the tonsil is formed by the superior constrictor and styloglossus muscles.

Solution to Question 2:

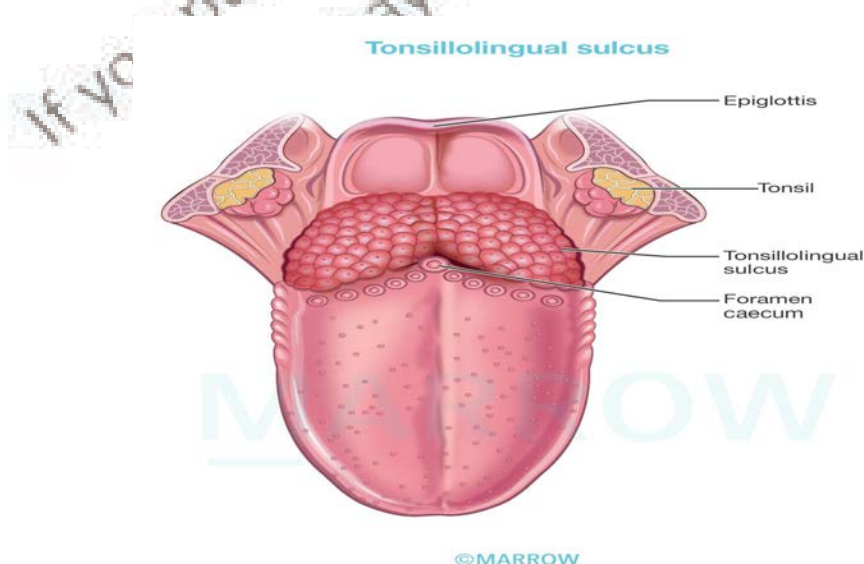
The growth of tonsils is rapid around the age of 5-6 years of age and usually reaches their maximum size by 12 years of age.

Tonsillar involution begins at puberty when the reactive lymphoid tissue begins to atrophy, and by old age, only a little tonsillar lymphoid tissue remains.

Solution to Question 3:

The area referred to here is the tonsillolingual sulcus which is known as the graveyard of ENT surgeons.

The palatine tonsil is separated from the tongue by a sulcus called the tonsillolingual sulcus, which may be the seat of carcinoma. This area is easily missed during oral examination hence known as the graveyard of ENT surgeons.

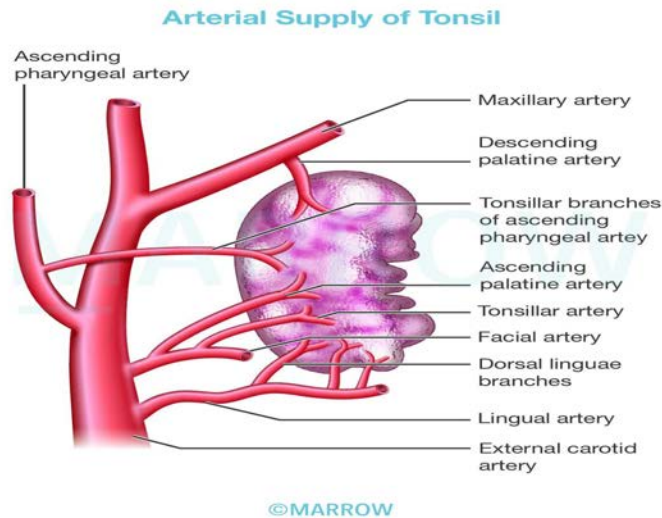


Solution to Question 4:

The main arterial supply of tonsil is by the tonsillar branch of the facial artery.

Five arteries supply the tonsil:

- Tonsillar branch of the facial artery (the main artery)
- Ascending palatine branch of the facial artery
- Ascending pharyngeal branch of the external carotid artery
- Dorsal lingual branches of the lingual artery
- Descending palatine branches of the maxillary artery



Solution to Question 5:

The lateral pterygoid muscle is not related to the bed of the tonsil and is the least likely structure to be injured in this scenario.

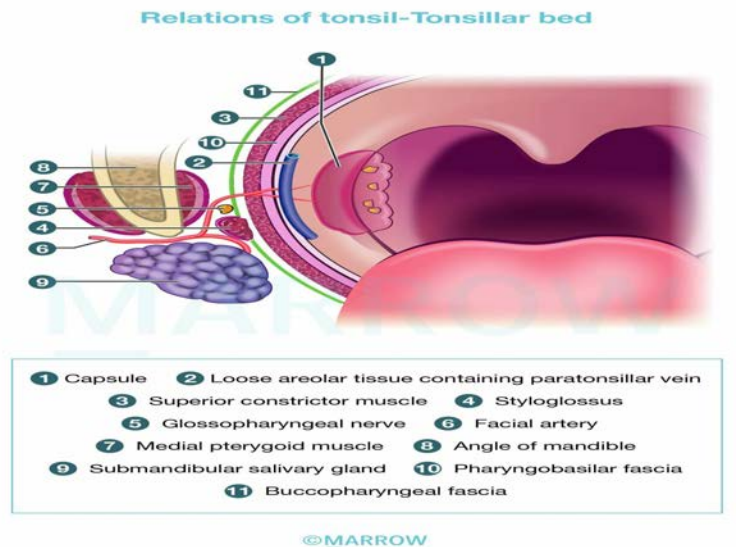
The bed of the tonsil is formed by the superior constrictor and styloglossus muscles.

The lower fossa is related to:

- The glossopharyngeal nerve
- The styloid process

Outside the superior constrictor, the tonsil is related to:

- The facial artery
- Submandibular salivary gland
- Posterior belly of the digastric muscle
- Medial pterygoid muscle
- The angle of the mandible



Solution to Question 6:

The given clinical scenario is suggestive of acute tonsillitis. The most common organism causing acute tonsillitis in children is β hemolytic streptococcus.

Other causative organisms include:

- Staphylococcus aureus
- Streptococcus pneumoniae
- Haemophilus influenzae

Solution to Question 7:

The image shows purulent inflammation of the tonsils suggestive of acute tonsillitis. Jugulo-digastric (tonsillar) lymph nodes will primarily enlarge in this condition.

The lymphatics from the palatine tonsils pierce the superior constrictor muscle and primarily drain into the upper deep cervical nodes (level II), particularly the jugulo-digastric (tonsillar) node situated below the angle of the mandible. So, it is the most likely group of lymph nodes to enlarge in case of acute tonsillitis.

Solution to Question 8:

The patient's history is suggestive of acute tonsillitis. The fibroid variety is a type of chronic tonsillitis and is not expected to be seen in this case.

Depending on the pathology within the tonsillar tissue, acute tonsillitis can be classified into the following types:

- Acute catarrhal or superficial tonsillitis: Here tonsillitis is a part of generalized pharyngitis and is primarily seen in viral infections.
- Acute follicular tonsillitis: The infection spreads into the crypts, which become filled with purulent material, presenting at the openings of crypts as yellowish spots.
- Acute parenchymatous tonsillitis: Here tonsil substance is affected. Tonsil is uniformly enlarged and red.
- Acute membranous tonsillitis: It is a stage ahead of acute follicular tonsillitis when exudation from the crypts coalesces to form a membrane on the surface of the tonsil.

Types of chronic tonsillitis:

- Chronic follicular tonsillitis
- Chronic parenchymal tonsillitis
- Chronic fibroid tonsillitis

Solution to Question 9:

The given clinical vignette is suggestive of acute tonsillitis. Excessive bleeding is not a known complication of this condition.

Complications of acute tonsillitis:

- Chronic tonsillitis - chronic infection may persist in lymphoid follicles of the tonsil in the form of microabscesses
- Peritonsillar abscess
- Parapharyngeal abscess
- Cervical abscess due to suppuration of jugulodigastric lymph nodes
- Acute otitis media
- Rheumatic fever - often seen in association with tonsillitis due to group A β -hemolytic streptococci
- Acute glomerulonephritis (rare)
- Subacute bacterial endocarditis - acute tonsillitis in a patient with valvular heart disease may be complicated by endocarditis. It is usually due to *Streptococcus viridans* infection

Solution to Question 10:

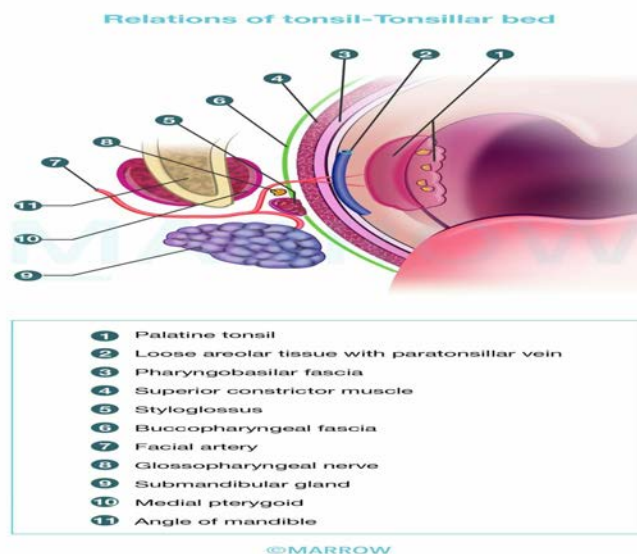
The tympanic branch of the glossopharyngeal nerve (cranial nerve IX) is responsible for referred otalgia from tonsillitis.

The tonsils and tonsillar fossa are supplied by the glossopharyngeal nerve. The glossopharyngeal nerve and the styloid process lie in relation to the lower part of the tonsillar fossa. Therefore, any irritation or pain can be referred to the ear along the tympanic branch of the glossopharyngeal (Jacobson's) nerve.

Causes of referred otalgia via the glossopharyngeal nerve include acute pharyngitis, tonsillitis, peritonsillar abscess, post-tonsillectomy, foreign bodies embedded in the base of the tongue or tonsillar area and an elongated styloid process (Eagle's syndrome).

The ear receives extensive sensory innervation via four cranial nerves (V, VII, IX, and X) and two spinal segments (C2 and C3).

The image below shows the anatomic relations of the tonsil and tonsillar bed:



Solution to Question 11:

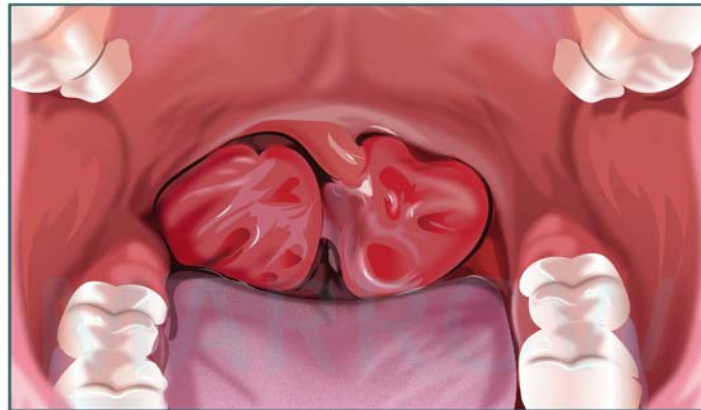
The given clinical presentation is suggestive of obstructive sleep apnea syndrome(OSAS) secondary to chronic parenchymatous tonsillitis.

In parenchymatous tonsillitis, the tonsils grow very large following repeated attacks of acute tonsillitis and lead to obstructive symptoms, including OSAS.

Types of chronic tonsillitis:

- Chronic follicular tonsillitis- tonsillar crypts are full of infected cheesy material, which shows on the surface as yellowish spots.
- Chronic parenchymatous tonsillitis- there is hyperplasia of lymphoid tissue. Tonsils are very much enlarged and may interfere with speech, deglutition, and respiration. Attacks of sleep apnoea may occur. Long-standing cases develop features of cor pulmonale.
- Chronic fibroid tonsillitis- tonsils are tiny but infected, with a history of repeated sore throats.

Chronic parenchymatous tonsillitis



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Solution to Question 12:

Müller's sign is the pulsation of uvula during systole seen in aortic regurgitation. It is not seen in patients with chronic tonsillitis.

The classical signs of chronic tonsillitis are:

- Irwin-Moore sign- tonsils are tiny, but pressure on the anterior pillar expresses frank pus or cheesy material.
- Bluish Flushing of anterior pillars compared to the rest of the pharyngeal mucosa is a vital sign of chronic tonsillar infection.
- Enlargement of jugulo-digastric lymph nodes even in the absence of acute infection is a reliable sign of chronic tonsillitis. During acute attacks, the nodes enlarge further and become tender.

Solution to Question 13:

The given clinical scenario is suggestive of a tonsillar cyst. It is a condition due to blockage of a tonsillar crypt and appears as a yellowish swelling over the tonsil. Very often, it is symptomless and can be easily drained.

Option A: Intratonsillar abscess is an accumulation of pus within the substance of the tonsil. It usually follows a blockage of the opening of the crypt in acute follicular tonsillitis. There is marked local pain and dysphagia. Tonsil appears swollen and red.

Option B: Tonsilloliths (calculus of the tonsil) are seen in chronic tonsillitis when its crypt is blocked with retention of debris. Inorganic salts of calcium and magnesium are then deposited leading to the formation of a stone. It may gradually enlarge and then ulcerate through the tonsil. It generally causes local discomfort or foreign body sensation.

Option D: In follicular tonsillitis, purulent material is present at the opening of the crypts as yellow dots. It presents with fever, dysphagia, and sore throat.

Solution to Question 14:

Ludwig's angina is not associated with the formation of a grey-white membrane. It is characterized by rapidly spreading infection of the sublingual space initially and later the submaxillary space, often dentogenic in origin.

D/D for membrane over tonsil:

- Streptococcus & other pyogenic organisms
- Diphtheria
- Infectious mononucleosis
- Vincent's angina
- Leukemia
- Candida
- Agranulocytosis
- Malignancy
- Aphthous ulcer
- Traumatic ulcer

Solution to Question 15:

Vincent's angina is caused by fusiform bacilli and spirochaetes, namely *Bacillus fusiformis* and *Borrelia vincenti*.

The condition presents with low-grade fever and sore throat, and a membrane over the tonsil. The membrane is easily removed to reveal irregular ulcers over the tonsil. Throat swab will show both the organisms typical of the disease, namely fusiform bacilli and spirochaetes.

Solution to Question 16:

The given clinical history and bull-neck appearance (image) point to the diagnosis of diphtheria. The diphtheritic membrane is firmly adherent and bleeds on removal.

A greyish-white membrane forms over the tonsils in the oropharynx and spreads to the soft palate and posterior pharyngeal wall. The membrane in diphtheria extends beyond the tonsils onto the soft palate and is dirty grey. It is adherent, and its removal leaves a bleeding surface. In the larynx, the diphtheritic membrane may cause airway obstruction.

Cervical lymph nodes, particularly the jugulodigastric, become enlarged and tender, sometimes presenting a bull-neck appearance. The patient is ill and toxæmic, but fever seldom rises above 38°C.

Solution to Question 17:

Hard palate paralysis is not seen in diphtheria.

The exotoxin produced by *C. diphtheriae* is toxic to the heart and nerves. It causes:

- Myocarditis
- Cardiac arrhythmias
- Acute circulatory failure

Neurological complications include paralysis of the soft-palate, diaphragm, and ocular muscles. In the larynx, the diphtheric membrane may cause airway obstruction.

Solution to Question 18:

The pseudomembrane on oral examination and bull neck appearance is characteristic of diphtheria. Sensitivity to diphtheria antitoxin is common and should be tested prior to the initiation of therapy.

Treatment is started on clinical suspicion without waiting for the culture report. The aim is to neutralize the free exotoxin still circulating in the blood and to kill the organisms producing this exotoxin.

Antitoxins:

- Dose- based on the site involved, duration, and severity of the disease
- If < 48 hours or just confined to tonsils: 20,000–40,000 units
- If > 48 hours or extensive membrane: 80,000–120,000 units
- Given by i.v. infusion in saline in about 60 min
- Sensitivity tested by a conjunctival or intracutaneous test with diluted antitoxin
- If a hypersensitivity reaction occurs, desensitization should be done

Antibiotics:

- Benzyl penicillin 600 mg 6 hourly for 7 days
- Erythromycin is used in penicillin-sensitive individuals (500 mg 6 hourly orally)

Solution to Question 19:

Acute tonsillitis is a contraindication for tonsillectomy and hence, cannot be posted for elective surgery.

Acute tonsillitis or any upper respiratory tract infection is a contraindication for tonsillectomy since there is an increased chance of hemorrhage and sepsis.

Indications of tonsillectomy are:

- Recurrent episodes of acute tonsillitis – most common indication

Seven or more episodes in 1 year, or

Five episodes per year for 2 years, or

Three episodes per year for 3 years, or

Two weeks or more of lost school or work in 1 year

- Febrile seizures due to fever in tonsillitis
- Peritonsillar abscess (quinsy) - recurrent episodes in adults and a single episode in children
- Hypertrophy of tonsils causing airway obstruction, difficulty in deglutition, interference with speech
- Unilateral enlargement of tonsil with suspicion of malignancy
- Chronic tonsillitis with bad taste or halitosis which is unresponsive to medical treatment (relative indication)

Solution to Question 20:

Diphtheria and streptococcal carriers are relative indications for tonsillectomy and the surgery can be safely performed in patients presenting with the same.

Contraindications for tonsillectomy:

- Haemoglobin level less than 10 g%.
- Presence of acute infection in upper respiratory tract, even acute tonsillitis. Bleeding is more in the presence of acute infection.
- Children under 3 years of age.
- Overt or submucous cleft palate.
- Bleeding disorders, e.g. leukaemia, purpura, aplastic anaemia or haemophilia.
- At the time of epidemic of polio.
- Uncontrolled systemic disease, e.g. diabetes, cardiac disease, hypertension or asthma.
- Tonsillectomy is avoided during the period of menses.

Solution to Question 21:

Bleeding from the surgical site within a period of 24 hours is categorised as a reactionary haemorrhage.

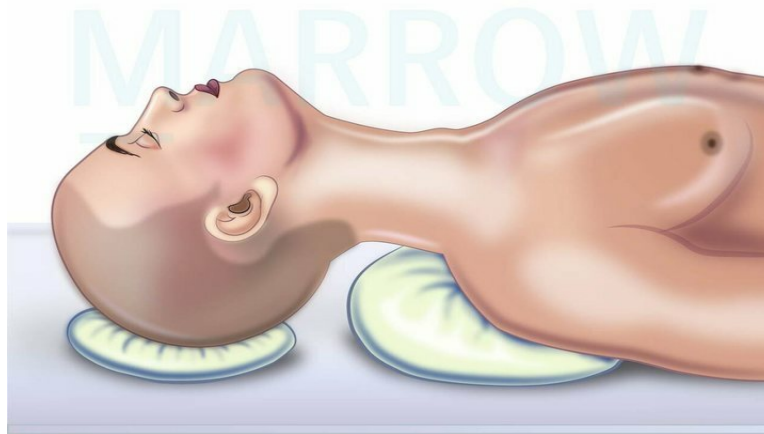
Primary haemorrhage - occurs at the time of operation.

Secondary haemorrhage - usually seen between the fifth to tenth postoperative days.

Solution to Question 22:

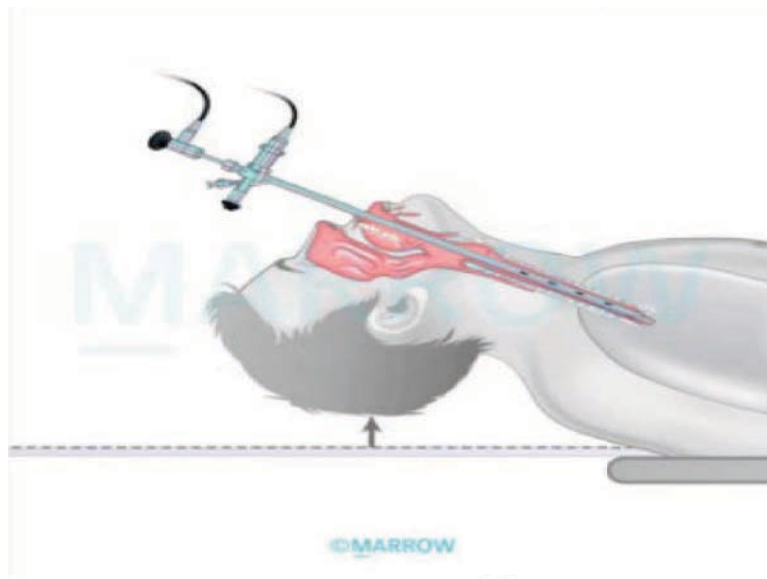
The given image shows Rose's position and it is not a suitable position to perform a direct laryngoscopy. The barking-dog position is used for direct laryngoscopy.

Rose's position: The patient lies supine with head extended by placing a pillow under the shoulders. It is followed in adenoidectomy, tonsillectomy, and tracheostomy.



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Barking-dog position: Patient lies supine. The Head is elevated by 10–15 cm by placing a pillow under the occiput or raising the operation table's head flap. The neck is flexed on the thorax and the head is extended on the atlanto-occipital joint.



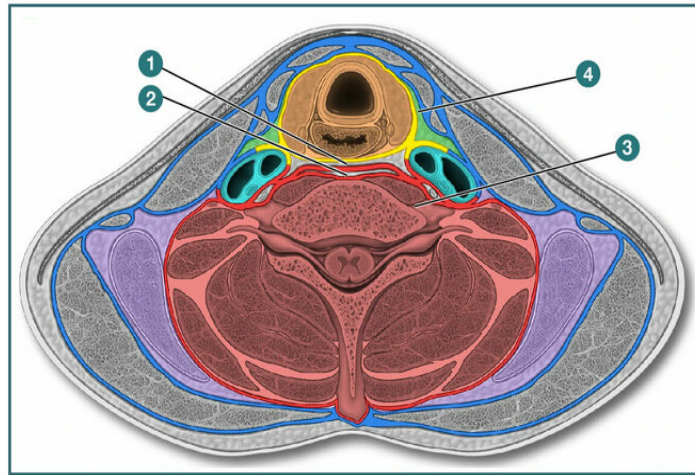
else,

Sold by @Itachi,
If you purchased this from some
you may have been scammed.

Abscesses of Pharynx

Question 1:

Which of the following marked structures is the space of Gillette?



- a) 1
- b) 2
- c) 3
- d) 4

Question 2:

An ENT surgeon is performing an incision and drainage for a retropharyngeal abscess due to suppurative lymphadenitis. Which among the following groups of lymph nodes is most likely to be involved?

- a) Rouvier node
- b) Delphian node
- c) Virchow's node
- d) Cloquet's node

Question 3:

A concerned mother brought her toddler to the ER with complaints of noisy breathing and the child refusing to feed. On examination, the child has adenoid facies and there is a bulge in the posterior pharyngeal wall lateral to the midline. Which of the following is a false statement regarding the disease described above?

- a) Suppuration of nodes of Rouvier can cause the above condition
- b) Caries of the cervical spine is the most common cause
- c) In adults similar condition may result from penetrating injury of the cervical esophagus
- d) Torticollis is one of the symptoms.

Question 4:

A 42-year-old HIV positive man presents with an acute onset high-grade fever with chills and shortness of breath. On evaluation, he's diagnosed to have a necrotising mediastinitis secondary to rupture of a retropharyngeal abscess. What is the likely route of spread of infection to the mediastinum?

- a) Between buccopharyngeal fascia and alar fascia.
- b) Between alar fascia and prevertebral fascia
- c) Between the capsule of tonsil and superior constrictor muscle
- d) Between prevertebral fascia and vertebrae

Question 5:

Given below is an X-ray of a child who presented with stridor. What is your diagnosis?



- a) Epiglottitis
- b) Retropharyngeal abscess
- c) Ludwig's angina
- d) TB spine

Question 6:

Which of the following options is synonymous with quinsy?

- a) Dental abscess.
- b) Peritonsillar abscess.
- c) Prevertebral abscess.
- d) Parapharyngeal abscess

Question 7:

A middle-aged diabetic man who had undergone a tooth extraction presents with fever and trismus. On examination, his posterior pillar is pushed medially and there is a swelling in the neck at the upper 1/3rd of the sternocleidomastoid muscle. What is the most likely diagnosis?

- a) Retropharyngeal abscess
- b) Parapharyngeal abscess
- c) Ludwig's angina
- d) Peritonsillar abscess

Question 8:

Which of the following is not synonymous with parapharyngeal abscess?

- a) Pterygomaxillary abscess
- b) Pharyngomaxillary abscess
- c) Lateral pharyngeal abscess
- d) Medial pharyngeal abscess

Question 9:

Which of the following is correct regarding a parapharyngeal abscess?

- a) Always spreads from the retropharyngeal space
- b) Usually self limited
- c) It has preponderance for male gender
- d) Oral antibiotics are used

Question 10:

Which of the following features will not be manifested in an infection of the post-styloid compartment of the parapharyngeal space?

- a) Trismus
- b) Vocal cord palsy
- c) Parotid swelling
- d) Horner's syndrome

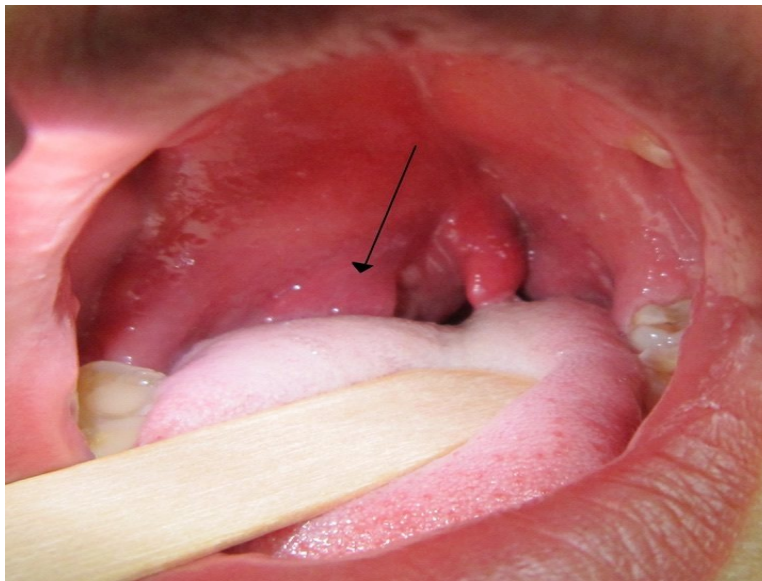
Question 11:

In which of the following clinical scenarios will you not perform an intraoral incision?

- a) A 21-year-old woman with fever, hot potato voice and bulging of soft palate and anterior pillar above the tonsil on the right side.
- b) A 50-year-old diabetic man with fever, neck swelling over upper 1/3rd of the left sternocleidomastoid and the tonsils on the same side pushed medially
- c) A 3-year-old child with fever and swelling in the posterior pharyngeal wall lateral to the midline
- d) A 50-year-old man with fever due to penetrating injury of oral mucosa by wood splinters

Question 12:

A 9-year-old boy comes with complaints of right ear pain, difficulty in opening the mouth, painful swallowing, and fever. The oral cavity examination reveals the following. The external facial examination is unremarkable. Which of the following is likely?



- a) Pharyngitis
- b) Parotid abscess
- c) Bezold's Abscess
- d) Quinsy

Question 13:

Which of the following statements about quinsy is false?

- a) Collection of pus between the capsule of tonsil and superior constrictor
- b) Presents with odynophagia and drooling of saliva
- c) Doesn't involve floor of mouth
- d) Commonly occurs bilaterally

Question 14:

A 16-year-old girl presented with complaints of fever, right-sided throat pain, and difficulty in swallowing. On examination, she had hot potato speech and halitosis. The right tonsil was not visualised due to oedematous pillars and the uvula was pushed to the left. How will you manage this patient?

- a) Antibiotics and analgesia
- b) Interval tonsillectomy
- c) Incision & drainage along with antibiotics

d) Hot tonsillectomy

Question 15:

In a patient with recurrent peritonsillar abscess, when is the ideal time to perform a tonsillectomy?

- a) 6 months
- b) 6 weeks
- c) 6 days
- d) Along with incision and drainage

Question 16:

A patient presents with complaints of pain while swallowing since 2 days. He also gives a history of toothache since 1 week. On examination, there is a swelling and erythema as in the figure below. What is your diagnosis?



- a) Ludwig's angina
- b) Vincent's angina
- c) Quinsy
- d) Parapharyngeal abscess

Question 17:

Which of the following is a false statement about Ludwig's angina?

- a) It involves both sublingual and submandibular space
- b) It is usually bilateral
- c) Dental infections are the most common cause
- d) H.influenza is the most common organism implicated.

Question 18:

A 23-year-old man presents to the OPD with odynophagia and swelling. He is diagnosed with Ludwig's angina. How might you describe the swelling?

- a) Tender fluctuant swelling in the floor of mouth
- b) Tender fluctuant swelling in the parotid region
- c) Woody hard swelling in the floor of mouth
- d) Woody hard swelling in the parotid region

Question 19:

A 50-year-old man with multiple comorbidities, post-op day 5 of major abdominal surgery has developed high-grade fever and swelling over the parotid area. On examination, there is tenderness, induration and pus exuding from the Stenson's duct on applying pressure over the parotids. Which of the following options is not a likely cause for the above condition?

- a) Dehydration
- b) Staphylococcus aureus
- c) Oral infections
- d) Excessive gland stimulation

Question 20:

A patient with a parotid abscess is planned for incision and drainage (I & D). Which of the following practices will you avoid in this patient?

- a) Put an incision that is similar to parotidectomy
- b) Blunt dissection of the pus pockets
- c) Dissection parallel to the facial nerve branches
- d) Wait for the swelling to become fluctuant before performing I & D

Answer Key

Question No.	Correct Option
1	a
2	a
3	b
4	b
5	b
6	b
7	b
8	d
9	c
10	a
11	b
12	d
13	d
14	c
15	b
16	a
17	d
18	c
19	d
20	d

Detailed Explanations

Solution to Question 1:

Option 1 is Gillette's space.

Retropharyngeal space is divided into two lateral compartments (spaces of Gillette) by a median fibrous raphe.

Option 2 - Danger space - between alar fascia anteriorly and prevertebral fascia posteriorly.

Option 3 - Prevertebral space.

Option 4 - Anterior cervical space.

Solution to Question 2:

Rouvier Nodes or retropharyngeal lymph nodes suppurate to form retropharyngeal abscess.

Each lateral compartment of retropharyngeal space contains lymph nodes that usually disappear at 3-4 years of age. Suppuration of these lymph nodes is the most common cause of retropharyngeal space abscess in children.

Option B - Delphian nodes are the pre-laryngeal lymph nodes.

Option C - Virchow's node is the left supraclavicular lymph node.

Option D - Cloquet's nodes are the deep inguinal lymph nodes.

Solution to Question 3:

The above clinical condition is suggestive of acute retropharyngeal abscess whereas caries of the cervical spine results in a chronic retropharyngeal abscess.

Acute retropharyngeal abscess is commonly seen in children below 3 years. It is the result of suppuration of retropharyngeal lymph nodes secondary to infection in the adenoids, nasopharynx, posterior nasal sinuses, or nasal cavity.

In adults, it may result from penetrating injury of the posterior pharyngeal wall or cervical esophagus.

Clinical features of a retropharyngeal abscess include

- Dysphagia and difficulty in breathing - as the abscess obstructs the air and food passages.
- Stridor and croupy cough
- Torticollis - the neck becomes stiff and the head is kept extended.
- A bulge in the posterior pharyngeal wall, usually seen on one side of the midline.

Chronic retropharyngeal abscess is the result of

- Caries of the cervical spine
- Tuberculous infection of retropharyngeal lymph nodes secondary to tuberculosis of deep cervical nodes.

Solution to Question 4:

The likely route of spread of infection is between the alar fascia anteriorly and the prevertebral fascia posteriorly. It is also called the danger space or alar space.

It lies just posterior to the retropharyngeal space. It extends from the skull base to the diaphragm. Infections from this space can spread to the mediastinum, and hence it is named the danger space.

Solution to Question 5:

The given image is suggestive of a retropharyngeal abscess. It can be diagnosed on a lateral x-ray of the soft tissues of the neck. The features include :

- Widening of the prevertebral soft tissue shadow
- Loss of cervical lordosis
- In an acute abscess, air-fluid levels can be seen.

Solution to Question 6:

Quinsy is synonymous with peritonsillar abscess.

It is the collection of pus between the fibrous capsule of the tonsil and the superior constrictor muscles of the pharynx.

Solution to Question 7:

The clinical scenario is suggestive of parapharyngeal abscess.

Parapharyngeal abscess presents with:

- Trismus due to spasm of medial pterygoid
- Tonsils pushed medially
- Swelling in the upper part of the sternocleidomastoid (space limited to the level of hyoid bone).

Option A - Retropharyngeal abscess presents as a bulge in the posterior pharyngeal wall on one side of the midline.

Option C - In Ludwig's angina, tonsils are not pushed medially.

Option D - Peritonsillar abscess doesn't produce swelling in the upper 1/3rd of the sternocleidomastoid.

Solution to Question 8:

Parapharyngeal abscess is not synonymous with medial pharyngeal abscess.

Parapharyngeal abscess is synonymous with pharyngomaxillary or lateral pharyngeal or pterygomaxillary abscess.

Solution to Question 9:

Parapharyngeal abscess has a preponderance for male gender.

Parapharyngeal abscess is an emergency condition. Infection can spread to the parapharyngeal space from any of the other deep neck spaces - peritonsillar, retropharyngeal and submandibular spaces. It commonly spreads from a dental infection, tonsillitis or peritonsillar abscesses. The patient is usually systemically unwell, with severe trismus and possibly airway compromise.

Initial broad-spectrum intravenous antibiotics should include cover against streptococci and anaerobic organisms.

Carotid sheath vessel thrombosis, necrotizing fasciitis and mediastinitis are some of the complications.

Marked neck swelling, airway compromise, and male gender appear to be predictive of higher-risk cases.

Solution to Question 10:

Trismus is not a feature of infection of the post-styloid compartment of the parapharyngeal space. Trismus is seen in pre-styloid abscess due to the involvement of medial pterygoid muscle.

Parapharyngeal space is divided into two compartments by the styloid process and its attachments. Clinical features depend on the compartment involved.

In a post-styloid abscess, there is a bulge of the pharynx behind the posterior pillar, accompanied by paralysis of cranial nerves IX, X (resulting in vocal cord paralysis), XI, XII, and the sympathetic chain, which can manifest as Horner's syndrome. Swelling in the parotid region may also be present. Notably, trismus, the inability to fully open the mouth, is usually absent in this type of abscess.

A pre-styloid abscess exhibits distinct features including the prolapse of the tonsil and tonsillar fossa, along with trismus caused by the spasm of the medial pterygoid muscle. Marked odynophagia is common. Additionally, an external swelling can be observed behind the angle of the jaw.

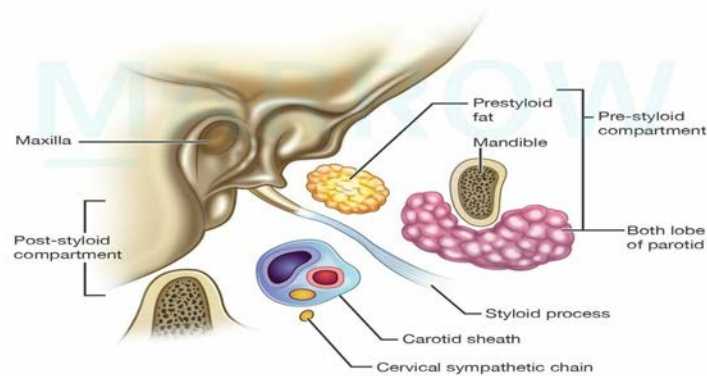
Fever, sore throat, torticollis (caused by spasm of the prevertebral muscles), and signs of systemic toxicity are common features.

Diagnosis can be established using a contrast-enhanced CT scan (CECT). If there is suspicion of thrombosis of the internal jugular vein or an aneurysm of the internal carotid artery, magnetic resonance arteriography (MRA) can be useful.

Patients require intravenous antibiotic therapy with amoxicillin-clavulanic acid, carbapenem antibiotics plus drugs that target anaerobic organisms, such as clindamycin or metronidazole. A parapharyngeal abscess is drained by an external horizontal incision made on the skin a few centimeters below the angle of the mandible.

The image attached below shows the components of both spaces:

Pre-styloid and Post-styloid components of Parapharyngeal space



©Marrow

Solution to Question 11:

An intraoral incision is not sufficient in the case of option B - parapharyngeal abscess. An external neck incision must be made always to ensure complete drainage of the abscess.

Option A- This is a case of quinsy as the child has odynophagia and a hot potato voice.

Option B - This is a case of parapharyngeal abscess as the swelling is in the upper 1/3rd of the neck and the tonsil of the same side is pushed medially

Option C- This is a case of retropharyngeal abscess since there is fever and swelling in the posterior pharyngeal wall lateral to the midline.

Option D- This is a case of Sublingual abscess involving the sublingual space which is caused by penetrating injuries of oral mucosa due to fishbone or wood splinters.

Indications for Incision and Drainage:

- Sublingual abscess: Always intraoral incision
- Parapharyngeal abscess: Always external neck incision.
- Peritonsillar abscess: Always intraoral.
- Retropharyngeal abscess:
- Intraoral - in rose position
- External - Dean approach
- Ludwig's angina:
- Intraoral—if the infection is still localized to sublingual space.
- External—if the infection involves submaxillary space.

Solution to Question 12:

The given clinical scenario describes a child with fever, otalgia, trismus, and odynophagia with a normal external facial examination. The oral cavity shows congested and edematous tonsil and tonsillar pillars with edematous uvula pushed to the opposite side. These findings are highly suggestive of quinsy (peritonsillar abscess).

It is the collection of pus in the peritonsillar space between the fibrous capsule of the tonsil and the superior constrictor muscles of the pharynx. Both aerobic and anaerobic organisms are seen.

Patient presents with fever, unilateral throat pain, odynophagia, hot potato voice, torticollis, foul breath, referred otalgia through glossopharyngeal nerve and trismus due to spasm of pterygoid muscles.

Oral cavity examination may show congested and edematous tonsils, pillars, and soft palate on the involved side. Edematous uvula pushed to the opposite side. Pus may be seen on the tonsillar region. Jugulodigastric lymph nodes may be enlarged.

The management of quinsy is by antibiotics and incision and drainage to drain the abscess followed by interval tonsillectomy 4 to 6 weeks later.

Rare complications seen are parapharyngeal abscess, edema of the larynx, septicemia, lung abscess, or jugular vein thrombosis.

Other options:

Option A: Pharyngitis presents with fever, throat pain, enlarged tonsils, and lymphoid follicles on the posterior pharyngeal wall.

Option B: Parotid abscess occurs due to the suppuration of parotid space. It presents with swelling, redness, firmness, and tenderness in the parotid area and at the angle of the mandible. It is usually unilateral, but bilateral abscesses can occur. Due to the thick capsule, fluctuation is often difficult to elicit.

Option C: Bezold abscess is a complication of acute otomastoiditis where the infection/pus breaks through the thin medial side of the tip of the mastoid and presents as a swelling in the upper part of the neck.

Solution to Question 13:

Peritonsillar abscess or quinsy is usually unilateral.

It is a collection of pus in the peritonsillar space which lies between the capsule of the tonsil and the superior constrictor muscle.

Usually, it is unilateral though occasionally bilateral abscesses are recorded.

It presents with marked odynophagia such that the patient cannot even swallow his saliva which dribbles from the angle of his mouth. The patient is usually dehydrated.

Muffled and thick speech, often called hot potato voice.

Solution to Question 14:

The above clinical scenario is suggestive of peritonsillar abscess or quinsy and management involves incision and drainage with antibiotics.

Management of quinsy

- Hospitalization
- Intravenous fluids to combat dehydration
- Antibiotics - Suitable antibiotics in large IV doses to cover aerobes and anaerobes
- Analgesics
- Drainage by needle aspiration or incision and drainage
- Tonsillectomy when there is airway obstruction or recurrence.

Incision and drainage of the abscess: A peritonsillar abscess is opened at the point of the maximum bulge above the upper pole of the tonsil or just lateral to the point of junction of the anterior pillar with a line drawn through the base of the uvula.

Solution to Question 15:

In a patient with recurrent peritonsillar abscess (quinsy), ideally, an interval tonsillectomy is done 4-6 weeks after the infection settles down.

Hot tonsillectomy: Earlier tonsillectomy was done during incision and drainage. But this is not followed now as it is dangerous and can lead to aspiration, bleeding & spread of infection.

Solution to Question 16:

The above clinical scenario is suggestive of Ludwig's angina.

Ludwig's angina is an infection of the submandibular space which lies between the mucous membrane of the floor of the mouth and tongue on one side and the superficial layer of deep cervical fascia extending between the hyoid bone and mandible on the other.

Solution to Question 17:

Mixed infection with aerobes and anaerobes is seen in Ludwig's angina. Among these alpha-hemolytic streptococci, staphylococci and Bacteroides groups are common.

Ludwig's angina is an infection of the submandibular space.

The space is divided into two compartments by the mylohyoid muscle:

- Sublingual compartment (above the mylohyoid).
- Submandibular and submental compartment (below the mylohyoid).

The two compartments are continuous around the posterior border of the mylohyoid muscle. Both compartments freely communicate in the midline with the opposite side. Therefore infection rapidly becomes bilateral.

The most common cause of Ludwig's angina is dental infections (accounts for 80%) of cases.

Solution to Question 18:

Ludwig's angina presents as a tender woody hard swelling on the floor of the mouth.

There can be marked difficulty in swallowing (odynophagia) with varying degrees of trismus. The tongue is progressively pushed upwards and backward, threatening the airway. Laryngeal edema may also be present.

Usually, there is cellulitis of the tissues rather than frank abscess. Incision and drainage show salted pork appearance.

Solution to Question 19:

The above clinical scenario is suggestive of a parotid abscess and excessive gland stimulation is not a cause for its formation.

Etiology of Parotid Abscess:

- Dehydration, particularly in postsurgical cases (5-7 post-op days) and debilitated patients, with stasis of salivary flow is the predisposing cause.
- Infection from the oral cavity travels via the Stenson's duct to invade the parotid gland (retrograde spread of infection). The most common organism is *Staphylococcus aureus* but *Streptococci*, anaerobic organisms, and rarely the gram negative organisms have also been cultured.

Solution to Question 20:

In a case of a parotid abscess, one must never wait for fluctuation or pus point to occur. This is because the superficial layer of parotidomasseteric fascia is tense and the fluctuation will be apparent only in later stages.

A parotid abscess refers to an infection and pus formation in the parotid space. It is caused most commonly by *Staphylococcus aureus*. Post-surgical cases and debilitated patients with stasis of salivary flow are at risk of developing parotid abscesses.

Patients typically present with unilateral swelling, redness, induration, and tenderness over the parotid gland and angle of the mandible. Dehydration and fever are also present. Diagnosis is by ultrasonography or a CT scan. Multiple small loculi of pus are seen in the parenchyma. Aspiration of abscess can be done to determine the culture and sensitivity of the infective agent.

Correction of dehydration, improving oral hygiene and promoting salivary flow. Intravenous antibiotics are also used in treatment.

Surgical drainage under local or general anesthesia is carried out by a preauricular incision as employed for parotidectomy. A skin flap is raised to expose the surface of the gland, and the abscess or abscesses are bluntly opened working parallel to the branches of the VIIth nerve. The skin incision is loosely approximated over a drain and allowed to heal by secondary intention.

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Nasopharyngeal Angiofibroma

Question 1:

What type of lesion is angiofibroma?

- a) Benign but locally aggressive
- b) Benign with no local invasion
- c) Premalignant
- d) Malignant

Question 2:

A patient presented with difficulty in breathing and swallowing. On examination, a painless nasopharyngeal mass was noted. Angiofibroma was suspected. Which of the following statements is false about this condition?



- a) More common in females
- b) Usually develops at puberty
- c) Hormone-dependent condition
- d) Biopsy is contraindicated

Question 3:

A 16-year-old boy presents to you with complaints of nasal obstruction and frequent nose bleeds. CT scan shows Holman-Miller sign. You suspect a benign tumour of the nasopharynx. What is its most common site of origin?

- a) Maxillary antrum
- b) Sphenopalatine foramen
- c) Anterior wall of the sphenoid bone
- d) Roof of nasopharynx

Question 4:

Which of the following features will not be seen in a patient with nasopharyngeal angiofibroma?

- a) Cranial nerve palsies
- b) Proptosis
- c) Rhinolalia aperta
- d) Conductive hearing loss

Question 5:

A mother brought her 16-year-old boy with a unilateral profuse nosebleed. After successful control of bleeding, a CT scan confirmed the diagnosis of juvenile angiofibroma. What is the reason for it to cause excessive bleeding?

- a) Invasion of surrounding vessels
- b) Lacks a capsule
- c) Vessels lack contractile component
- d) Tumor outgrows blood supply

Question 6:

Which of the following deformity is likely to be seen in a patient with juvenile nasopharyngeal angiofibroma?

- a) Gorilla-like facies
- b) Frog-like facies
- c) Bird-like facies

- d) Lion-like facies

Question 7:

A teenage boy presents to you with recurrent epistaxis and hearing loss. On examination, he has a frog-face deformity and conductive hearing loss. Which of the following is the investigation of choice to diagnose the condition?

- a) NCCT scan
- b) MRI scan
- c) CECT scan
- d) Carotid angiography

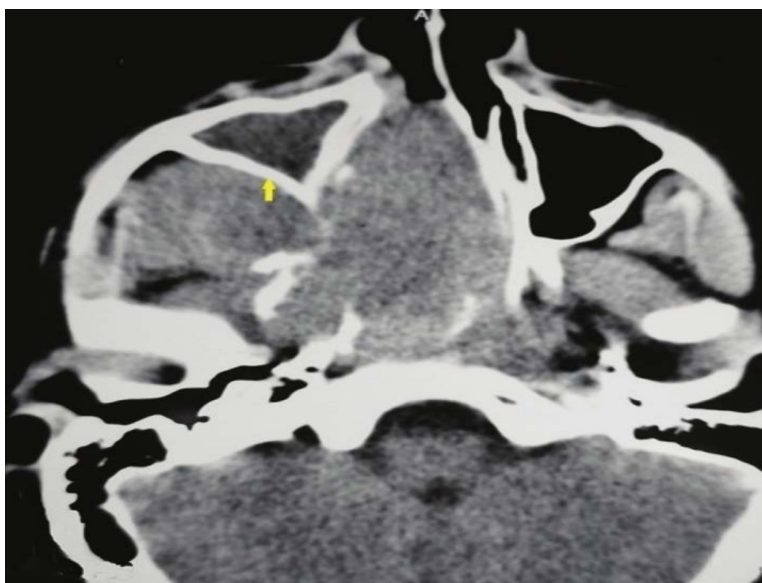
Question 8:

In a suspected case of nasopharyngeal angiofibroma, CECT is performed. Which of the following signs will help diagnose this condition?

- a) Tram-track sign
- b) Antral sign
- c) Stripe sign
- d) Eye-of-a-tiger sign

Question 9:

Imaging of a patient with recurrent epistaxis shows the following finding. Which of the following statements is false about his condition?



- a) Commonest benign tumor of nasopharynx
- b) Tumor does not have a capsule
- c) Digital palpation should never be done in OPD
- d) Recurrence after excision is rare

Question 10:

A teenager has a mass in his nasopharynx that has minimally spread into the sphenoid sinus with no lateral extension. Which stage of nasopharyngeal angiofibroma could this be?

- a) Stage IA
- b) Stage IB
- c) Stage IIA
- d) Stage IIB

Question 11:

A young boy with a history of frequent nosebleeds and unilateral nasal blockage. CT scan shows Holman- Miller sign. It also shows the presence of a mass in the nasopharynx that extends into and occupies the pterygopalatine fossa. What is the best treatment for this patient?

- a) Surgical management
- b) Hormonal therapy
- c) Radiotherapy

d) Chemotherapy

Question 12:

Radiotherapy is indicated when the spread of angiofibroma extends to which site?

- a) Infratemporal fossa
- b) Pterygopalatine fossa
- c) Orbit
- d) Middle cranial fossa

Answer Key

Question No.	Correct Option
1	a
2	a
3	b
4	c
5	c
6	b
7	c
8	b
9	d
10	b
11	a
12	d

Detailed Explanations

Solution to Question 1:

Angiofibroma is a histologically benign tumor but locally aggressive. It is a rare tumor, though it is the commonest of all benign tumors of the nasopharynx.

Angiofibroma arises from the sphenopalatine foramen. Recurrent spontaneous epistaxis is the most common presentation of angiofibroma.

The investigation of choice is a CT scan. There is anterior bowing of the maxillary sinus's posterior wall, which is referred to as the Holman-Miller sign/ antral sign.

Surgery is the treatment of choice. Radiotherapy is used for tumors with intracranial extension. Chemotherapy is used in combination with surgery or as monotherapy when surgery is contraindicated.

Solution to Question 2:

The image shows a nasopharyngeal mass suggestive of juvenile nasopharyngeal angiofibroma (JNA), which is causing difficulty in breathing and swallowing. It is more common in males.

The tumor is predominantly seen in adolescent males in the second decade of life. It is thought to be testosterone dependent. Patients have a hamartomatous nidus of vascular tissue in the nasopharynx which is activated to form angiofibroma when male sex hormones appear.

A biopsy is contraindicated due to the risk of severe bleeding. The vessels lack contractile components and hence bleeding cannot be controlled with adrenaline.

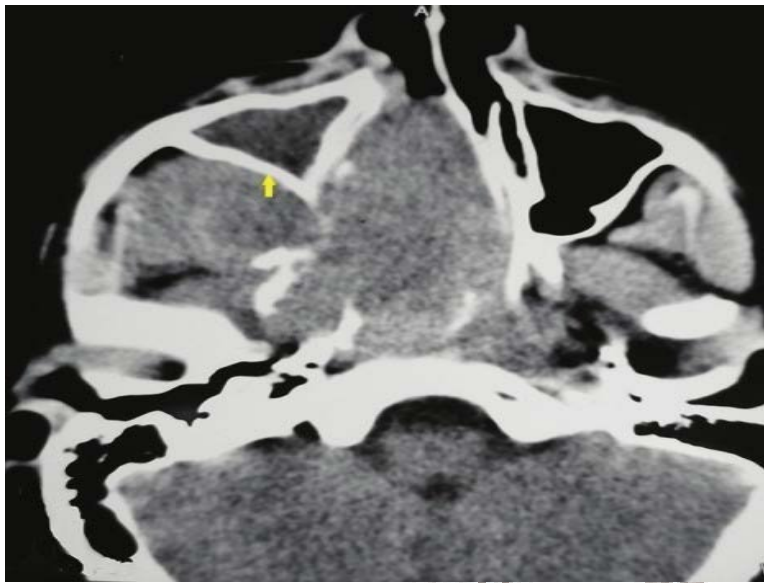
Solution to Question 3:

The given clinical scenario along with a CT scan showing the Holman-Miller sign is suggestive of the diagnosis of juvenile nasopharyngeal angiofibroma (JNA). It most commonly arises from the sphenopalatine foramen.

From here the tumour grows into the:

- Nasal cavity
- Nasopharynx
- Pterygopalatine fossa
- Pterygomaxillary fossa
- Infratemporal fossa

Given below is CT showing the anterior bowing of the posterior wall of the maxillary sinus- Holman-Miller sign.



Solution to Question 4:

Nasopharyngeal angiofibroma presents with rhinolalia clausa (hyponasal speech), not rhinolalia aperta (hypernasal speech).

The patients present with progressive nasal obstruction and hyponasal speech due to mass in the postnasal space. Profuse, recurrent, and spontaneous epistaxis is the most common presentation. The patients may be markedly anemic due to repeated blood loss.

Conductive hearing loss and otitis media with effusion occur due to obstruction of the eustachian tube.

It can spread to the orbit giving rise to proptosis and broadening of the nasal bridge, giving rise to frog face deformity. Other features include swelling of the cheek and cranial nerve palsies (II, III, IV, and VI).

Solution to Question 5:

Angiofibroma is made of vascular and fibrous tissues. Vessels of angiofibroma lack contractile components, they are just endothelium-lined spaces with no elastic or muscle coat.

This accounts for severe bleeding as the vessels lose the ability to contract. Hence the bleeding cannot be controlled by the application of adrenaline. Therefore, the biopsy is contraindicated.

Solution to Question 6:

Nasopharyngeal angiofibroma when spreads to the orbit, lead to proptosis and broadening of the nasal bridge, giving rise to frog-face deformity.

Option A: Gorilla-like facies are seen in acromegaly.

Option C: Bird-like facies are seen in the Pierre-Robin sequence.

Option D: Lion-like facies are seen in Paget's disease of bone when craniofacial bones are involved.

Solution to Question 7:

The given clinical scenario is suggestive of nasopharyngeal angiofibroma. Investigation of choice to diagnose this condition is contrast enhancement CT scan (CECT).

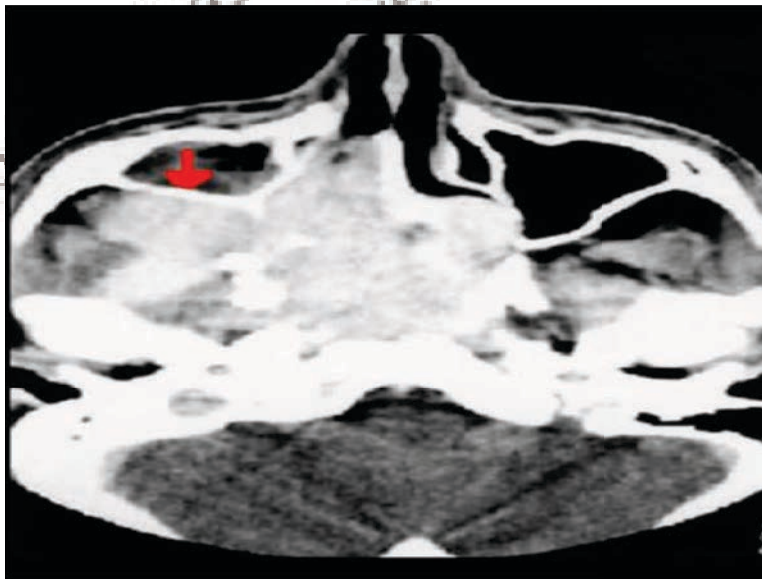
MRI scan can be used complementary to CT to see soft tissue extensions of the tumor.

Carotid angiography shows the extent of the tumor, its vascularity, and feeding vessels which mostly come from the external carotid system. Embolization of vessels can be done at this time to decrease bleeding at operation.

A biopsy is contraindicated in this condition as it can lead to massive hemorrhage.

Solution to Question 8:

Anterior bowing of the posterior wall of the maxillary sinus in CECT of head and neck is called an antral sign or Holman-Miller sign, which is pathognomonic for angiofibroma.



Option A: Tram track sign is used to differentiate optic nerve sheath meningiomas from optic gliomas. Optic nerve sheath meningiomas will show a positive tram-track sign.

Option C: Stripe sign/ tigroid pattern represents a specific pattern of demyelination, with sparing of perivascular white matter. It is seen in metachromatic leukodystrophy (MLD).

Option D: Eye-of-a-tiger sign represents marked low signal intensity of the globi pallidi on T2W MRI images.

Solution to Question 9:

The image shows anterior bowing of the posterior wall of the maxillary sinus, which is called an antral sign or Holman-Miller sign. It is pathognomonic for angiofibroma. Recurrence of juvenile nasopharyngeal angiofibroma is common.

It is a rare tumor but is the commonest benign tumor of the nasopharynx. It commonly arises from the sphenopalatine foramen. It does not have a capsule.

Digital palpation should never be done until at the time of operation, as it can lead to severe hemorrhage.

Solution to Question 10:

The mass has spread into the sphenoid sinus with no lateral extension, so it's Stage 1B according to the Radkowski classification.

Multiple classification systems exist for the staging of juvenile nasal angiofibroma, of which the most commonly used ones are the Fisch and Radkowski classification systems:

Staging of juvenile nasopharyngeal angiofibroma (Radkowski, et al):

Fisch staging system of juvenile nasopharyngeal angiofibroma:

Juvenile nasopharyngeal angiofibroma (JNA) is a highly vascular and locally aggressive tumor seen almost exclusively in adolescent males, associated with significant morbidity and occasional mortality.

Note: The Fisch system is considered to be the more robust and practical of the two.

Stage	Spread
IA	Limited to the nose and/or nasopharyngeal vault
IB	Extension into ≥ 1 sinus
IIA	Minimal extension into the pterygopalatine fossa
IIB	Full occupation of pterygopalatine fossa with or without erosion of orbital bones
IIC	Infratemporal fossa with or without cheek or posterior to pterygoid plates

Stage	Spread
IIIA	Erosion of skull base-minimal intracranial
IIIB	Erosion of skull base-extensive intracranial with or without cavernous sinus

Type	Details
1	Tumour limited to the nasopharyngeal cavity; bone destruction negligible or limited to the sphenopalatine foramen
2	Tumour invading the pterygo palatine fossa or the maxillary, ethmoid, or sphenoid sinus with bone destruction
3	A: Tumour invading the infratemporal fossa or orbital region without intracranial involvement B: Tumour invading the infratemporal fossa or orbital region with intracranial extradural (parasellar) involvement
4	A: Intracranial intradural tumour without infiltration of the cavernous sinus, pituitary fossa, or optic chiasm B: Intracranial intradural tumour with infiltration of the cavernous sinus, pituitary fossa, or optic chiasm

Solution to Question 11:

The clinical picture suggests that of stage IIB nasopharyngeal angiofibroma. Surgery is the mainstay of treatment of angiofibroma for stage I and stage II tumors.

Radiotherapy is indicated in stage III where there is an intracranial extension or when the tumor is unresectable.

Chemotherapy can be used for very aggressive recurrent tumors and residual lesions.

Stages of nasopharyngeal angiofibroma:

Hormonal therapy was used in the past, but not anymore.

Stage	Spread
IA	Limited to the nose and/or nasopharyngeal vault
IB	Extension into ≥ 1 sinus
IIA	Minimal extension into the pterygopalatine fossa
IIB	Full occupation of pterygopalatine fossa with or without erosion of orbital bones
IIC	Infratemporal fossa with or without cheek or posterior to pterygoid plates
IIIA	Erosion of skull base-minimal intracranial
IIIB	Erosion of skull base-extensive intracranial with or without cavernous sinus

Solution to Question 12:

Radiotherapy is used for intracranial extension of disease when tumour derives its blood supply from the internal carotid system. Recurrent angiofibromas have also been treated by radiotherapy.

Nasopharyngeal Carcinoma

Question 1:

Which of the following viruses is related to etiology of nasopharyngeal carcinoma?

- a) Human papillomavirus
- b) Epstein-Barr virus
- c) Herpes virus
- d) Adenovirus

Question 2:

What is nasopharyngeal carcinoma also known as?

- a) Indian tumor
- b) Chinese tumor
- c) Guangdong tumor
- d) Kandahar tumor

Question 3:

Which of the following is not associated with a high incidence of nasopharyngeal carcinoma in high-risk populations of China?

- a) Firewood
- b) Preserved fish
- c) Vit C deficiency
- d) Human papillomavirus

Question 4:

A patient with a family history of nasopharyngeal carcinoma presents with nasal discharge and a painless lump in the upper neck. On posterior rhinoscopy, where will you most commonly expect to find a mass originating from?

- a) Nasal septum

- b) Lateral wall of nose
- c) Fossa of Rosenmuller
- d) Sphenopalatine foramen

Question 5:

What is the most common presenting feature of nasopharyngeal carcinoma?

- a) Nasal obstruction
- b) Epistaxis
- c) Painless cervical lymphadenopathy
- d) Headache

Question 6:

A housewife from rural Tripura with a past history of infectious mononucleosis presents with complaints of double vision and a neck swelling that has progressively increased in size. She also gives a history of nasal discharge and is annoyed that her husband is recently complaining of a change in her voice. What is the most likely speech pattern observed here?

- a) Rhinolalia clausa
- b) Rhinolalia aperta
- c) Puberphonia
- d) Hot potato voice

Question 7:

A 45-year-old Chinese woman presents with left-sided cervical lymphadenopathy, nasal obstruction, and left-sided hearing loss. What is the most likely cause for hearing loss in this case?

- a) Serous otitis media
- b) Temporal bone metastasis
- c) Inner ear hydrops
- d) Middle ear infiltration

Question 8:

You have sent a patient with nasopharyngeal carcinoma and hearing loss for tympanometry. Which of the following tympanometry curves do you expect to find?

- a) Type C
- b) Type B
- c) Type As
- d) Type Ad

Question 9:

You are an ENT resident on a clinical elective in Guangzhou. You receive a patient with painless cervical lymphadenopathy, conductive hearing loss, palatal asymmetry, and facial pain. Which of the following do you see here?

- a) Samter's triad
- b) Trotter's triad
- c) van der Hoeve syndrome
- d) Kartagener syndrome

Question 10:

Which of the following branches of the trigeminal nerve is involved in Trotter's triad?

- a) Ophthalmic
- b) Maxillary
- c) Mandibular
- d) Greater auricular

Question 11:

A known case of right-sided nasopharyngeal carcinoma now presents with drooping of eyelids, reduced sweating on the right side of the face, and constricted pupil. Involvement of which of the following structures leads to these features?

- a) Sympathetic chain
- b) Cranial nerve V
- c) Cranial nerve VI
- d) Jugular foramen

Question 12:

Which of the following WHO categories does differentiated non-keratinizing squamous cell carcinoma of the nasopharynx belong to?

- a) Type 1
- b) Type 2
- c) Type 3
- d) Type 4

Question 13:

A patient with persistent cervical lymphadenopathy now came with spontaneous epistaxis. Nasal endoscopy showed a mass arising from the fossa of Rosenmuller. Biopsy from the mass revealed a keratinizing squamous cell carcinoma. What is the treatment of choice in this case?

- a) Surgery
- b) Radiotherapy
- c) Chemotherapy
- d) Surgery and radiotherapy

Answer Key

Question No.	Correct Option
1	b
2	c
3	d
4	c
5	c
6	a
7	a
8	b
9	b

10	c
11	a
12	b
13	b

Detailed Explanations

Solution to Question 1:

Epstein–Barr Virus (EBV) is closely associated with nasopharyngeal cancer (NPC).

EBV antigens are currently being used to screen high-risk populations. The antigens being used are

- Viral capsid antigen (VCA) - highly sensitive for NPC
- Early antigen (EA) - highly specific for NPC.

Solution to Question 2:

Nasopharyngeal carcinoma is also known as Guangdong tumor. This is because the Southern Chinese province of Guangdong has the highest prevalence of the tumor.

Chinese, especially from southeastern China (Canton), have a higher genetic susceptibility to nasopharyngeal cancer (NPC). Chinese emigrants are also genetically more susceptible, however, the susceptibility falls in subsequent generations.

Other South-East Asian countries such as Indonesia, Malaysia, and people of Mongoloid origin are also more susceptible.

Solution to Question 3:

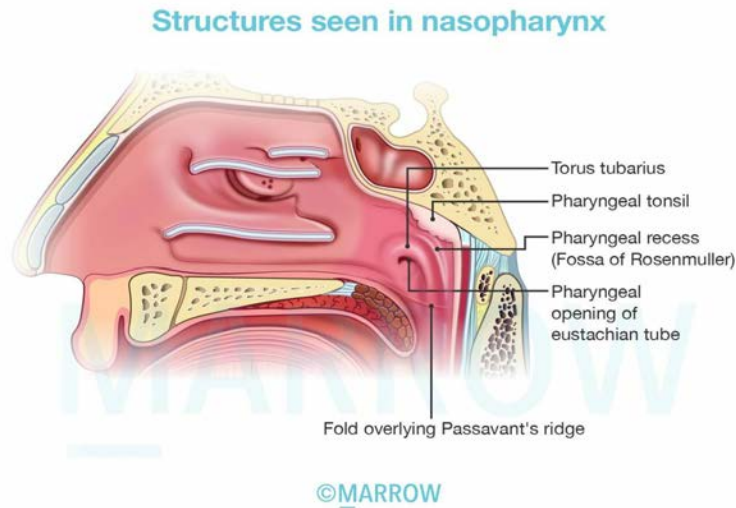
Human papillomavirus is not associated with a high incidence of nasopharyngeal carcinoma in high-risk populations of China. EBV (Epstein Barr virus) causes nasopharyngeal carcinoma. HPV(Human papillomavirus) causes inverted papilloma of the nose.

High-risk populations such as the Chinese have a very high incidence of nasopharyngeal carcinoma. The following factors have been implicated in the pathogenesis:

- Burning of incense or wood (polycyclic hydrocarbons)
- Consumption of preserved salted fish (nitrosamines)
- Vitamin C deficient diet (vitamin C blocks nitrosification of amines and is thus protective)

Solution to Question 4:

The most common site of origin for nasopharyngeal carcinoma is the fossa of Rosenmüller.



Solution to Question 5:

Painless cervical lymphadenopathy is the most common (60-90%) presenting feature of nasopharyngeal carcinoma followed by-

- Hearing loss
- Nasal obstruction
- Epistaxis
- Cranial nerve palsies -Cranial nerves V and VI are the most commonly involved among all the cranial nerves.
- Headache
- Earache
- Neck pain
- Weight loss

Solution to Question 6:

The given case scenario of exposure to EBV, progressive cervical lymphadenopathy, nasal discharge, diplopia (involvement of cranial nerve VI), and change in voice is suggestive of nasopharyngeal carcinoma. The most likely speech pattern, in this case, is rhinolalia clausa or denasal speech, due to obstruction of the nasopharynx by the tumor.

Rhinolalia clausa is the lack of nasal resonance for words resonated in the nasal cavity and is usually seen due to blockage of nose or nasopharynx

Other nasal symptoms of nasopharyngeal carcinoma are:

- Nasal obstruction
- Nasal discharge
- Epistaxis

Option B- Rhinolalia aperta (hypernasal voice) is seen in Velopharyngeal insufficiency.

Option C- Puberphonia is the persistence of prepubertal voice into adolescence or adulthood.

Option D- Muffled and thick speech, often called "hot potato voice" is seen in peritonsillar abscess or quinsy.

Solution to Question 7:

The given clinical scenario is suggestive of nasopharyngeal carcinoma. Hearing loss, in this case, is due to serous otitis media.

Serous otitis media is caused due to mechanical obstruction of the Eustachian tube by the tumor. The presence of unilateral serous otitis media in an adult should raise suspicion of nasopharyngeal growth.

Solution to Question 8:

Type B tympanometry curve is expected in nasopharyngeal carcinoma. Due to mechanical obstruction of the Eustachian tube by the tumor, serous otitis media occurs, causing conductive hearing loss.

Solution to Question 9:

The given clinical scenario is suggestive of Trotter's triad/Sinus of Morgagni syndrome/Trotter's syndrome seen in nasopharyngeal carcinoma.

Trotter's triad consists of:

- Conductive hearing loss due to Eustachian tube obstruction.
- Ipsilateral temporoparietal neuralgia due to involvement of mandibular branch of the trigeminal nerve (CN V) which presents as facial pain.
- Palatal paralysis due to involvement of vagus nerve (CN X) which presents as palatal asymmetry.

Option A - Samter's triad seen in ethmoidal polyps consists of nasal polypi, asthma, and Aspirin intolerance

Option C - van der Hoeve syndrome is seen in otosclerosis consists of osteogenesis imperfecta, otosclerosis, and blue sclera

Option D - Kartagener syndrome consists of bronchiectasis, sinusitis, situs inversus, and ciliary dyskinesia.

Solution to Question 10:

The mandibular branch of the trigeminal nerve is involved in Trotter's triad. Trotter's triad seen in nasopharyngeal carcinoma. It consists of:

- Conductive hearing loss due to Eustachian tube obstruction.
- Ipsilateral temporoparietal neuralgia due to involvement of mandibular branch of the trigeminal nerve (CN V) which presents as facial pain.
- Palatal paralysis due to involvement of vagus nerve (CN X) which presents as palatal asymmetry.

Solution to Question 11:

Involvement of the sympathetic chain causes these features. The given clinical scenario is suggestive of Horner's syndrome secondary to nasopharyngeal carcinoma involving the cervical sympathetic chain.

Horner's syndrome consists of ptosis, miosis, and ipsilateral anhidrosis.

Option B - Cranial nerve V (Trigeminal nerve) involvement presents as ipsilateral temporoparietal neuralgia.

Option C - Cranial nerve VI (Abducens nerve) involvement causes ophthalmoplegia

Option D - Jugular foramen involvement causes paralysis of cranial nerves IX, X, and XI, causing jugular foramen syndrome, also seen in acoustic neuroma and glomus jugulare.

Solution to Question 12:

Differentiated non-keratinized squamous cell carcinoma is categorized as Type 2.

WHO Classification based on Histopathology:

Note- The latest WHO classification has combined types 2 and 3 into one category:

Non-keratinizing Squamous cell carcinoma which is further subclassified into differentiated and non-differentiated types.

Type	Histopathological category
Type I	Keratinizing Squamous cell carcinoma (SCC)

Type	Histopathological category
Type II	Non-keratinizing Squamous cell carcinoma

Basaloid squamous cell carcinoma is a rare subtype of epithelial cancer seen in the nasopharyngeal area. It was included alongside the above classification from 2005. It is known for its aggressive clinical progression and poor chances of survival.

Solution to Question 13:

The given clinical scenario is suggestive of nasopharyngeal carcinoma. The treatment of choice is radiotherapy.

Nasopharyngeal carcinoma is extremely radiosensitive, hence treatment of local and regional disease is invariably radiotherapy.

Stage I and II are treated by radiotherapy alone.

Stage III and IV are treated with radiotherapy and chemotherapy. Chemotherapy can be neoadjuvant, concurrent, or adjuvant to radiotherapy.

Anatomy and Physiology of Larynx

Question 1:

At what level does the adult larynx end?

- a) Lower border of C3
- b) Lower border of C4
- c) Lower border of C5
- d) Lower border of C6

Question 2:

What is the average final anteroposterior diameter of glottis in women?

- a) 12 mm
- b) 26 mm
- c) 20 mm
- d) 24 mm

Question 3:

What is the primary function of the larynx?

- a) Phonation
- b) Respiratory passage
- c) Protection of lower airways
- d) Cough reflex

Question 4:

The presence of which structure is responsible for infants to be able to breathe simultaneously while suckling milk?

- a) Small tongue
- b) Small soft palate

- c) High larynx
- d) Small pharynx

Question 5:

Which embryological structure develops into the epiglottis?

- a) 4th pharyngeal pouch
- b) 2nd arch
- c) Hypobranchial eminence
- d) 3rd arch

Question 6:

Which of the following comprises the larynx?

- a) 2 paired and 2 unpaired cartilages
- b) 3 paired and 3 unpaired cartilages
- c) 2 paired and 3 unpaired cartilages
- d) 3 paired and 2 unpaired cartilages

Question 7:

The lateral x-ray of an elderly man shows calcification of the laryngeal cartilages. Which of the following is unlikely to be involved?

- a) Thyroid
- b) Cricoid
- c) Arytenoid
- d) Epiglottis

Question 8:

Which of the following cartilage has a signet ring shape?

- a) Thyroid
- b) Cricoid

- c) Epiglottis
- d) Arytenoid

Question 9:

The skeletal remains of a person were found in a nearby forest. The thyroid cartilage was identified and the thyroid angle measures 90 degrees. The deceased person can be deemed as a

- a) Child
- b) Man
- c) Woman
- d) Cannot say

Question 10:

Which of the following is called cartilage of Santorini?

- a) Epiglottis
- b) Arytenoid
- c) Corniculate
- d) Cuneiform

Question 11:

An ENT surgeon is operating on the larynx of a patient and he approaches the arytenoid cartilage. Which of the following is false regarding the attachments of the arytenoid cartilage?

- a) Base - cricoid cartilage
- b) Vocal process - vocal ligament
- c) Apex - intrinsic muscles of larynx
- d) Apex - corniculate cartilages

Question 12:

Which of the following is not involved in the closure of glottis?

- a) Posterior cricoarytenoids

- b) Lateral cricoarytenoids
- c) Cricothyroid
- d) Thyroarytenoids

Question 13:

An opera singer presents with difficulty raising the pitch of her voice while singing. Which of the following muscles is likely to be affected?

- a) Posterior cricoarytenoid
- b) Lateral cricoarytenoid
- c) Cricothyroid
- d) External part of thyroarytenoid

Question 14:

You are the ENT resident examining a patient in the wards. On asking him to swallow you notice the larynx moves upwards. Which of the following muscles is not responsible for this movement?

- a) Thyrohyoid
- b) Stylohyoid
- c) Sternohyoid
- d) Digastric

Question 15:

Which of the following is an intrinsic membrane of the larynx?

- a) Cricothyroid
- b) Cricotracheal
- c) Thyrohyoid
- d) Hyoepiglottic

Question 16:

Following a road traffic accident, a 30-year-old man presents to the emergency department with maxillofacial trauma and inability to open the mouth. To establish an airway, an emergency laryngotomy is performed. Where is the opening made?

- a) Cricotracheal membrane
- b) Cricothyroid membrane
- c) Thyrohyoid membrane
- d) Cricothyroid muscle

Question 17:

Which of the following is not seen as part of the larynx?

- a) Vestibule
- b) Ventricle
- c) Labyrinth
- d) Saccul

Question 18:

A surgeon is worried about the spread of laryngeal carcinoma to the pre-epiglottic space. Which of the following structures do not border this space?

- a) Hyoepiglottic ligament
- b) Suprahyoid epiglottis
- c) Thyroid lamina
- d) Thyrohyoid membrane

Question 19:

A 35-year-old smoker presents with hoarseness of voice. On laryngoscopy, there is edema of the Reinke's space. Where is this space located?

- a) Glottis
- b) Supraglottis
- c) Subglottis
- d) Paraglottic space

Question 20:

You are performing a direct laryngoscopy procedure to enter the larynx and visualise the structures it contains. Which of the following structures form the inlet of the larynx?

- a) Ventricular fold
- b) Aryepiglottic fold
- c) Vocal cord
- d) Glossoepiglottic fold

Question 21:

Which of the following muscle causes opening of laryngeal inlet?

- a) Vocalis
- b) Thyroepiglotticus
- c) Cricothyroid
- d) Thyroarytenoid

Answer Key

Question No.	Correct Option
1	d
2	b
3	c
4	c
5	c
6	b
7	d
8	b
9	b
10	c
11	c
12	a
13	c
14	c

15	a
16	b
17	c
18	b
19	a
20	b
21	b

Detailed Explanations

Solution to Question 1:

In adults, the larynx ends at lower border of C6. It extends from C3 to C6 vertebra.

Solution to Question 2:

The average anteroposterior diameter of the glottis is 26mm in women and 36 mm in men.

Glottis, also known as rima glottidis, is the narrowest part of the laryngeal cavity. It is formed by:

- Anterior 2/3rd by membranous cords
- Posterior 1/3rd by vocal processes of arytenoids.

Solution to Question 3:

The primary function of the larynx is to protect the airway from ingested food and aspirated materials.

This is done by the sphincteric action of the larynx.

Other important functions are:

- Respiration
- Phonation
- Building up of intra-thoracic pressure.

Solution to Question 4:

The high position of the larynx in infants allows simultaneous breathing while feeding.

The infant's larynx is positioned high in the neck, level of the glottis being opposite to C3 or C4 at rest and reaches C1 or C2 during swallowing. The high position allows the epiglottis to meet the soft palate and make a nasopharyngeal channel for nasal breathing during suckling.

The milk feed passes separately over the dorsum of the tongue and sides of the epiglottis, thus allowing breathing and feeding to go on simultaneously.

Solution to Question 5:

The epiglottis develops from the hypobranchial eminence which is a part of the 4th arch.

Solution to Question 6:

The larynx has 3 unpaired and 3 paired cartilages.

Unpaired:

- Thyroid
- Cricoid
- Epiglottis

Paired:

- Arytenoid
- Corniculate
- Cuneiform.

Solution to Question 7:

Epiglottis is unlikely to be calcified as it is made up of elastic cartilage.

Only hyaline cartilages can undergo ossification (calcifications on x-ray). It occurs between the ages of 25-65y. These calcifications may be confused with foreign bodies of esophagus or larynx on x-rays.

The hyaline laryngeal cartilages are:

- Thyroid
- Cricoid
- Most of arytenoid.

The elastic laryngeal cartilages are:

- Epiglottis
- Corniculate

- Cuneiform
- Tip of arytenoid.

Solution to Question 8:

Cricoid cartilage resembles a signet ring. It has a wide posterior lamina and a thin anterior arch.

Solution to Question 9:

The thyroid angle measures 90 degrees (right-angled) in men.

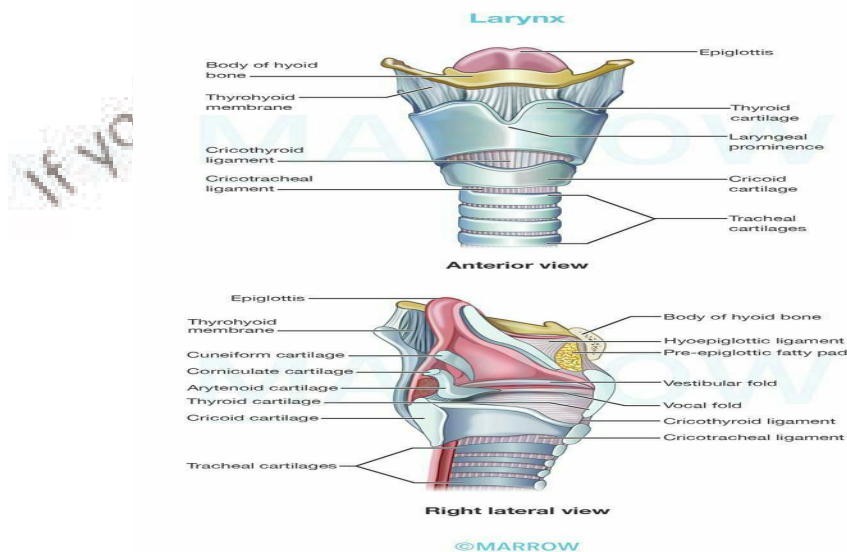
It is the angle between the two thyroid lamina in the midline. It depends on gender:

- Men: 90 degrees (forms the prominent adam's apple)
- Women: 120 degrees.

Solution to Question 10:

Corniculate cartilage is also known as the cartilage of Santorini.

Cuneiform cartilage is also known as the cartilage of Wrisberg.



Solution to Question 11:

The apex of the arytenoid supports the corniculate cartilages.

The following are the attachments of the arytenoid:

- Base - cricoid
- Vocal process - vocal ligament
- Muscular process - intrinsic muscles of larynx
- Apex - corniculate cartilages.

Solution to Question 12:

The posterior cricoarytenoid opens the glottis as it is the only abductor of the vocal cord. All other muscles close the glottis.

Solution to Question 13:

Cricothyroid is likely to be affected.

Cricothyroid and vocalis (internal part of thyroarytenoid) are the tensors of the vocal cord which help in raising the pitch of the voice.

The external laryngeal nerve is the smaller, external branch. It descends on the larynx, beneath the sternothyroid muscle, to supply the cricothyroid muscle. The external branch functions to tense the vocal cords by activating the cricothyroid muscle, and increasing pitch.

Solution to Question 14:

The sternohyoid is not responsible for upward movement (elevation). It is a depressor of the larynx.

The extrinsic muscles of the larynx can be elevators or depressors.

Elevators:

Primary elevators act directly as they are attached to the thyroid cartilage and include

- Stylopharyngeus
- Salpingopharyngeus
- Palatopharyngeus
- Thyrohyoid

Secondary elevators act indirectly as they are attached to the hyoid bone and include

- Mylohyoid (main)
- Digastric
- Stylohyoid

- Geniohyoid

Depressors:

- Sternohyoid
- Sternothyroid
- Omohyoid

Easy to remember: Check the other attachment(non-laryngeal side). If it's above the level of the hyoid, the muscle is an elevator. If it's below the level of the hyoid, it's a depressor.

Solution to Question 15:

The cricothyroid membrane is an intrinsic membrane of the larynx as it connects laryngeal structures to each other.

Extrinsic membranes and ligaments connect laryngeal structures with surrounding structures outside the laryngeal framework like the trachea or hyoid. These are:

- Cricotracheal membrane
- Thyrohyoid membrane
- Hyoepiglottic ligament.

Intrinsic membranes and ligaments connect laryngeal structures with each other. These are:

- Cricovocal membrane
- Quadrangular membrane
- Cricothyroid ligament
- Thyroepiglottic ligament.

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Solution to Question 16:

In laryngotomy, an opening is made on the cricothyroid membrane.

Laryngotomy is also called cricothyrotomy or mini-tracheostomy. It is done as an emergency life-saving procedure to buy some time to allow a patient to be carried to the operation theatre. Once stable, it is replaced by intubation or tracheostomy.

Most of the laryngeal foreign bodies are arrested above the vocal cords, i.e. above the middle of the thyroid cartilage. Thus, an effective airway can be provided by piercing the cricothyroid membrane.

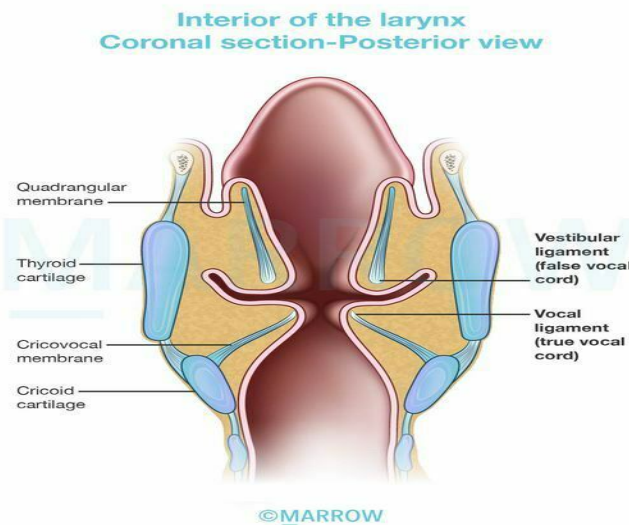
Solution to Question 17:

Labyrinth is seen in inner ear and not in larynx.

The 2 mucosal folds within the larynx divide the larynx into 3 cavities:

- Vestibule: Above the false/vestibular folds
- Ventricle: Between the vestibular and vocal/true folds
- Subglottis: Below the true folds

Saccule an extension of ventricle is present in the paraglottic space.



Solution to Question 18:

Suprahyoid epiglottis does not form a boundary of the pre-epiglottic space. The infrahyoid epiglottis forms the posteromedial border.

Two potential spaces within the larynx provide an avenue for the spread of tumors:

- Pre-epiglottic space
- Paraglottic space.

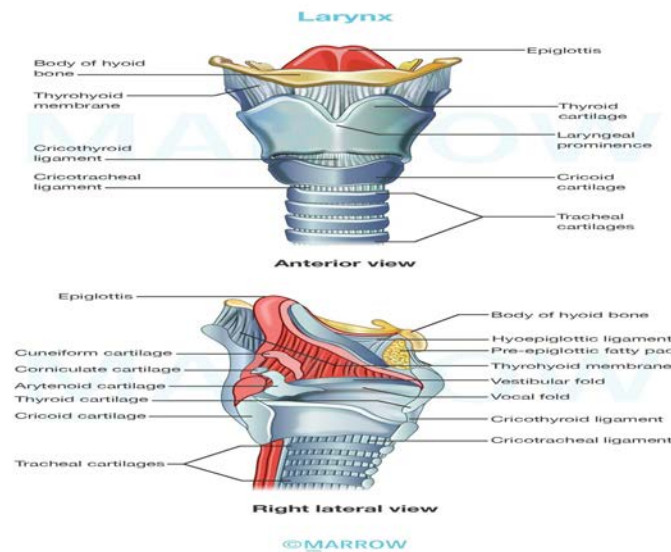
The pre-epiglottic space is bounded:

- Superiorly by the hyoepiglottic ligament
- Anteriorly by the thyroid and thyrohyoid membrane
- Posteromedially by the infrahyoid epiglottis and quadrangular membrane.

The paraglottic space is bounded:

- Laterally by thyroid
- Medially by ventricle and quadrangular membrane

- Inferomedially by conus elasticus
- Posteriorly by pyriform sinus.

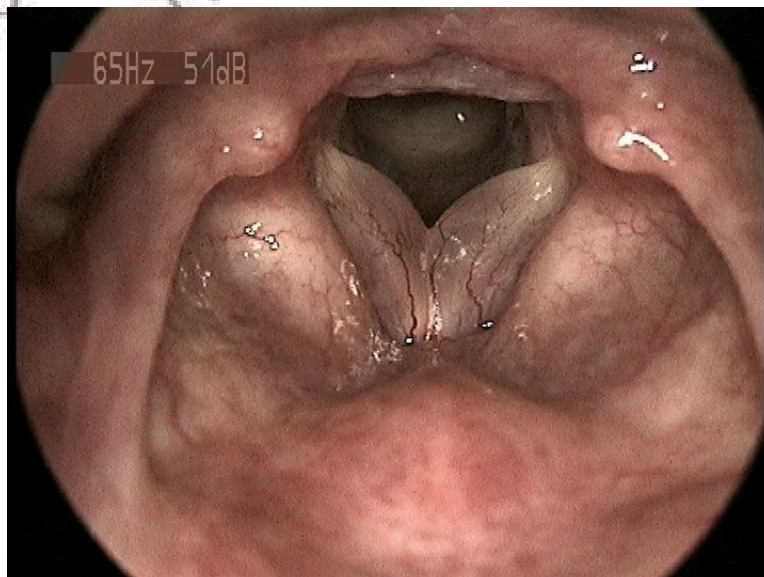


Solution to Question 19:

Reinke's space is seen in the glottis. It is a potential space underneath the epithelium of vocal cords.

It contains scanty subepithelial connective tissue. Edema of this space is commonly seen in smokers and is called Reinke's edema or smoker's larynx.

The laryngoscopy image below shows diffuse bilateral edema of the vocal folds or Reinke's edema.



Solution to Question 20:

Aryepiglottic folds form the inlet of the larynx.

The laryngeal inlet is an oblique opening with the following boundaries:

- Anterior: Free margin of the epiglottis
- Posterior: Interarytenoid fold
- Sides: Aryepiglottic folds.

Solution to Question 21:

Thyroepiglotticus causes the opening of the laryngeal inlet

The intrinsic muscles of the larynx acting on the laryngeal inlet are:

- Openers of laryngeal inlet
- Thyroepiglottic
- Closers of laryngeal inlet
- Interarytenoid (oblique part)
- Aryepiglottic (posterior oblique part of interarytenoids).

Stridor and Congenital Conditions of Larynx

Question 1:

Inspiratory stridor is found in what kind of lesions:

- a) Supraglottic
- b) Subglottic
- c) Tracheal
- d) Bronchus

Question 2:

What is the most common cause of congenital stridor?

- a) Laryngeal web
- b) Laryngomalacia
- c) Subglottic hemangioma
- d) Laryngeal cyst

Question 3:

What is congenital laryngeal stridor also known as?

- a) Laryngomalacia
- b) Laryngeal web
- c) Laryngeal papilloma
- d) Quinsy

Question 4:

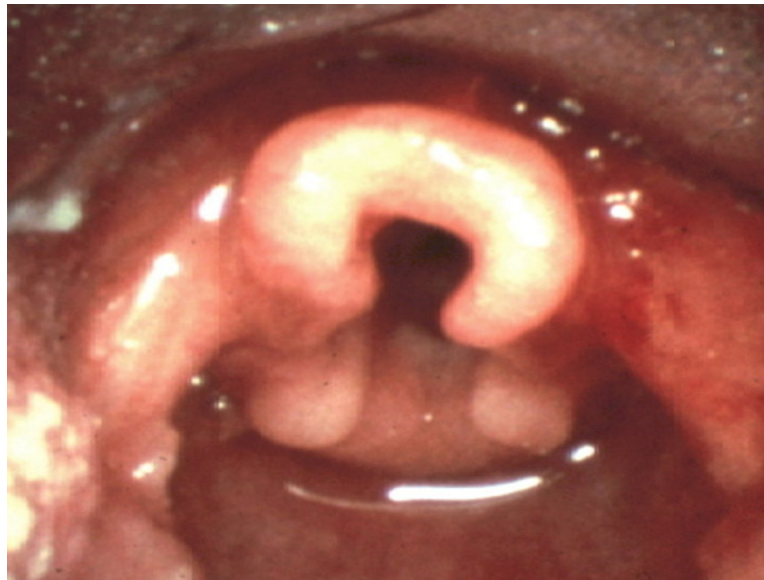
A 1-week old infant has been diagnosed to have laryngomalacia. Which of the following statements is false about this condition?

- a) Most common cause of stridor in newborn
- b) Elongated omega shaped epiglottis

- c) Symptoms decrease in prone position
- d) It has a poor prognosis

Question 5:

A worried mother brings her 2-week old infant to you with complaints of noisy breathing that worsens during feeding. On direct laryngoscopy, the following is seen. What is the preferred treatment for this condition?



- a) Steroid therapy
- b) Supraglottoplasty
- c) Conservative treatment
- d) Tracheostomy

Question 6:

A 5-year-old child is brought to you with complaints of noisy breathing. On performing direct laryngoscopy, subglottic stenosis is noted. Which grading system will you use to report the extent of the stenosis?

- a) Hyams grading
- b) House Brackmann grading
- c) Myer Cotton's grading
- d) Modified Mallampati grading

Question 7:

A 45-year-old man presents to the ENT OPD with a swelling on the right side of his neck. The swelling moves with deglutition. On examination, the swelling is compressible and there is a gurgling sound heard on compression. What is his diagnosis?

- a) Branchial cleft cyst
- b) Laryngocele
- c) Cystic hygroma
- d) Thyroglossal duct cyst

Question 8:

You are the ENT resident performing a direct laryngoscopy procedure on a patient and report the presence of a laryngocele. The given condition is due to an enlarged diverticulum from which part of the larynx?

- a) Vocal cords
- b) Saccule of ventricle
- c) Anterior commissure
- d) Posterior commissure

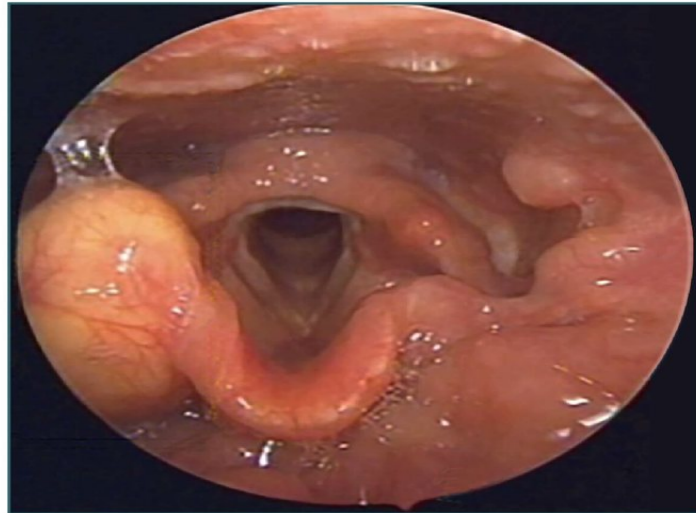
Question 9:

A mother brings her 4-month-old infant to the ENT department with complaints of difficulty in breathing. On examination, you notice biphasic stridor. Rigid bronchoscopy shows subglottic hemangioma. Which of the following is not included in the treatment of this condition?

- a) Steroids
- b) LASER Excision
- c) Laryngofissure
- d) Embolization

Question 10:

A 30-year-old singer complains of a change in her voice. Direct laryngoscopy procedure shows the presence of laryngeal cysts. Which part of the larynx does the cyst arise from?



- a) Aretenoids
- b) Aryepiglottic folds
- c) Epiglottis
- d) Pyriform sinus

Question 11:

Identify the lesion of vocal cord in the image given below :



- a) Reinke's edema
- b) Malignancy

- c) Tracheomalacia
- d) Laryngeal papilloma

Question 12:

A 5 year old with gradually progressive hoarseness of voice over three months and respiratory distress for 2 weeks. Diagnosis is

- a) Croup
- b) Respiratory papillomatosis
- c) Vocal nodule
- d) Acute epiglottitis

Answer Key

Question No.	Correct Option
1	a
2	b
3	a
4	d
5	c
6	c
7	b
8	b
9	d
10	b
11	d
12	b

Detailed Explanations

Solution to Question 1:

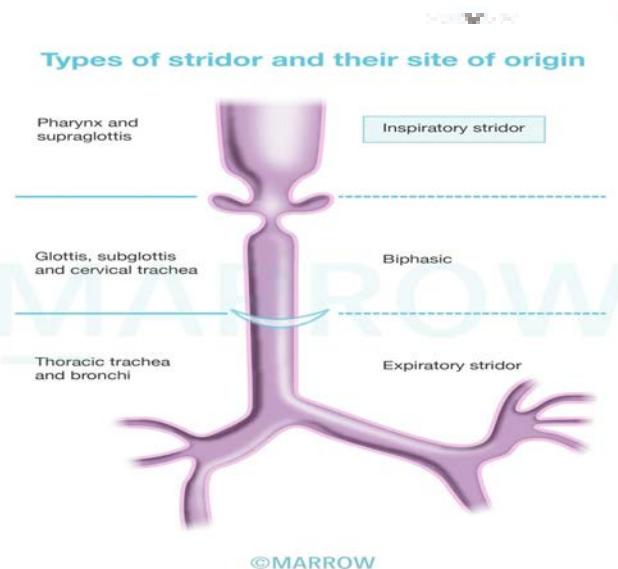
Inspiratory stridor is seen in supraglottic lesions. It is often produced in obstructive lesions of supraglottis or pharynx, e.g. laryngomalacia or retropharyngeal abscess.

Expiratory stridor is produced in lesions of thoracic trachea, primary and secondary bronchi such as bronchial foreign body, and tracheal stenosis.

Biphasic stridor is seen in lesions of glottis, subglottis, and cervical trachea like laryngeal papillomas, vocal cord paralysis, and subglottic stenosis.

Stridor is an audible respiratory noise that gets produced by turbulent airflow through an obstructed upper airway. It is an inspiratory sound that may have an expiratory component. It is distinct from stertor which is a snoring-like noise produced from an obstructed nasopharynx or oropharynx.

The intensity and pitch of sounds originating from the respiratory system vary based on their site of origin within the respiratory tract and may be heard with or without a stethoscope. Usually, the pitch of the sound increases, and the intensity decreases as one goes lower into the respiratory tract.

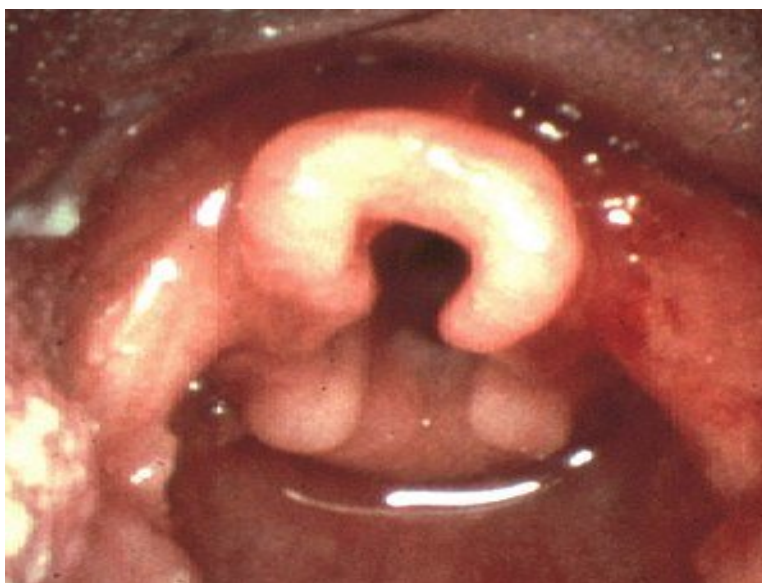


Solution to Question 2:

Laryngomalacia is the most common congenital cause of stridor.

It is characterized by excessive flaccidity of the supraglottic larynx which is sucked in during inspiration producing stridor and sometimes cyanosis.

It presents just after birth and subsides by 2 years of age. Stridor improves (decreases) in the prone position, worsens during feeding. The cry is normal. On examination, omega-shaped epiglottis is seen.



Solution to Question 3:

Laryngomalacia is also called congenital laryngeal stridor. It is the most common cause of stridor in newborns.

Solution to Question 4:

Laryngomalacia has an excellent prognosis.

Laryngomalacia is the most common cause of stridor in newborns.

It presents just after birth and subsides by 2 years of age. Stridor improves (decreases) in the prone position, worsens during feeding. The cry is normal. On examination, omega-shaped epiglottis is seen

Solution to Question 5:

The clinical picture and omega-shaped epiglottis are suggestive of laryngomalacia. Conservative treatment is preferred for laryngomalacia as this condition self-corrects by two years of age.

Tracheostomy may be required for some cases of severe respiratory obstruction. Supraglottoplasty is required in severe laryngomalacia. Signs of severe laryngomalacia include:

- Difficulty feeding
- Increased effort in breathing
- Poor weight gain
- Pauses in the breathing

- Frequent spitting up

Solution to Question 6:

Myer Cotton's grading is used for subglottic stenosis.

It is the abnormal thickening of cricoid/fibrous tissue proliferation below cords. It causes the airway to narrow and presents with dyspnea, biphasic stridor and a normal cry.

Diagnosis:

- In full-term infants, subglottic diameter ≤ 4 mm.
- In premature infants, subglottic diameter ≤ 3 mm.

The Myer Cotton's grading system allows to stratify the stenosis and plan for treatment:

- Grade 1: $\leq 50\%$ stenosis
- Grade 2: 51-70% stenosis
- Grade 3: 71-99% stenosis
- Grade 4: No lumen detected

Other options

- Option A - Hyams grading is a predictive tool used in olfactory neuroblastoma.
- Option B - House Brackmann grading is used as a facial nerve grading system.
- Option D - Modified mallampati classification is used to assess difficult airway

Solution to Question 7:

On compressing the swelling, there is a gurgling/hissing sound known as Bryce's sign. The given clinical picture with a positive Bryce sign is suggestive of laryngocele.

Laryngocele is an air-filled cavity within the larynx developing from the saccule. The patient presents with a neck swelling that is soft and compressible.

Options A: Branchial cleft cysts does not move with deglutition.

Option C: Cystic hygroma does not move with deglutition and usually presents before 2 years.

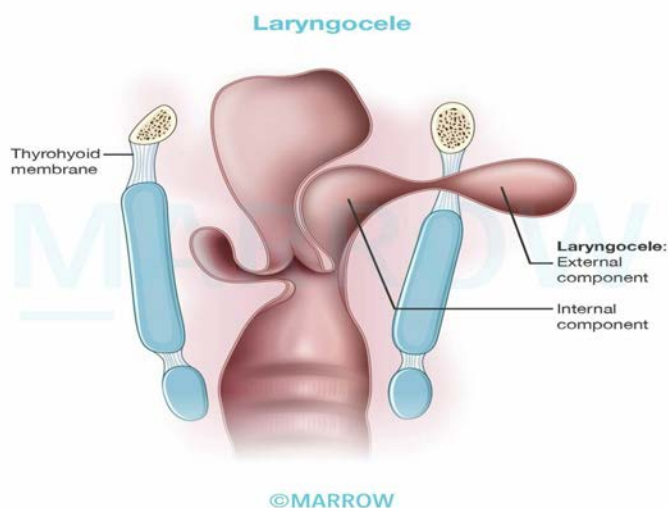
Option D: Thyroglossal cyst presents as a midline swelling.

Solution to Question 8:

Laryngocele arises from the saccule of the ventricle. It is an abnormal dilation of the laryngeal saccule and extends beyond the thyroid cartilage and the ventricle.

It is seen in trumpet players, glass-blowers, or weight lifters due to raised transglottic air pressure. It can be internal, external, or combined. External laryngocele presents as a reducible swelling in the neck which increases in size on coughing or performing Valsalva maneuver.

Internal laryngocele is managed by marsupialization. External laryngocele is managed by excision (external/LASER).



Solution to Question 9:

The given clinical picture is of congenital subglottic hemangioma. Embolization is not preferred in congenital laryngeal conditions. All congenital laryngeal conditions tend to regress with growth.

The principal management is conservative as many hemangiomas involute spontaneously. Depending on the status of the airway, treatment can involve:

- To wait and watch
- Steroid therapy
- Tracheostomy and observation

If the condition does not regress with time, the modalities of treatment include:

- LASER excision
- Laryngofissure and excision
- -pexy and -plasty procedures (epiglottopexy, supraglottoplasty, etc)

Solution to Question 10:

Laryngeal cysts arise in the aryepiglottic folds.

These are benign conditions that need no intervention unless they are large. Large cysts can cause airway obstruction/change in voice. These require immediate aspiration followed by deroofing.

Solution to Question 11:

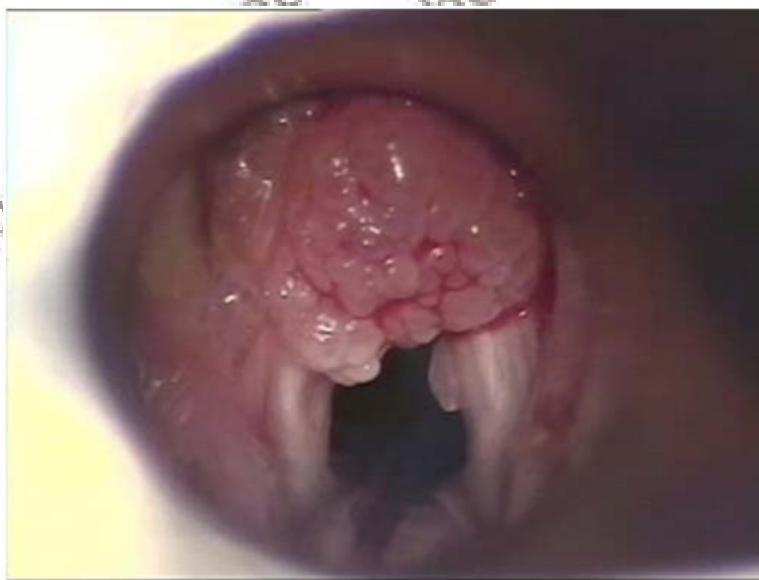
The image showing grape-like projections, involving the anterior part of the vocal cords depicts a laryngeal papilloma.

Laryngeal papilloma is a benign tumor of the larynx caused by the human papillomavirus (HPV type 6 and 11). They can occur in children, most commonly around 3 -5 years of age (juvenile papillomatosis), as well as in adults. They involve the supraglottic and glottic regions of the larynx. Seeding of the lesions can occur during invasive laryngeal procedures, such as intubation. They have a high recurrence rate and very low potential for malignancy.

Adult-onset laryngeal papillomatosis is less common and usually seen in males aged 30-50 years. It is less aggressive and involves the anterior part of the vocal cords. The treatment is the same as juvenile papillomatosis.

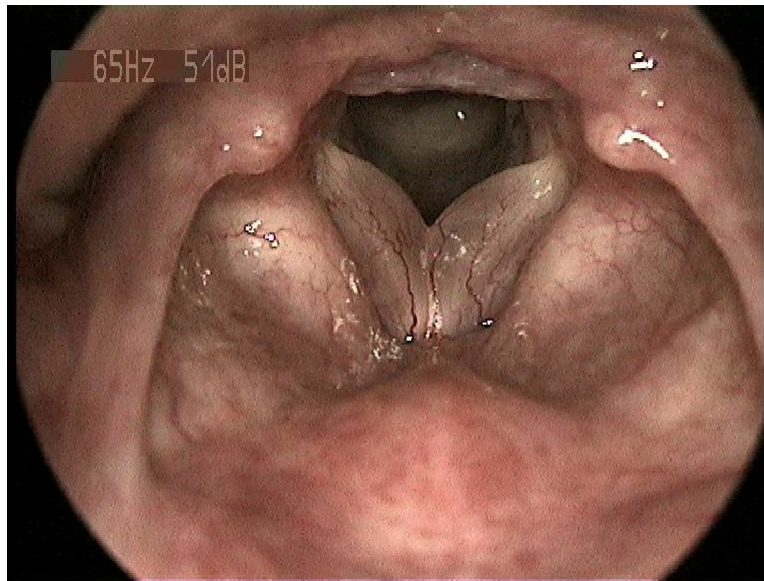
Patients present with hoarseness, aphonia, dyspnea, or stridor. Laryngoscopy shows pinkish to white grape-like projections that can be sessile or pedunculate with a visible central vascular core. The diagnosis is confirmed by direct laryngoscopy and biopsy. Treatment consists of micro-laryngoscopy and CO₂ laser excision.

The image below depicts laryngeal papillomatosis on laryngoscopy:



Reinke's edema (Option A) is a collection of fluid in the subepithelial space of Reinke. It generally occurs in patients with vocal abuse or smoking. The vocal cords will show diffuse, symmetrical swelling.

The laryngoscopic image below shows Reinke's edema:



Solution to Question 12:

The given clinical scenario of a child presenting with chronic hoarseness (3 months) of voice and respiratory distress for 2 weeks suggests the diagnosis of respiratory papillomatosis (juvenile papillomatosis).

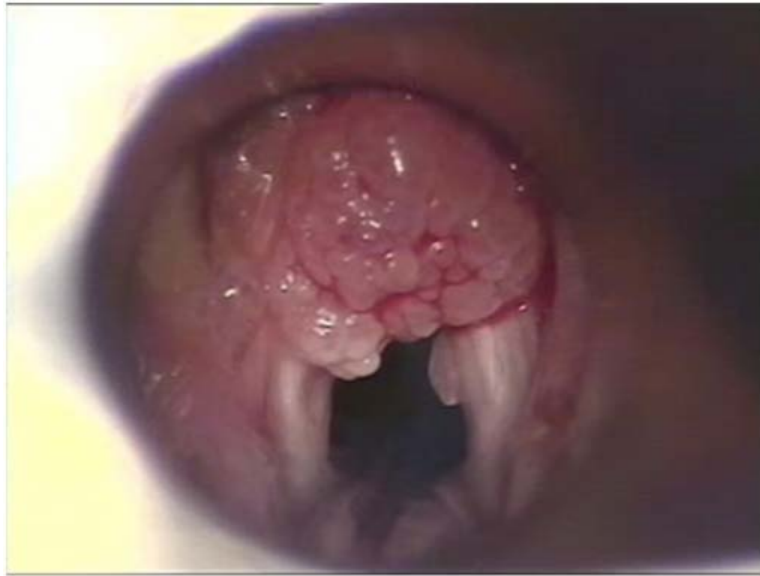
Croup (Option A) and acute epiglottitis (Option D), which cause acute upper respiratory tract obstruction, are ruled out due to the chronic nature of the symptoms. In croup, the upper airway obstruction becomes apparent after 1 to 3 days of low-grade fever, rhinorrhea, and pharyngitis. On the other hand, in epiglottitis, a child experiences respiratory distress within hours of developing a fever.

Vocal nodules (Option C) are not typically associated with respiratory distress.

Respiratory papillomatosis is the most common respiratory tract neoplasm in children. They are simple warts (benign tumors) caused by human papillomavirus types 6 and 11 commonly seen in children aged 3 to 5 years.

The course involves remissions and exacerbations of recurrent papillomas, mostly affecting the glottic area (vocal cords), and supraglottis. The patient presents with progressively worsening hoarseness, sleep-disordered breathing, dyspnea, stridor, and severe airway obstruction if left untreated.

Diagnosis confirmed by direct laryngoscopy, which shows grape-like projections (shown below), and biopsy. Treatment consists of micro laryngoscopy and CO₂ laser excision. Antivirals like interferon alpha2a, ribavirin, cidofovir, and acyclovir are used as adjunct therapies.



Other options:

Option A: Croup (laryngotracheobronchitis) is caused by the parainfluenza virus. The patient initially presents with an upper respiratory tract infection (rhinorrhea, pharyngitis, cough, low-grade fever) followed by a characteristic barking cough, hoarseness, and inspiratory stridor.

Option C: Vocal nodule (singer's or screamer's nodule) occurs as a result of chronic vocal abuse. They appear symmetrically at the junction of the anterior one-third and posterior two third of the vocal cord. The patient presents with chronic hoarseness and voice fatigue.

The image below shows symmetric vocal nodules:



Option D: Acute epiglottitis is caused by streptococcus pyogenes, pneumococci, Hemophilus influenzae, and staphylococcus aureus. It is characterized by an acute rapidly progressive course of high-grade fever, dyspnea, and stridor.

Voice and Speech Disorders

Question 1:

Which of the following voice changes is seen in dysphonia plica ventricularis?

- a) High pitched voice in adolescent boys
- b) Hoarseness due to plication of epiglottis
- c) Low pitched voice due to use of vestibular folds
- d) Weak fatiguable voice in TB larynx

Question 2:

Which of the following statements is false about spasmodic dysphonia?

- a) Only a local laryngeal disorder
- b) May be associated with other dystonias
- c) Abductor type produces breathy voice and adductor type produces strangled voice
- d) Botulinum toxin is used in treatment

Question 3:

A 17-year-old girl is brought to the ENT OPD by her father who says his daughter has lost her ability to speak following the death of her mother. Her comprehension is normal but she is mute. Which of the following statements is false about her condition?

- a) The vocal cords are abnormal
- b) Vocal cords fail to adduct on phonation but adduct during coughing
- c) Aphonia is sudden and unaccompanied by laryngeal symptoms
- d) Treatment includes reassurance of this patient

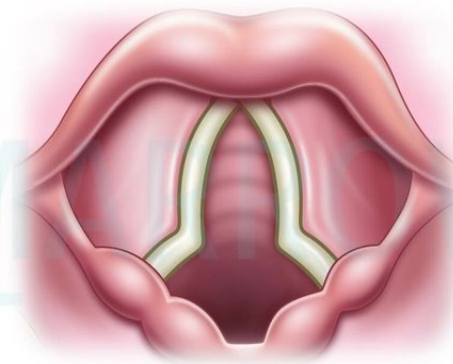
Question 4:

A 16-year-old boy complains of being bullied in school for his high-pitched voice. A Gutzmann's pressure test is done which helps you to diagnose his condition. What condition does he have?

- a) Dysphonia plica ventricularis
- b) Habitual dysphonia
- c) Spasmodic dysphonia
- d) Puberphonia

Question 5:

A 45-year-old teacher comes to you with complaints of dryness, tightness, and soreness in the throat. He says these symptoms appear after long hours of talking in class. He is also unable to hold long conversations. On direct laryngoscopy, the following is seen. What is his possible diagnosis?



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- a) Habitual dysphonia
- b) Spasmodic dysphonia
- c) Phonasthenia
- d) Laryngeal pseudosulcus

Question 6:

You perform a direct laryngoscopy procedure on a patient and notice a pseudosulcus in the larynx. What is this condition due to?

- a) Laryngopharyngeal reflux
- b) Vocal abuse
- c) Tuberculosis larynx

d) Lupus larynx

Question 7:

A 20-year-old woman presents to you with a common cold. Due to this, her voice has changed. Which of the following conditions does not cause a similar voice change?

- a) Adenoid hypertrophy
- b) Nasal polyp
- c) Large nasopharynx
- d) Nasopharyngeal mass

Question 8:

What type of speech pathology will be seen in this condition?



- a) Rhinolalia clausa
- b) Rhinolalia aperta
- c) Puberphonia
- d) Androphonia

Question 9:

Wagner and Grossman theory is related to _____

- a) Facial palsy
- b) Vocal cord palsy
- c) Palatal palsy
- d) Hypoglossal palsy

Question 10:

Which of the following nerves provide the sensory supply to the larynx below the vocal cords?

- a) Recurrent laryngeal nerve
- b) Internal laryngeal nerve
- c) External laryngeal nerve
- d) Inferior laryngeal nerve

Question 11:

Which of the following muscles is not supplied by the recurrent laryngeal nerve?

- a) Posterior Cricoarytenoid
- b) Lateral Cricoarytenoid
- c) Cricothyroid
- d) Interarytenoid

Question 12:

While assisting a direct laryngoscopy procedure being done on a patient, you notice that the vocal cords are in a slightly abducted position. Which intrinsic muscle of the larynx is responsible for this?

- a) Cricothyroid
- b) Posterior cricoarytenoid
- c) Lateral cricoarytenoid
- d) Interarytenoid

Question 13:

You are a junior resident in the ENT department asked to examine a patient's larynx. The patient has complete paralysis of vocal cords following a total thyroidectomy procedure. What is the expected position of the vocal cords?

- a) Median
- b) Paramedian
- c) Intermediate
- d) Full abduction

Question 14:

What is the most common cause for bilateral recurrent laryngeal nerve paralysis?

- a) Malignancies
- b) Surgical trauma
- c) Tuberculosis
- d) Idiopathic

Question 15:

Which of the following does not cause bilateral recurrent laryngeal nerve palsy?

- a) Thyroid surgery
- b) Thyroid malignancy
- c) Cervical lymphadenopathy
- d) Ortner's syndrome

Question 16:

What is the most common inflammatory cause for vocal cord palsy?

- a) Skull base granuloma
- b) Suppurative thyroiditis
- c) Pulmonary tuberculosis
- d) Jugular vein thrombophlebitis

Question 17:

Which of the following nerves is more prone to injury, causing vocal cord paralysis?

- a) Left recurrent laryngeal nerve
- b) Right recurrent laryngeal nerve
- c) Left superior laryngeal nerve
- d) Right superior laryngeal nerve

Question 18:

A 45-year-old woman has undergone a total thyroidectomy procedure and has been shifted to the post-op ward. 2 hours later she develops stridor and dyspnea, but her voice is normal. Which of the following vocal cord lesions is the most likely cause for her condition?

- a) Unilateral abductor paralysis
- b) Bilateral abductor paralysis
- c) Unilateral complete palsy
- d) Bilateral complete palsy

Question 19:

A patient in the ICU is aphonic but can maintain respiration. He has a history of recurrent chest infections. Which of the following nerve palsies would you suspect?

- a) Unilateral recurrent laryngeal nerve palsy
- b) Unilateral superior laryngeal nerve palsy
- c) Bilateral recurrent laryngeal nerve palsy
- d) Bilateral complete palsy

Question 20:

A 25-year-old singer has undergone a subtotal thyroidectomy for thyroid nodules. She presents to you one week later with complaints of difficulty in raising the pitch of her voice when she tries to reach high notes. There is no difficulty in her breathing and her voice is otherwise normal. Paralysis of which muscle is responsible for her condition?

- a) Posterior cricoarytenoid
- b) Cricothyroid
- c) Lateral cricoarytenoid

d) Interarytenoid

Question 21:

A patient has been posted for type 1 thyroplasty surgery. What does this procedure involve?

- a) Lateralisation of vocal cord
- b) Medialisation of vocal cord
- c) Shortening of vocal cord
- d) Lengthening of vocal cord

Answer Key

Question No.	Correct Option
1	c
2	a
3	a
4	d
5	c
6	a
7	c
8	b
9	b
10	a
11	c
12	b
13	c
14	b
15	d
16	c
17	a
18	b
19	d
20	b
21	b

Detailed Explanations

Solution to Question 1:

Dysphonia plica ventricularis is a low-pitched voice produced by the vestibular folds.

It can either be due to functional voice misuse or due to organic lesions like fixation of the true vocal cord. It is characterized by a low-pitched, rough and strained voice.

Treatment includes voice therapy and psychological counseling.

Solution to Question 2:

Spasmodic dysphonia is not only a local laryngeal disorder. It is associated with many neurological conditions too.

There are 2 types of spasmodic dysphonia:

Adductor dysphonia

- The adductor muscles of the larynx go into spasm causing vocal cords to go into adduction. The voice becomes strained/strangled.
- Treatment consists of botulinum toxin injections in the thyroarytenoid muscles on one or both sides to relieve the spasm.

Abductor dysphonia

- It is due to the spasm of the posterior cricoarytenoid muscle and thus the glottis is kept open.
- The patient gets a breathy voice or breathy breaks in voice.

Treatment consists of botulinum toxin injections in posterior cricoarytenoid muscles. Patients who do not respond to toxin injection can be treated by type 1 thyroplasty or fat injection.

Solution to Question 3:

The given clinical picture represents that of functional aphonia. In this condition, vocal cords are normal.

Functional aphonia or hysterical aphonia is mostly seen in emotionally labile females in the age group of 15-30 years. Aphonia is usually sudden and unaccompanied by other laryngeal symptoms.

Treatment is to reassure the patient of normal laryngeal function and psychotherapy.

Solution to Question 4:

The given clinical scenario with a positive Gutzmann's pressure test is suggestive of puberphonia.

It is a condition seen in emotionally labile adolescent males. They continue to speak in the prepubertal high pitched voice.

Gutzmann's pressure test is positive when pressing the thyroid prominence in a backward and downward direction relaxes the overstretched cords and produces a low tone voice. The treatment is speech therapy.

Solution to Question 5:

Phonasthenia is the condition seen in this patient. The clinical scenario of voice overuse and keyhole appearance of glottis point to this diagnosis. Here, both thyroarytenoid and interarytenoid muscles are involved.

Phonasthenia is the weakness of voice due to fatigue of laryngeal muscles. This leads to a phonatory gap. The shape of the gap depends on the muscle involved:

- Thyroarytenoid alone: elliptical space
- Interarytenoid alone: triangular space
- Both thyroarytenoid and interarytenoid: keyhole appearance



Solution to Question 6:

Laryngopharyngeal reflux is the cause of the pseudosulcus of the larynx. Pseudosulcus of vocal cords is a depression in the vocal cord mucosa due to chronic irritation from the reflux.

Vocal Sulci:

Types	True vocal sulci	Pseudosulcus	
Etiology	Sulcus vocalis	Sulcus vergetur e	Laryngopharyngeal reflux
Epidermal cysts	Developmental defect		
Treatment	PhonosurgeryBreak the fibrous bands underneath	Treat acid reflux	

Solution to Question 7:

Common cold leads to rhinolalia clausa. A large nasopharynx doesn't cause rhinolalia clausa but instead rhinolalia aperta.

Rhinolalia clausa or hyponasality is a lack of nasal resonance for words that are resonated in the nasal cavity. Words with M, N, NG sounds like B, D, G. It is due to the blockage of the nose or nasopharynx.

Common causes of rhinolalia clausa are:

- Common cold
- Nasal allergy
- Nasal polypi
- Nasal growth
- Adenoids
- Nasopharyngeal mass
- Familial speech pattern.

Solution to Question 8:

Rhinolalia aperta or hypernasality of speech will be seen in this condition of cleft palate.

It is seen when certain words which have little nasal resonance are resonated through the nose. The defect is the failure of the nasopharynx to cut off from the oropharynx or abnormal communication between the oral and nasal cavities.

Common causes are:

- Velopharyngeal insufficiency
- Congenitally short soft palate
- Submucous palate
- Large nasopharynx
- Cleft of soft palate

- Paralysis of soft palate
- Post adenoidectomy
- Oronasal fistula
- Familial speech pattern
- Habitual speech pattern.

NOTE:

- Clausa = Close; closed nose and nasopharynx
- Aperta = Open/aperture; abnormally open nose and nasopharynx

Solution to Question 9:

Wagner and Grossman hypothesis

- Explains the position of the cord in recurrent laryngeal nerve (RLN) palsy.
- States that "The complete paralysis of the recurrent laryngeal nerve results in paramedian cord due to the adduction by intact cricothyroid muscle".
- Since cricothyroid is supplied by SLN, it is spared and causes mild adduction.

Solution to Question 10:

Recurrent laryngeal nerve provides the sensory supply to the larynx below the vocal cords.

Above the vocal cords, the larynx is supplied by the internal laryngeal nerve- a branch of the superior laryngeal nerve.

Solution to Question 11:

Cricothyroid is not supplied by the recurrent laryngeal nerve. It is supplied by the superior laryngeal nerve.

All the other muscles are supplied by the recurrent laryngeal nerve.

Solution to Question 12:

Posterior cricoarytenoid is the only intrinsic muscle of the larynx that causes abduction of the vocal cords.

Solution to Question 13:

Intermediate (cadaveric) position is seen in complete paralysis of the vocal cord.

- This is a neutral position of the cricoarytenoid joint.
- Abduction and adduction take place from this position.
- It is not always seen in cadavers (misnomer).

Solution to Question 14:

Surgical trauma is the most common cause of bilateral recurrent laryngeal nerve paralysis.

The incidence of permanent RLN paralysis is approximately 1% to 1.5% for total thyroidectomy and less for near-total procedures.

Note: Previously, extra laryngeal malignancies had been the most common cause which included lung (50%), esophagus, thyroid, and larynx.

Solution to Question 15:

Ortner's syndrome is a rare cause of left (unilateral) recurrent laryngeal nerve palsy due to compression by enlarged cardiac structures such as the left atrium or pulmonary artery.

Causes of recurrent laryngeal nerve palsy:

- Thyroid surgery
- Neck trauma
- Thyroid malignancy
- Cervical lymphadenopathy.

Though the risk of bilateral nerve involvement is much less than unilateral, it is possible. Depending on whether the palsy is unilateral or bilateral, findings will differ.

Terminologies in vocal cord palsy:

- Pure RLN palsy: abductor palsy (the only abductor, posterior cricoarytenoid is affected)
- Pure SLN palsy: rare (only cricothyroid, one of the adductors and the main tensor is affected)
- Complete paralysis: both SLN and RLN are involved.

Solution to Question 16:

Pulmonary tuberculosis is the most common inflammatory cause of vocal cord palsy.

Lesions include:

- Apical

- Mediastinal scarring
- Mediastinal lymph nodes.

Solution to Question 17:

The left recurrent laryngeal nerve is more prone to injury, causing vocal cord paralysis. The reasons being:

- It is longer as it loops around the arch of the aorta to climb back to the larynx.
- It is in association with more structures in the mediastinum.

Other options:

- Right recurrent laryngeal nerve loops around the subclavian artery and therefore it's shorter and safer.
- Unilateral paralysis of the superior laryngeal nerve is rare and they are usually a part of a combined paralysis.

Solution to Question 18:

The above clinical picture of stridor and dyspnea with a normal voice is most likely due to bilateral abductor paralysis (B/L RLN injury). It is the most dangerous vocal cord lesion.

All the other options will have an abnormality in the voice. Normal voice is seen only in B/L RLN injury.

In bilateral abductor paralysis,

- Both the cords lie in the median or paramedian position, the airway is inadequate causing dyspnea and stridor. This may lead to death if untreated.
- The voice is good.
- Dyspnea and stridor become worse on exertion or during an attack of acute laryngitis.

Solution to Question 19:

The clinical presentation is that of bilateral complete palsy (SLN + RLN).

Both cords are in a cadaveric position, the patient has aphonia but respiration is maintained. Aphonia is suggestive of bilateral complete palsy, unilateral palsy presents with hoarseness of voice. Sensations are lost from supraglottis (supplied by the internal branch of SLN), the patient will develop aspiration and multiple chest infections.

Solution to Question 20:

Paralysis of cricothyroid is responsible for the difficulty in raising the pitch of her voice.

Cricothyroid is the main tensor of the vocal cords. It is supplied by the external laryngeal nerve (branch of the superior laryngeal nerve).

Solution to Question 21:

Medialization of the vocal cord is done in type 1 thyroplasty.

Note to remember: MLSL

Type	Thyroplasty	Uses
I	Medialization	In Complete palsy
II	Lateralization	In RLN palsy
III	Shortening (relax the vocal cord)	To reduce pitch
IV	Lengthening (tighten the vocal cord)	To increase pitch

Laryngeal Carcinoma

Question 1:

What is the false statement regarding carcinoma larynx?

- a) Squamous cell is the most common histological type
- b) Glottis is the most common site
- c) More common in females
- d) Tobacco is an important causative factor

Question 2:

Which of the following cancers presents with hoarseness of voice as its earliest symptom?

- a) Glottic carcinoma
- b) Supraglottic carcinoma
- c) Subglottic carcinoma
- d) Nasopharyngeal carcinoma

Question 3:

Which of the following is not the reason behind the development of hoarseness of voice in laryngeal carcinoma?

- a) Infiltration of thyroarytenoid muscle
- b) Recurrent laryngeal nerve involvement
- c) Cricoarytenoid fixation
- d) Phonotrauma

Question 4:

A 60-year-old man is diagnosed with carcinoma limited to the glottis of the larynx. Which of the following lymph nodes would be initially involved?

- a) Pretracheal
- b) Prelaryngeal

- c) Upper jugular
- d) None of the above

Question 5:

Which of the following laryngeal carcinomas has maximum lymph node involvement?

- a) Glottic carcinoma
- b) Subglottic carcinoma
- c) Supraglottic carcinoma
- d) All have equal chances of lymph node involvement

Question 6:

A patient has been diagnosed with laryngeal carcinoma limited to the supraglottis. Which of the following subsites will not be affected?

- a) Epiglottis
- b) False cords
- c) Aryepiglottic folds
- d) Pyriform sinus

Question 7:

A 55-year-old man is diagnosed with supraglottic carcinoma. Which of the following lymph node levels is the earliest to be involved?

- a) Level I and II
- b) Level II and III
- c) Level III and IV
- d) Level II and VI

Question 8:

Which of the following statements is false regarding supraglottic carcinoma?

- a) Epiglottis, false vocal cords are most commonly involved

- b) Lymphatic spread to upper and middle jugular lymph nodes
- c) Hoarseness of voice is an early symptom
- d) Bilateral nodes seen in epiglottic cancer

Question 9:

A patient has been posted for a surgical procedure to excise cancerous lesions from the vocal cords. Which laser is most commonly used?

- a) Nd Yag
- b) Holmium
- c) CO₂
- d) Diode

Question 10:

A 50-year-old man presents to you with complaints of weight loss, hoarseness of voice, and difficulty in breathing. On direct laryngoscopy, a mass involving both the vocal cords with extension to the subglottis is seen. Cords are immobile and there is no lymph node involvement. A biopsy confirms the diagnosis of glottic carcinoma. What is the next step in the management of this case?

- a) Transoral microsurgery
- b) Chemotherapy and radiotherapy
- c) Transoral microsurgery and radiotherapy
- d) Total laryngectomy

Question 11:

In which of the following conditions would you not recommend a total laryngectomy as the treatment?

- a) Glottic carcinoma with invasion of the thyroid cartilage
- b) Glottic carcinoma involving the posterior commissure
- c) Transglottic tumours with fixation of the vocal cords
- d) Tumour involving both vocal cords with normal mobility

Question 12:

In the oncology ward, four patients have been diagnosed with laryngeal carcinoma. Which of the following patients would you advise for transoral laser microsurgery as the first choice?

- a) A patient with vocal cord palsy, with subglottic extension
- b) A patient with tumor limited to one cord, with normal mobility
- c) A patient with tumor invading the thyroid cartilage
- d) A patient with tumor limited to larynx, with fixed cord

Question 13:

Which of the following is an indication of high tracheostomy?

- a) Vocal cord palsy
- b) Tracheomalacia
- c) Foreign body obstruction
- d) Suspicion of Laryngeal carcinoma planned for surgery later

Answer Key

Question No.	Correct Option
1	c
2	a
3	d
4	d
5	c
6	d
7	b
8	c
9	c
10	b
11	d
12	b
13	d

Detailed Explanations

Solution to Question 1:

Laryngeal carcinoma is more common in males.

Laryngeal cancer

- Male:Female ratio is 10:1
- Risk factors: Smoking tobacco and alcohol
- 90-95% are squamous
- Most common site: Glottis followed by supraglottis and subglottis.

Solution to Question 2:

Glottic carcinoma presents with hoarseness of voice as its earliest symptom. Lesions of the vocal cords affect its vibratory capacity. Therefore they are diagnosed early and have a better prognosis.

Supraglottic and subglottic growths are often silent. Hoarseness is a late symptom.

Solution to Question 3:

Phonotrauma is not the reason behind the development of hoarseness of voice in laryngeal carcinoma. It occurs due to overuse of voice and is associated with the development of benign lesions such as vocal nodules and cysts.

Hoarseness in supraglottic cancer and subglottic cancer indicates spread of disease to:

- Vocal cords with infiltration of the thyroarytenoid muscle
- Involvement of recurrent laryngeal nerve
- Cricoarytenoid joint involvement.

Hoarseness in glottic carcinoma is early because lesions of the cord affect its vibratory capacity.

Solution to Question 4:

Tumours limited to the glottis usually do not spread to any lymph node station as there are no lymphatics in the vocal cords.

However, if the tumour invades the supra- or the infra- glottic spaces, lymph node spread can occur.

Squamous cell carcinoma is the most common malignant tumour of the larynx. Glottis is the most common site of cancer.

Solution to Question 5:

Supraglottic carcinoma has the highest chance of lymphatic spread in the larynx.

Lymph node enlargement is often a presenting complaint as it occurs early in the course of the disease.

Solution to Question 6:

The Pyriform sinus is not a subsite of the supraglottis.

Epiglottis, false cords, aryepiglottic folds (medial wall of the pyriform sinus), and arytenoids are the subsites of supraglottis.

Aryepiglottic folds are shared between the supraglottis and hypopharynx.

Solution to Question 7:

Level II (upper jugular) and III (middle jugular) lymph nodes are the most common and earliest nodes involved in supraglottic carcinoma.

Solution to Question 8:

Hoarseness of voice is a late symptom of supraglottic carcinoma.

The most common site of supraglottic carcinoma is the epiglottis followed by false cords and then aryepiglottic folds. Nodal metastases occur early to the upper and middle jugular lymph nodes. Bilateral metastases can be seen in cases of epiglottic cancer. Supraglottic growths are often silent as hoarseness is a late symptom.

Solution to Question 9:

Endoscopic resection with CO₂ laser is most commonly used in laryngeal surgeries.

Carcinoma of the mobile membranous vocal cords can be precisely excised with the CO₂ laser under the microscope. It gives the same result as being traditionally treated with radiotherapy. Laser excision has the advantage of lower cost, lower duration of treatment, and morbidity.

Solution to Question 10:

The clinical picture of glottic carcinoma involving immobile vocal cords with extension to the subglottis suggests that the tumour belongs to the T3 stage. The treatment of choice is concurrent chemotherapy and radiotherapy.

TNM classification of glottic carcinoma (American Joint Committee on Cancer, 2002)

Treatment of laryngeal cancers:

TLM: Transoral laser microsurgery; RT: Radiotherapy; OPL: Open partial laryngectomy; CT: Chemotherapy; CRT: Chemoradiotherapy, TL: Total laryngectomy.

Stage	Description
T1	Tumour limited to vocal cord (s) with normal mobility
T1a	Tumour limited to one vocal cord
T1b	Tumour involves both vocal cords
T2a	Tumour extends to supraglottis and/or subglottis, with normal vocal cord mobility
T2b	Tumour extends to supraglottis and/or subglottis, with impaired vocal cord mobility
T3	Tumour limited to the larynx with vocal cord fixation and/or invades paraglottic space and/or minor thyroid cartilage erosion
T4a	Tumour invades thyroid cartilage and/or invades tissue beyond the larynx
T4b	Tumour invades prevertebral space, encases carotid artery or invades mediastinal structures

T stage	Treatment
T1a	TLM > RT
T1b (mid-cord lesion)	RT/TLM > OPL
T2a	RT/TLM > OPL
T2b	RT ± CT > TLM
T3	CRT
T4a	TL > CRT

T stage	Treatment
T4b	Palliative management

Solution to Question 11:

Tumour limited to both vocal cords with normal mobility can be classified as stage T1b. Treatment will include radiotherapy or transoral laser microsurgery. There is no indication for a total laryngectomy.

Total laryngectomy is indicated in:

- All T4 lesions
- Invasion of thyroid or cricoid cartilage
- Bilateral arytenoid cartilage involvement
- Lesions of the posterior commissure
- Failure after radiotherapy or conservation surgery
- Transglottic cancers with cord fixation.

It is contraindicated in patients with distant metastasis.

Treatment of T3 cancers is controversial, open vertical hemilaryngectomy and more extensive partial laryngectomies are used in carefully selected cases. Total laryngectomy or CRT is recommended for bulky T3 tumors or tumors not suitable for conservation laryngeal surgery.

Solution to Question 12:

Among the given options, a patient with tumor limited to one cord, with normal mobility will be advised for transoral laser microsurgery as the first choice. TNM staging in this patient is T1a.

Option A: TNM staging here is T2b and the patient is advised for chemoradiotherapy as the first choice.

Option C: TNM staging here is T4a and the patient is advised for total laryngectomy as the first choice.

Option D: TNM staging here is T3 and the patient is advised for chemoradiotherapy as the first choice.

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T3	CRT
T4a	TL > CRT
T4b	Palliative management

Solution to Question 13:

Laryngeal carcinoma is an indication of high tracheostomy. It is done at the level of the 1st tracheal ring and an opening is made above the isthmus of the thyroid gland.

The disadvantage of a high tracheostomy is subglottic stenosis at the level of the cricoid and perichondritis of the cricoid. However, in laryngeal carcinoma this would be a temporary measure as the definitive management is total laryngectomy.

Tracheostomy involves making an opening and placing a stoma in the anterior wall of the trachea.

Functions:

- Provides an alternative pathway for breathing by bypassing the obstruction.
- Improves alveolar ventilation by reducing dead space by 30-50% and reducing airway resistance
- Uses a cuff to protect the airways from secretions and blood
- Allows suction of secretions in the setting of the impaired cough reflex.
- Allows positive pressure respiration
- Allows administration of anesthesia

Indications:

- Respiratory obstruction due to
- Infections (laryngo-tracheo-bronchitis, epiglottitis)
- Trauma due to external injury, mandibular or maxillofacial fractures
- Neoplasms
- Foreign bodies or edema
- Bilateral abductor paralysis
- Congenital anomalies
- Retained secretions
- Inability to cough due to coma, spasm, or paralysis of respiratory muscles
- Aspiration of pharyngeal secretions
- Painful cough
- Respiratory insufficiency due to emphysema, chronic bronchitis

Instruments

Question 1:

A surgeon is to perform a mastoidectomy procedure for a patient with CSOM. He uses the following instrument during the procedure. What is the name of the given instrument?



- a) Mollison's mastoid retractor
- b) Lempert's endaural retractor
- c) Myringotome
- d) Jansen's self-retaining mastoid retractor

Question 2:

The following instrument is used for:



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- a) Septoplasty
- b) Myringoplasty
- c) Myringotomy
- d) Adenoidectomy

Question 3:

A 25-year-old patient is to undergo mastoid surgery. Which of the following instruments will not be present in the operation theatre?

- a) Lempert's curette
- b) Lempert's endaural retractor
- c) Lempert's endaural speculum
- d) Lempert's trocar

Question 4:

A patient is undergoing a mastoidectomy procedure. During the procedure, a surgeon uses the following instrument to explore the mastoid air cells. Name the given instrument

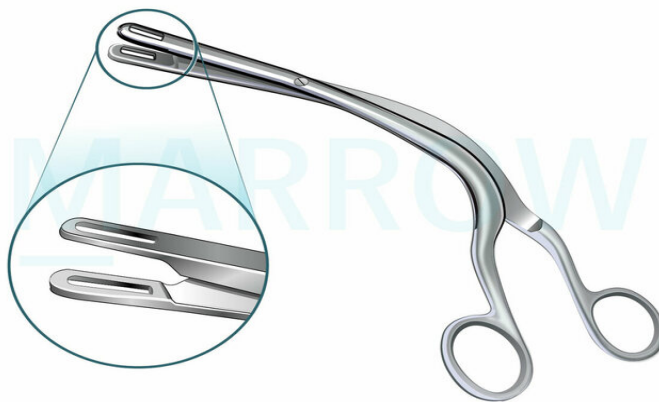


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- a) MacEwen's curette
- b) Volkmann's curette
- c) McElroy curette
- d) Mastoid gouge

Question 5:

Which procedure is the following instrument not used in?



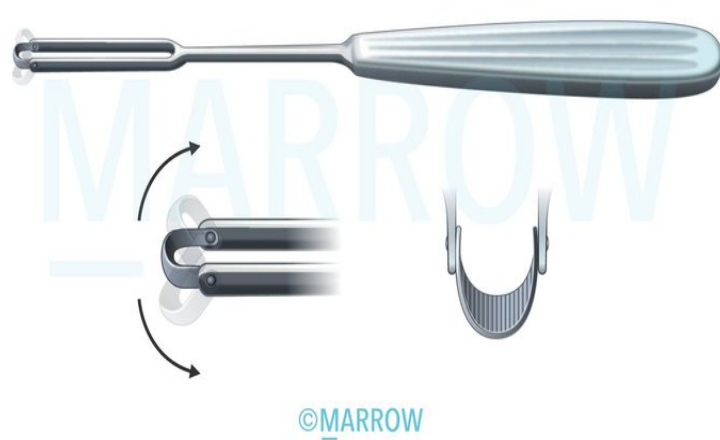
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- a) Functional endoscopic sinus surgery (FESS)
- b) Caldwell–Luc operation

- c) Submucosal resection
- d) Polypectomy

Question 6:

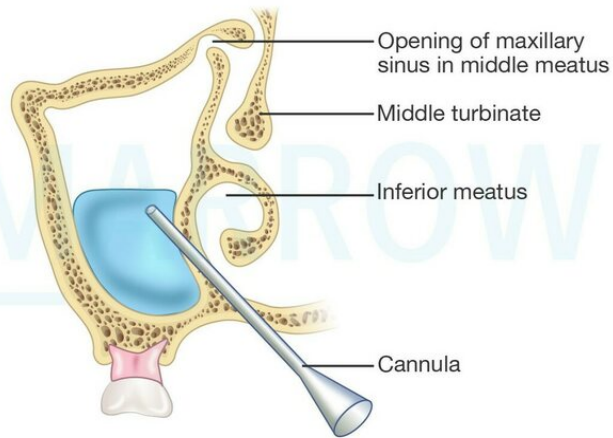
The following instrument is:



- a) Ballenger swivel knife
- b) Killian's nasal gouge
- c) Eves' tonsil snare
- d) Leighton tonsillotome

Question 7:

A 35-year-old man with chronic sinusitis undergoes the following procedure to relieve him of his symptoms. Which of the following instruments is required to carry out this procedure?



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- a) Tilley–Lichtwiz trocar
- b) Tilley's harpoon
- c) Nasal snare
- d) Tilley's antral burr

Question 8:

Given below is an instrument used for irrigation. In which of the following procedures will it not be used?



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- a) Intranasal antrostomy

- b) Caldwell–Luc operation
- c) Trephination
- d) Both A and B

Question 9:

A 20-year-old woman has complaints of reduced hearing on her right side. You perform a diagnostic test using the following instrument. What is the name of this instrument?



- a) Digital audiometer
- b) Tympanometer
- c) Rhinomanometer
- d) BERA

Question 10:

An objective diagnostic test is performed on a patient who complains of loss of hearing on his right side. The following instrument is used to perform this test. What is this instrument known as?



- a) Digital audiometer
- b) Tympanometer
- c) Speech audiometer
- d) BERA

Answer Key

Question No.	Correct Option
1	a
2	c
3	d
4	a
5	a
6	a
7	a
8	c
9	a
10	b

Detailed Explanations

Solution to Question 1:

The above image shows Mollison's mastoid retractor. It is used in mastoidectomy procedures to retract soft tissues after incision and elevation of flaps. It is self-retaining and hemostatic.

Shown below is Jansen's self-retaining mastoid retractor.



Solution to Question 2:

The given instrument with an arrowed blade and angled shank is a myringotome. It is used for myringotomy.

Solution to Question 3:

Lempert's trocar does not exist.

Option A: Lempert's curette (scoop) is used for removal of bony septa and granulations in mastoid surgery.

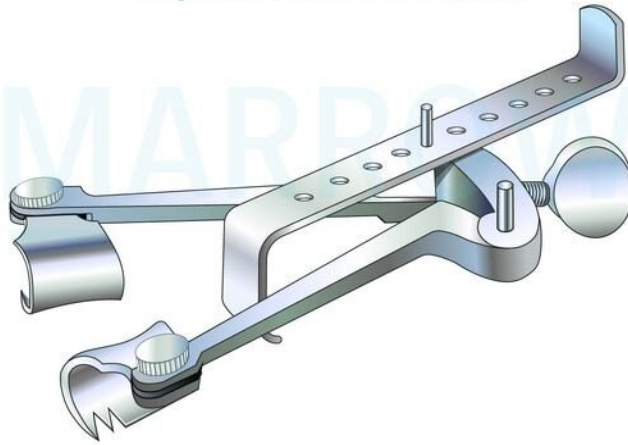
Lempert's curette



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Option B: Lempert's endaural retractor is used for endaural approach to ear surgery. It has two lateral blades which retract the flaps and a third central blade with holes. The central blade retracts the temporalis muscle. The central blade can be fixed to the body of the retractor by its hole.

Lempert's endaural retractor



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Option C: Lempert's endaural speculum is used to spread open the meatus when giving the local injection or making an endaural incision.

Lempert's endaural speculum



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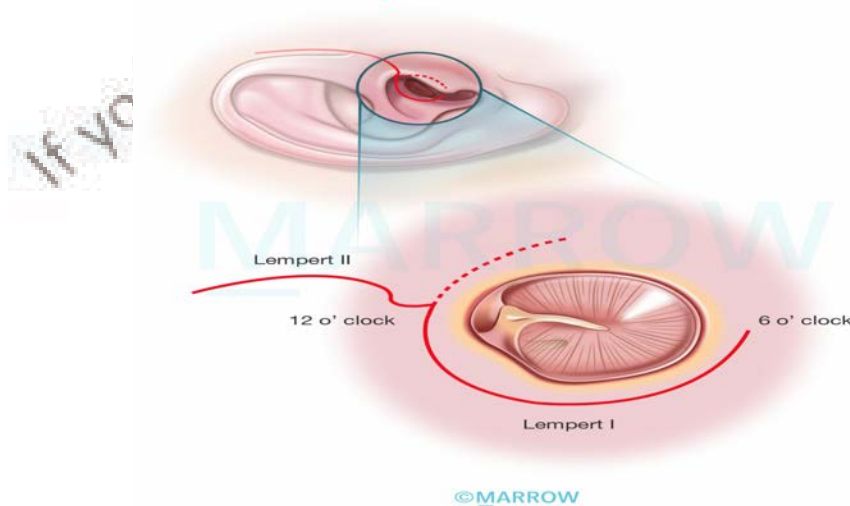
Lempert's incision is used in endaural approach. It consists of two parts:

Lempert I: It is a semicircular incision, made from 12 o'clock to 6 o'clock position in the posterior meatal wall at the bony-cartilaginous junction.

Lempert II: Starts from the first incision at 12 o'clock and then passes upwards in a curvilinear fashion between tragus and the crus of helix. It passes through the incisura terminalis and thus does not cut the cartilage.

Both mastoid and external canal surgery can be done.

Lempert's endaural incision



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Solution to Question 4:

The above image is a MacEwen's curette and cell seeker. During mastoid surgeries, one end is used to explore the air cells and the other end is used to curette the intervening septa and

granulations.

Note: MacEwan's (suprameatal) triangle is an important landmark to locate the mastoid antrum in mastoid surgery. It is bounded by-

- Temporal line
- The posterosuperior segment of the bony external auditory canal
- The line drawn as a tangent to the external canal

Solution to Question 5:

The image shows Luc's forceps. It is not used in FESS.

Luc's forceps is used in:

- Caldwell- Luc operation - to remove mucosa
- Submucosal resection- to remove bone or cartilage
- Polypectomy- to grasp and avulse polyps
- To take a biopsy from the nose or throat

Solution to Question 6:

The image is of a Ballenger swivel knife. It is used in the removal of septal cartilage in SMR (Submucosal resection) operation.

The blade of the knife revolves automatically and changes direction when cutting the cartilage anteroposteriorly downwards and posteroanteriorly. Different sizes of blades are present. It can be used to harvest septal cartilage as a graft for reconstruction.



Solution to Question 7:

The procedure is proof puncture (antral lavage) which uses Tilley-Lichwitz trocar and cannula. They are used for the puncture of the lateral wall of the inferior meatus at a point 1.5-2.0cm from the anterior end of the inferior turbinate, near the attachment of concha with the lateral wall.

Tilley-Lichwitz trocar and cannula:



Solution to Question 8:

The instrument is Rose's sinus douching cannula. It is used in irrigation of the maxillary sinus, which should already have a nasoantral window due to intranasal antrostomy or Caldwell Luc operation. The direction of the tip is indicated by the hook outside.

Note:

Trephination (also known as trepanning or burr holing) is a surgical intervention where a hole is drilled, incised or scraped into the skull using simple surgical tools. Trephination of frontal sinus is done for acute frontal sinusitis if there is persistence or exacerbation of pain or pyrexia in spite of medical treatment for 48 h or if the lid swelling is increasing and threatening orbital cellulitis.

Solution to Question 9:

The image is of a Digital audiometer.

An audiometer is an electronic device that produces pure tones, which can be increased or decreased in 5 dB steps. The amount of intensity that has to be raised above the normal level is a measure of the degree of hearing loss at that frequency. This is charted in the form of a graph

called an audiogram.

The difference in the thresholds of air and bone conduction (A-B gap) is a measure of the degree of conductive deafness.

Uses:

- To measure the threshold of hearing of air and bone conduction
- Helps to predict speech reception threshold
- To diagnose the type of hearing loss and measure the degree
- For the prescription of hearing aids
- Helps to find the degree of handicap for medicolegal purposes
- A record can be kept for future reference.

Solution to Question 10:

The image is of a tympanometer. It is used in evaluating the status of the middle ear.

It is based on a simple principle that when a sound strikes the tympanic membrane, some of the sound energy is absorbed while the rest is reflected. A stiffer tympanic membrane would reflect more sound energy than a compliant one.

By charting the compliance of the tympano-ossicular system against various pressure changes, different types of graphs called tympanograms are as follows:

